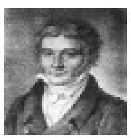
Lean Six Sigma Black Belt training programme

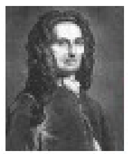
Lean Six Sigma

Quality Gurus

Evolution of Six Sigma



1736: French mathematician Abraham de Moivre publishes an article introducing the normal curve.



1818: Gauss uses the normal curve to explore the mathematics of error analysis for measurement, probability analysis, and hypothesis testing.

1896: Italian sociologist Vilfredo Alfredo Pareto introduces the 80/20 rule and the Pareto distribution in *Cours d'Economie Politique.*

1924: Walter A. Shewhart introduces the control chart and the distinction of special vs. common cause variation as contributors to process problems.

1941: Alex Osborn, head of BBDO Advertising, fathers a widely-adopted set of rules for "brainstorming".

1949: U. S. DOD issues Military Procedure MIL-P-1629, *Procedures for Performing a Failure Mode Effects and Criticality Analysis.*

1960: Kaoru Ishikawa introduces his now famous cause-and-effect diagram.

1970s: Dr. Noriaki Kano introduces his two-dimensional quality model and the three types of quality.

1986: Bill Smith, a senior engineer and scientist introduces the concept of Six Sigma at Motorola

1994: Larry Bossidy launches Six Sigma at Allied Signal.

1995: Jack Welch launches Six Sigma at GE.

Source: Slideshare.net

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quality Gurus

Dr. Walter Shewhart (1891-1967)

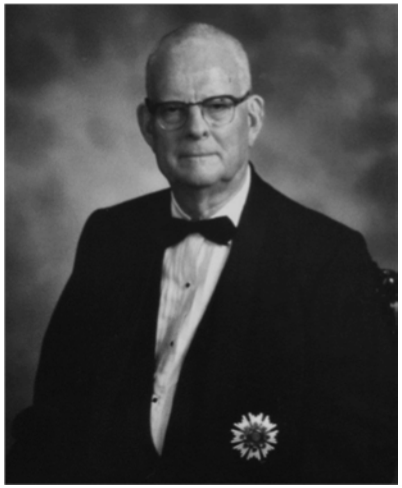
- Known for framing the problems of failures in terms of "assignable causes" and "chance cause" variation.
- Known for the introduction of the SPC – control charts as a tool for distinguishing between assignable and chance cause variation.
- Invented control charts which are widely used across industries to monitor processes and to determine when there are changes in a process.
- Known for the introduction of the continuous improvement cycle – Plan –Do –Check –Act (PDCA)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quality Gurus

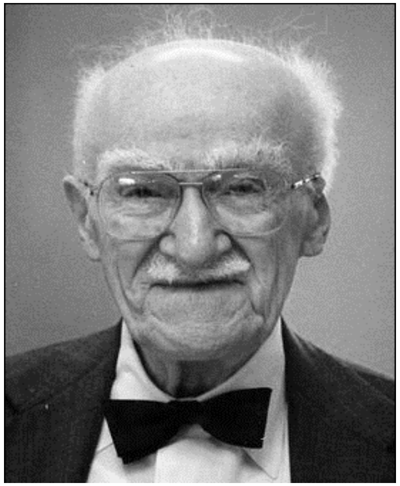
Dr W Edward Deming (1900-1993)

- Made a significant contribution to Japan's reputation for innovative, high quality products and for its economic power.
- Championed the work of Walter Shewhart including statistical process control, operational definitions and "Shewhart Cycle" which had evolved into PDSA (Plan-Do-Study-Act).



Dr. Joseph M Juran (1904-2008)

- Made a significant contribution to Japan's reputation for innovative, high quality products and for its economic power.
- Known for Juran Trilogy – *quality planning, quality control and quality improvement*
- First to apply the work of Vilfredo Pareto to quality issues - "*vital few and trivial many".*



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quality Gurus

Philip B Crosby (1928-2001)

Philip Crosby is known for his four absolutes of quality management

- Quality means conformance to requirements
- Quality comes from prevention
- Quality performance standard is zero defects
- Quality measurement is the price of non-conformance

Dr. Kaoru Ishikawa (1915-1989)

- Considered as a key figure in the development of quality initiatives in Japan, particularly the quality circle.
- Best known for the Ishikawa or fishbone or cause and effect diagram often used in the industrial processes analysis.
- Translated, integrated and expanded the management concepts of W. Edwards Deming and Joseph M. Juran into the Japanese system.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Six Sigma History

Evolution of Six Sigma

What does Quality Mean?

- Detecting and correcting mistakes in the product such that it meets compliance standards.

OR

- Preventing defects in the first place through manufacturing controls and product design such that it meets performance standards.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Exercise – Quality Inspection

Read the passage given below. The sixth letter of the alphabet is a defect. Report the number of defects:

"The necessity of training farm hands for first-class farms in the fatherly handling of farm livestock is foremost in the minds of farm owners. Since the forefathers of the farm owners trained the farm hands for first-class farms in the fatherly handling of farm livestock, the farm owners feel they should carry on with the family tradition of training farm hands of first- class farms in the fatherly handling of farm livestock because they believe it is the basis of good fundamental farm management."

Why do you think everyone of us have got different values?

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Evolution of Six Sigma

"The real problem at Motorola is that our quality stinks"

……1979, Art Sundry

"A product found defective and corrected during manufacturing had high probability of failing during early use by customer"

……1985, Bill Smith

- Late 1970s - Motorola started experimenting with problem solving through statistical analysis
- Motorola started Six Sigma approach to achieve it's one of the top ten corporate goal of improving the quality by ten times within five years in 1981.
- The term "Six Sigma" was coined by Bill Smith, an engineer with Motorola
- 1987 - Motorola officially launched it's Six Sigma program as follows:
  - Improve quality 10 times by 1989
  - Improve quality 100 times by 1991
  - Achieve six sigma (3.4 DPMO) performance by 1992
  - Motorola won the first Malcolm Baldridge National Quality Award in 1988.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Origin of Six Sigma

Beginning of a new era at Motorola..

- Improve the quality
- Lower production cost
- Lower production time
- Focus on how the product was designed and made

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Growth of Six Sigma - GE

- Jack Welch launched Six Sigma at GE in Jan,1996
- 1998/99 - Green Belt exam certification became the criteria for management promotions at GE
- 2002/03 - Green Belt certification became the criteria for promotion to management roles at GE
- Scope of six sigma initiative has changed from 'manufacturing' to the entire business– service, product design and innovation.

Source: Internet

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Focus of Six Sigma

- Reduce Variation
- Reduce Defects
- Delighting Customer
- Reduce Cost
- Reduce Cycle Time

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Sigma Levels and PPM

Six Sigma	Part per Million (PPM)
2σ	3,08,000
3σ	66,800
4σ	6,210
5σ	234
6σ	3.4

Mumbai Dabbawallas performance levels exceed Six Sigma limits

Six Sigma - Its significance

99 % Good (2.8 Sigma)	99.99967 % Good (6 Sigma)
20,000 lost articles of mail/ hour	Seven articles lost per hour
Unsafe drinking water for almost 15 minutes each day	One unsafe minute every seven months
5,000 incorrect surgical operations per week	1.7 incorrect operations per week
Two short or long landings at most major airports each day	One short or long landing every five years
200,000 wrong drug prescriptions each year	68 wrong prescriptions per year
No electricity for almost seven hours each month	One hour without electricity every 34 years

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

What is Six Sigma?

Sigma is a letter in the Greek alphabet

Metric

A **metric** that demonstrates quality levels at 99.99967 per cent performance for processes

Benchmark

A **benchmark** for product and process capability on a quality basis

Tool

A practical application of statistical **'tools'** to help define, measure, analyze, improve, and control the processes

Commitmen

A **commitment** to customers to offer the highest quality, reduced cost products

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Multiple meanings of Sigma

σ

Standard deviation

- The Greek symbol ‘sigma’ which means standard deviation.
- Is a measure of variation

Process capability

- A statistical measure of a process’s ability to meet customer requirements (CTQs).
- Process Sigma= 6σ equates to 3.4 defects per million opportunities

Management philosophy

- View processes/measures completely from a customer point of view
- Continual improvement
- Integration of quality and daily work
- Satisfying customer needs profitably

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Six Sigma Team

Six Sigma Team

- Apex Council
- Champion or Sponsor
- Process owner
- Master Black Belt
- Black Belt
- Green Belt
- Team Members

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Apex Council

Top Management:

- Accountable for Six Sigma business results
- Develop a strong case for Six Sigma
- Plan and actively participate in implementation
- Create a vision and market “change”; Become a powerful advocate
- Set clear (SMART) objectives
- Hold itself and others accountable
- Demand specific measures of results
- Communicate results (including setbacks)
- Helping to quantify the impact of Six Sigma efforts on bottom line.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Champion or Sponsor

Senior Manager:

- Oversees a Six Sigma project
- Is accountable to the Apex Council
- Sets rationale and goal for project
- Be open to changes in project definitions
- Find resources (time, support, money) for team
- Help the team overcome roadblocks; smoothen implementation
- Focus on data-driven management
- Identify and recruit other key players
- Assist in identifying and developing training materials

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Process Owner

Functional Head:

- Implements solutions through Black Belts and project teams
- Provide resources and helps resolve conflicts
- Accountable to the Apex Council
- Owns end-to-end process
- Sets goals for projects
- Project review: timeline and project is on track
- Responsible for holding the gains.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Master Black Belt

Six Sigma Coach:

- Advise and mentor Black Belts and teams
- Communicate with champions and apex council
- Establish and adhere to a schedule for projects
- Deal with resistance to Six Sigma
- Resolve team conflicts
- Estimate, measure, and validate savings
- Gather and analyze data on team activities
- Plan and execute training
- Help teams promote and celebrate successes
- Document overall progress of Six Sigma.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Black Belt

Facilitator:

- Six Sigma implementation experts with the ability to develop, coach, and lead multiple cross-functional process improvement teams
- Use tools to quickly and efficiently drive improvement
- Facilitate to keep team focused on the project objective
- Ensure that the Six Sigma methods are followed
- Help teams learn and understand Six Sigma tools and techniques through regular project reviews
- Responsible for the ultimate success of the project
- Trains and develops Green Belts
- Spread Six Sigma awareness throughout the organization.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Green Belt

Project Team Leaders:

- Lead and Execute Six Sigma as part of their daily jobs
- Keep the project team focused on the project goal
- Extract equal participation from all team members. Counsel non participating team members and motivate them to participate
- Ensure discipline of Team Meetings is followed and that every meeting starts with an Agenda. Ensure MOM is distributed the same day
- Regularly follow up with team members to ensure that assigned tasks are completed on time
- Manage conflicts and seek intervention of Process Owner / Champion if necessary
- Dual responsibility of being process experts as well as trained resource on Six Sigma methods and quality tools

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

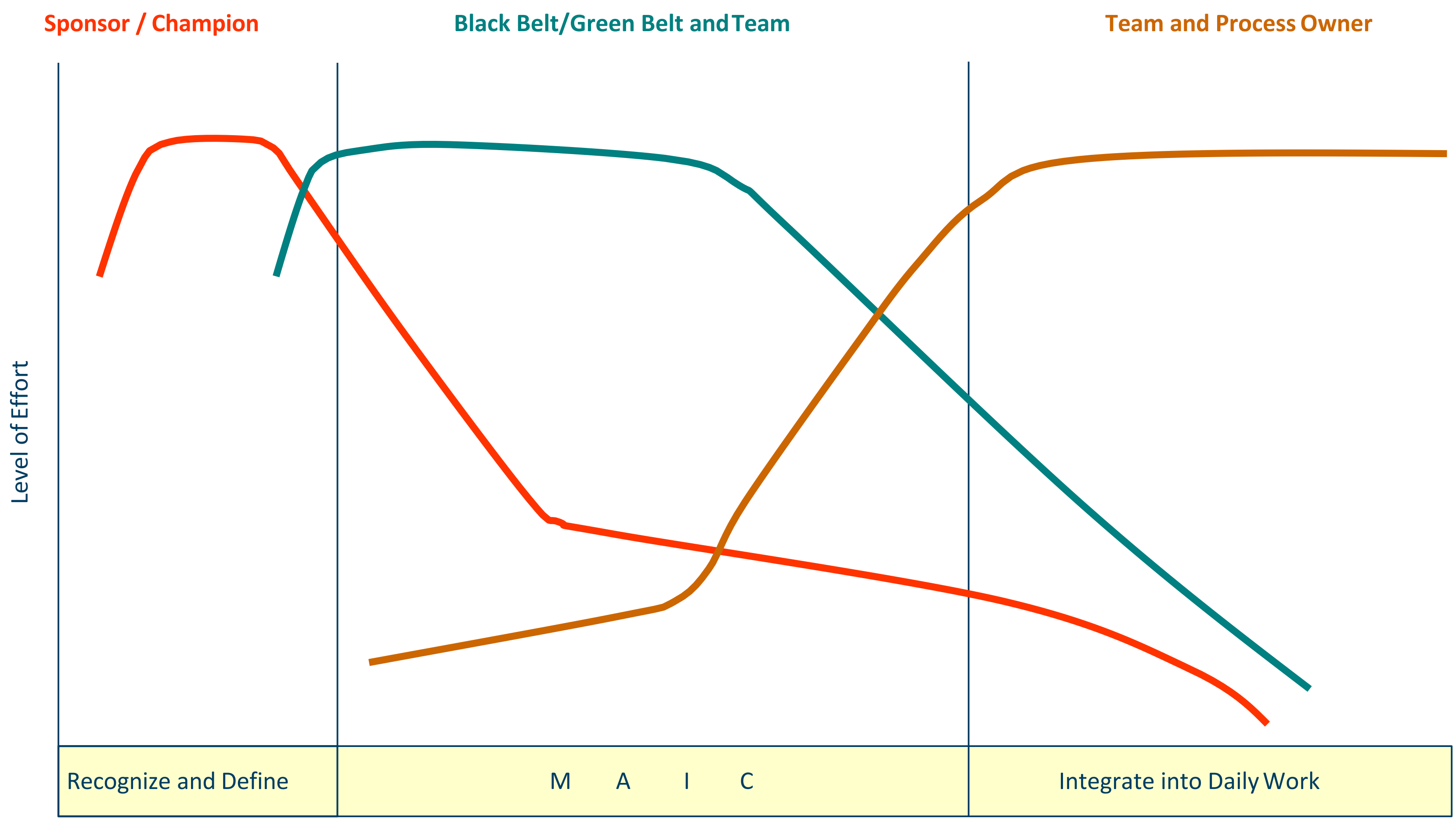
Team Members

Process Experts:

- Team members are vital for success
- Good knowledge of product, process, customer
- Willing to work in teams
- Time to work on projects
- Active in Data collection
- Responsible for improvement
- High Participation.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Typical Project Life Cycle and Effort



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Six sigma team roles and responsibilities

Champion

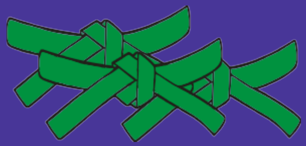
- Leadership: overall initiative
- Project funding and resources
- Felicitate with rewards and recognition

Master Black Belt

- Strategize on Six Sigma
- Mentor Black Belts
- Train resources on Six Sigma

Black Belt

- Lead multiple Six Sigma project teams
- Provide insights on data collection and analysis
- Advise on the appropriate tools for each phase of DMAIC



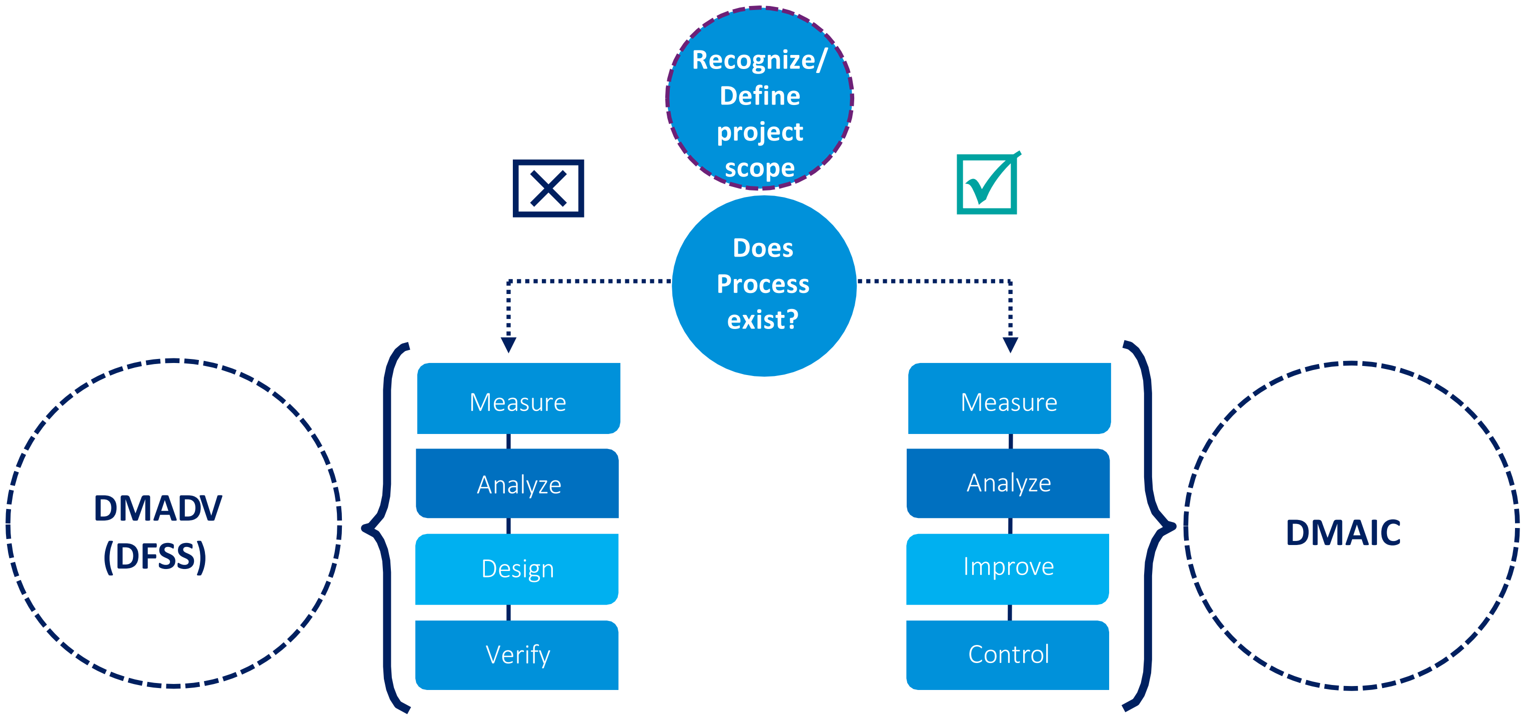
Green Belt

- Learn / Use Six Sigma Tools
- Work on Six Sigma Projects
- Integrate Six Sigma with their daily jobs

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

DMAIC vs DMADV

DMAIC Vs. DMADV



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

DMAIC methodology

Define
Selection of performance characteristics critical in meeting customer requirements

Measure
Creation and validation of a measurement system

Analyze
Identification of sources of variation from the performance objectives

Improve
Discovery of process relationships and establishment of new procedures

Control
Monitoring of implemented improvements to maintain gains and help ensure corrective actions are taken when necessary

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

DMADV methodology

Define
Initiate, scope and plan the project

Measure
Understand customer needs and specify the CTQs

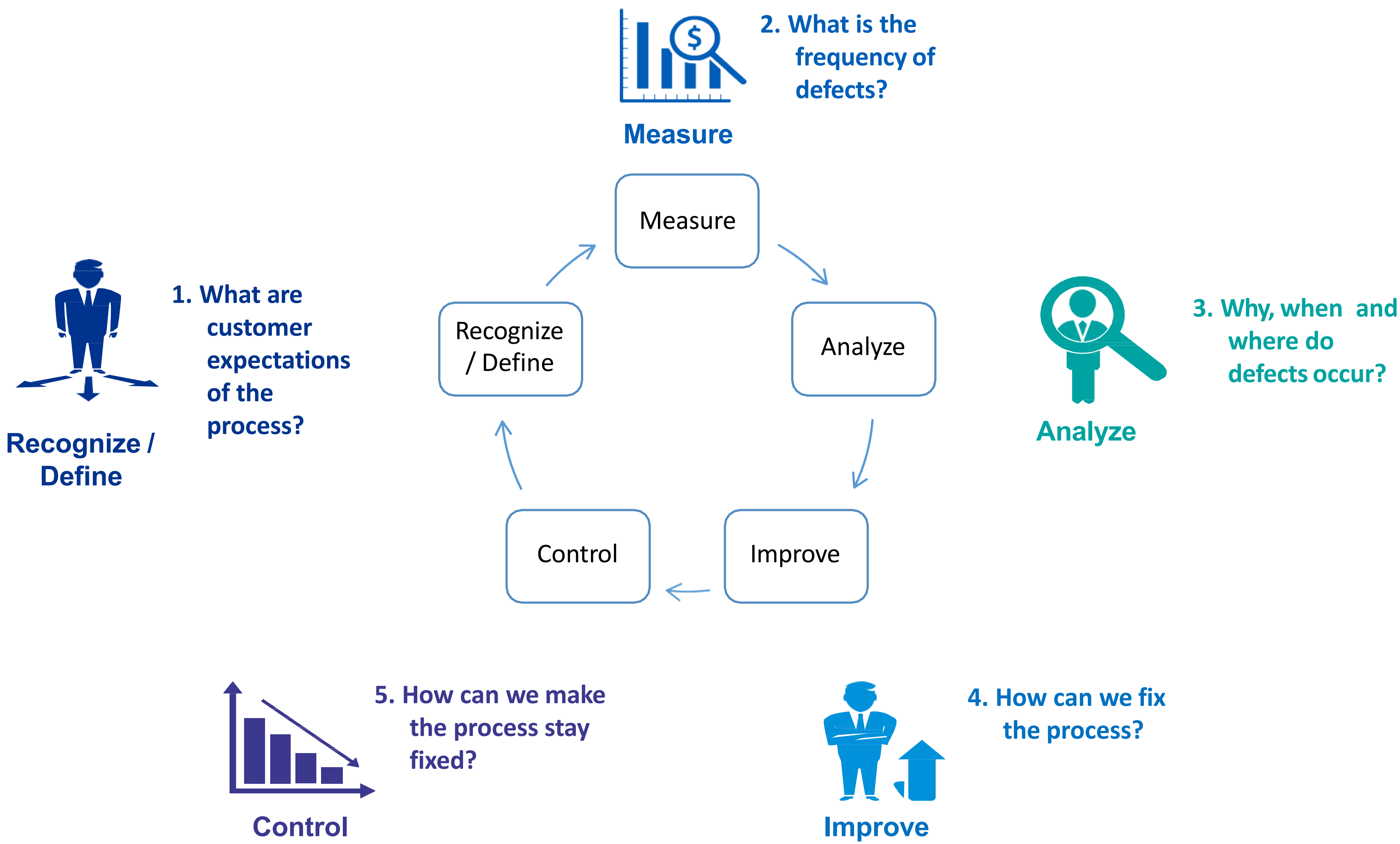
Analyze
Develop design concepts and high-level design

Design
Develop detailed design and control/test plan

Verify
Test design and implement full-scale processes

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

DMAIC approach



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Introduction to Lean

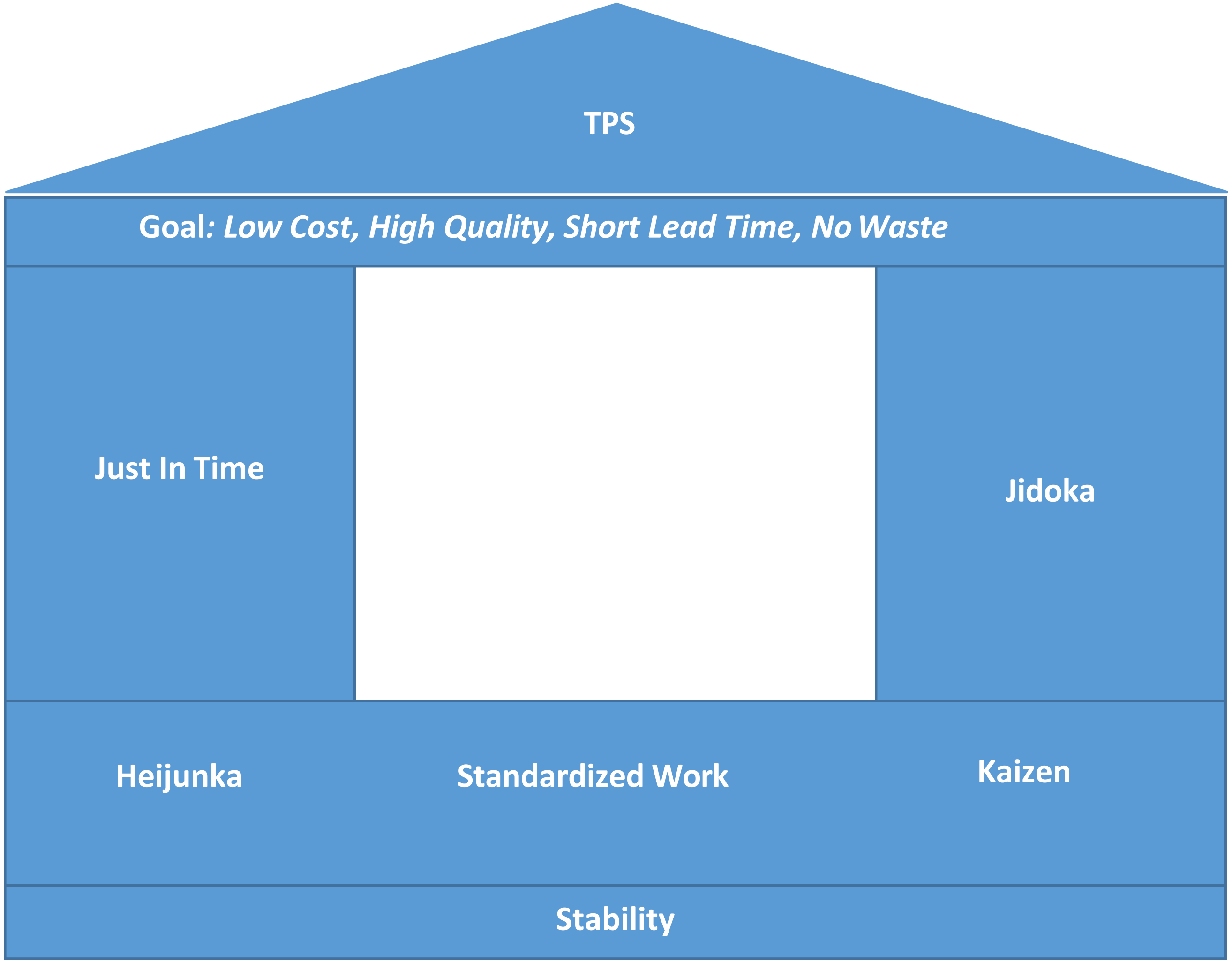
Toyota Production System (TPS)

Toyota Production System: An Overview

- Back in 1940s, Toyota based their production system on moving assembly line
- Post World War II, when Japanese industry was decimated, the Toyoda family decided to start their automotive company called Toyota
- Toyota Production System (TPS) was developed by Taiichi Ohno and Eiji Toyoda between 1948-1975
- In line with their new venture, Taiichi Ohno was given the assignment of scaling up Toyota automobiles to new heights in line with the American companies
- Ohno drew lots of ideas from the West and a lot of experimentation to ultimately develop TPS

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Origin of Lean: Toyota Production System



KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

TPS Pillars

1. Just In Time (JIT):

- JIT is one of the pillars of TPS
- JIT refers to a system of production that makes and delivers just what is needed, just when it is needed, and just in the amount needed
- JIT aims for complete elimination of waste to achieve the best possible quality, lowest possible cost and use of resources, and the shortest possible production and delivery lead times

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

TPS Pillars

2. Jidoka

- Jidoka is one of the two pillars of TPS
- Jidoka highlights the causes of problems
- Also referred as *autonomation*, i.e. automation with human intelligence. As it gives an equipment the ability to distinguish between good and bad parts without operator intervention.
- Here, the machine detects a problem and communicates it via problem display board
- The operators can confidently continue performing work at another machine, as well as easily identify the root cause and prevent recurrence
- Once the root cause is rectified, the improvements are incorporated in the standard workflow

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

TPS Pillars

3. Heijunka

- Heijunka, in Japanese refers to levelization
- Leveling the type and quantity of production over a fixed period of time
- Heijunka enables organization meet demand while reducing wastes in the entire value chain
- For instance: Assume that a toy producer receives orders for 500 of the same toy per week: 200 orders on Monday, 100 on Tuesday, 50 on Wednesday, 100 on Thursday, and 50 on Friday. Instead of trying to meet demand in sequence of the orders, the toy producer would use heijunka to level demand by producing an inventory of 100 toys near shipping to fulfill Monday's orders. Every Monday, 100 toys will be in inventory. The rest of the week, production will make a 100 toys per day – a level amount.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

TPS Pillars

4. Standardized Work

- The foundation of daily operation in TPS is Standardized Work
- Standardized Work sets up the best work sequence for each process. Once the best sequence has been determined, it is always repeated in exactly the same way, thereby avoiding wastes, errors and sustaining quality

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

TPS Pillars

5. Kaizen

- Kaizen is a Japanese word for the philosophy that defines management's role in continuously encouraging and implementing small improvements involving everyone.
- It is the process of continuous improvement in small increments that make the process more efficient, effective, under control, and adaptable.
- It focuses on simplification by breaking down complex processes into their sub processes and then improving them

(Refer Kaizen in detail under the Lean Management Tools)

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Goal of TPS

- Design out overburden (muri), inconsistency (mura), and eliminate waste (muda)
- Enable the production process as flexible as possible
- Follow 5s at every step of the production process
- Participation of all resources in the work process
- Elimination of problems or imperfection so as to reduce inventory

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Outcome of TPS

- Reduction in lead time
- Improvement of quality
- Practiced by one of the ten largest companies across the world
- Faster response
- Lower cost

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

History of Lean

Lean History

Lean, similar to Six Sigma is a process improvement that is solely based on the fundamental goal of waste elimination and flow maximization.

In other words, meeting customer requirements faster, quicker and better, that is in a more effective and efficient way.

Lean manufacturing, as management philosophy, came mostly from the Toyota Production System (TPS). The term “lean” was first introduced in article “Triumph of the Lean Production System” written by John Krafcik in 1988.

Kiichiro Toyoda, founder of Toyota Motor Corporation, discovered many problems in their manufacturing process. In 1936 his processes hit new problems and he developed the “Kaizen” improvement teams. Toyota’s view is that the main method of lean is not the tools, but the reduction of three types of waste:

- muda (“non-value-adding work”)
- muri (“overburden”)
- mura (“unevenness”)

Taiichi Ohno, considered to be the father of the Toyota Production System, was instrumental in improving overall customer value by focusing on reduction of the process wastes at Toyota.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Lean History

Three men were especially prominent in creating the Toyota Production System (TPS) Lean Enterprise:

Sakichi Toyoda

Sakichi Toyoda was the inventor of Automatic Looms who founded the Toyota Group

Kiichiro Toyoda

Kichiiro Toyoda was the son of Sakichi Toyoda. He travelled to USA to study the Ford Motors operation system and adapted that to Toyota Production

Taiichi Ohno

Taiichi Ohno rose to become the Executive Vice President of Toyota. He was inspired by the operations of Walmart and created the Pull System

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Assembly Line Production

The Ford Model T was an American car built between 1908 and 1928 by the Ford Motor Company of Detroit, Michigan. It is one of the most important cars in history because it was one of the first cars to be sold for very little money, making it easy for people to travel from place to place.

Henry Ford's revolutionary advancements in assembly-line automobile manufacturing made the Model T the first car to be affordable for a majority of Americans.

1909

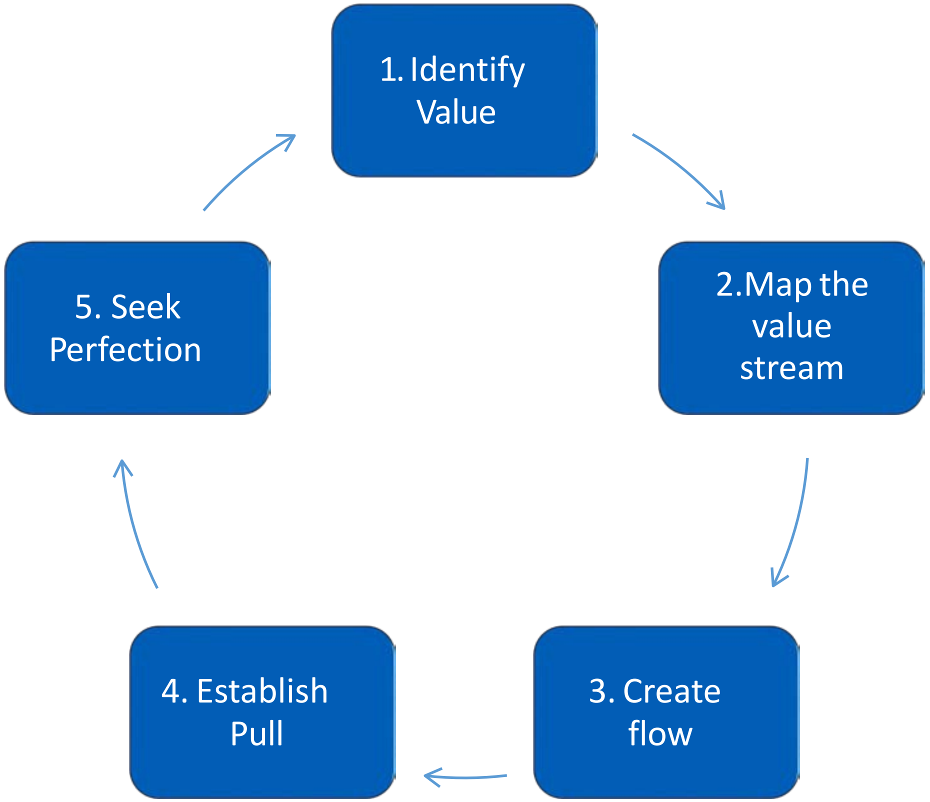
1928

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Lean Principles

Principles of Lean Thinking

The Lean Enterprise Institute (LEI), founded by James P.Womack and Daniel T.Jones in 1997, introduced the five key lean principles: value, value stream, flow, pull, and perfection.



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Principle 1 - Identify Value

What is value?

Who defines value?

According to Womack and Jones – Value is expressed in terms of a specific product or service (or both) which meets customers' need at a specific price at a specific time.

The key question to be asked here is – What is the timeline? What is the price point? What are the other requirements that must be met?

Account opening takes 3 to 4 days

→

Introduction of instant account opening with online documentation

Providing value to the customers is the only reason for the existence of our business.

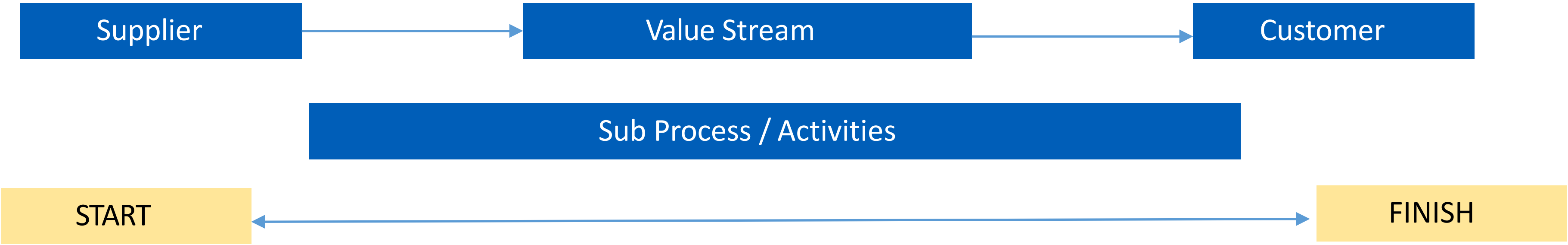
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Principle 2 - Map the value stream

According to Womack and Jones – Mapping the value stream is a step in taking a specific product or service (from scratch) to its final recipient i.e.the end customer.

What is a Value Stream?

- Value stream map is a sequence of steps taken to create product / service to the end customer
- Value stream mapping identifies the Value Added (VA) / Non-Value Added (NVA) / Value Enablers (VE) steps in a process and quantifies time spent on eachstep
- Eliminate the NVA's that contribute to the highest time to the process.



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

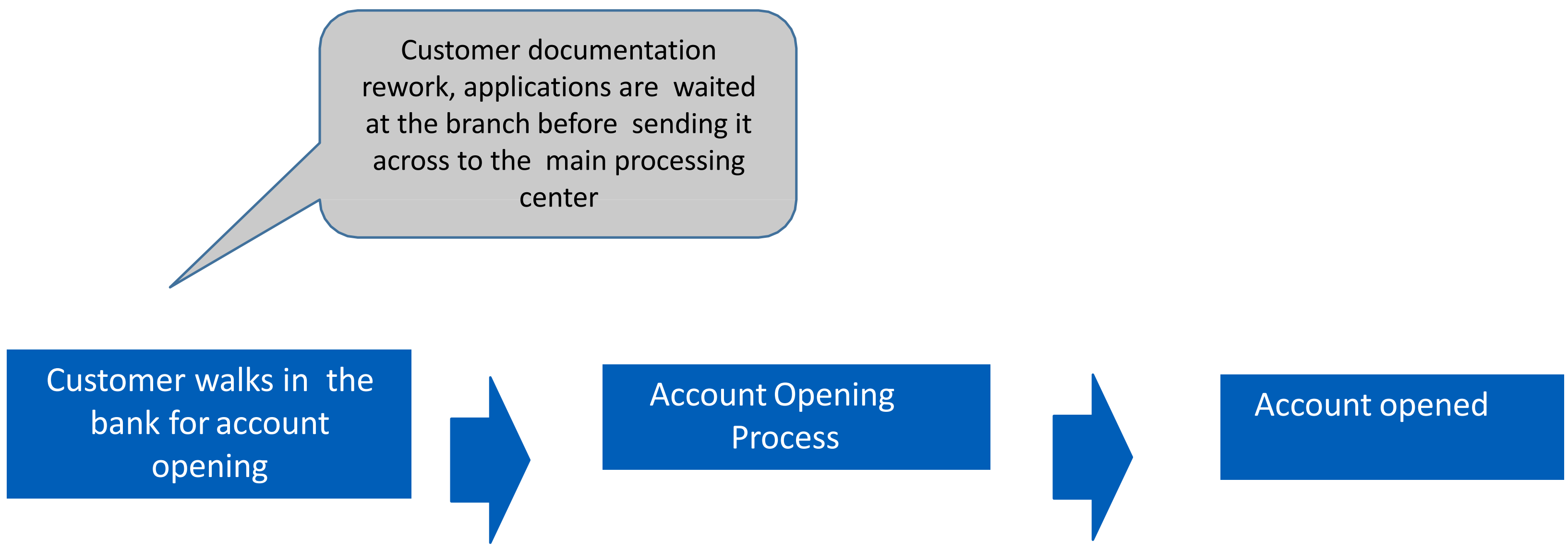
Principle 3 – Create Flow

What is a Flow?

It is a movement of the product or service from the supplier to the end customer

Womack and Jones advises – *"make the value creating steps occur in tight sequence, such that the product or service flow smoothly towards the end customer."*

- The rationale behind flow is that the product or service flow smoothly in the value chain without interruptions
- Waiting time and hand-offs are eliminated from the process flow

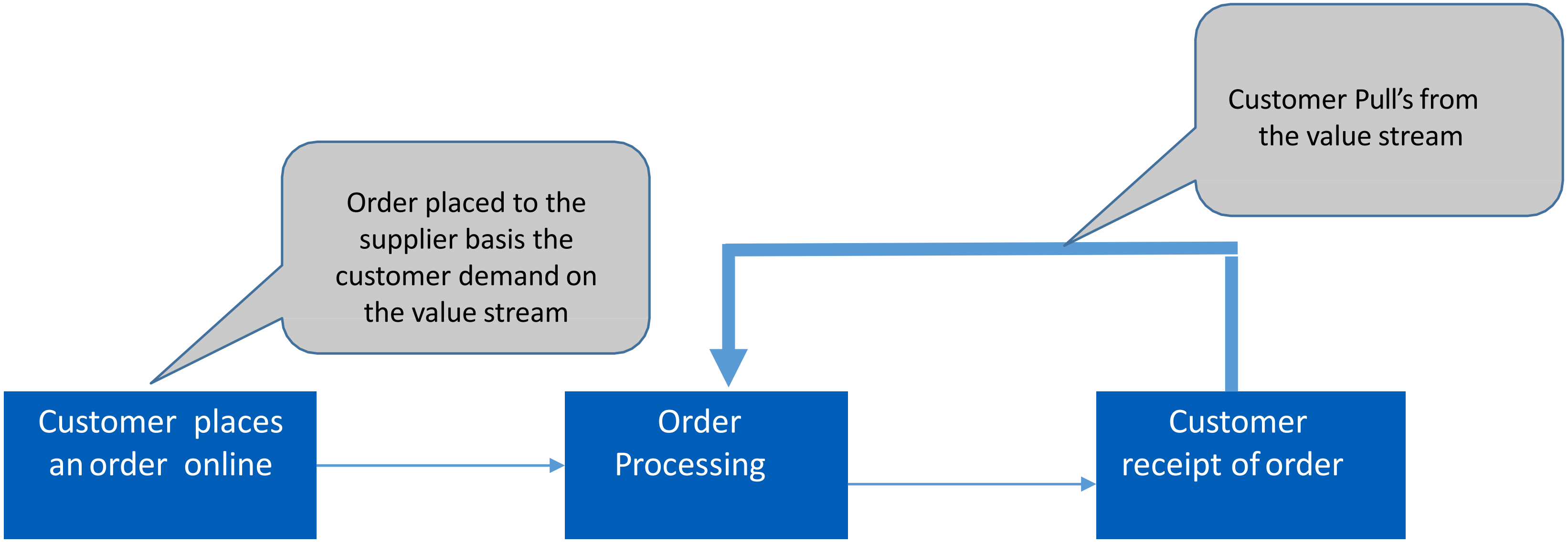


© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Principle 4 - Establish Pull

Pull is the movement of product or service from the supplier to the customer only when the customer needs it.

- Making or processing the product only when there is a customer demand
- Create a “Just in Time” manufacturing or delivery
- Reduced storage, stockpile, inventory



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Principle 5 – Seek Perfection

Perfection or Continual Improvement is a never ending journey towards delivering world class product or service to your end user or customer.

- Mistake proof the process
- Include process improvement as part of the organization culture
- Continuously keep fine tuning the process
- Restart from 1 to 4

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Recognize Phase

Recognize Phase

Strategies for understanding customer requirements:

- Balance Scorecard (BSC)
- Cost of poor Quality (COPQ)
- Voice of Customer (VOC)
- Quality Function Deployment (QFD)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Balanced Scorecard (BSC)

What is Balanced Scorecard?

- Balanced Scorecard is a performance management system for implementing strategic planning and achieving continuous improvement at all levels of an organization.

- BSC enables an organization to measure what is critical to an organization i.e. value drivers

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Balanced Scorecard: An overview

- Balanced Scorecard as a concept was first published by Robert Kaplan and David P Norton in 1992, later followed by a book in 1996
- BSC broke the traditional performance measurement that focused only on external accounting data
- BSC enables management to provide a balanced approach with a mix of both financial and non-financial measures each compared to a 'target' value within a single concise report

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Why Balanced Scorecard?

- BSC increases focus on strategy and results
- Improves organization's performance by measuring what really matters Aligns
- organization's strategy with employees on a daily basis
- Focuses on key drivers for future performance
- Improves communication of the organization's vision and strategy Prioritize
- projects, products, and services

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Balanced Scorecard Model

The Balanced Scorecard model views the organization from four perspectives viz; Financial, Internal Business Process, Learning & Growth, and Customer

Financial

In order to succeed financially, how do we appear to our shareholders?

Internal Business Processes

In order to satisfy our shareholders and customers, what business process do we excel at?

Vision & Strategy

Customer

In order to achieve our vision, how do we appear to our customers; what is important to our customers?

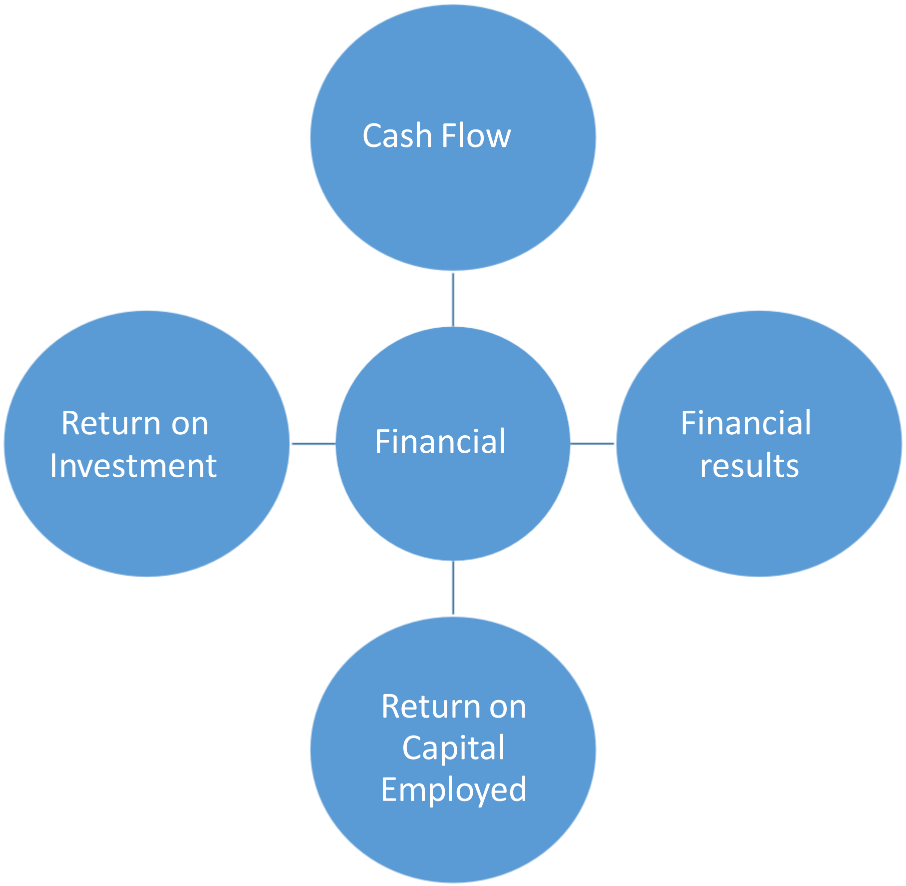
Learning & Growth

In order to achieve our vision, how do we create value and innovate ourselves?

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

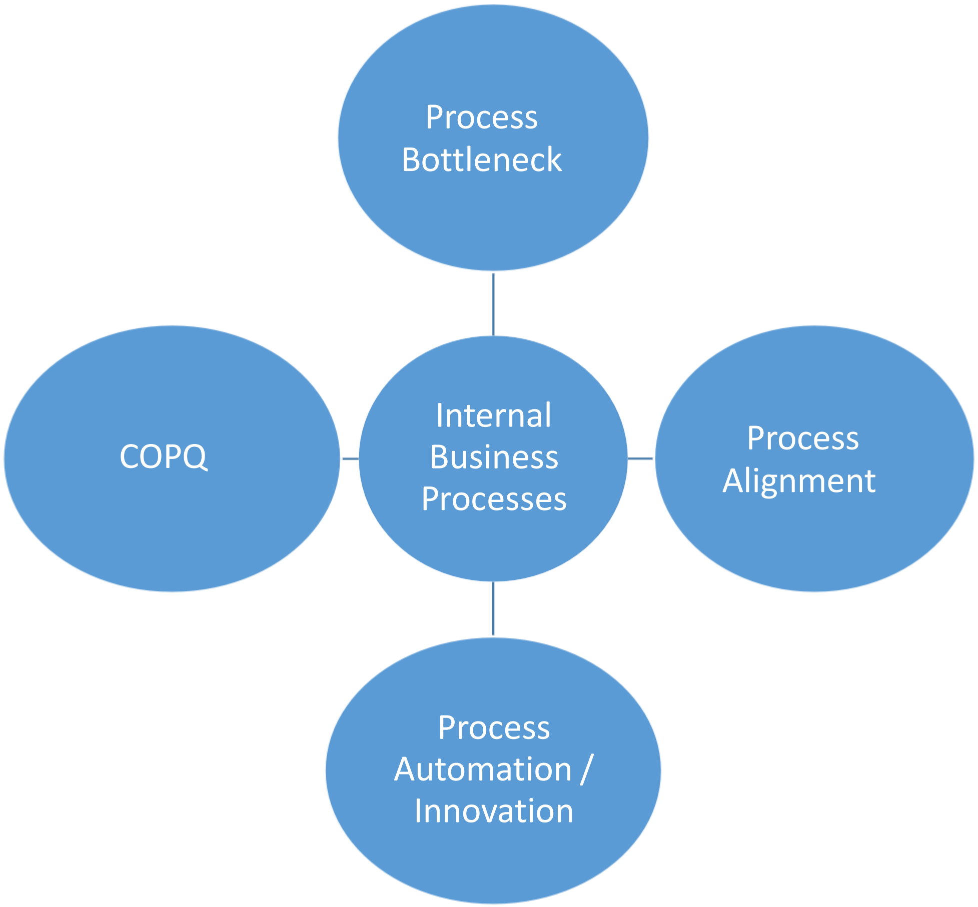
Balanced Scorecard Model - Financial Perspective



This perspective views organizational financial performance and the use of financial resources

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

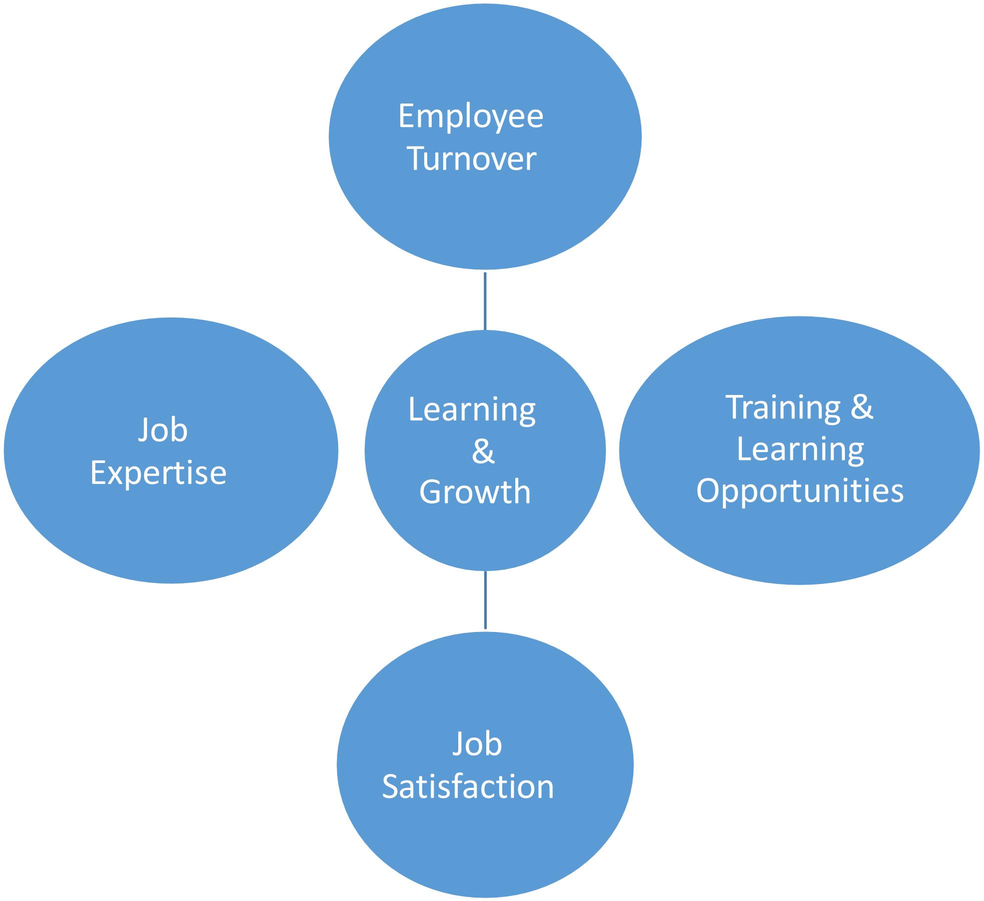
Balanced Scorecard Model - Internal Process Perspective



This perspective views organizational performance through the lenses of the quality and efficiency related to our product or services or other key business processes

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

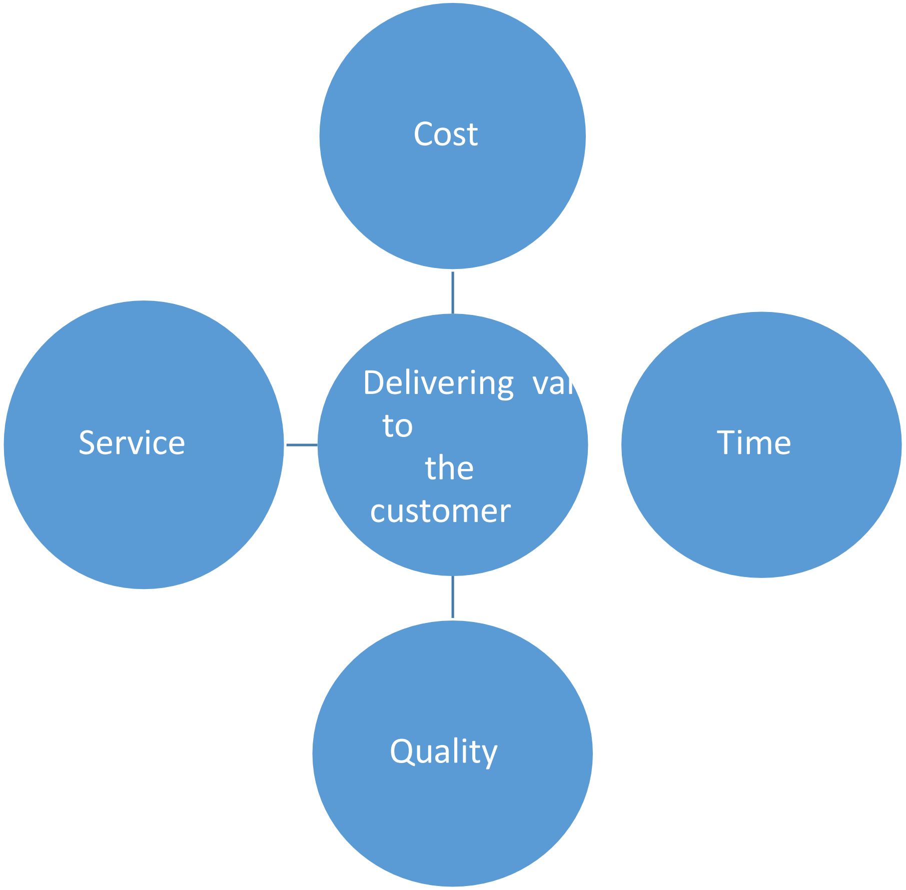
Balanced Scorecard Model - Learning & Growth Perspective



This perspective views organizational performance through the lenses of human capital, infrastructure, technology, culture and other capacities that are key to breakthrough performance

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Balanced Scorecard Model - Customer Perspective



This perspective views organizational performance from the point of view the customer or other key stakeholders that the organization is designed to serve

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Balanced Scorecard: Advantages

- Aligns the organization’s vision and mission to business strategy Improves
- internal and external communication
- Monitors organization performance
- Enables better decision making
- Provides the management with a comprehensive picture of the processes

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Balanced Scorecard: Disadvantages

- The BSC relies heavily on a well define strategy and understanding of linkages between strategic objectives and metrics. Without the same, the implementation could fail

  Time consuming
- Does not provide recommendations High
- implementation cost
-

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Cost of Poor Quality (COPQ)

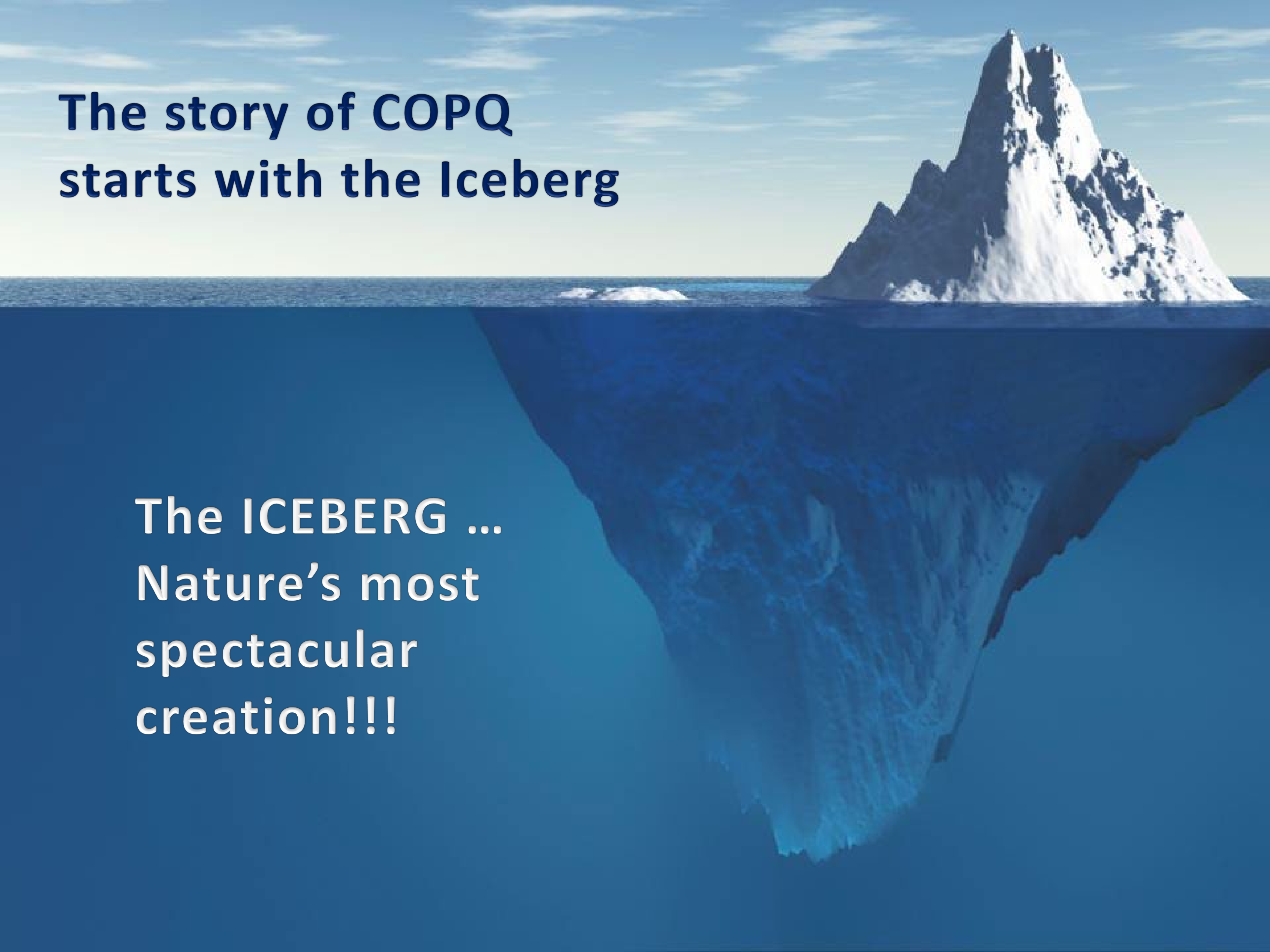
What is Cost of Poor Quality?

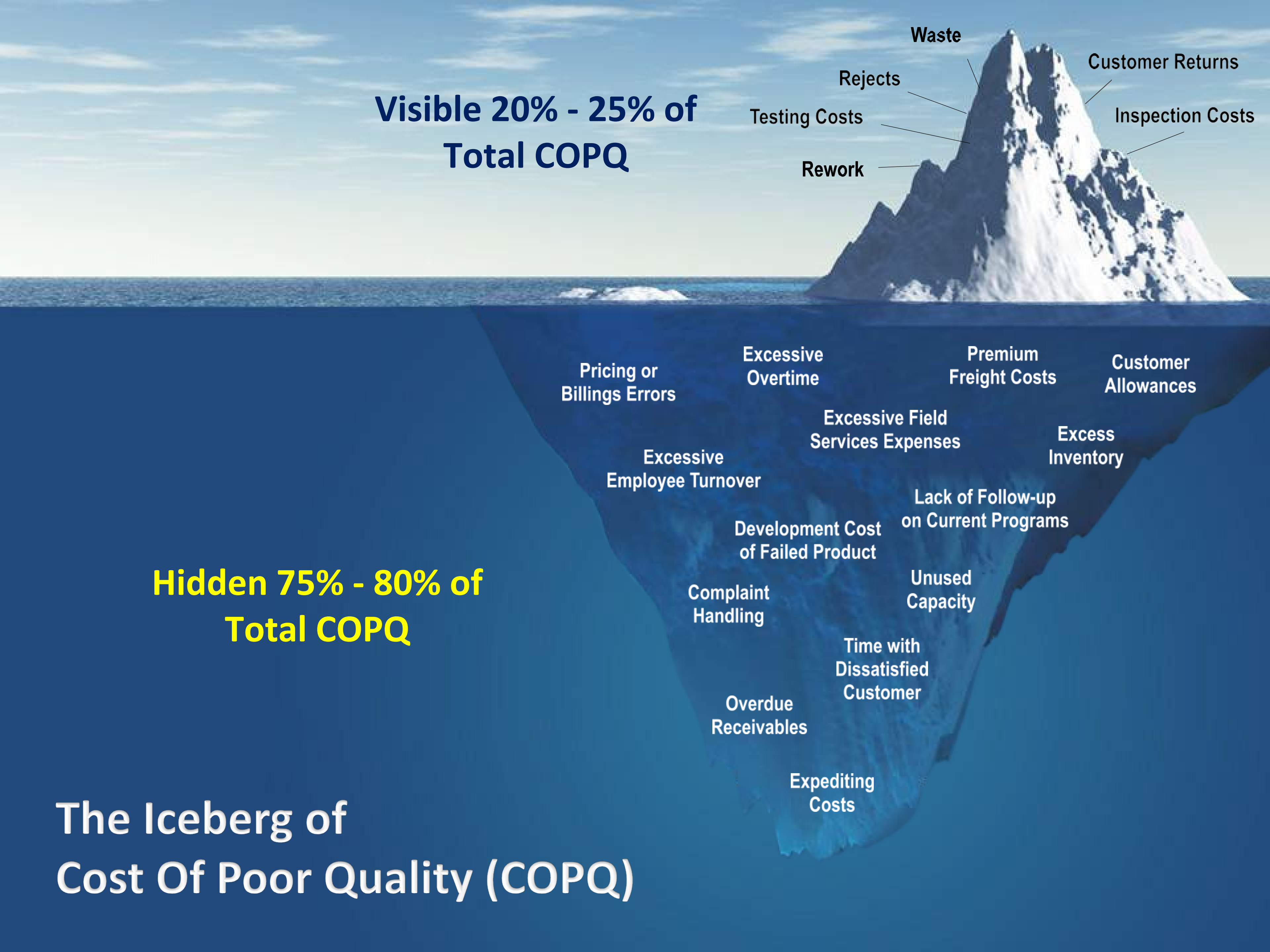
- Costs incurred due to not having 100% confidence that product or process quality is perfect all the time.

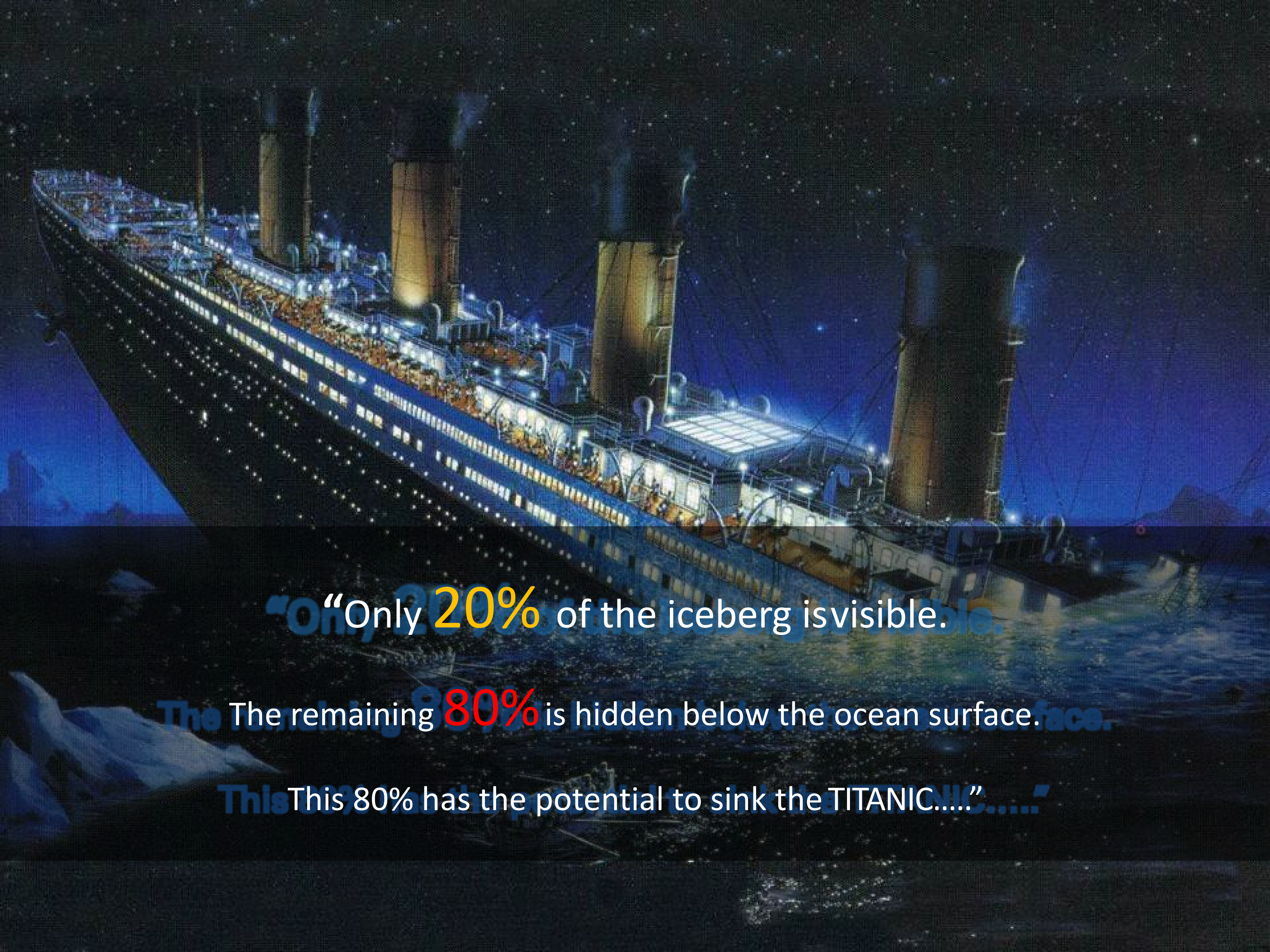
- In other words, COPQ can be defined as costs that would disappear if there were no defects

ACTUAL COST – IDEAL COST = COPQ

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.


The story of COPQ
starts with the Iceberg
The ICEBERG …
Nature’s most
spectacular
creation!!!


Visible 20% - 25% of
Total COPQ
Waste
Rejects
Testing Costs
Rework
Customer Returns
Inspection Costs
Pricing or
Billings Errors
Excessive
Overtime
Premium
Freight Costs
Customer
Allowances
Excessive Field
Services Expenses
Excess
Inventory
Excessive
Employee Turnover
Lack of Follow-up
on Current Programs
Development Cost
of Failed Product
Unused
Capacity
Complaint
Handling
Time with
Dissatisfied
Customer
Overdue
Receivables
Expediting
Costs
Hidden 75% - 80% of
Total COPQ
The Iceberg of
Cost Of Poor Quality (COPQ)


“Only 20% of the iceberg isvisible.
The remaining 80% is hidden below the ocean surface.
This 80% has the potential to sink the TITANIC.....”

Cost of Poor Quality

Prevention Cost (Cost of Conformance)

- Quality education
- Process design
- Defect cause removal
- Process change
- Quality audit
- Preventive maintenance

Appraisal Cost (Cost of Conformance)

- Test
- Measurements
- Evaluations and assessments
- Problem analysis
- Detection
- Inspection
- Test equipment maintenance

COPQ

Internal Failure Cost (Cost of Non Conformance)

- Re-inspection and retest
- Scrap
- Rework
- Repairs
- Service
- Defect removal
- Lost production

External Failure Cost (Cost of Non Conformance)

- Returned products
- Legal exposure and costs
- Liability and damage claims
- Poor availability
- Malfunction
- Replacement
- Poor safety
- Complaint administration and warranty

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Hard Saves vs Soft Saves

Six Sigma is primarily aimed at reducing spending by eliminating defects and inefficiencies.

Two types of savings categories:

- Hard Savings

The absolute removal of spending

- Soft Savings

The removal of spending from a specific process step, but the spending still takes place and the resources are not applied to another productive use.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Hard Saves vs Soft Saves

The Cost of Poor Quality (COPQ) is used to:

- Develop SOW (Define phase)
- Estimate the potential savings the project could yield (Measure phase)
- Quantify the actual savings once improvements have been made (Validate phase)

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Cost of Poor Quality (COPQ)

COPQ Simulation Exercise

Team Training, Inc (TTI) provides training services to a wide variety of service and manufacturing companies. The person who answers the phone at TTI doesn't have the time or knowledge to answer questions from callers. He transfers each caller to the appropriate person. Often, the "appropriate person" is not in the office. Some clients or potential clients hang up rather than leave voice mail messages because they need information right away.

TTI's marketing representative prepares a contract to be signed by the client requesting the training. Several iterations of the contract are often needed until, finally, the contract matches the customer's needs. Every version of the contract is proofread by a TTI clerk before mailing it to the client.

Occasionally, a TTI representative misinterprets a client's needs. In these cases, the training misses the mark and the client is not pleased.

It is difficult to reach people at the client organizations. So, the TTI materials representative estimates the number of course manuals needed for a given training session. Of course, she over-estimates to make sure everyone has a manual. Rush print jobs and overnight shipments of manuals are routine.

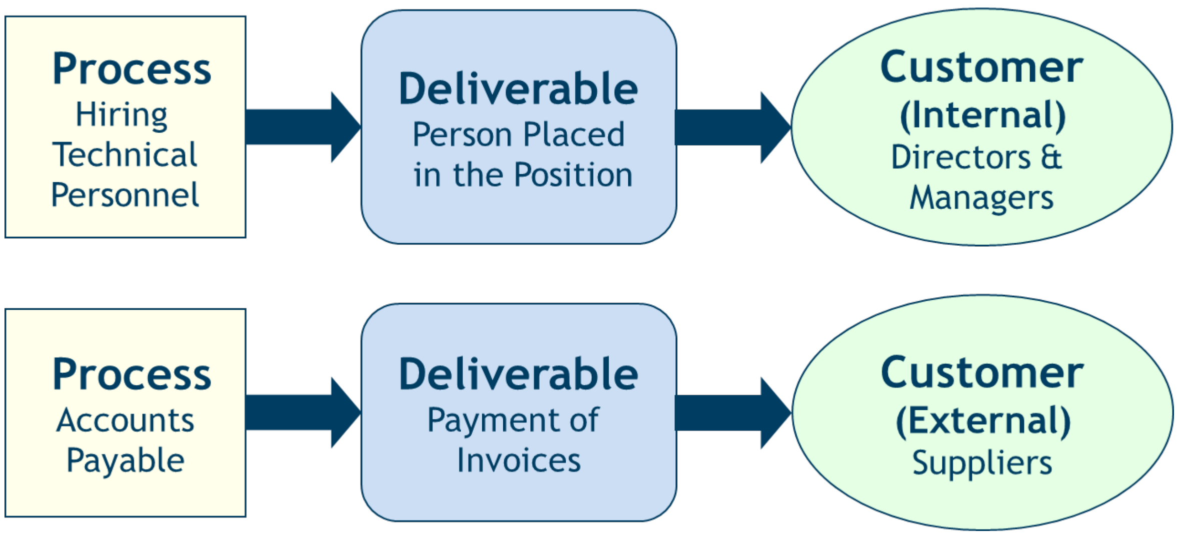
List the Failure Costs

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Voice of Customer (VOC)

Who is your customer?

- Define products or services provided to customer
- Identify related process
- Are your customers – External and (or) Internal ??



Customers can be Internal or External

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

What does your customer need?

Before you approach to a business problem, ask these questions:

- How does the customer view my process?
- What does the customer look at to measure my performance? What
- does the customer need from me to fulfill his process?

The approach towards any problem must be ***"Outside-In",*** that is view the problem from customer's perspective.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Translating VOC into CTQs

VOC

CTQs

Identify customer segments that need to be targeted → Gather verbatim VOC and determine service quality issue → Translate to needs statement and develop a CTQ - project Y metric output characteristic

CTQs: Critical To Quality Characteristics

A specific measurable attribute of the output that is a key requirement for customer satisfaction

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Gather Voice Of Customer (VOC)

Key Considerations In Collecting Customer Data:

- Collector's bias may affect what is heard
- What contact/relationship do you have with the customer? What
- are your time constraints?
- What budget is available?
- How much certainty to do you need to move forward with the project? Ensure
- customer expectations are aligned with our intentions/actions

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Gather Voice Of Customer (VOC)

Identify customer segments that need to be targeted

How?

Gather verbatim VOC and determine service quality issue

What?

- List your customers
- Define customer segments
- Narrow the list of customers
- Gather VOC

- Review existing VOC data
- Decide on what to collect
- Use appropriate tools to gather VOC
  - Surveys
  - Interviews
  - Be a customer
  - Focus group
  - Customer observation
  - Listening posts
  - Competitive comparison
- Collect data

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Voice of Customer (VOC)

Quality Function Deployment (QFD)

Origin of Quality Function Deployment

- Quality Function Deployment (QFD) is a structured approach to defining customer needs or requirements and translating them into specific plans to produce products to meet those needs.
- The "voice of the customer" is the term to describe these stated and unstated customer needs or requirements.
- Quality Function Deployment(QFD) is a system for designing a product or service based on customer demands that involves all members of the producer or supplier organization
- QFD originated in Bridgestone Tires and Mitsubishi Heavy Industries in late 1960s and early 1970s when they used quality charts that take customer requirements into account in their product design process
- In Japanese, 'deployment' refers to an extension or broadening of activities and hence 'Quality Function Deployment' means the responsibilities for producing a quality item must be assigned to all parts of a corporation.
- In 1978 Yoji Akao and Shigeru Mizuno published the first work on this subject showcasing how design considerations could be deployed to every element of competition

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quality Function Deployment - Multiple Definitions

- Quality Function Deployment also known as QFD is a technique that believes in designing the products and services to reflect the desire and tastes of customers

- QFD is a systematic way of documenting and breaking down customer needs into manageable and actionable detail.

- QFD is a method for translating customer requirements into an appropriate company program and technical requirements at each phase of the product realization cycle.

- It is an orderly process for determining critical quality characteristics

"The key to improving quality is linking the design of products or services to the processes that produce them"

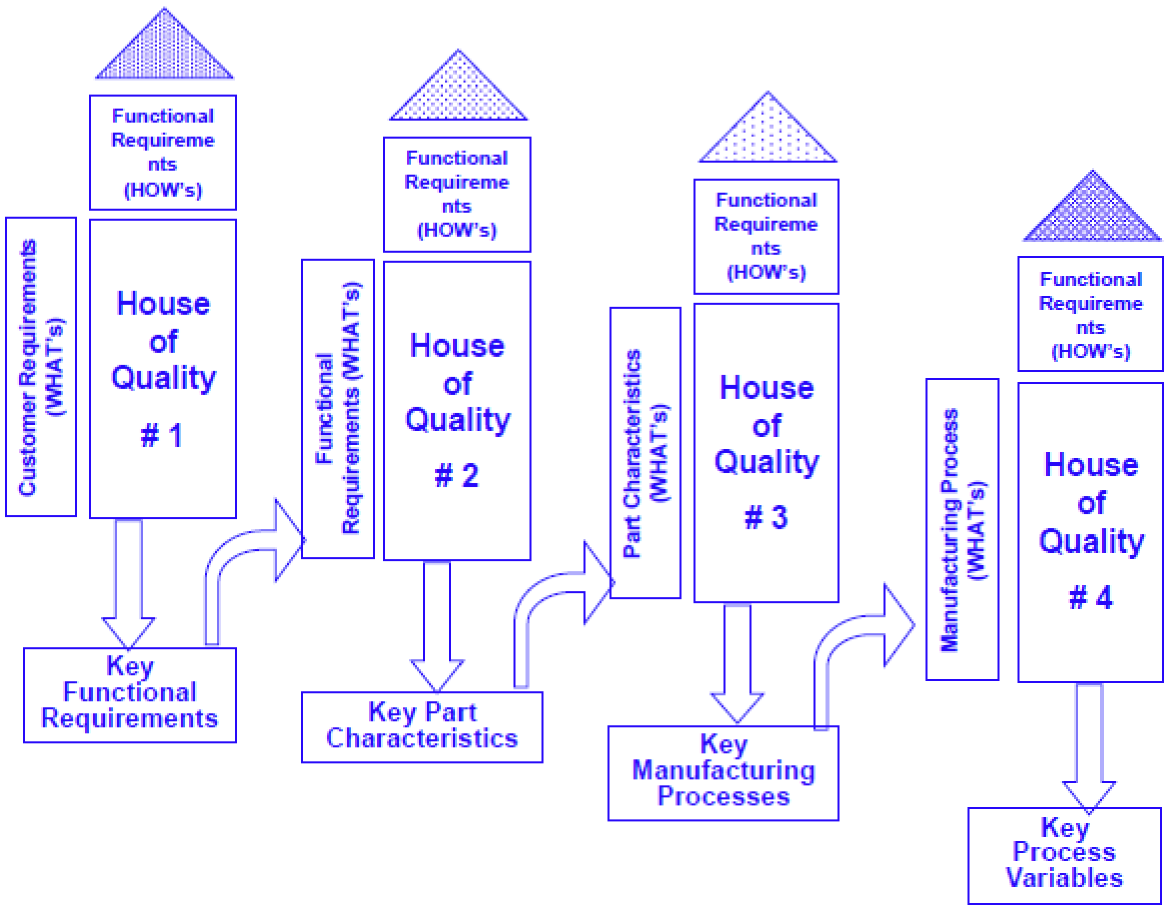
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Why Quality Function Deployment?

- QFD is a proven technique that can reduce start up time and costs
- QFD reduces the uncertainty involved in product and process design
- QFD is a technique that promotes cross-functional teamwork
- QFD is a planning methodology that organizes relevant information to facilitate better decision making
- QFD provides a better tracking system for future design or process improvements

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

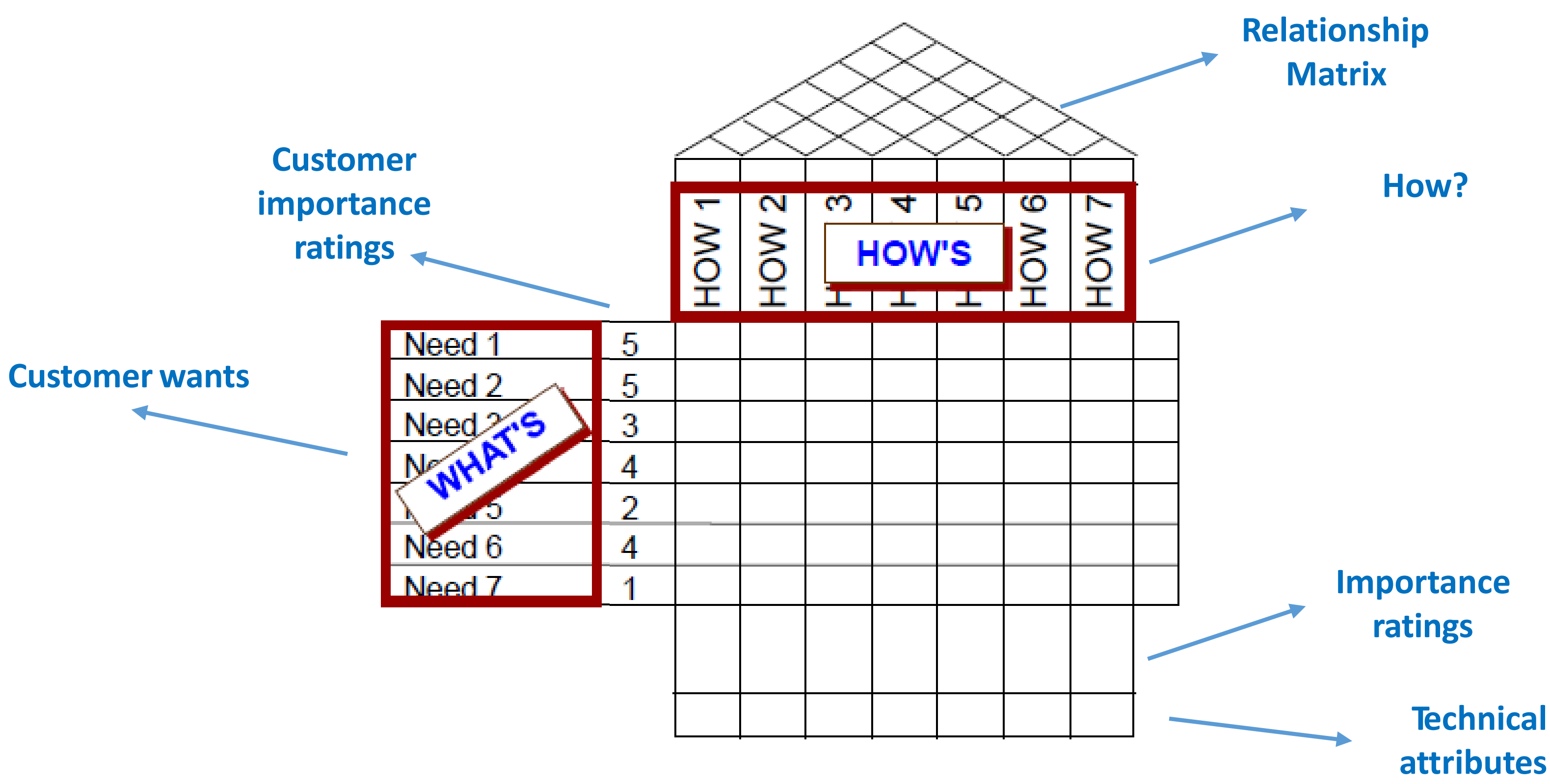
How does Quality Function Deployment work?



A product feature or process step that must be controlled to guarantee that you deliver what the customer wants

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quality Function Deployment: Structure



QFD, is also known as House of Quality due to its structure like a building block and roof

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quality Function Deployment: Structure

1. Establishes the customer needs (What’s)
2. Ranks the importance – how important are the “What’s” to the customer?
3. Establish the “How’s” – How to satisfy customer needs?
4. Create a correlation between the “What’s” and “How’s” (H-Strong, M-Medium, L-Weak)
5. Evaluate the impact of each function / process / product/ service on the customer wants
6. Establish a importance rating for each “What”
7. Evaluate competing products
8. Establish targets for each “How / What” correlation
9. End the QFD with the self assessment for each How and What

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Quality Function Deployment: Example

Let us look at building a House of Quality for a Pizza business.

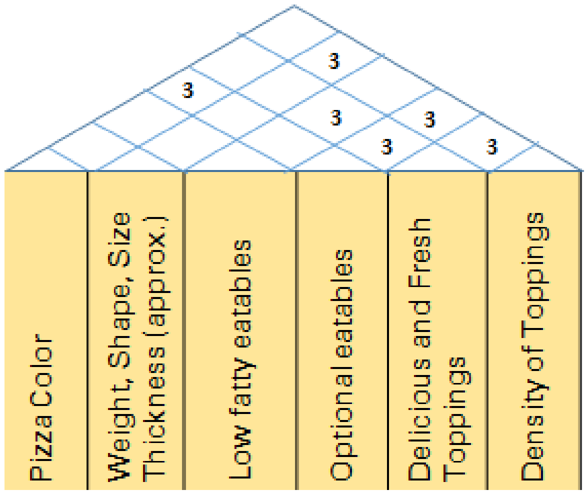
1. Let us list out basic customer wants:
   - Good taste
   - Low price
   - Low fats and healthy
   - Appearance
   - Fresh and hot delivery
   - Good texture

2. Basis customer wants, determine the importance ratings (on a scale of 1-10)
   - Good taste (9)
   - Low price (7)
   - Low fats and healthy (8)
   - Appearance (7)
   - Fresh and hot delivery (8)
   - Good texture (8)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quality Function Deployment: Example

3. Establish “How” to satisfy customer wants:
   - Delicious fresh toppings
   - Appropriate weight, size, thickness and shape
   - Low fatty eatables
   - Pizza Color
   - Variety and density of toppings



4. Establish correlation between “What” and “How”: Here we have used the scale of (1,3,6)
   1 being the lowest and 6 being the highest.

5. Evaluate the impact of each function / process / product/ service on the customer wants

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Quality Function Deployment: Example

6. Establish importance rating for each "what"

How's?	Ratings	Total
Pizza Color	3*8	24
Appropriate weight, size, thickness and shape	6*9 + 3*7	75
Low fatty eatables	6*8 + 3*8	72
Optional eatable	7*1 + 3*8	31
Delicious fresh toppings	3*8 + 3*7	45
Variety and density of toppings	9*1 + 3*8	33

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quality Function Deployment: Example

7. Evaluate competing products

Here we are considering our competition two pizza companies i.e. Company A and Company B
(G = Good, F = Fair, P = Poor)

Customer Needs	Importance Ratings	Company A	Company B
Good Taste	9	G	F
Low Price	7	P	P
Good Texture	8	G	G
Low fats and healthy	8	F	F
Fresh & Hot Delivery	8	G	G
Appearance	7	G	F

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quality Function Deployment: Example

Customer Needs	Importance Ratings	Pizza Color	Weight, Shape, Size Thickness (appropr.)	Low fatty eatables	Optional eatables	Delicious and Fresh Toppings	Density of Toppings	Company A	Company B
Good Taste	9		6				1	G	F
Low Price	7				1			P	P
Good Texture	8	3				3		G	G
Low fats and healthy	8			6				F	F
Fresh & Hot Delivery	8			3	3		3	G	G
Appearance	7		3			3		G	F
Importance ratings		24	75	72	31	45	33		

The Weight, Shape, Size and Thickness of pizza is importance to the customer

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Benefits of Quality Function Deployment

The results we can expect by carrying out the QFD studies are many:

- Better understanding of customer needs
- Improved organization on development projects
- Improved introduction to production
- Fewer design changes late in development
- Fewer manufacturing start-up problems
- Reputation for being serious about quality
- Increased business
- Documented product definition based on customer requirements

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Define Phase

Define Phase - Roadmap

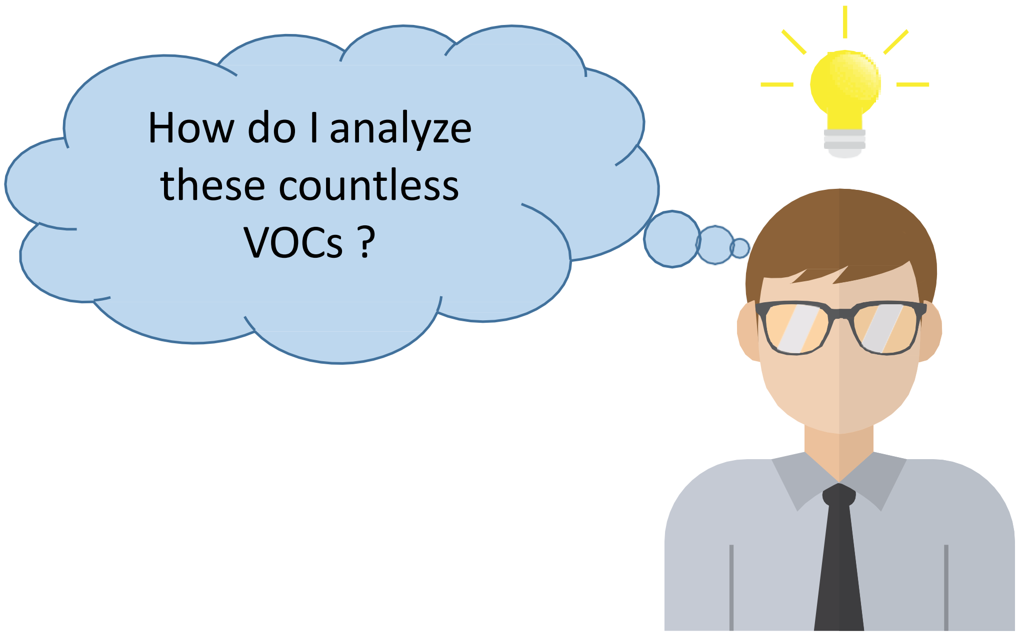
Voice of Customer (VOC) → Determine Critical To Quality (CTQ) → Develop Project Charter → Six Sigma Team (organization) → Map the process

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Voice of Customer (VOC)

Affinity Diagram

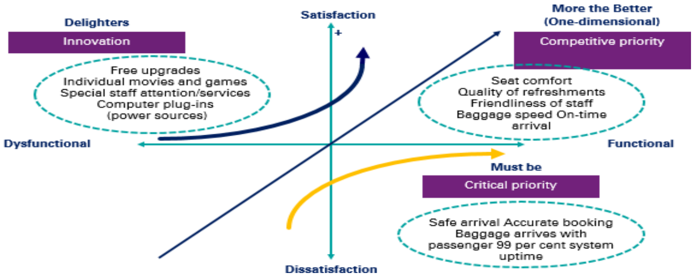
Tools for analyzing VOC



- Affinity Diagram
- Kano Model

Flexible product	Easy process	Availability	Personal interface	Advice/consulting
Low interest rate	Easy application	Will come to my facility	Knowledgeable reps	Knows about my finances
Variable terms	Easy access to capital	Available outside normal business hours	Professional	Knows about my business
All charges clearly stated	Quick decision	Available when I need to talk	Friendly	Makes finance suggestions
Pay back when I want	Can apply over phone	Responsive to my calls	Make me feel comfortable	Cares about my business
No prepayment penalties/ charges	Know status of loan during application	Talk to one person	Patient during process	Has access to experts
Pre-approved credit	Know status of loan (post-approval)	Will come to my facility	Knowledgeable reps	Provides answers to questions
Variable terms	Preference if bank customer	Available outside normal business hours	Professional	Calls if problems arise

Organize VOC into broad categories



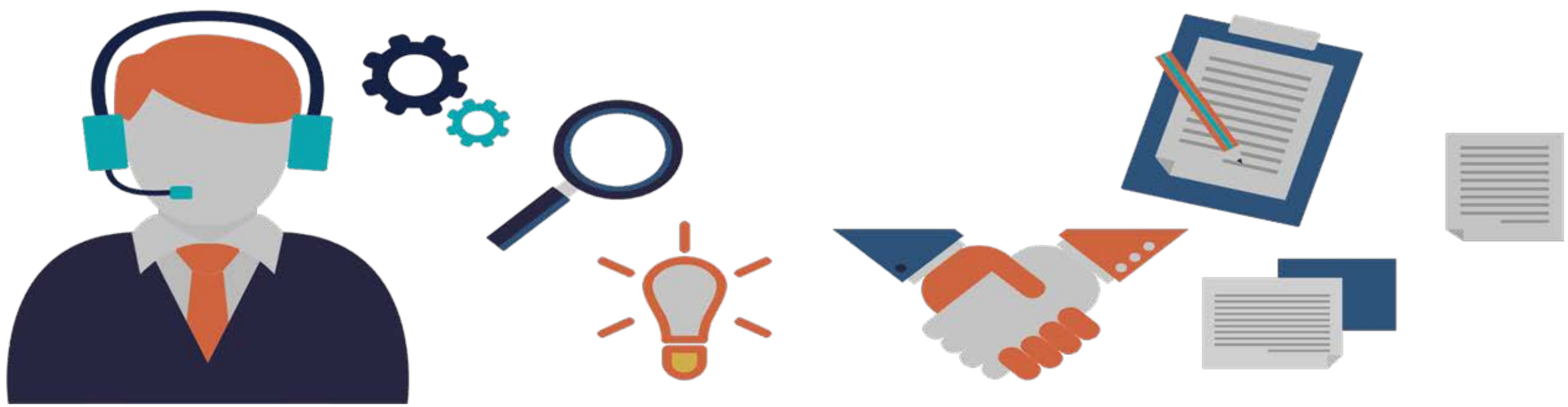
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Voice of Customer (VOC)

Affinity Diagram

Affinity Diagram

- Record each VOC on a post it note in bold letters
- Without talking sort the ideas simultaneously as a team into 5 –10 related groupings For each
- grouping create summary or header cards using consensuses
- Draw the final affinity diagram connecting all finalized header cards with their grouping



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Affinity diagram – An Example

Flexible product	Easy process	Availability	Personal interface	Advice/consulting
Low interest rate	Easy application	Will come to my facility	Knowledgeable reps	Knows about my finances
Variable terms	Easy access to capital	Available outside normal business hours	Professional	Knows about my business
All charges clearly stated	Quick decision	Available when I need to talk	Friendly	Makes finance suggestions
Pay back when I want	Can apply over phone	Responsive to my calls	Make me feel comfortable	Cares about my business
No prepayment penalties/ charges	Know status of loan during application	Talk to one person	Patient during process	Has access to experts
Pre-approved credit	Know status of loan (post-approval)	Will come to my facility	Knowledgeable reps	Provides answers to questions
Variable terms	Preference if bank customer	Available outside normal business hours	Professional	Calls if problems arise

Organize VOC into broad categories

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Voice of Customer (VOC)

Kano Model

Kano Model

Purpose:

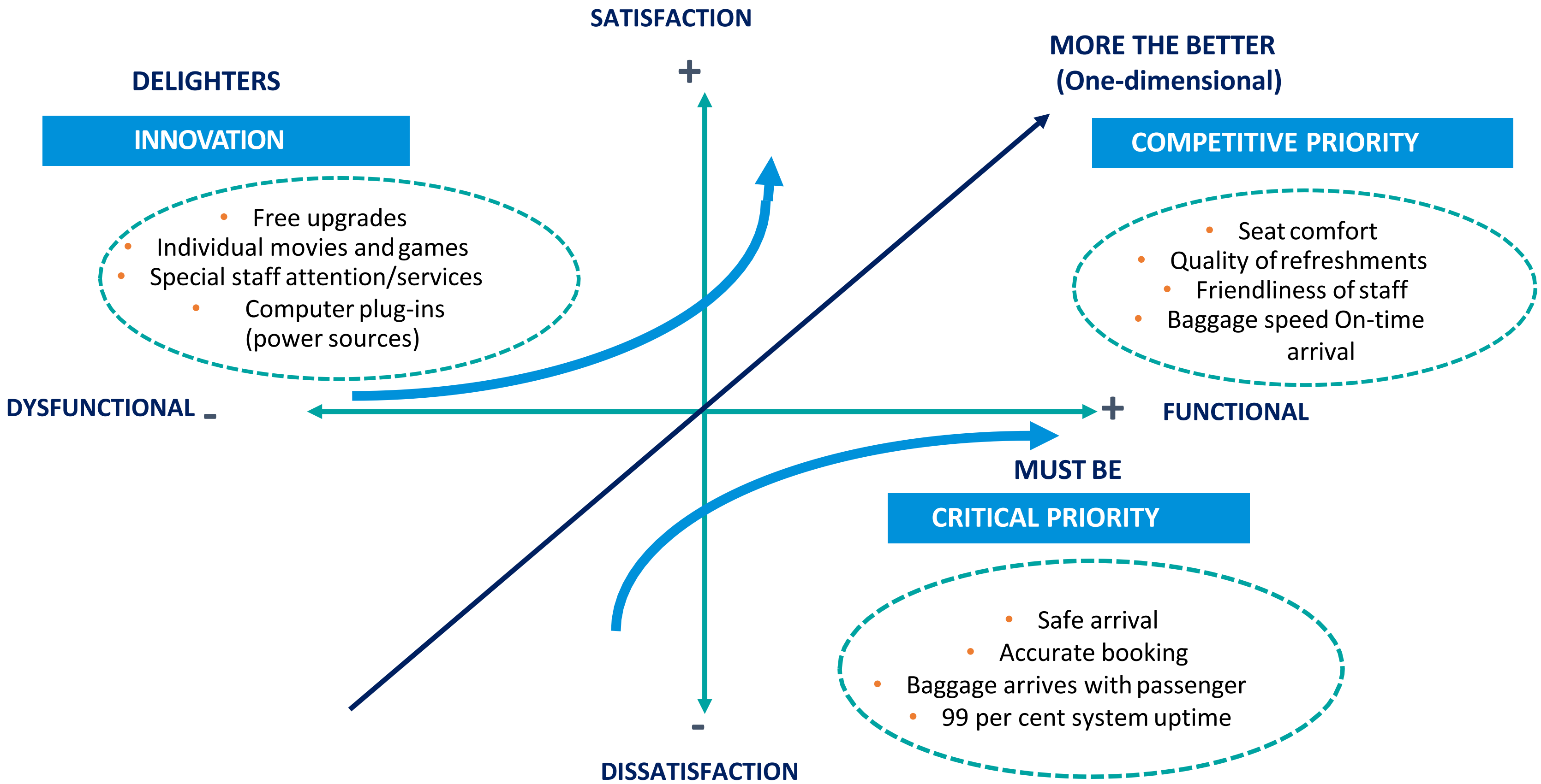
- To identify & prioritize the full range of the customers needs
- Kano model helps to describe which needs, if fulfilled contribute to customer dissatisfaction neutrality or delight
- Kano Model Identifies
    - Must be needs - Critical to customer expectation
    - More is better – Critical to customer satisfaction
    - Delighter – Converting wants to needs

How to built ?

- Gather sorted customer needs Classify
- the needs into 3 Categories
    - Must be
    - More the better
    - Delighters
- If there is insufficient data to enable the classification, collect addition data on VOC
- Prioritize the customer needs to develop the CTQ

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Prioritizing VOC for CTQ Identification - Kano model



KANO MODEL HELPS TO PRIORITIZE OUR EFFORTS TOWARDS SATISFYING CUSTOMERS

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Determine Critical To Quality (CTQ)

Example: Translating VOC to CTQs

Verbatim

- ‘You take too much time in getting back to me!’
- ‘These forms are too cumbersome!’

Specific need

- Quick response
- User friendly forms

CTQs

- Process turn around time not more than 10 minutes.
- Form < 2 pages and < 10 minutes to complete

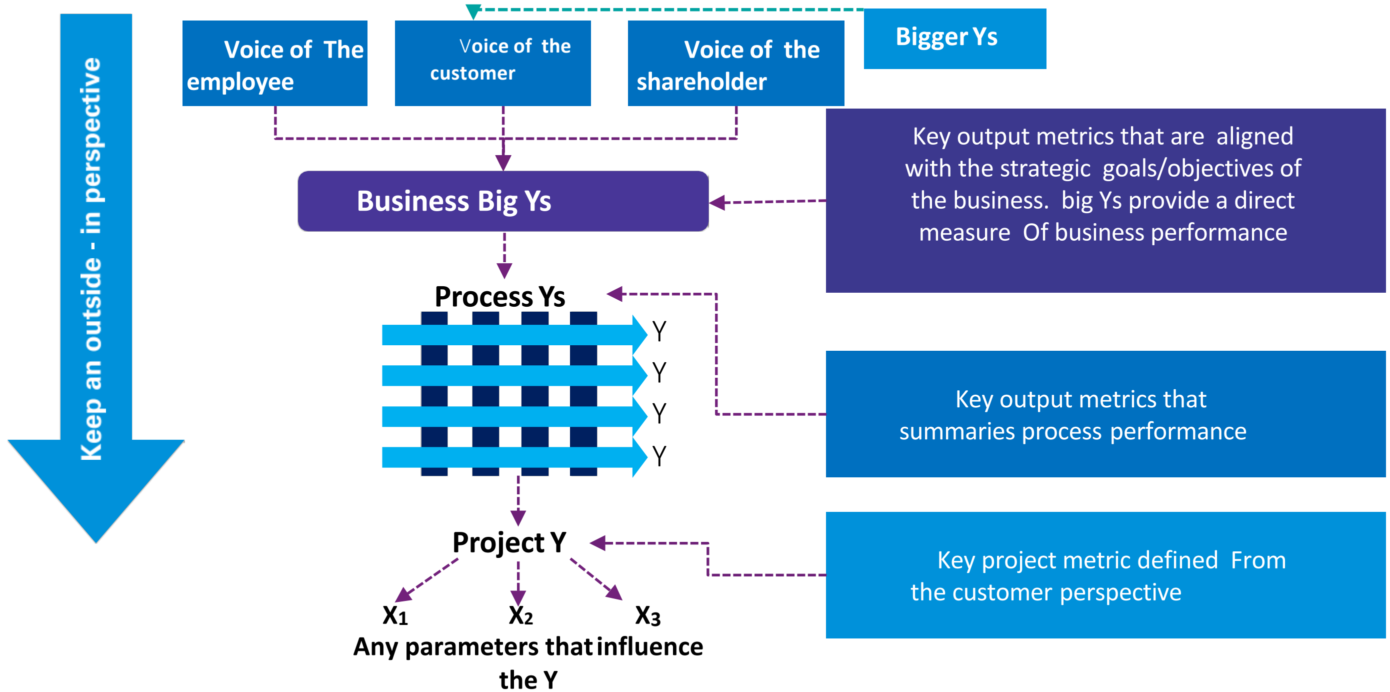
Validate CTQ with customer

What gets measured gets managed… help ensure measurable CTQs

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Drivers of project selection



Strong linkage between projects and big Y is important

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Develop Project Charter

What is a Charter?

One of the most important things necessary to get a team started on a footing is a charter

A Charter:

- Clarifies what is expected of the project
- Keep the team focused
- Keeps the team aligned with organizational priorities
- Transfers the project from the Champion to the Improvement Team
- Used as a tool by the Apex Council to review project progress

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Business case

Why is this project worth doing?

Why is it important to do now?

What are the consequences of **not** doing this project?

What activities have higher or equal priority?

How does it fit with business initiatives and targets?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Problem Statement

The purpose of the Problem Statement is to describe what is wrong

Description of the "pain"

- What is wrong or not meeting our customer's needs?
- When and where does the problem occur?
- How big is the problem?
- What's the impact of the problem?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Problem Statement

Key Points / Potential Pitfalls

- Is the problem based on observation (fact)
- Does the problem statement prejudge a root cause?
- Can data be collected by the team to verify and analyze the problem?
- Is the problem statement too narrowly or broadly defined?
- Is a solution included in the statement?
- Is the statement blaming any person or function?
- Would customers be happy if they knew we were working on this?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Problem Statement - Examples

Example 1

Poor Statement

Because our customers are dissatisfied with our service, they are late paying their bills.

Improved Statement

In the last 6 months (when) 20% of our repeat customers were over 60 days late (what) paying our invoices (where). The current rate of late payments represents 30% of our outstanding receivables (how big). This negatively affects our operating cash flow (impact)

Example 2

Poor Statement

In the year 2019-2020, the rework in machined castings was very high due to poor inspection practices (blame). Rework lead to delay in delivery of the equipment and underutilization of capacity.

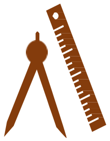
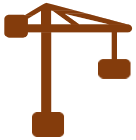
Improved Statement

In the year 2019-2020 (when), the rework in machined castings (what) was in excess of 3600 manhours (how big) for the production of 700 m/cs (where). This high quantum of rework lead to delayed delivery of equipment as well as underutilization of capacity (impact). This problem if not addressed immediately will lead to potential loss of business and reputation.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Goal Statement

- The Goal Statement defines the team's improvement objective
- Define the improvement the team is seeking to accomplish
- Must not assign blame, presume cause, or prescribe solution!
- Goal Statement has four parts:
  - Start with a verb (e.g., reduce, eliminate, control, increase)
  - Focus of project (cycle time, accuracy, etc.)
  - Has a definite Target (by 50%, by 75%)
  - Has a definite deadline (completion time)

Specific	Measurable	Achievable	Realistic	Time bound
				

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Project Scope and Milestones / Project Plan

Project Scope

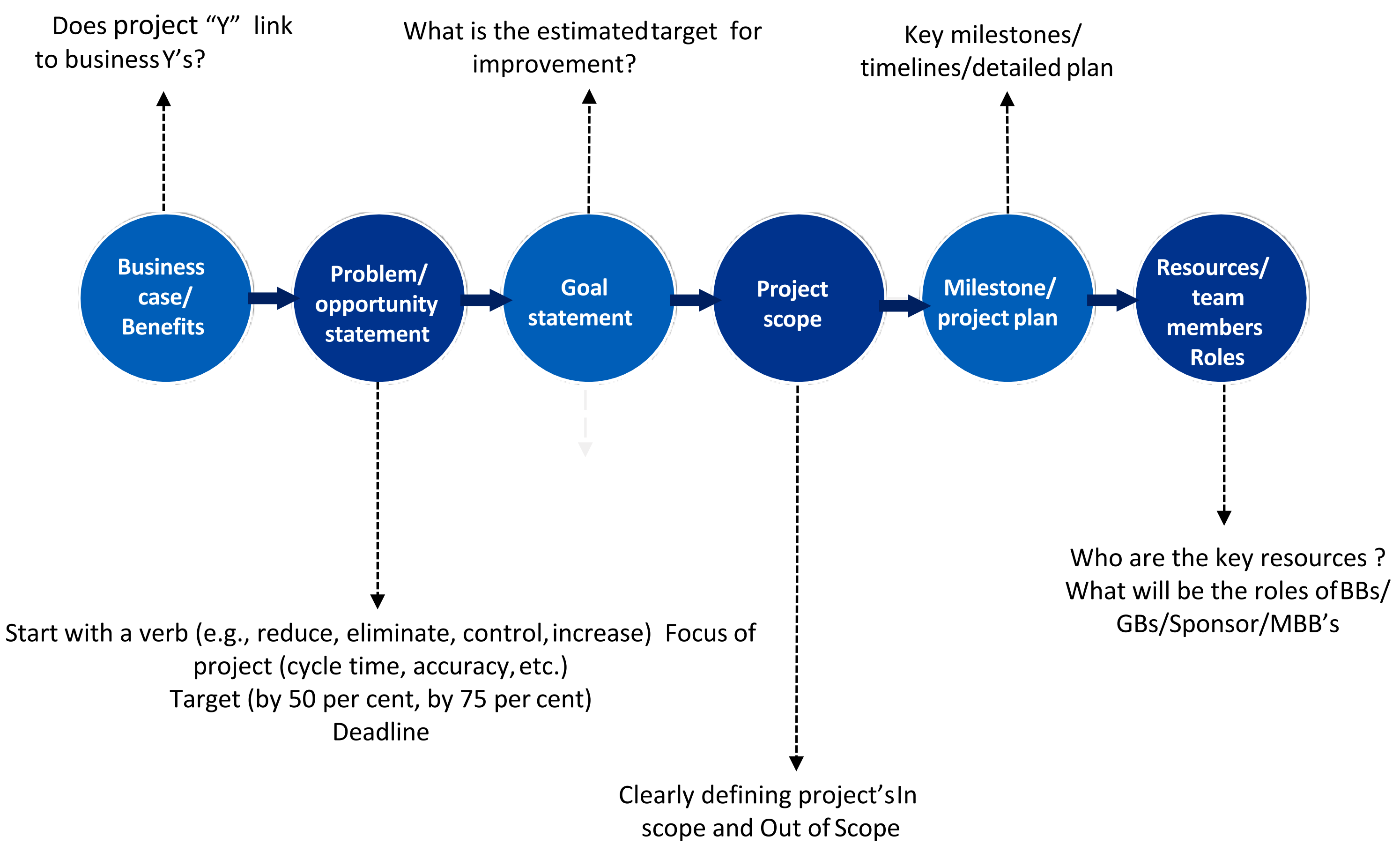
- What process will the team focus on?
- What are the boundaries of the process we are to improve? (Start and End points of the process)
- What (if anything) is out of bounds for the team?
- What (if any) are the possible constraints?
- What is the time commitment expected of team members?
- What will happen to our ‘regular jobs’ while we are doing the project?

Milestones (Project Plan)

- It is a detailed project plan with key steps and target completion dates
- Tied to phases of DMAIC process, with defined tollgate reviews
- Aggressive and Realistic (no ready-made solution)
- Documented, shared with all project team members and Champion, and updated regularly

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Elements of a project charter



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Sample project charter

Project Name	Improving Accuracy of Payments Processing		
Process name	Accounts Payables Process for Malaysia	Project Number	
Sub Process Name	Black Belt	H T	
	Sponsor	H K	

Business Case	Problem/Opportunity Statement	Goal Statement
The Malaysian team is presently handling the trade creditor payments for Bank of Malaysia. Average lines processed being in the range of 15,000 to 16,000 per month. Many a time, there are few errors which if passed on to the customers which may have a financial impact thus incurring a monetary loss and high Customer dissatisfaction. Such errors also have an impact on the TAT as it results in rework.	The data for the month of October 2002 shows that 65 errors which were corrected during processing and there were seven errors identified by the Customers which led to Customer Complaints during the month.	To reduce the number of errors to less than 10 errors (both first pass and second pass) per month by end of 31st December 2002

Project Scope Includes	Project Excludes
Includes vendor payments, staff claims.	Project Excludes Fixed Assets batch run, Daily Batch run and Account reconciliations.

Project Milestones

Start Date: 28-Oct-02 | End Date: 31-Dec-02

	Target Date	Actual Date	TG Review Date
Define	01-Nov-02	01-Nov-02	01-Nov-03
Measure	16-Nov-02	29-Nov-02	29-Nov-03
Analyze	23-Nov-02	29-Nov-02	29-Nov-03
Improve	30-Nov-02	31-Jan-03	31-Jan-03
Control	07-Dec-02	28-Feb-03	26-Mar-03

Project CTQ Details

Process CTQs	Unit	USL	Target
Accuracy	#	10/month	5/month
		Current	Target
	sigma	5.04	5.29

Project Comments

Project Charter Sign Off (With date)

Sponsor	
GB/BB	
BB/QL	

Project Translation Plans	Project Benefits Estimate
Possible to roll out the model to other AP shops in BOM	

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Sample project charter

Roles and responsibilities					
Name	Approver	Resource	Member	Interested party	Time commitment
Jack	X				2 hours a month
Harry	X				2 hours a month
Louis		X			2 hours a month
Amanda			X		4 hours a week
Jude			X		4 hours a week
Elsa			X		4 hours a week
Anna				X	1 hour a month

Estimated benefits (Enter details for columns whichever are applicable)						
Hard gains	Unit	Amount	Assumptions	Soft gains	Unit	Assumptions
				End customer satisfaction		
				Sigma increase		
				Employee satisfaction		
				Shareholder satisfaction		

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Create a Project Charter

Work out session

Great project

- **Be clearly bound with defined goals**
  - If it looks too big, it is
- **Be aligned with critical business issues and initiatives**
  - It enables full support of business
- **Be felt by the customer**
  - There should be a significant impact
- **Work with other projects for combined effect**
  - Global or local 'beta themes'
- **Show improvement that is locally actionable**
- **Relate to your day job**

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Project selection

Success factors

- Project scope is manageable
- Project has identifiable defect
- Project has identifiable impact
- Adequate buy-in from key stakeholders

To be successful

- Set up project scope charter and have it reviewed
- Measure where defects occur in the process
- Assess and quantify potential impact up-front
- Perform stakeholder analysis

Common pitfalls

- Resourcing of project is inadequate
- Duplicating another project
- Losing project momentum
- Picking the easy X, not the critical X

Avoiding pitfalls

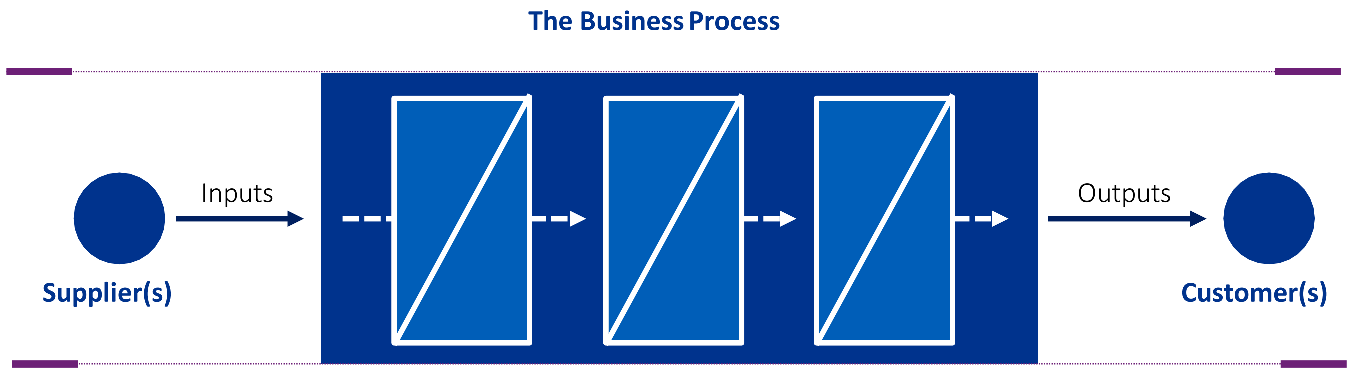
- Identify and get committed resources up-front
- Research database and translate where possible
- Set up milestones and communications plan

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Map the Process

Process definition and elements

Process is a collection of activities that takes one or more **inputs** and transforms them into **outputs** that are of value to the **customer**



Types of Process Maps

- SIPOC (Supplier-Input-Process-Output-Customer) (L1)
- Sub Process Map (L3,L4,L5)
- Value Stream Mapping (L1,L2)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Map the Process

SIPOC

SIPOC

Supplier: The provider of inputs to your process

Input: Materials, resources or data required to execute your process

Process: A collection of activities that takes one or more kinds of input and creates output that is of value to the customer

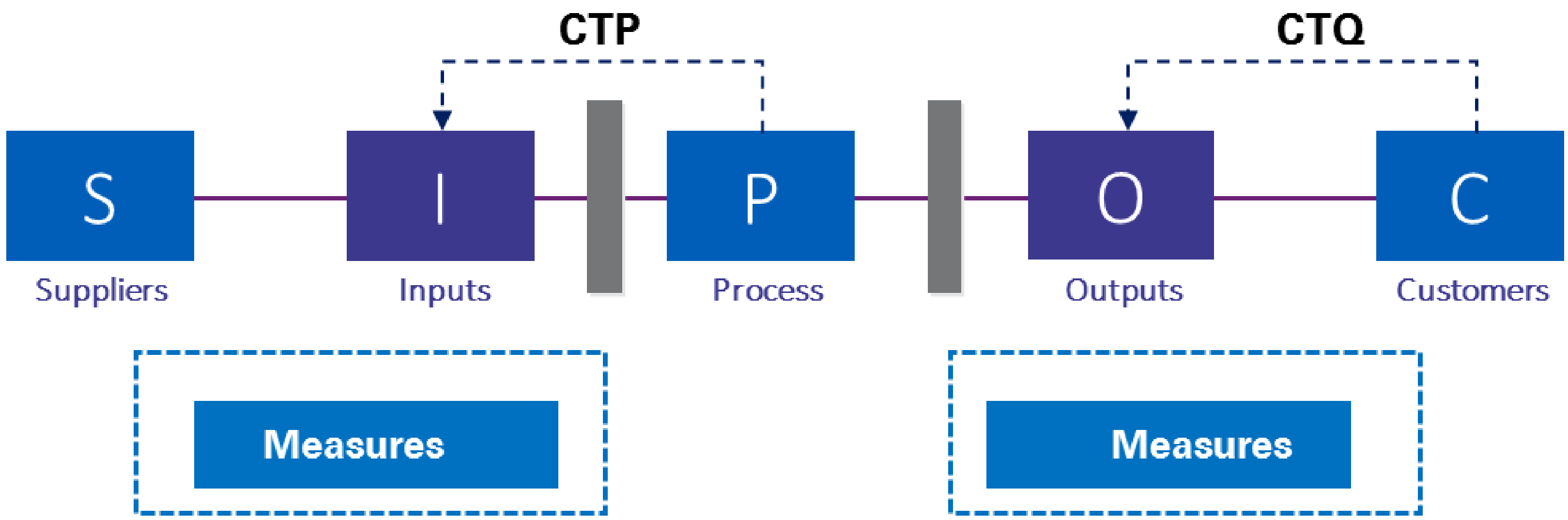
Output: The products or services that result from the process

Customer: The recipient of the process output – may be internal or external

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

SIPOC



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

SIPOC Example

Let us look at an example of Purchase Order (P.O.) Requisition to final Approval

SUPPLIER	INPUT	PROCESS	OUTPUT	CUSTOMER
S	I	P	O	C
User Department	Stock List	**Generate Purchase Requisition**	Purchase Requisition	User Department HOD
User Department HOD	1. Purchase Requisition 2. Budget	**Approve Purchase Requisition**	Approved Purchase Requisition	Procurement
Procurement	1. Approved Purchase Requisition 2. List of Approved Vendors	**Raise Request for Quotation**	Request for Quotation	Approved Vendors
Approved Vendors	Request for Quotation	**Receive Quotations**	Vendor Quotations	Procurement
1. User Department 2. Procurement	Vendor Quotations	**Techno Commercial Evaluation of Received Quotations**	Shortlisted Quotations	Procurement
Procurement	Shortlisted Quotations	**Select Preferred Vendor**	Quotation of Preferred Vendor	Procurement
Procurement	Quotation of Preferred Vendor	**Prepare Purchase Order**	Purchase Order	Signing Authority
Signing Authority	Purchase Order	**Approve Purchase Order**	Approved Purchase Order	Preferred Vendor

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Map the Process

Sub Process Mapping

Sub-process mapping

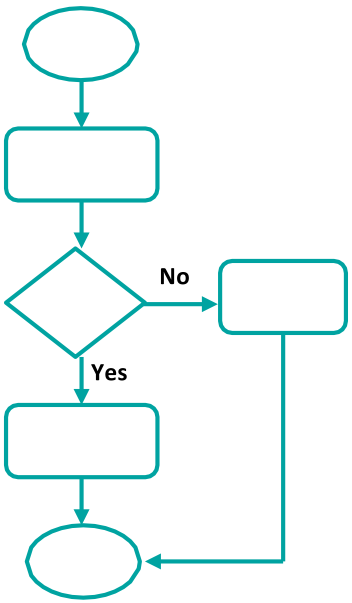
Here are some guidelines on building a sub process map. These are not absolute – but they should help you avoid some of the pitfalls of process mapping:

- Focus on 'As is' – To find out why problems are occurring in a process, you need to concentrate on how it's working now
- Clarify boundaries – If you're working from a well-done high level map, this should be easy. If not, you will need to clarify start and stop points
- Brainstorm Steps – It is usually much easier to identify the steps before you try to build the map
- Starting each step description with a verb (e.g., 'collate orders'; 'review credit data') helps you focus on action in the process
- Who does the step is best left in parentheses (or left out) – you want to avoid equating a person with the process step

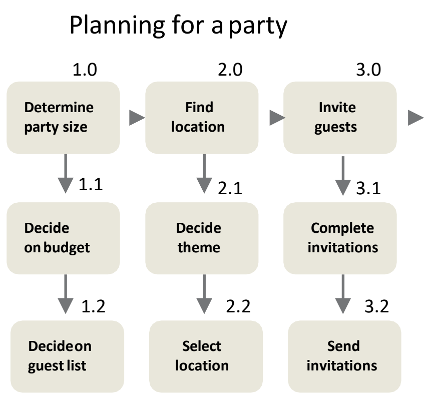
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Examples: Sub-process mapping

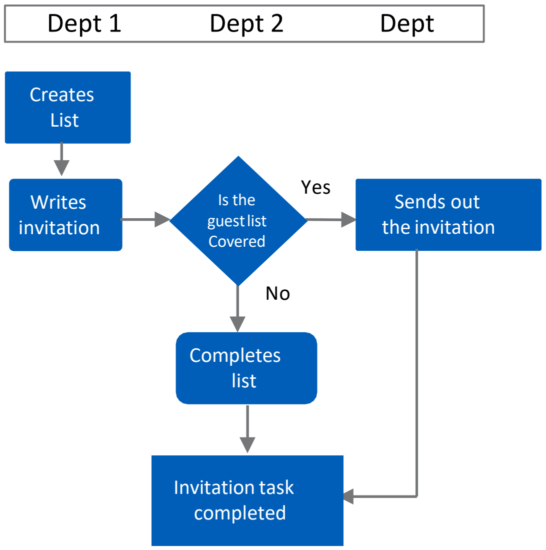
Process flowchart



Top down flow chart

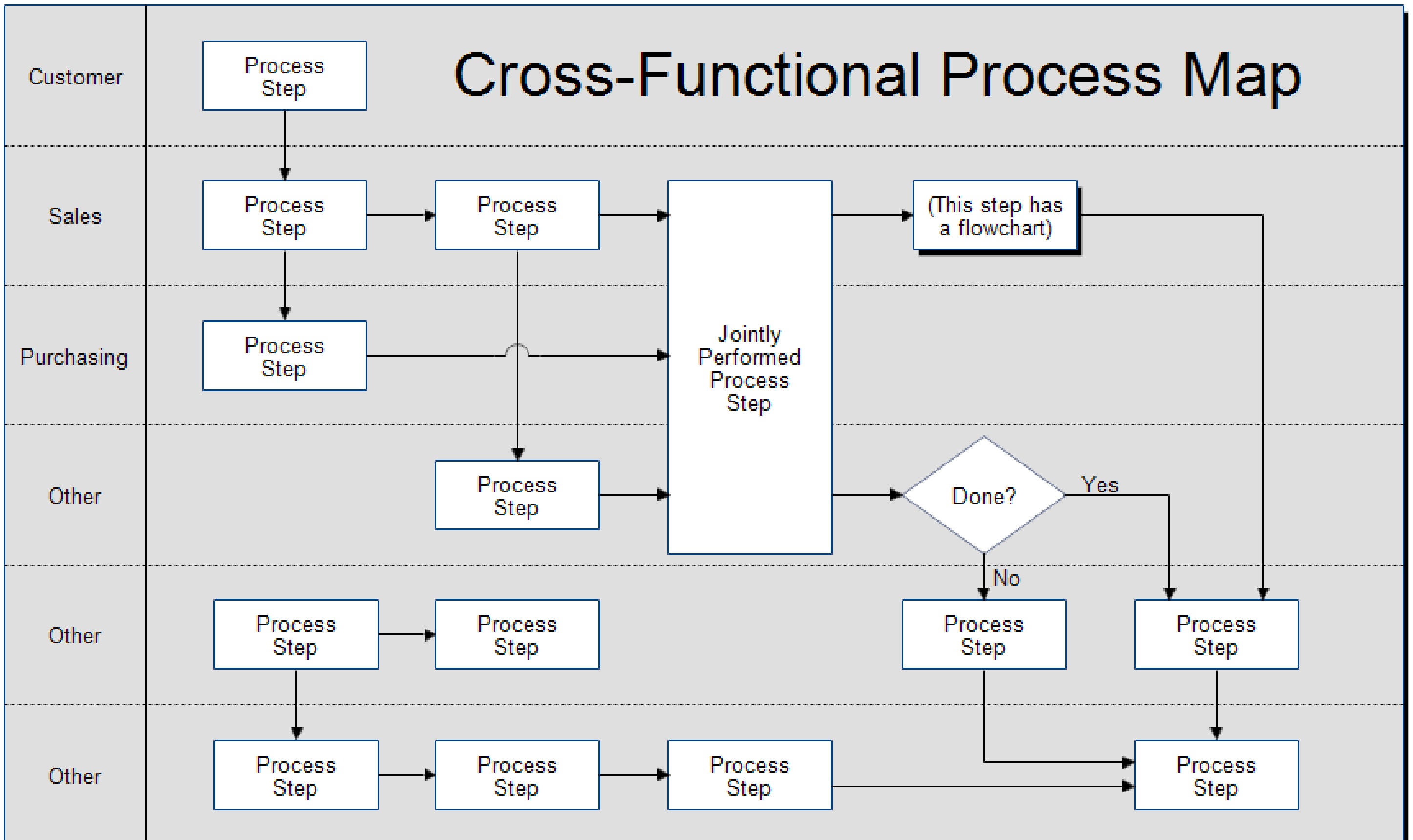


Deployment or cross-functional map/ swim-lane flowchart



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Examples: Cross Functional Process mapping



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Process mapping guidelines

- Define business process to be reviewed – name it – agree on beginning and end of process– bound it.

- Refer to CTQ work to identify primary outputs, the customers who receive them, and the customers' CTQs
    - Use nouns for outputs (e.g., sales call, proposal, etc.)
    - Use adjectives for CTQs (.e.g., timely, knowledgeable, accurate)
- Identify the process steps using brainstorming and affinity techniques
    - Write large one step per card
    - Do not try to establish order
    - All steps should begin with a verb
    - Do not discuss process steps in detail
- Use brainstorming and affinity techniques to identify critical inputs which affect the quality of the process
- For each critical input, identify the 'supplier' who provides it
- Validate to be sure the map represents the situation as it really is today (the 'as is' map)– not how you think it is, or how it should be

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Group Exercise

- Earlier in define, you developed a high-level or SIPOC process map. By looking at a process from a ‘big picture’ perspective
- You evaluated customer needs and supplier inputs, and determined initial measurement objectives
- Now, you will look in more detail at the sub processes defined in the SIPOC map
- Sub process maps provide specifics on the process flow that you can then analyze using several useful techniques.
- Choose what to sub process map by determining which of the major steps in the SIPOC have the biggest impact on the output (Ys).
- The block (or blocks) selected is the one on which you create a sub process map – using it to understand how and why it impacts the output
    - If the output is a time measure, which of the blocks consumes the largest portion of total time, or which one has the most variation or delays?
    - If the output is a cost measure, which of the blocks adds the most cost?
    - If the output is a function measure, which block has the most errors or problems?
- Like working with a puzzle, you begin to assemble the pieces of an area on which it makes sense to focus our efforts

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Points to Remember

Points to Remember:

- People who work on the process know it the best. Involve people who know (focuson) the ‘as Is process’
- Decide, clarify and agree upon process boundaries
- Use group activities like brainstorming
  - Use verb - noun format (e.g., Prepare contract not contracting)
  - Do not aim at the person taking care of the activity
- Respect the boundaries
- Do not start ‘problem solving’
- Validate and refine before analyzing

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Map the Process

Value Stream Mapping

What is a value stream?

- A value stream is all the activities required to bring a service/product from a customer request to fulfillment/completion.
- All the activities involved in creating value for the customer
- The process starts with raw material or information and ends with the end customer.
- The process involves functions both internal and external to the firm.
- Activities can be described as value added (VA) or non-value added (NVA)

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Key Terms

- **Value-added (VA):** This step in the process adds form, function, and value to the end product and for the customer. These are activities that the customer is willing to pay for.

- **Non-Value-Added (NVA):** This step does not add form, function, or assist in the finished goods manufacturing of the product

- **Non-Value-Added-But-Necessary (Value Enabled):** This step does not add value, but is a necessary step in the final value-added product

- **Cycle Time (CT):** Cycle time is the average time taken to produce single unit. It is derived from total units produced over a set value of time. It is also sometimes expressed as the interval between two consecutive units.

- **Lead Time (LT):** Lead time is the time it takes for one unit to progress through the entire operation from start to end.

- **Takt Time (TT):** Takt time is the time available for producing each unit based on customer demand. *TT = Net Operating Time / Customer Demand*

- **Process Time (PT):** Process Time is the actual time taken to complete one step / activity in the process. Since in the VSM, we map the Value Stream, this time is also called **Value Time**.

- **Process Cycle Efficiency (PCE):** This signifies the overall efficiency of a process. It is the ratio of Process / Value Time Vs Lead Time. *PCE = Process or Value Time / Lead Time*

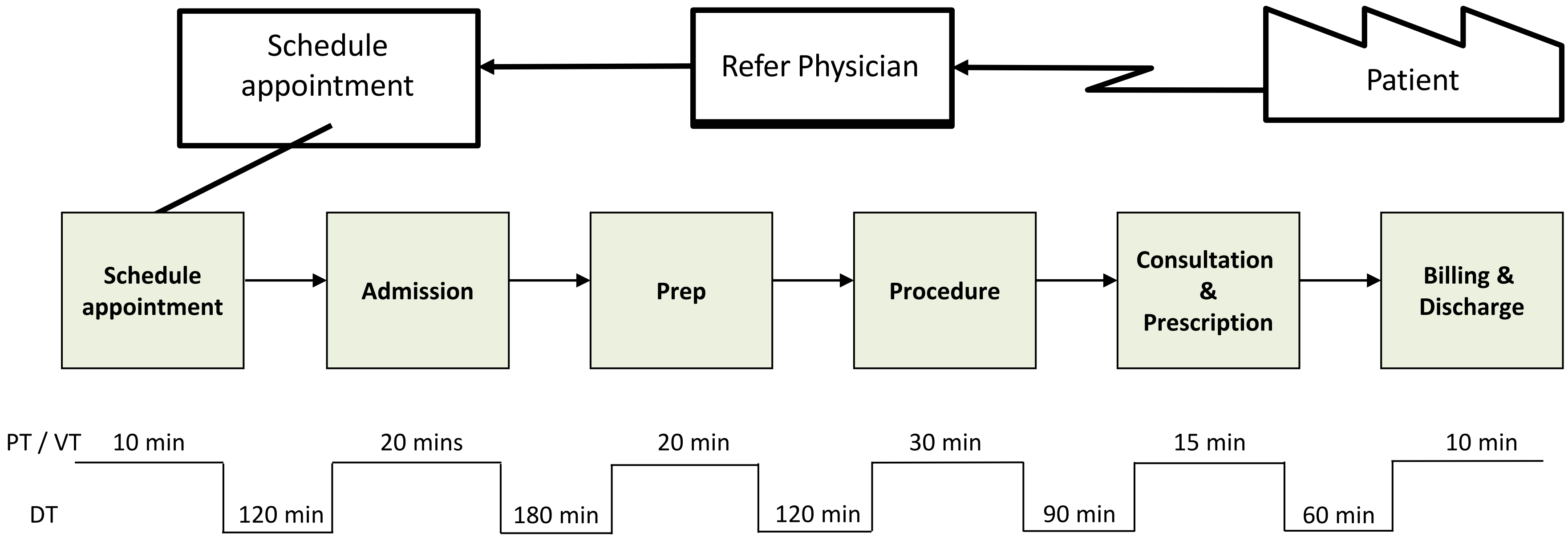
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Why map the value stream?

- Enables to visualize the process / production flow
- Allows to see waste in the system
- Prevents focusing on large improvement opportunities with little impact
- Creates framework for designing complete system
- Demonstrates interaction between information and material flow
- Enables to know the Lead Time and Cycle Time for the process

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Value Stream Mapping - Example

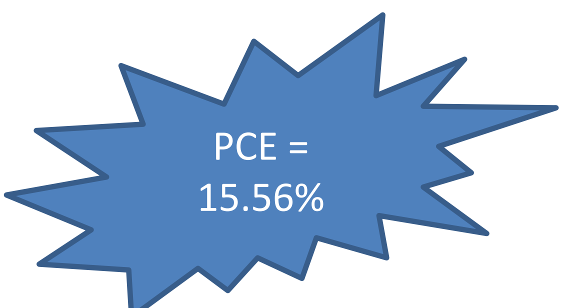


Process Time / Value Time = 10 min + 20 mins + 20 mins + 30 mins + 15 mins + 10 mins = 105 mins

Delay Time = 120 mins + 180 mins + 120 mins + 90 mins + 60 mins = 570 mins

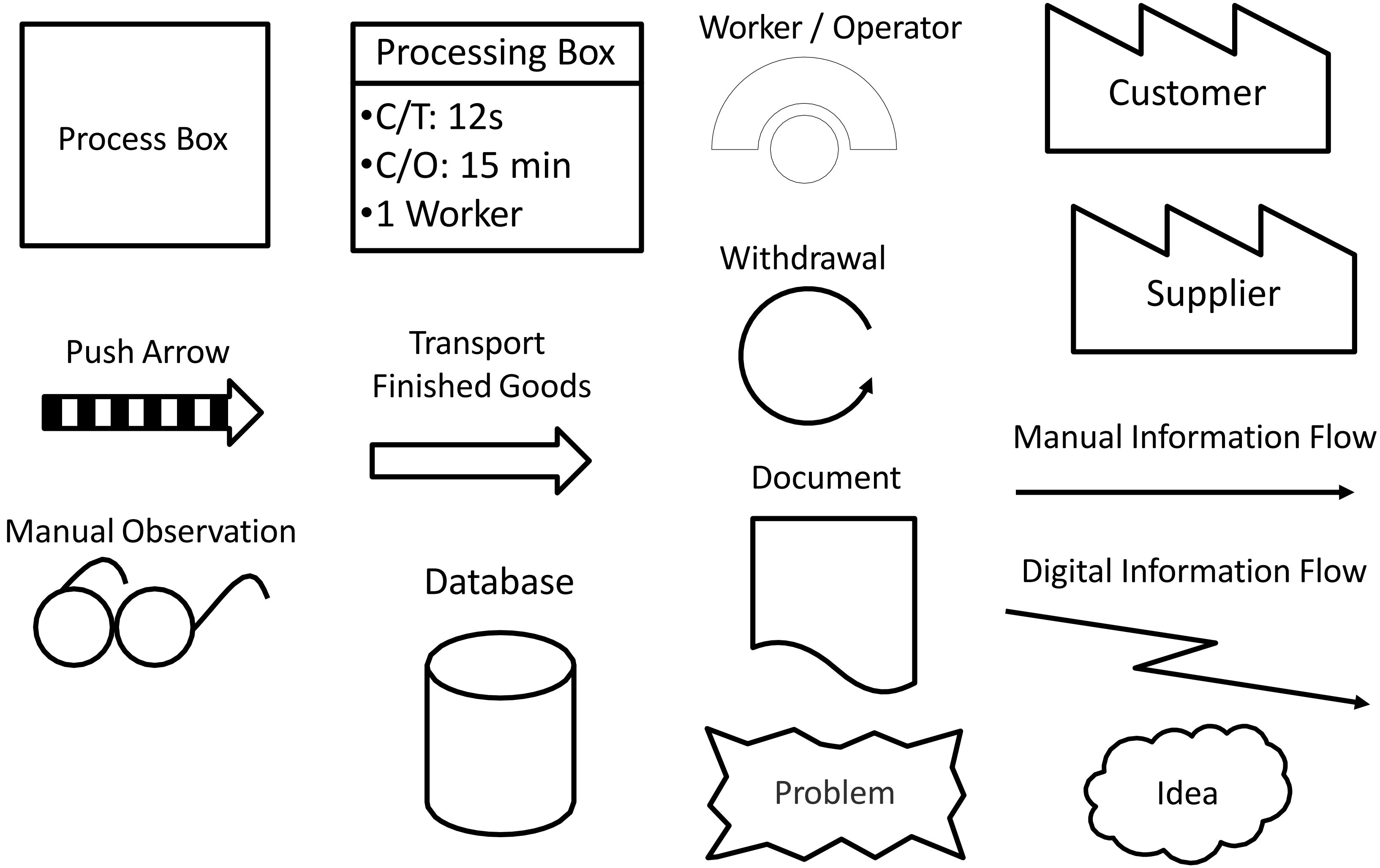
Lead Time = PT + DT = 675 mins

PROCESS CYCLE EFFICIENCY = $\dfrac{\textbf{Total Value Time}}{\textbf{Total Lead Time}}$



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Value Stream Mapping - Symbols



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Points to Remember

- Start with the customer – information flow
- Identify the product or service that is being worked on
- Determine your process steps from cradle to grave
- Identify the time it takes to perform the task without delays (starting or within the process) or interruptions within the total cycle time
- Identify and quantify the time it takes to perform the task including delays and interruptions– lead time (LT = CT + delays)
- Investigate the causes of the waste between processes – what are the barriers to flow?
- Calculate total processing time (cycle time) versus total lead time (throughput/turnaround)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Map The Process

Value Stream Mapping - Simulation

VSM Simulation Exercise

This case exercise is intended to develop the skill of reading and comprehending a complex customer input on the processes they follow, from where a BB is supposed to decipher and create an AS-IS flow map. Then deliberate on the same to identify the value, wastes and then evolve a possible, FUTURE state map and if time permits an IDEAL state map as well.

Setting is such that the roles are assumed to be at different corners of the office and all transmissions are physical exchanges and manual clerical job. Best possible effort needed at different locations are provided in seconds within brackets. All invoices must be out of office by the best time of 09:30AM before the morning Gemba rounds of Manager starts.

Post creation of the maps at different stages, it is expected that the professional must assess the lean efficiency metrics such as VA, NVA, Total Turn Around Time, and Takt time by simulating for 5 minutes assuming arrival pattern is in series with a fixed time interval to create a complete case work up. This is intended to be performed individually and then presented to coordinator.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Case Study

Bharat Paper Co. Ltd is into manufacturing and distribution of newsprint. Customer generates PO at a rate of 1 per 30 seconds and first note arrives at 0900AM. The delivery operator who receives the order adds time of collection, serial number at 09:03AM (90 seconds) and passed on to the incoming analyst who receives at 09:08AM who adds the locator code (70 seconds) from the code catalogue which takes up to 09:10AM.

Incoming analyst pass the same to delivery operator for verifying locator code (150 seconds), where it remains up to 09:15AM minutes. The filled up delivery note from delivery operator goes to incoming analyst in a different department requiring 2 minutes to reach and there adds distance code based on locator code and at 09:18AM (60 seconds) it passes to delivery operator where it remains till 09:20AM from where distance fee analyst takes possession of the same on which he creates a distance fee request slip based on distance code (180 seconds) which takes up to 09:25AM and returns to delivery operator for verification to identify issues in locator code, where it gets holed up till 09:30 AM and then passes on to customer accounts manager for verification that takes till 09:38AM (60 seconds).

Customer accounts manager calculates the total in invoice log and adds distance fee to invoice before both papers passed on to delivery operator (60 seconds). Delivery operator affixes validation stamp on invoice and without delay transmits the invoice to outward analyst who validates the invoice (45 seconds) by 09:45AM and sends to customer where customer logs the time of receipt at 09:50AM.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Inference from Define Phase

At the end of the Define phase, we should be able to identify the problem from all the following perspectives:

- Define the problem statement
- Identify your customers
- Identify the CTQ
- Create high-level process map
- Identify team members and business functions required
- Develop a project charter

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Measure Phase

Measure Phase - Roadmap

Types of Data + Data Distribution → Develop sampling strategy → Introduction to Minitab → Develop a Data Collection Plan → Validate Measureme nt system → Determine process capability

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Types of Data

Types of Data - Discrete and Continuous

Discrete (Attribute) Data

- Binary (Yes/No, Defect/No Defect)
- Ordered categories (1-5)
- Counts

Examples

- Number of incomplete applications
- Percent of responding with a “5” on survey
- Number of Green Belts trained

Continuous (Variable) Data

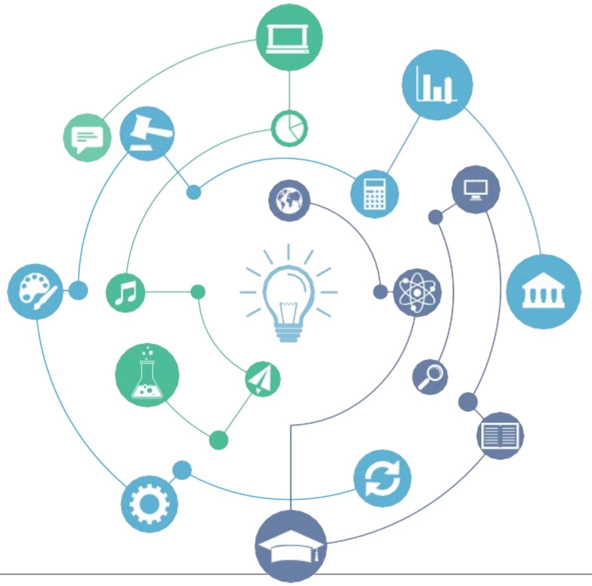
- Can be broken down into increments
- Infinite number of possible values

Examples

- Cycle time (measured in days, hours, minutes, etc.)
- Weight (measured in tons, pounds, etc.)

Why Data Type Important ?

- Choice of data display and analysis tools
- Amount of data required: continuous data often requires a smaller sample size than discrete data
- Information about current and historical process performance



KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Exercise - Type of data

1. Percent defective parts in hourly production
2. Percent cream content in milk bottles
3. Time taken to respond to a request
4. Number of blemishes per square yard of cloth, where pieces of cloth may be of variable size
5. Daily test of water acidity
6. Number of accidents per month
7. Number of defective parts in lot of size 100
8. Length of screws in samples of size ten from production lot
9. Number of employees who took leave in the last 5 years

KPMG

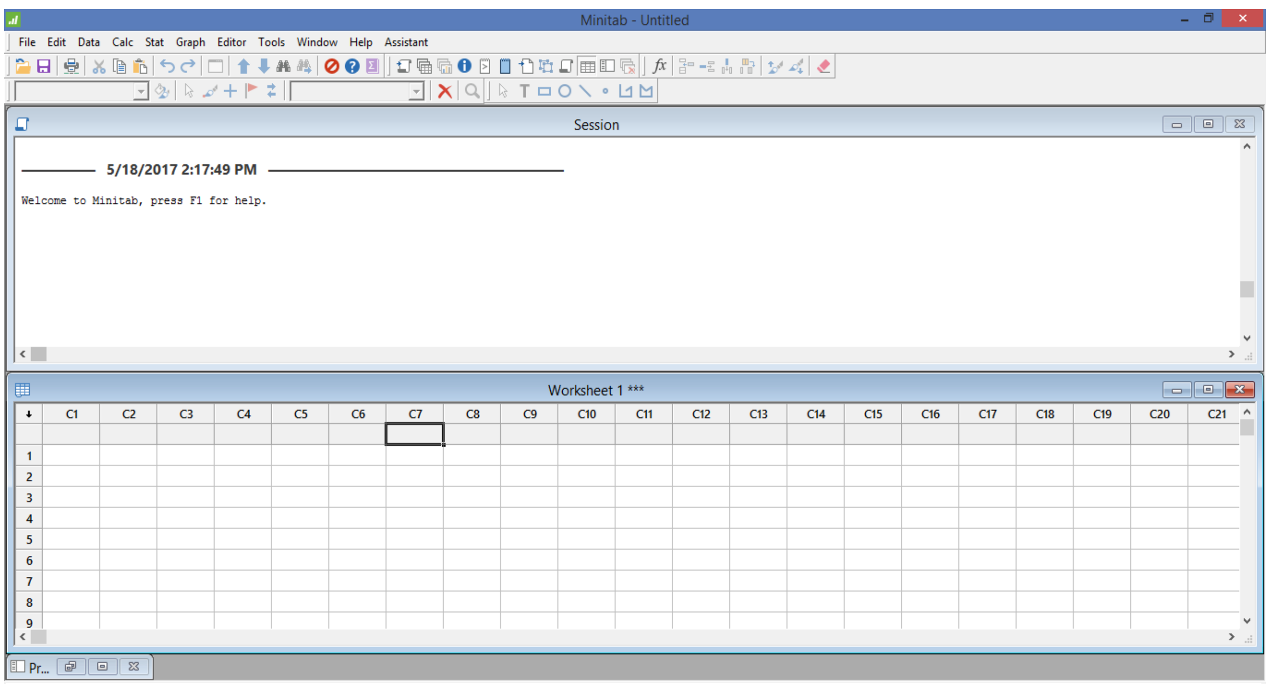
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Introduction to Minitab

Introduction to Minitab

- Minitab is a statistical software that is widely used across businesses and industries to solve or analyze complex statistical problems

- Minitab provides convenient features that streamline your workflow, a comprehensive set of statistics for exploring your data, and graphs for communicating your success.



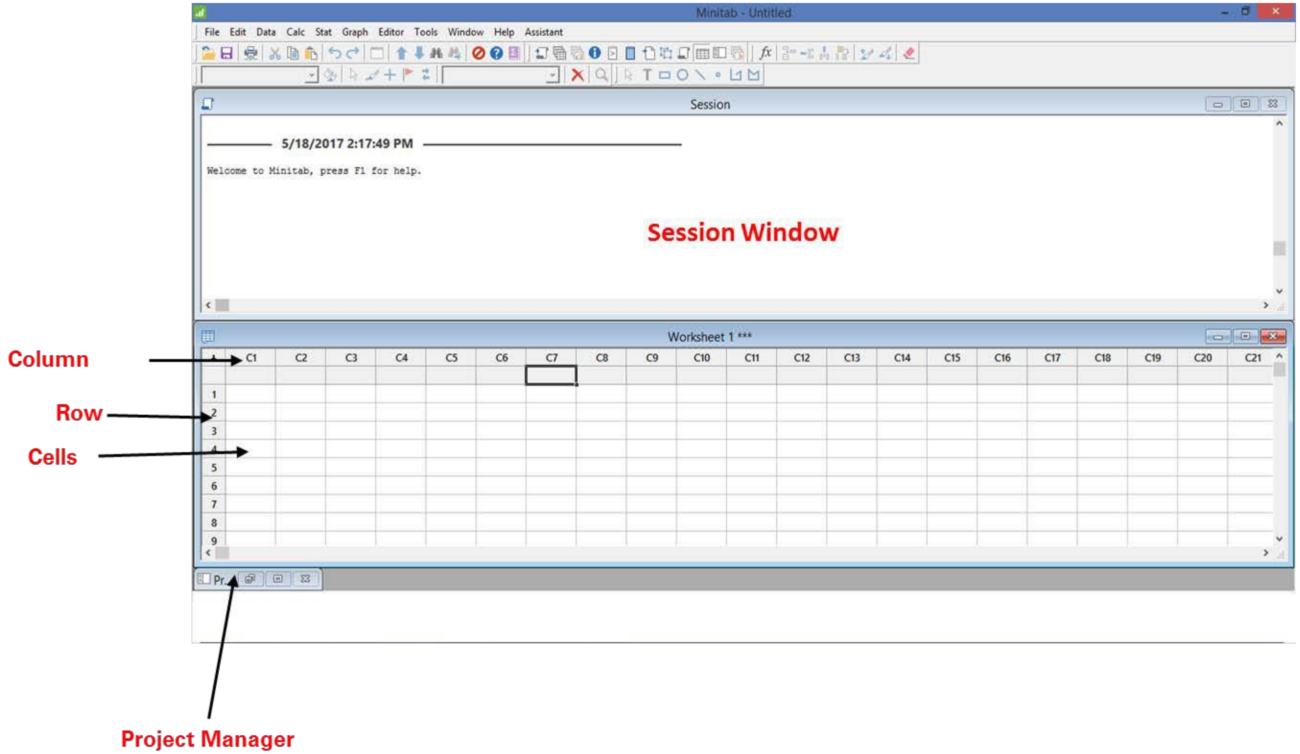
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Introduction to Minitab

1. **Session Window:** The Session window displays the results of your analyses in text format. Also, in this window, you can enter session commands instead of using Minitab's menus.

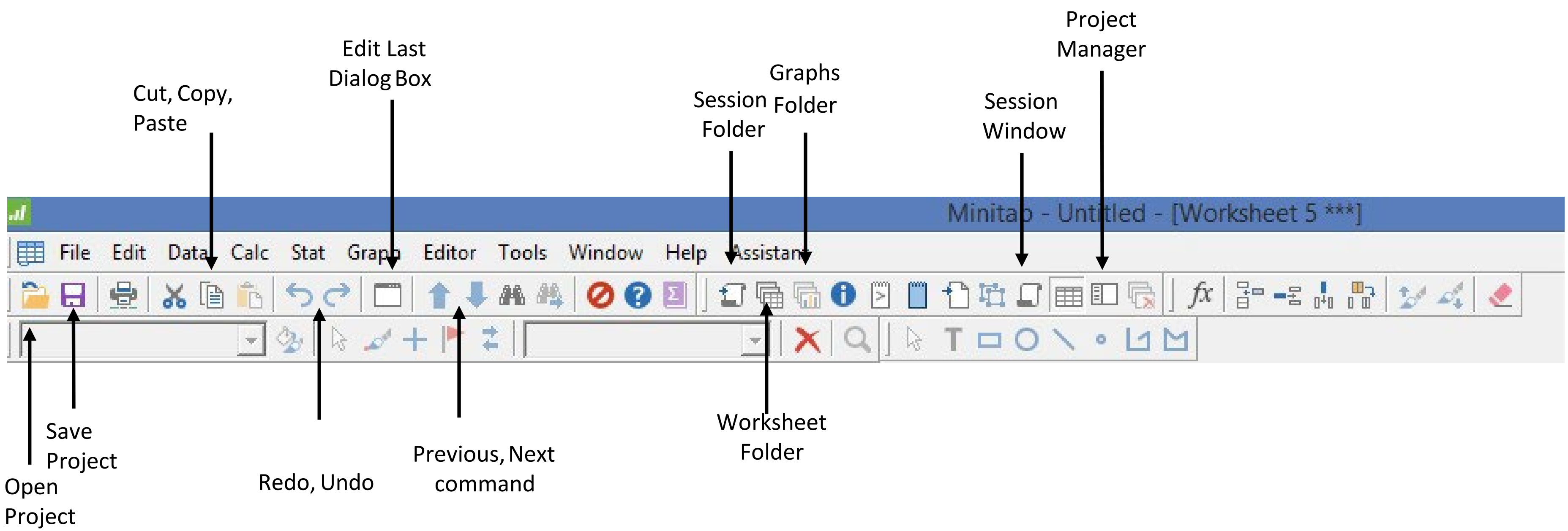
2. **Worksheet:** The worksheet, which is similar to a spreadsheet, is where you enter and arrange your data. You can open multiple worksheets.

3. **Project Manager**: The third window, the Project Manager, is minimized below the worksheet.



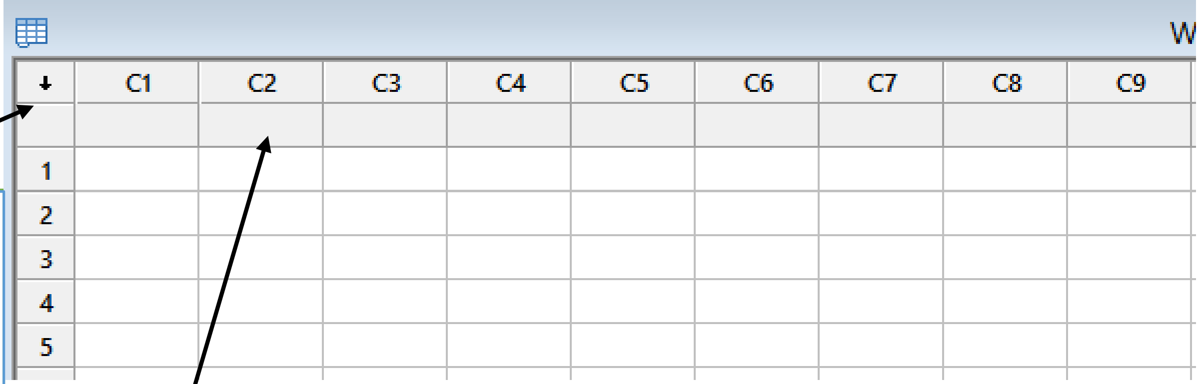
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Window Elements



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Worksheet Elements



The Data Entry Arrow

There is a data entry arrow above row 1, which indicates the direction the cursor will move after the "Enter" key is pressed

Column Name Row

The column name row is located just above row 1of the worksheet. Column names can be up to 31 characters and may contain space

Remember
When data is entered into Minitab, the program will read it as one of the three formats:

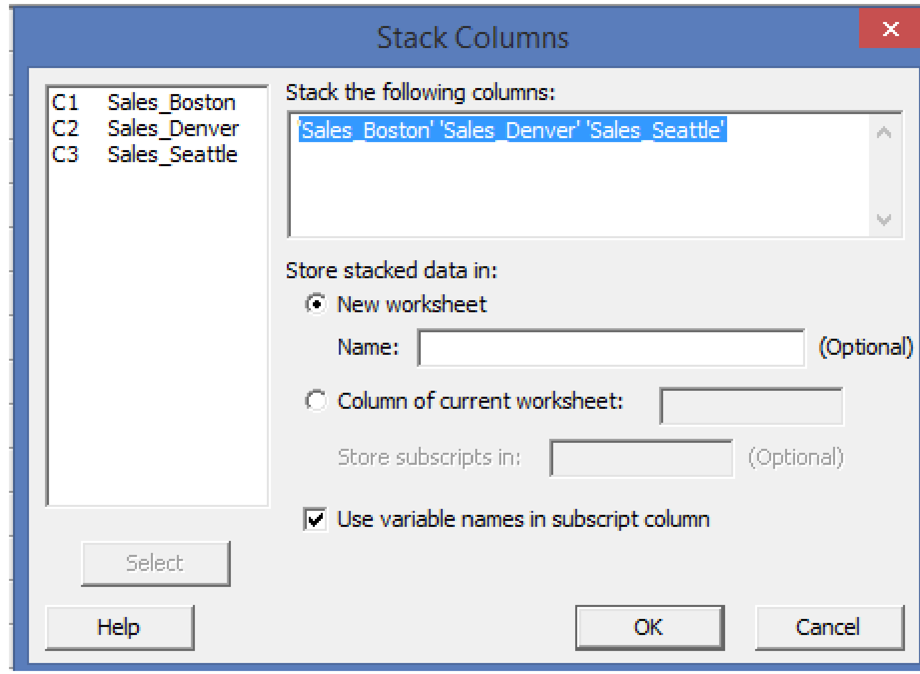
- Numeric
- Text
- Date / Time

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Combine data into single column

Stack Data

C1	C2	C3
Sales_Boston	Sales_Denver	Sales_Seattle
36	52	63
32	46	71
35	51	68
29	50	66



- Stack Data

- Go to Minitab
- Select: Data>Stack Columns
- Double click C1, C2 and C3 and the data is put in the variables box
- Click OK

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Combine data into single column

Stack Data

C1-T	C2
Subscripts	
Sales_Boston	36
Sales_Boston	32
Sales_Boston	35
Sales_Boston	29
Sales_Denver	52
Sales_Denver	46
Sales_Denver	51
Sales_Denver	50
Sales_Seattle	63
Sales_Seattle	71
Sales_Seattle	68
Sales_Seattle	66

- Minitab can stack data from to multiple columns to single column

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

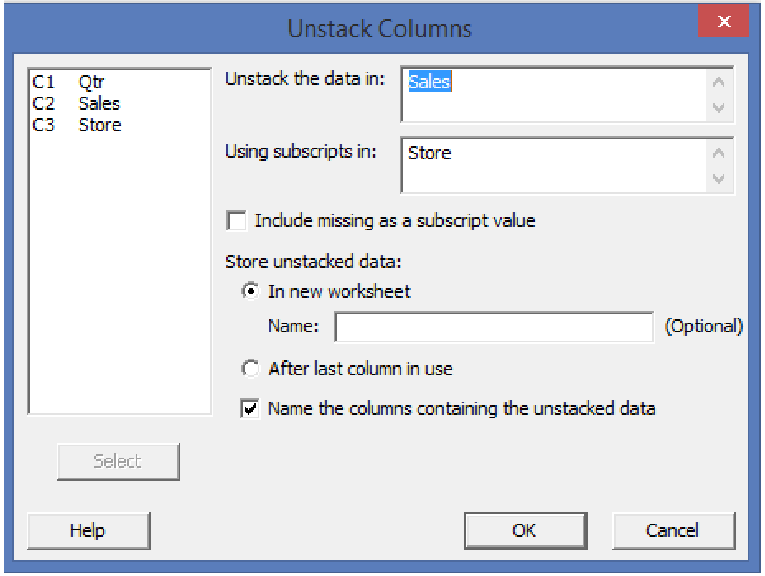
Distribute data into different columns

Unstack Data

Qtr	Sales	Store
1	52	Denver
1	36	Boston
1	63	Seattle
2	46	Denver
2	32	Boston
2	71	Seattle
3	51	Denver
3	35	Boston
3	68	Seattle
4	50	Denver
4	29	Boston
4	66	Seattle

Unstack Data

- Go to Minitab
- Select: Data>Unstack Columns
- Double click C2 and C3 and the data is put in the variables box
- Click OK



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Distribute data into different columns

Unstack Data

C1	C2	C3
Sales_Boston	Sales_Denver	Sales_Seattle
36	52	63
32	46	71
35	51	68
29	50	66

- Minitab can unstack data from single column to multiple columns

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Format for import to Minitab

When a data is imported to Minitab, it expects the data to fulfil following conditions for a proper interpretation:

- Unlike MS Excel, Minitab does not have blank rows between the column name and row name
- There are no total rows in the worksheet
- Columns must not contain special characters or symbols

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Import

At time, you may have to enter or import data into a Minitab worksheet before you start an analysis.

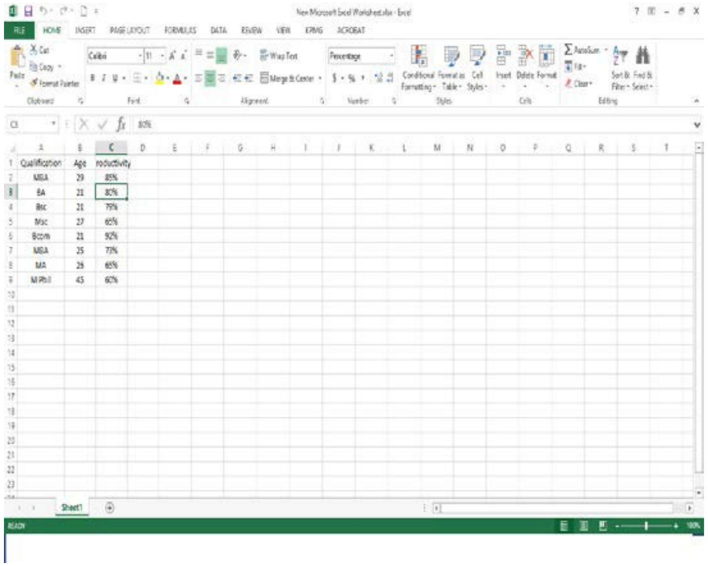
You can enter data in a Minitab worksheet in the following ways:

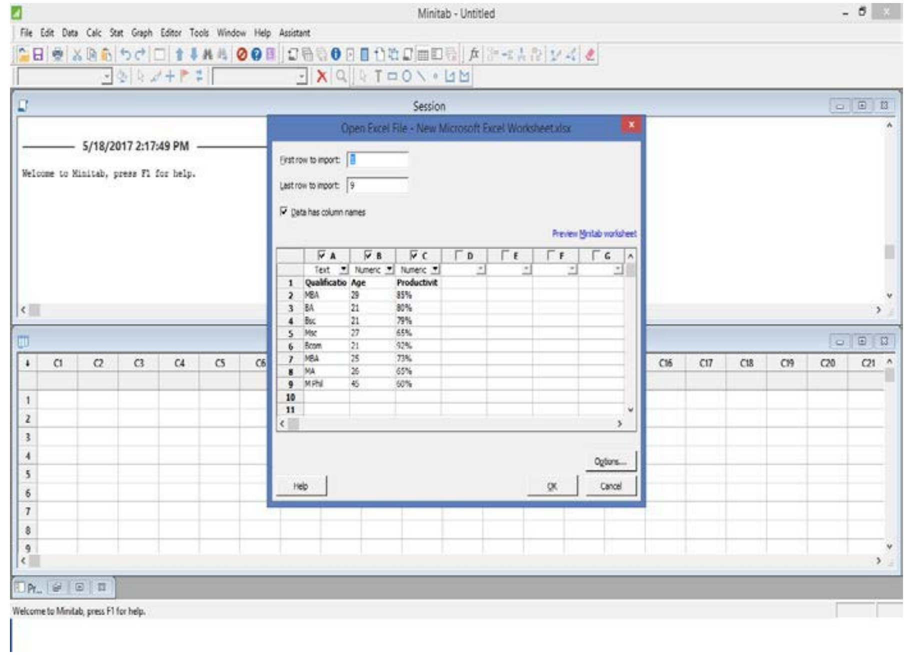
- Type the data directly into the worksheet.
- Copy and paste the data from other applications.
- Import the data from Microsoft Excel files or text files.

After your data are in Minitab, you might need to edit cells or reorganize columns and rows to prepare the data for analysis. Some common manipulations are stacking, specifying column names, and editing data values.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Import Excel file to Minitab

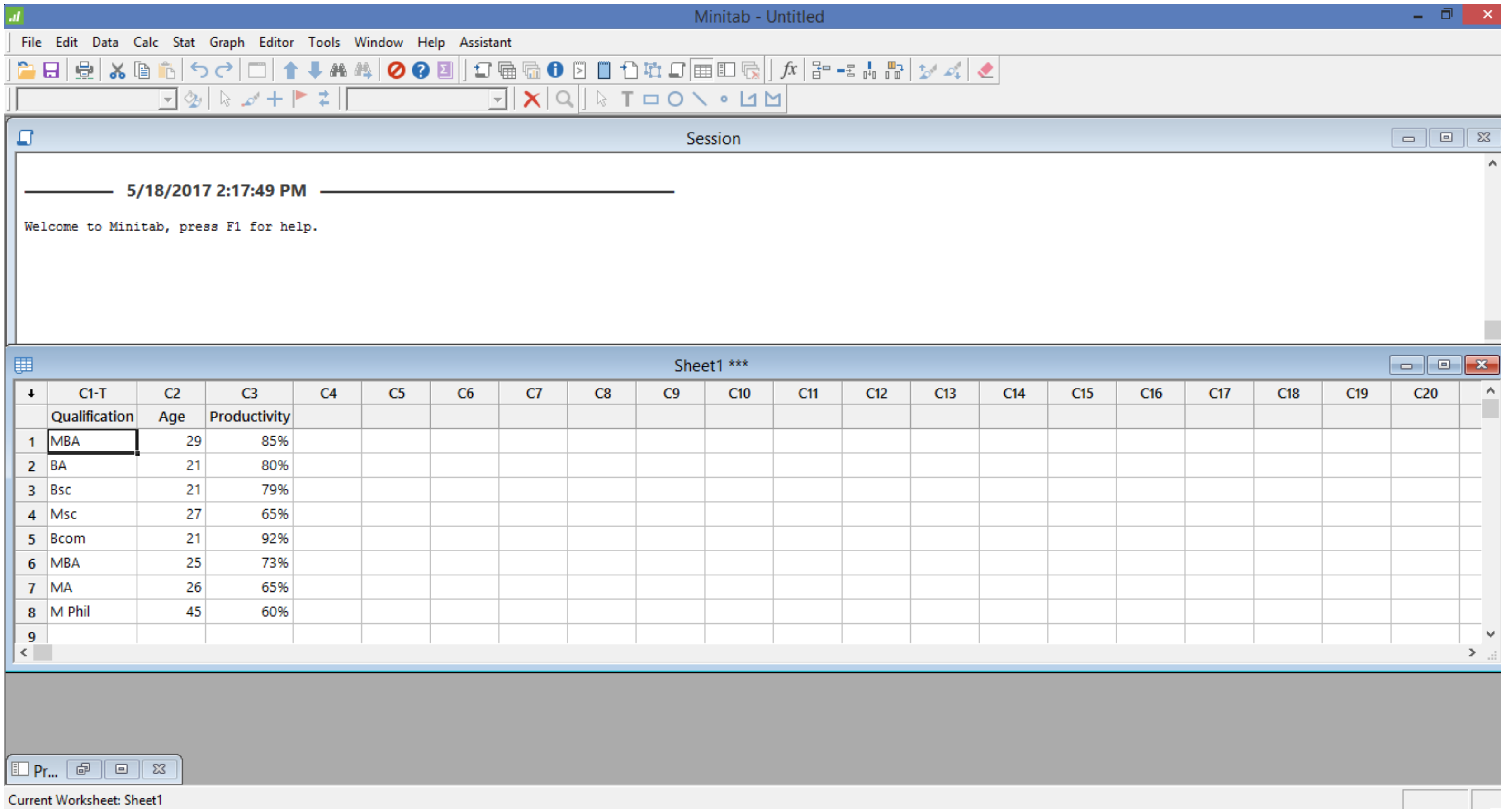




- Export MS Excel file to Minitab.
- Open the excel file from Minitab and clickok

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Import Excel file to Minitab



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Projects and Worksheets

In a project, you can perform analyses, and generate graphs. Projects contain one or more worksheets.

Project (.MPJ) files store the following items:

- Worksheets
- Graphs
- Session window output
- Session command history
- Dialog box settings
- Window layout
- Options

Worksheet (.MTW) files store the following items:

- Columns of data
- Constants
- Matrices
- Design objects
- Column descriptions
- Worksheet descriptions

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Types of Distribution

What is Probability?

- Probability is a chance of occurring an event.
- Probability is quantified as a number between 0 and 1, where, 0 indicates impossibility and 1 indicates certainty.
- The higher the probability of an event, the more certain that the event will occur.

•Probability of an event is the ratio of chance favoring the event by total possible event For instance:

$$\textit{Probability of an event} = \frac{\textit{Chances favoring the event}}{\textit{Total possible events}}$$

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Fundamentals of Probability

An Event: An event is one or more of the possible outcomes of doing some things.

For instance: If we toss a coin, getting a tail is an event, and getting a head is another event.

An Experiment: An experiment is an activity that produces an event. For

instance: Tossing a coin, Drawing a card from a deck of cards.

Sample Space: The set of all possible outcomes of an experiment is called the sample space for the experiment.

For instance: In a coin toss experiment, sample space is {head and tail}.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Fundamentals of Probability

- Two events are said to be *mutually exclusive* if one and only one of them can take place at a time. In tossing a coin, only Head or Tail can occur

- When outcome of one event does not influence the outcome of another event, the two events are called independent events. In tossing a coin, 1st tossing and 2nd tossing are independent

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Fundamentals of Probability

Let us assume tossing of a coin is an experiment.

Here,

Sample Space S = {head, tail}; where

Event 1- getting the head;

Event 2- getting the tail;

What is the probability of getting a head?

Probability (p) (getting head) = 1/2

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Types of Distributions

- **Discrete:** Applied to variables with specific outcomes; for example - heads or tails, success or failures, conforming or non-conforming

- **Continuous:** Represent populations that can have infinitely many values usually within a finite range ; for example – length, height, weight, time

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Distribution

Characteristics of variable data

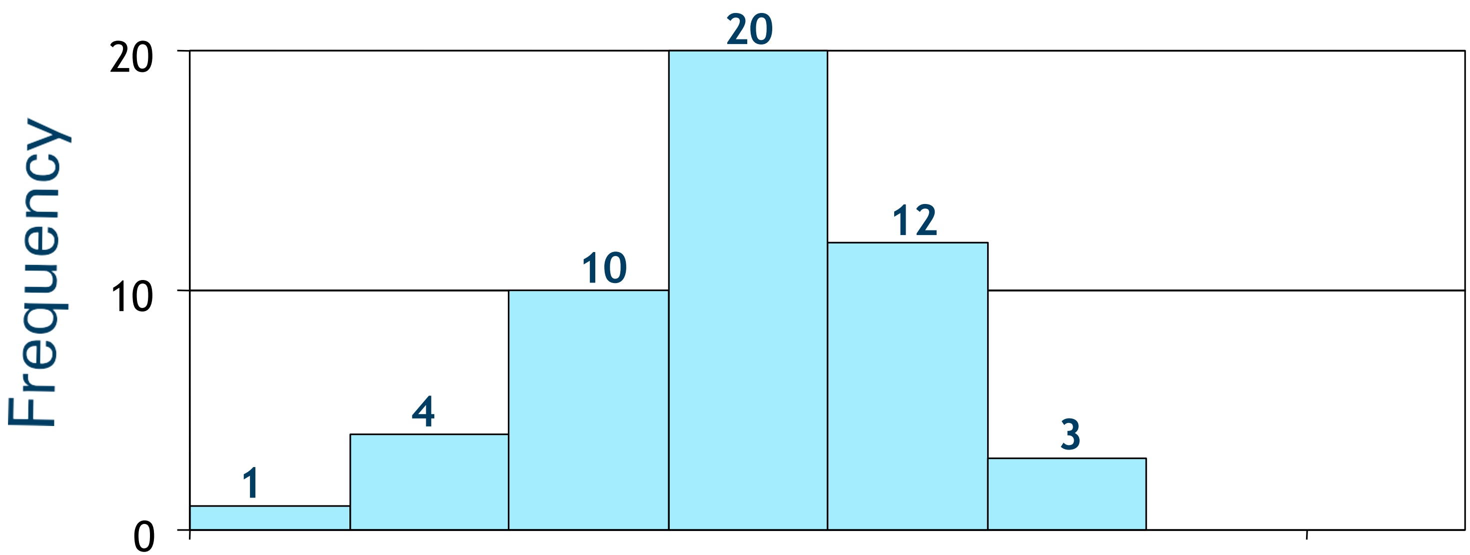
The following characteristics of a data set can provide considerable insight:

- Shape (Histogram)
- Central Tendency (Mean, Median, Mode)
- Variation (Range, Standard Deviation, Variance)
- Distribution (Normal and Skewed Distributions)
- Graphical Analysis (Histogram, Box Plot)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Histograms

- A histogram is a frequency polygon in which data are grouped into classes.
- The height of each bar shows the frequency in each class.



KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Example: Histogram

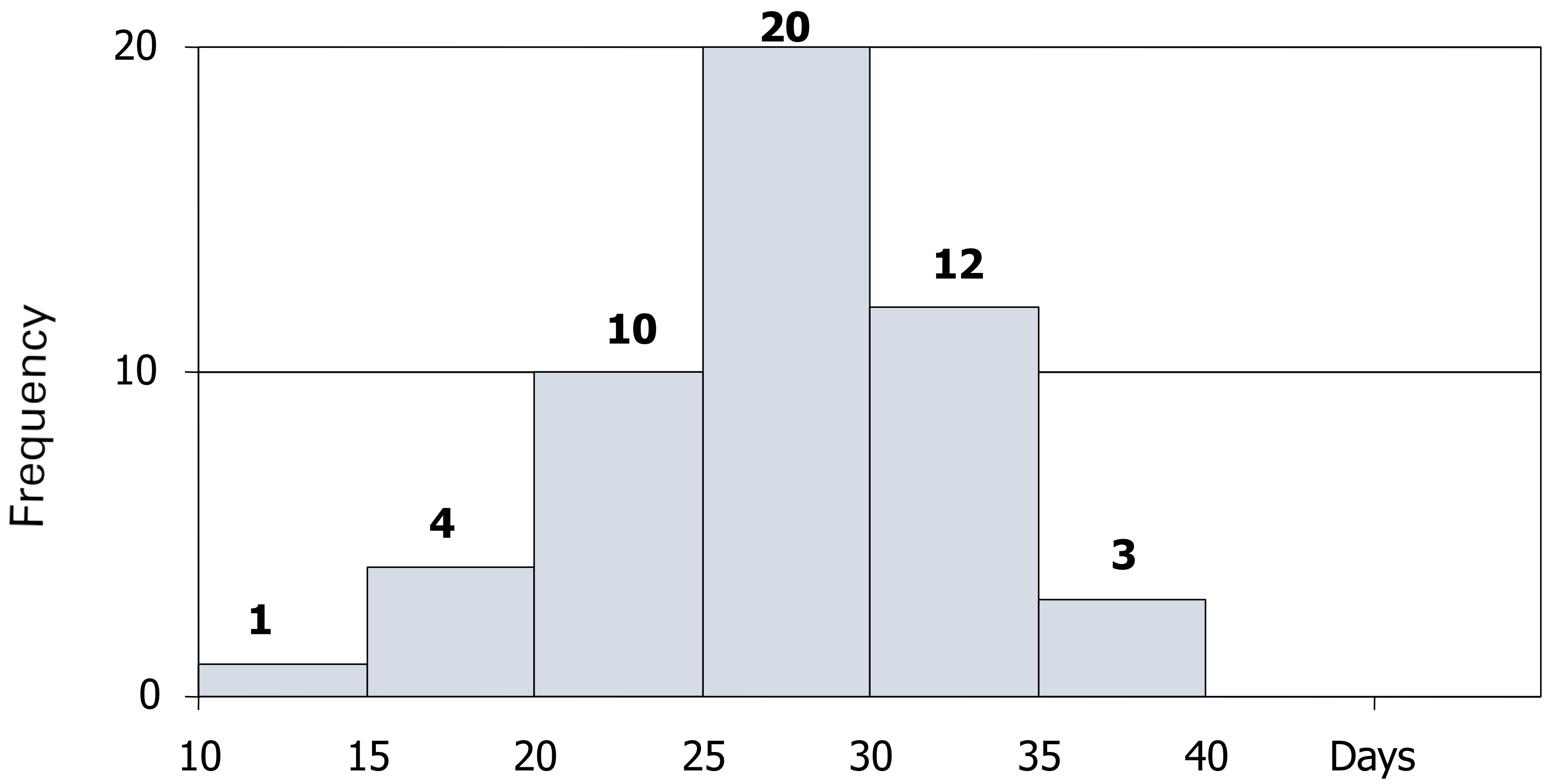
For each of 50 placements, the time (in days) it took to place a person in the position was recorded.

22	26	30	19	22	31	34	29	28	18
16	22	31	24	26	36	28	33	36	24
26	27	35	14	26	30	33	26	31	36
28	33	18	26	29	30	22	30	24	31
27	21	28	35	32	28	33	28	23	25

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Histograms

A histogram is a bar graph in which data are grouped into classes. The height of each bar shows how many data values fall in each class.



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Histograms

- When creating a histogram, the data must be properly grouped in order to understand the shape of the data distribution.
- For the given sample size, the right number of classes should be used.

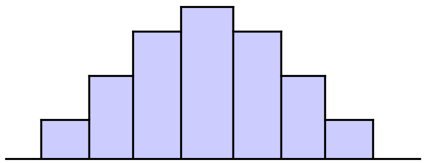
Number of Data Points	Number of Classes
Under 50	5-7
50 – 100	6-10
100 – 250	7-12
Over 250	10-20

In Minitab, refer the file ‘PotatoChip.MTW’ to create a histogram.

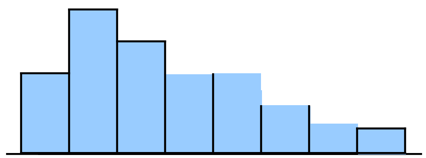
What might you conclude from the histogram?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

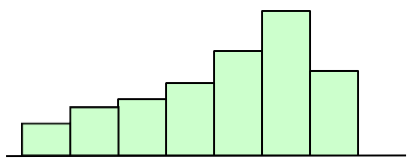
Shapes of Data Sets



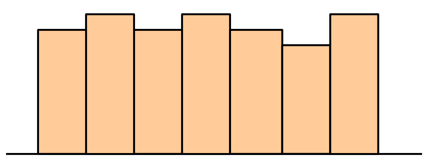
Bell Shape – The Normal Distribution



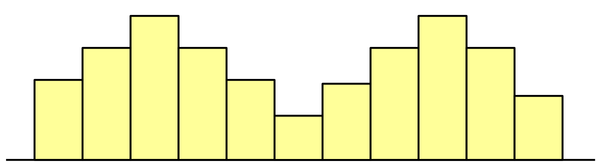
Right Skewed (Positively Skewed)



Left Skewed (Negatively Skewed)



Uniform Distribution



Bimodal Distribution

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Measures of Central tendency

Placement Time for an Analyst's Positions (in days) 22, 26,

26, 31, 33, 37, 37, 42, 52, 52, 52, 57, 59

Mean or Average
The sum of the values in a data set divided by the number of values.

$\overline{X}$ = 40.5 days

Mode
The most frequently occurring data value.

Mode = 52 days

Median
The middle observation in the data set that has been arranged in ascending or descending order.

Median = 37 days

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Measures of Dispersion

Placement Time for an Analyst's Positions (in days) 22, 26, 26, 31, 33, 37, 37, 42, 52, 52, 52, 57, 59

Range
The largest data value minus the smallest data value:

Range = Max - Min

Range = 37

Variance (s^2)
The average deviation of all data values from the mean.

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$s^2 = 162.67$

Standard Deviation (σ)
The square root of variance is standard deviation.

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

$\sigma = 12.74$

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Standard Deviation

Essentially, the standard deviation is representation of the deviation of individual data values from the sample mean.

Why Standard Deviation?

- Unlike the range, the standard deviation takes into account all the data values inthe sample.
- Unlike the variance, the standard deviation has the same units of measurement as the original data.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Normal Distribution

- Normal or Gaussian distribution is a descriptive model that describes real world situations.
- It is defined as a continuous frequency distribution of infinite range (can take any values not just integers as in the case of Binomial and Poisson distribution).
- This is the most important probability distribution in statistics and important tool in data analysis.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

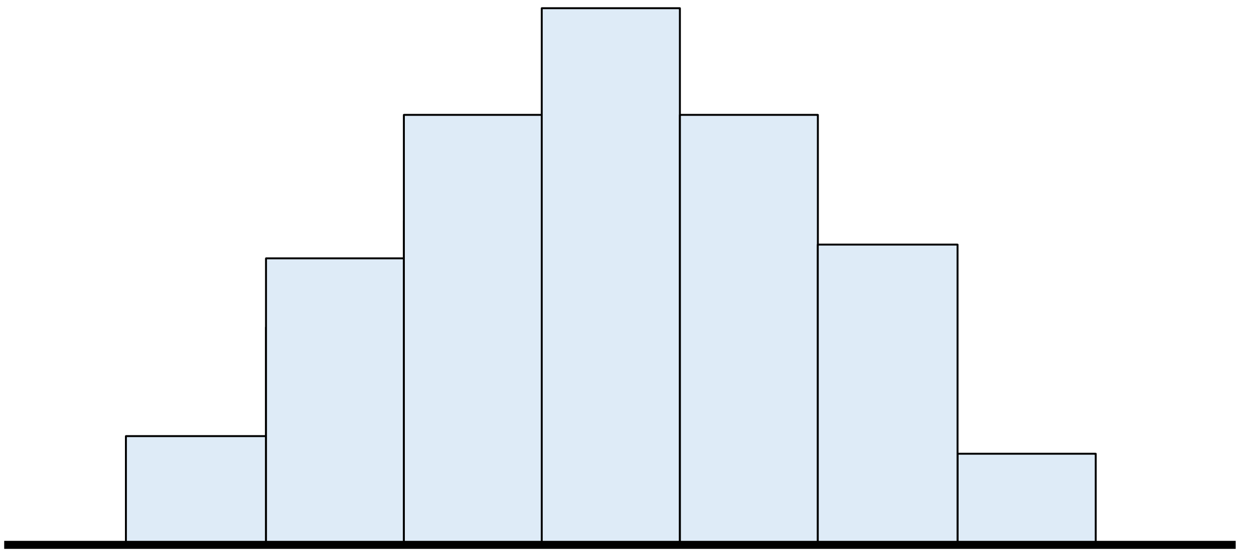
Characteristics of Normal Distribution

- It links frequency distribution to probability distribution
- Has a bell shape curve and is symmetric
- It is symmetric around the mean: two halves of the curve are the same (mirror images),
- In a perfectly centered Normal Distribution; mean = median = mode
- The total area under the curve is 1 (or 100%)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

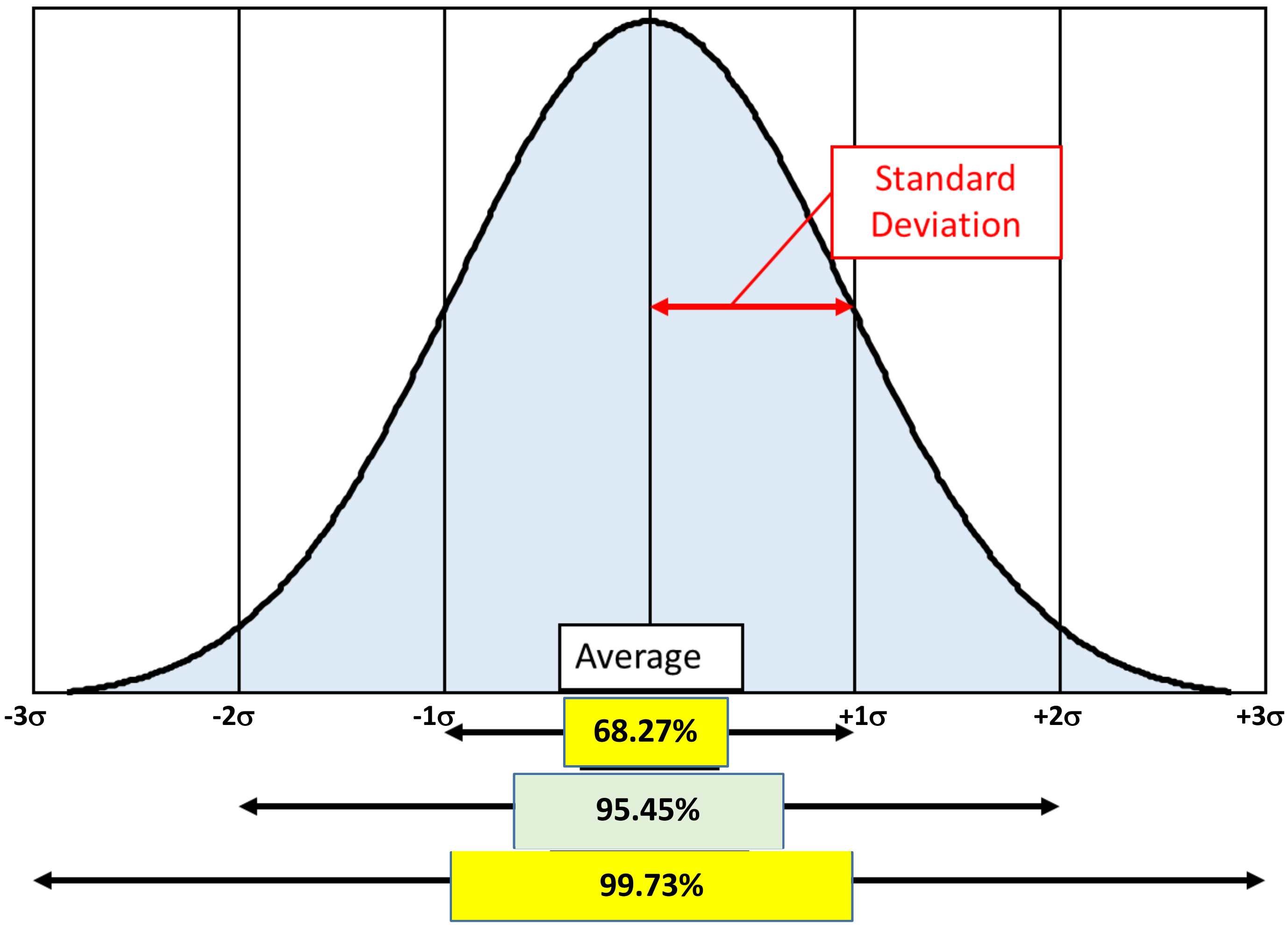
Normal Distribution

Since many process outputs have this shape, the properties of the normal curve canbe used to make predictions about the process population.



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Properties of Normal Distribution



KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Test for Normality

A normal curve originates from a histogram. A histogram is a frequency distribution chart showing the number of times a given value of the parameter we are trying to measuring occurs.

Minitab uses the Anderson-Darling test to determine if a set of data can be treated as normal data.

Interpreting the P-value:

The P-value is the probability of getting the particular sample if the population is normal.

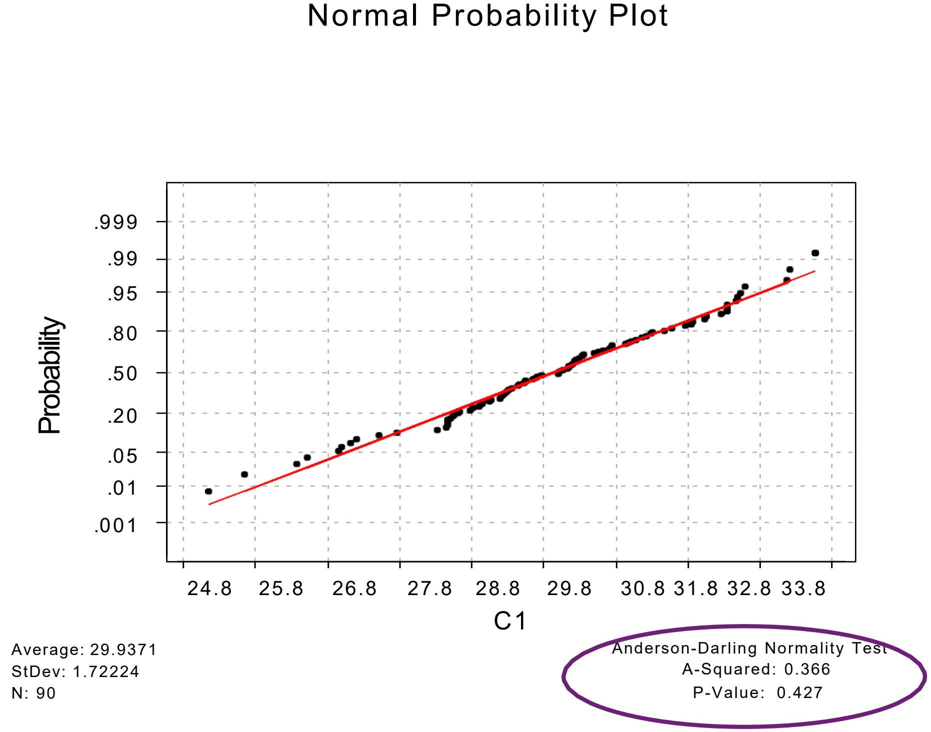
P-value < 0.05 means that the chance of getting this sample from a normal population is very small (less than 5%).

Stat > Basic Statistics >Normality Test

Null Hypothesis: Data is Normal

Alternate Hypothesis: Data is not Normal

Since the P-value > 0.05 we conclude that the data falls a normal distribution



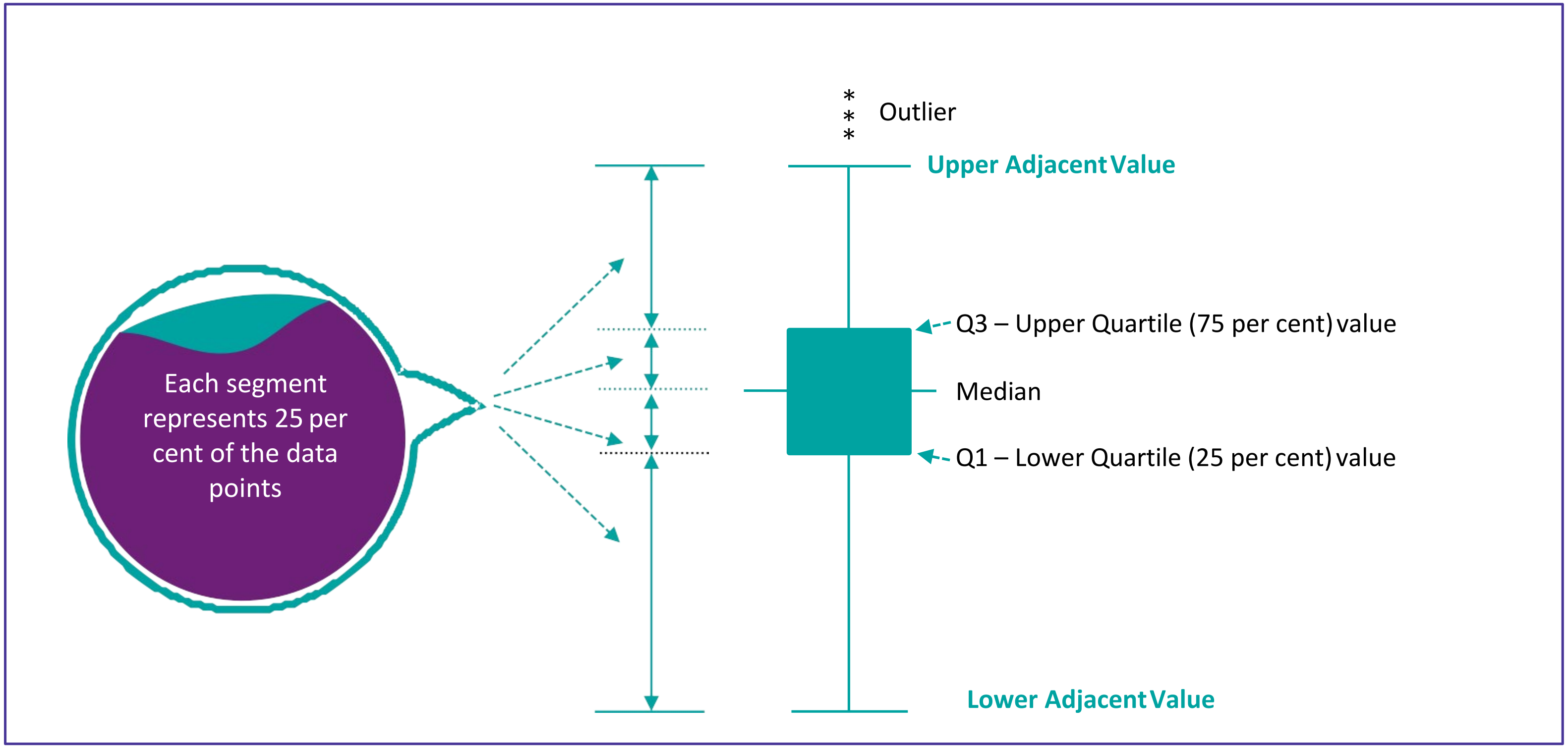
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Box plot

- Box Plot is a graphical tool to display central tendency (median) and dispersion (range)
- Box Plot enables to understand the distribution of data (quartiles)
- Box Plot gives the location of data
- Box Plot enables to get a quick comparison of two or more processes
- Box Plot is usually used at the initial stages of data analysis
- Box Plot indicates imminent instability in the process, through illustration of outliers

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Box plot



We can calculate the Inter Quartile Range (IQR): Q3 – Q1

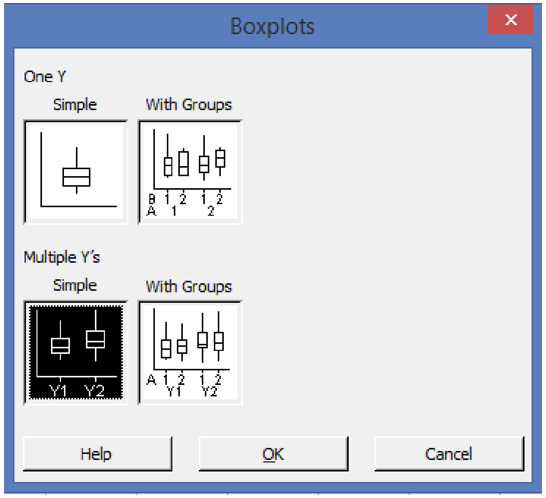
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

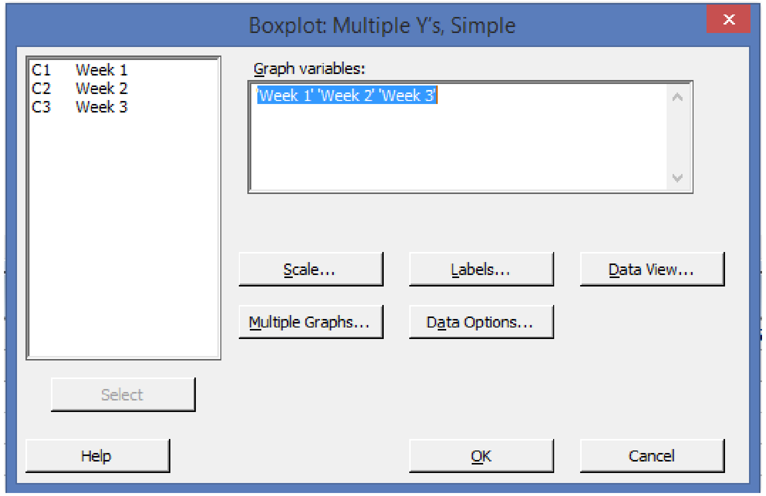
Things to look for in a Box plot

- Are the boxes about equal or different sizes?
- Do the groups appear normal or skewed?
- Are there any outliers?

Refer the data sheet – Pipe.MTW in Minitab

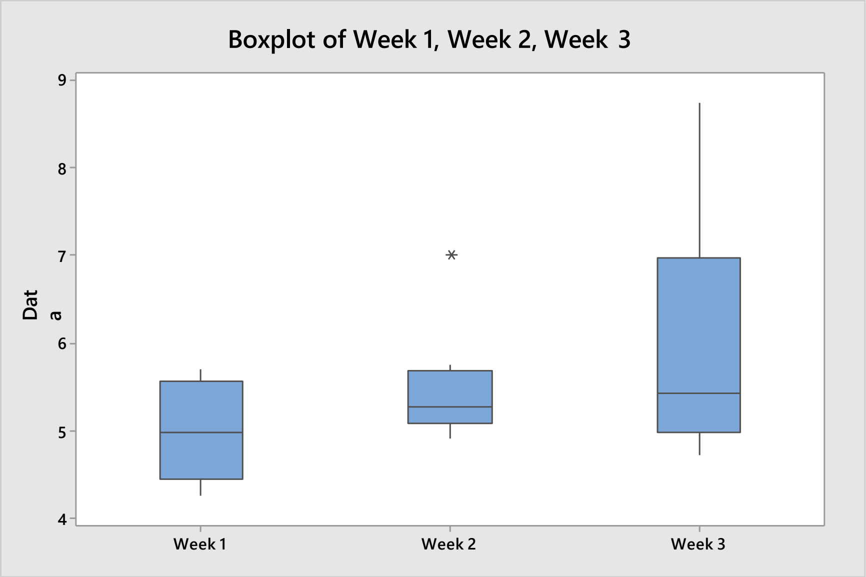
Choose Graph> Box Plot > Multiple Y's Simple





© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Box Plot using Minitab



Interpretation:

- Week 1 median is 4.985, and the interquartile range is 4.4525 to 5.5575.
- Week 2 median is 5.275, and the interquartile range is 5.08 to 5.6775. An outlier appears at 7.0.
- Week 3 median is 5.43, and the interquartile range is 4.99 to 6.975. The data are positively skewed.
- The medians for the three weeks are similar. However, during Week 2, an abnormally wide pipe was created, and during Week 3, several abnormally wide pipes were created.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Distribution - Discrete Data

What is Defective?

- A defective is a non-conforming unit.
- Examples would be a late payment, rejected trim cover, rejected foam pad, a missed putt in golf.
- p charts are designed for binomial data (0 or 1, Pass or Fail, Good or Bad).

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

What is a Defect?

- A defect is a non-conformity.
- Examples would be scratches on a trim cover, errors on an invoice, the number of tears on foam pad, a number of shots out of bounds per round of golf.
- u charts are based on the Poisson distribution (Defects per Unit, Number of accidents per 1000 hours).

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Discrete Distribution

- Binomial Distribution
- Poisson Distribution

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Binomial Distribution

Binomial is the name of the Mathematician who discovered Binomial Distribution Conditions

to Apply Binomial Distribution

- Events must be independent.
- Two possibilities only for any outcome.
- Probability of success + Probability of failure =1
- Total number of trials is the sample size, n.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Poisson Distribution

- Poisson distribution named after French mathematician Siméon Denis Poisson
- It is a discrete probability distribution that is primarily directed at populations with rare events
- Used as basis for control charts that deal with non-conformities

Basic assumptions

- Non-conformities should be independent and random
- Potential number of non-conformities must be quite large (infinite) and the actual number quite small
- Usually reserved for situations where the non-conformity rate is very small

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Develop Sampling Strategy

Sampling: Overview

- The Six Sigma team would always face a question 'How much data do we need to have a valid sample?' Though an important part of data collection is to obtain a sample of reasonable size, it is one of many questions to be addressed during the planning and development of a data collection strategy. Sample size is just one aspect of a valid data collection activity
- The validity of the data is impacted by many things: For example, operational definitions, data collection procedures and recording
- In process improvement, there are a number of questions to keep in mind relative to sampling:
- Is the data representative of the situation or is bias possible?
- Why am I sampling? To improve or control a process or to describe some characteristic of a population?
- What are the key considerations for either a process or population situation?
- What is the approach to sampling (e.g., random, systematic, etc.) and approximately how many to sample

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Population and Sampling

An entire set of items is called the *Population*.

What is Sampling?

- The small number of items taken from the population to make a judgment of the population is called a *Sample*.
- The numbers of samples taken to make this judgment is called *Sample size*.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Sampling Strategy

1. **Random Sampling**

Samples drawn at random.

That is, at any point in time, each unit in a “lot” has an equal chance of being the next unit selected for the sample.

Example: Cycle tyres produced in the assembly line have a random check of 10 per cent (QC) on daily production

2. **Stratified Sampling**

An attempt to draw the sample proportionately over the full operating range of the process.

For example: various batches of material; small and large contracts; all three shifts

Example: Every third tyre used with an aircraft is cut open and checked for quality

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Sampling Methods

1. **Simple Radom Sampling:** Every unit has the same chance of being selected.

Example: Survey across organization to know –"What percentage of employees have visited the intranet in last seven days. Select employees from Population at random And collect data.

2. **Stratified Random Sampling:** Random sampling from proportional subgroups of the population

Example: Average cycle time for LC issuance process of different countries. Each country is a strata (segment). Collect random data from each strata

3. **Systematic Sampling:** Includes every *k th* unit. The formula is

k = N/n (where N is population size and n is the sample size)

Example: Suppose you want to sample 10 houses from a street of 150 houses. 150/10=15, so every 15th house is chosen

4. **Rational subgroups:** Subgrouping is the process of putting measurements into meaningful groups to better understand the important sources of variation.

Example: While studying the arrival rate of documents as dispatched by the customer. The entire day is split up into quadrants, rational being the arrival rate of documents is similar within each quadrant and different between quadrants

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

When to Sample?

When to sample?

- Collecting all the data is impractical
- High cost implications due to population study
- Time availability
- Data collection can be a destructive process (crash testing of cars)
- When measuring a high-volume process

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Sampling bias

- Sampling must be representative to enable solid conclusions.
- The data have to represent the population or process.
- There should be no systematic (non-random) difference between the data you collect or do not collect.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

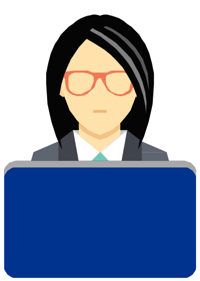
Determining the Sample Size

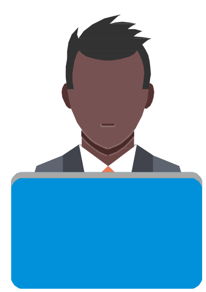
Three metrics that determine the sample size

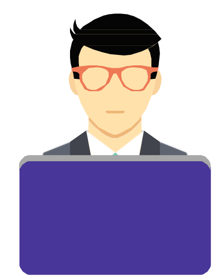
- **Level of confidence (Zc):**
  - “How confident I am that the result represents the true population”
  - The team needs to be sure of the adequate representation of the population with the sample chosen
  - For higher levels of confidence the sample size increases

- **Precision (Δ)**
  - “How accurate is the result or what are the errors or uncertainty in my result?”
  - Precision improves as sample size increases
  - Higher the precision, larger is the sample size required

- **Standard deviation of the population (σ)**
  - How much variation is in the total data population?
  - As standard deviation increases, a larger sample size is needed to obtain reliable results







Sample size is not dependent on the ‘populationsize’

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Determining the Sample Size - Continuous Data

The formula for calculating the sample size for continuous data is

$$n = \left[\frac{Z_c \sigma}{\Delta} \right]^2 \qquad n = \left[\frac{1.96 \sigma}{\Delta} \right]^2$$

Where:

n = minimum sample size

σ = estimate of standard deviation of the population

Δ = level of precision desired from the sample in units of proportion

1.96 = value of Zc at 95% confidence

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Determining the Sample Size - Example

Estimate the sample size required for correctly determining the average duration (AHT) of incoming phone calls with 3% precision. Historical data for the population shows a typical standard deviation of 1 minute.

$$n = \left[\frac{1.96\,\sigma}{\Delta} \right]^2$$

$$n = \left[\frac{1.96 * 1}{0.03} \right]^2 \qquad n = \left[\frac{1.96}{0.03} \right]^2$$

Thus Sample Size = n = 4270 samples

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Determining the Sample Size - Attribute Data

For Discrete Proportion data

$$n = \left[\frac{1.96}{\Delta} \right]^2 P(1-P)$$

Where:

n = minimum sample size

P = estimation of the proportion of the population or process which is defective

Δ = level of precision desired from the sample in units of proportion

1.96 = value of Zc at 95% confidence

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Determining the Sample Size - Example

Suppose you want to estimate within 2% of customers who will buy a new product. You're guessing that 50% of them will buy. What sample do you need?

$$n = \left[\frac{1.96}{\Delta}\right]^2 P(1-P)$$

$$n = \left[\frac{1.96}{0.02}\right]^2 0.5\,(1-0.5)$$

Thus Sample Size = n = 2400 samples

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Determining the Sample Size - Example

How many customers do you need to sample for your estimate to be within 4%

$$n = \left[\frac{1.96}{\Delta} \right]^2 P(1-P)$$

$$n = \left[\frac{1.96}{0.04} \right]^2 0.5\ (1-\ 0.5)$$

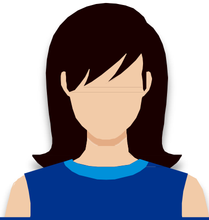
$$n = 625$$

625 customers needed

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Develop a Data Collection Plan

What is Segmentation and Stratification?



Segmentation

A process used to divide a large group of data into smaller, logical categories for analysis. Segmentation is commonly used by us in our day to day business to understand and interpret information

- Example: Segmentation of customers based on the cities

Stratification

A process which uses summary metrics (central tendency, dispersion) to make the decision about when to separate different processes for continued analysis

Unlike segmentation, stratification involves the uses data rather than just 'information'. Values of the central tendency and dispersion are used to stratify the given data set.

- Example : Stratification of the customers based on the business volumes they provide us

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data collection plan

Why collect data?

- The purpose of the data collection exercise

What to collect ?

- Identify measures
- Define operational definitions
- What data to be collected

How to collect?

- Formulate data collection plan
- Sampling strategy
- Pilot data collection plan

Collect data

- Pilot and validation collection plan
- Train data collectors

Ensure consistency & stability

- Develop measurement system analysis
- Test and validate
- Monitor and improvise

Word of caution: Ineffective data leads to ineffective conclusions

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data collection plan

- Data collection occurs multiple times throughout DMAIC. The data collection plan described here can be used as the guide for data collection. This help us ensure that we collect useful, accurate data that is needed to answer our process questions

- It is important to be clear about the data collection goals to ensure the right data is collected. If your data is in the wrong form or format, you may not be able to use it in your analysis

- Operational definitions help to guide thinking on what properties will be measured and how they will be measured. There is no single right way to write an operational definition. There is only what people agree to for a specific purpose. The critical factor is that any two people using the operational definition will be measuring the same thing in the same manner.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Operational Definition

What is an Operational Definition?

An operational definition is a clear, concise description of a measurement and the process by which it is to be collected.

Purpose of Operational Definition:

- To remove ambiguity: Everyone has a consistent understanding
- To provide a clear way to measure the characteristic
  - Identifies what to measure
  - Identifies how to measure it
  - Makes sure that no matter who does the measuring, the results are consistent

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data collection plan - Example

A format of an excel spread sheet that can be used for creating data collection plans forour projects. The sample data collection plan below is for the project CTQ, in this example, number of rejections.

However one should remember that based on C-E diagram and the SIPOC all those Xs which the team feels to have a influence over the Y should also be included in the data collection plan.

Example of a data collection plan

					Sampling			Reporting Plan				
Measure	**Operational Definition**	**Target**	**Method/ Source**	**Unit of Measur**	**Sample**	**Collection Frequency**	**Sample Size**	**Reported On**	**Reported Frequency**	**Reporting to**	**Responsi-bility**	**Time Frame**
No. of Rejections	Any Transaction getting rejected in the Misc Step Processing & appearing in the GAMX Rejection Report & GMC Transmission Rejection Message	0.10%	GAMX Rejection Report & GMC Transmission Rejection Message	Nos.	No Sampling. Entire Population data collected	Daily	6803	Every Monday	Daily	Manager	Processor of Misc Step	1 Month

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Validate Measurement System

Validate Measurement System

Why do we need to validate Measurement System?

- To verify the adequacy of measurement system for Y when establishing process baseline
- To verify the adequacy of measurement system when verifying causes
- To verify the adequacy of measurement system when verifying solutions.
- To verify the adequacy of measurement system when controlling the X's

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Validate Measurement System - Objective

To determine what percentage the Total Observed Variation is due to Measurement Device and Measurement Method in addition to the true Part to Part variation.

To statistically verify that the current measurement system provides:

- Unbiased results
- Minimal variability within the measurement system
- True representative values of the factors being measured

Key Points

- There is no perfect measurement system. All
- measurement systems contain variation.
- Gage system error within a measurement system is the sum of:
  - *Bias (Accuracy)*
  - *Stability*
  - *Sensitivity*
  - *Repeatability*
  - *Reproducibility*

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Types of Measurement Errors - Key Definitions

- ***Accuracy:*** *The difference between the observed average of measurements and the true average of the items measured.*

- ***Repeatability:*** *The variation in measurements obtained with a gage when used several times by one operator while measuring the identical characteristic on the same sample piece*

- ***Reproducibility:*** *The variation in the average of measurements taken by different operators using the same gage while measuring the identical characteristic on the same pieces.*

- ***Stability:*** *The variation in the average of at least two sets of measurements obtained with a gage as a result of time on the same pieces.*

- ***Sensitivity:*** *The ability of the measuring instrument to detect the smallest unit of change in measured value.*

KPMG

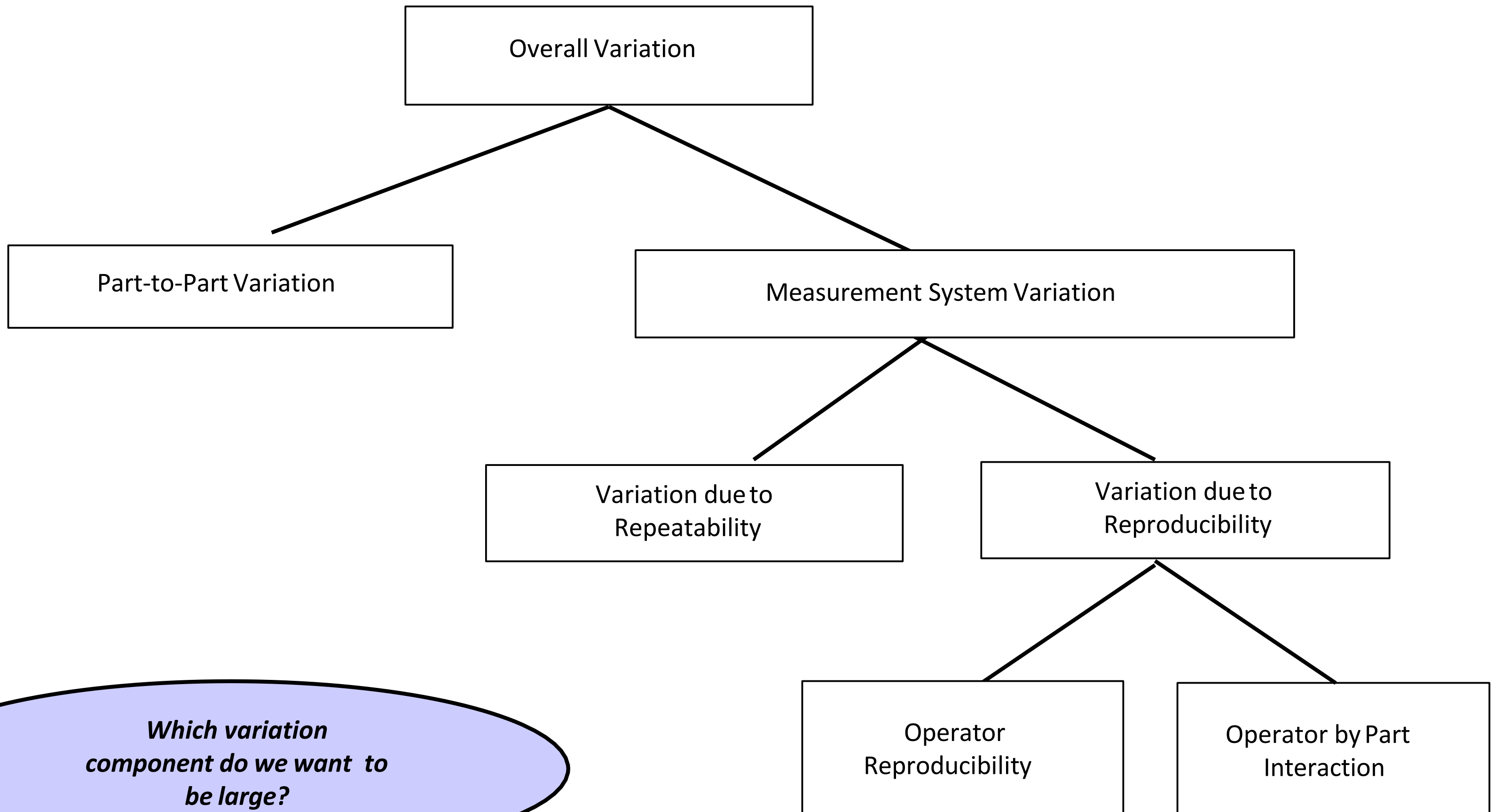
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Key Points

- Gage R&R Studies are a method to quantify the repeatability and reproducibility of a measuring system.
- Gage R&R studies are conducted to evaluate a Gage's suitability for a defined purpose.
- Accuracy and Stability are addressed by calibration.
- Sensitivity is addressed through ensuring correct Least Count of measuring instrument.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Breaking down overall variation



Which variation component do we want to be large?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Validate Measurement System - Methods

Two methods for Validating Measurement System:

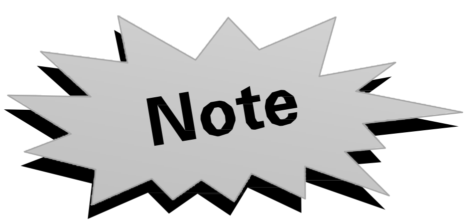
- Variable Gage R&R
- Attribute Agreement Analysis

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Variable Gage R&R

- At least two operators (persons doing the measuring) should participate. Two or three operators are typical.
- At least 10 parts should be measured. The same characteristic is measured on each part. These are 10 units of the same type product that represent the full range of manufacturing variation.
- Each operator needs to measure each part two or three times. Parts should be measured in random order.
- Parts should be masked so that operator does not realize that he / she is measuring the same part number of times.



It is very important that an operator not be aware of his or her earlier measurement when doing a repeat measurement on the same part.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Variable Gage R&R - Acceptability Criteria

Study Variation (SV%)	% Contribution	Acceptable Criteria
0% to 10%	Less than 1%	Good Measurement System
10% to 30%	1% to 9%	Conditionally Accepted (Depending on the Criticality of the Application)
Greater than 30%	Greater than 9%	Not Acceptable

Number of Distinct Categories	Less than 5	Not Acceptable

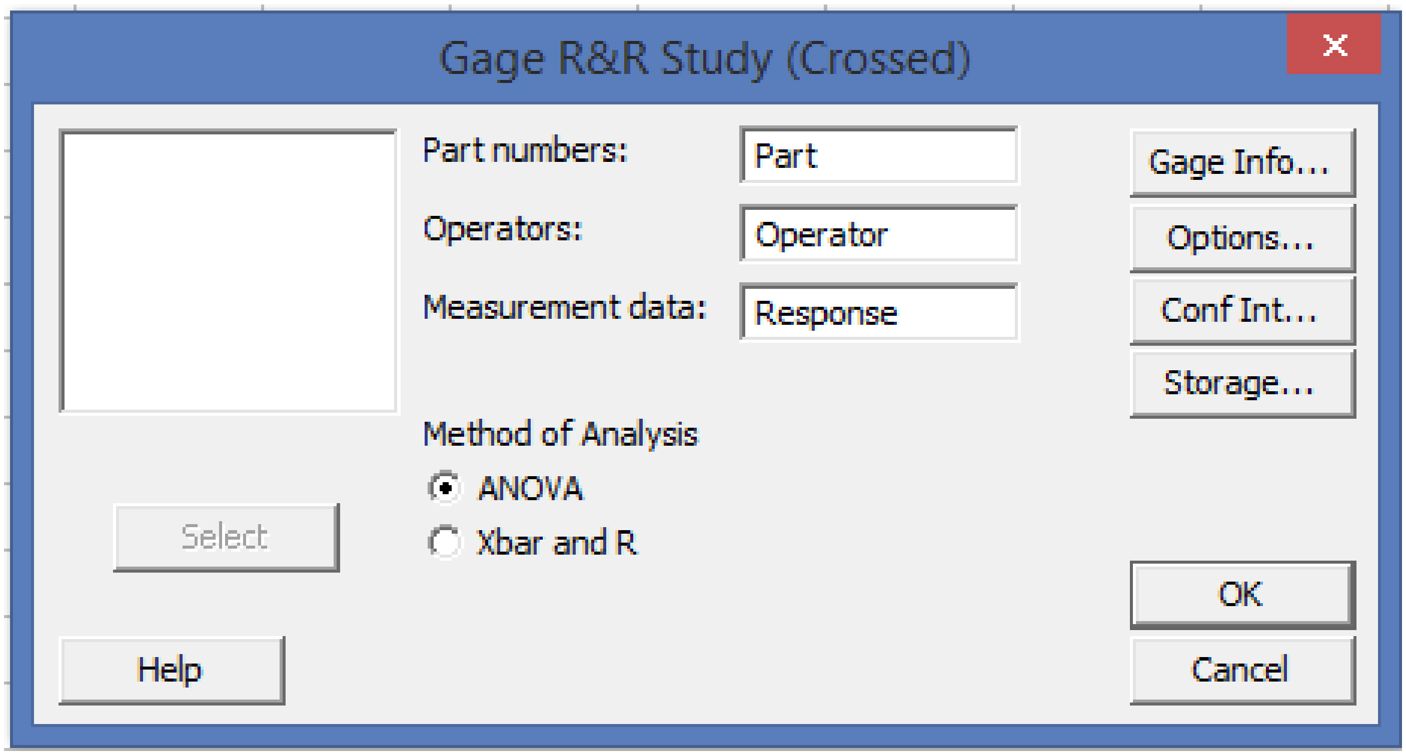
KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Variable Gage R&R using Minitab

Refer the data sheet – Thickness. MTW in Minitab

Choose Stat > Quality Tools > Gage Study > Gage R&R Study (Crossed).



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Variable Gage R&R using Minitab

Results for: Thickness.MTW

Gage R&R Study - ANOVA Method
Two-Way ANOVA Table With Interaction

Source	DF	SS	MS	F	P
Part	9	2.05871	0.228745	39.7178	0.000
Operator	2	0.04800	0.024000	4.1672	0.033
Part * Operator	18	0.10367	0.005759	4.4588	0.000
Repeatability	30	0.03875	0.001292		
Total	59	2.24913			

α to remove interaction term = 0.05

Gage R&R

Source	VarComp	%Contribution (of VarComp)
Total Gage R&R	0.0044375	**10.67**
Repeatability	0.0012917	3.10
Reproducibility	0.0031458	7.56
Operator	0.0009120	2.19
Operator*Part	0.0022338	5.37
Part-To-Part	0.0371644	89.33
Total Variation	0.0416019	100.00

Measurement System does not meet the criteria as the SV% and Contribution % exceed the criteria

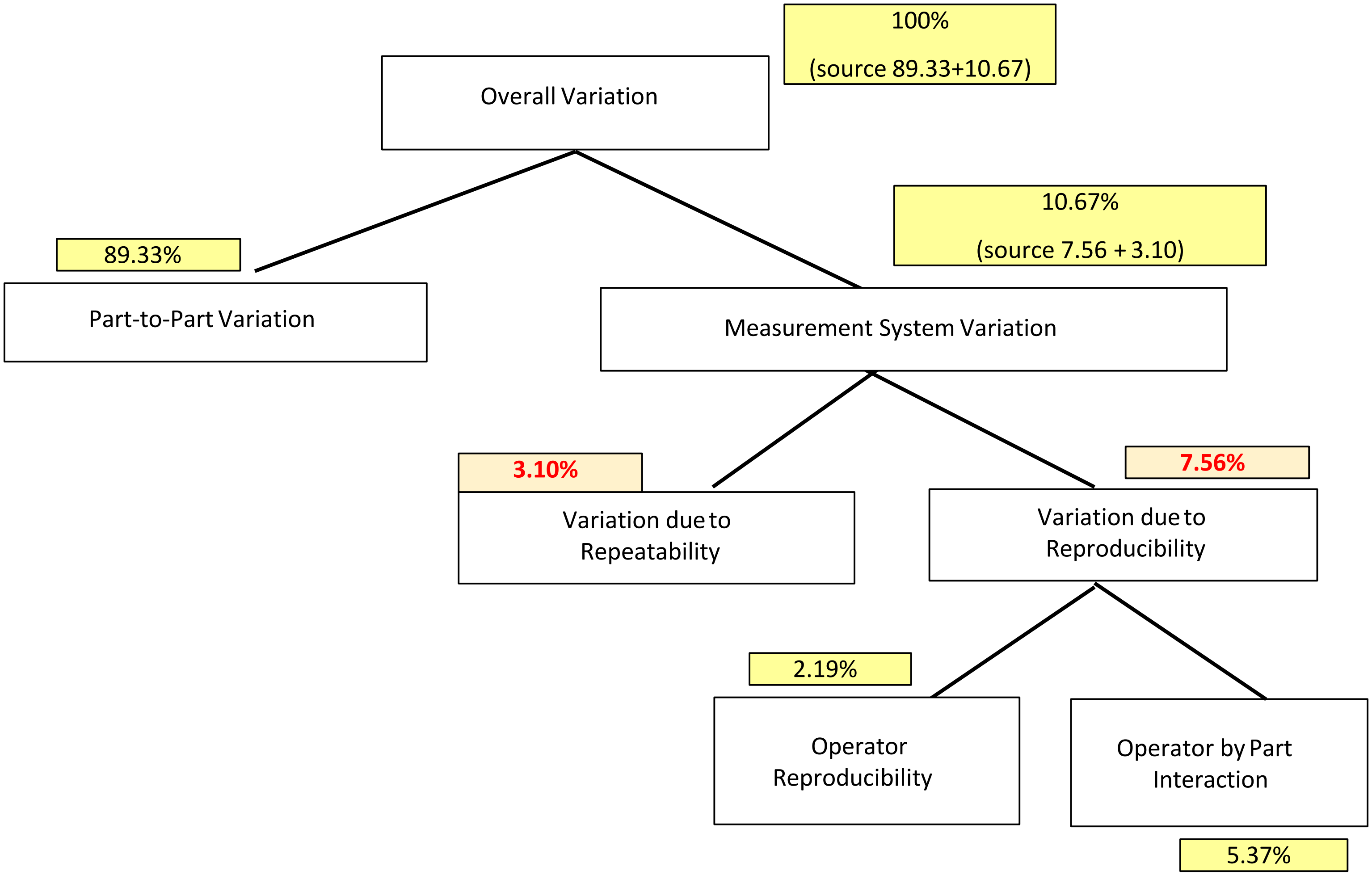
Source	StdDev (SD)	Study Var (6 × SD)	%Study Var (%SV)
Total Gage R&R	0.066615	0.39969	**32.66**
Repeatability	0.035940	0.21564	17.62
Reproducibility	0.056088	0.33653	27.50
Operator	0.030200	0.18120	14.81
Operator*Part	0.047263	0.28358	23.17
Part-To-Part	0.192781	1.15668	94.52
Total Variation	0.203965	1.22379	100.00

Number of Distinct Categories = 4

The number of distinct categories is less than 5, therefore it is not acceptable.

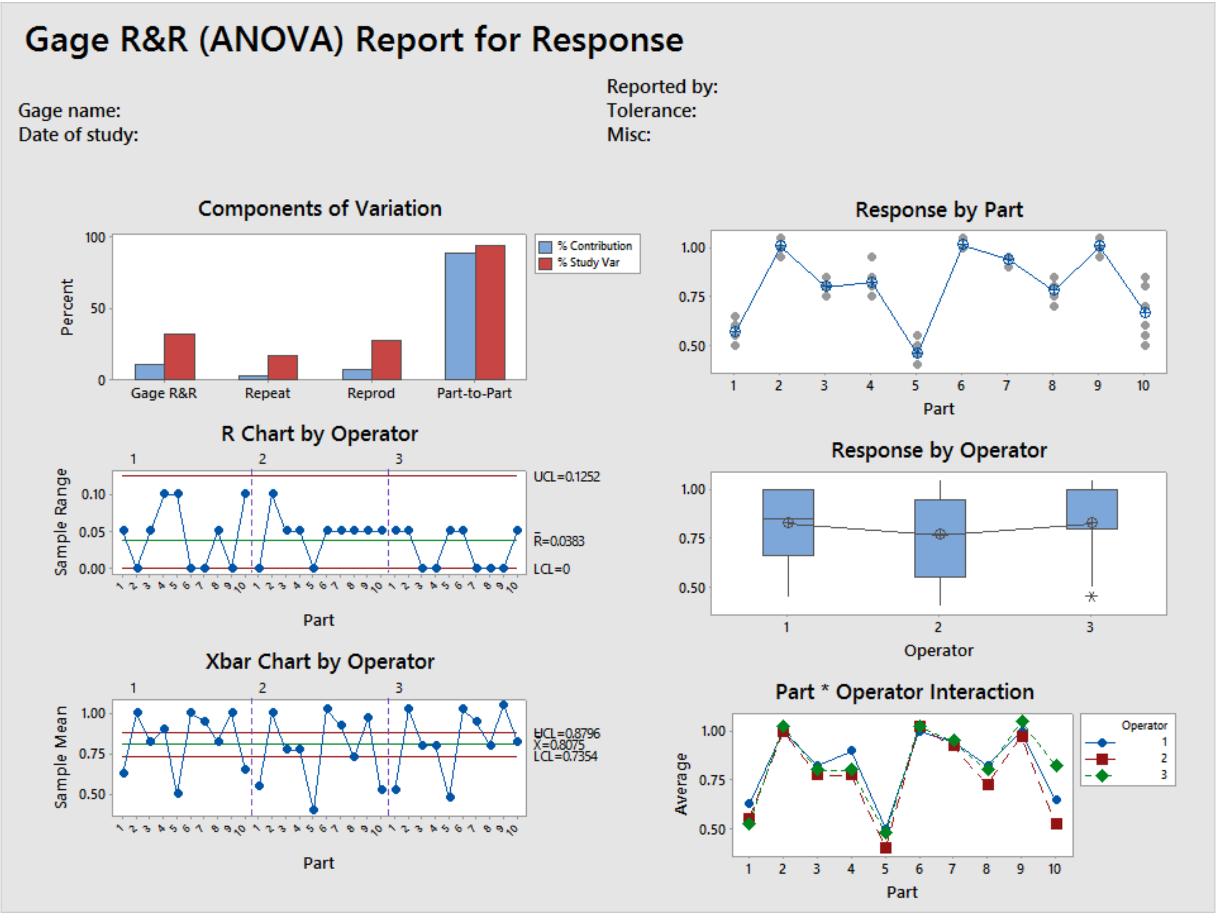
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Breaking down overall variation



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Variable Gage R&R using Minitab



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis

- It is also important to have good repeatability and reproducibility when obtaining attribute data.

- If one operator, for example, decides a unit has an "appearance" defect and another operator concludes the same unit has no defect, then there is a problem with the measurement system.

- Similarly, the measurement system is inadequate when the same person draws different conclusions on repeat evaluations of the same unit of product.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis

- An attribute measurement system compares each part to a standard and accepts the part if the standard is met.
- The screen effectiveness is the ability of the attribute measurement system to properly discriminate good from bad.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis – Acceptability Criteria

While 100% is the most desirable result in Attribute Agreement Analysis , the following guidelines are frequently used:

Kappa Percentage	Guideline
0.90 to 1.00	Acceptable
0.80 to 0.90	Marginal
Less than 0.80	Not Acceptable

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab

Let us consider an situation, wherein you need to:

1. Select a minimum of 30 parts from the process. These parts should represent the full spectrum of process variation (good parts, defective parts, borderline parts).

2. An "expert" inspector performs an evaluation of each part, classifying it as "Good" or "Not Good."

3. Independently and in a random order, each of 2 or 3 operators should assess the parts as "Good" or "Not Good."

4. Enter the data into Minitab to quantify the effectiveness of the measurement system.

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab

		Appraiser 1		Appraiser 2	
Sample	Attribute	First	Repeat	First	Repeat
1	G	G	G	G	G
2	G	G	G	G	G
3	G	G	G	G	G
4	G	G	G	G	G
5	G	G	G	G	G
6	G	NG	G	G	G
7	G	G	G	G	G
8	G	G	G	G	G
9	NG	G	G	NG	NG
10	NG	NG	NG	G	G
11	G	G	G	G	G
12	G	G	G	G	G
13	NG	NG	NG	NG	NG
14	G	G	G	G	G
15	G	G	G	G	G
16	G	G	G	G	G
17	NG	NG	NG	NG	NG
18	G	G	G	G	G
19	G	G	G	G	G
20	G	G	G	G	G

In this example:

- 20 samples are selected
- The 20 samples represent the range of parts seen in production (usually from obviously good to obviously bad)
- An “expert” establishes the “Attribute” column
- Two operators evaluate each sample twice in random order to eliminate bias

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab

		Appraiser 1		Appraiser 2	
Sample	Attribute	First	Repeat	First	Repeat
1	G	G	G	G	G
2	G	G	G	G	G
3	G	G	G	G	G
4	G	G	G	G	G
5	G	G	G	G	G
6	G	**NG**	**G**	G	G
7	G	G	G	G	G
8	G	G	G	G	G
9	NG	G	G	NG	NG
10	NG	NG	NG	G	G
11	G	G	G	G	G
12	G	G	G	G	G
13	NG	NG	NG	NG	NG
14	G	G	G	G	G
15	G	G	G	G	G
16	G	G	G	G	G
17	NG	NG	NG	NG	NG
18	G	G	G	G	G
19	G	G	G	G	G
20	G	G	G	G	G

- Examining the data, there is only one case where an appraiser failed to agree with his or her previous reading.
- This occurs in appraiser 1's data set on sample #6.
- In the other 39 pairs of data, both appraisers agree with their previous readings.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab

		Appraiser 1		Appraiser 2	
Sample	Attribute	First	Repeat	First	Repeat
1	G	G	G	G	G
2	G	G	G	G	G
3	G	G	G	G	G
4	G	G	G	G	G
5	G	G	G	G	G
6	G	**NG**	**G**	G	G
7	G	G	G	G	G
8	G	G	G	G	G
9	NG	**G**	**G**	NG	NG
10	NG	NG	NG	**G**	**G**
11	G	G	G	G	G
12	G	G	G	G	G
13	NG	NG	NG	NG	NG
14	G	G	G	G	G
15	G	G	G	G	G
16	G	G	G	G	G
17	NG	NG	NG	NG	NG
18	G	G	G	G	G
19	G	G	G	G	G
20	G	G	G	G	G

- Examining the data, there are three rows where an appraiser did not completely agree with the attribute set by the expert.
- So, appraiser 1 agreed with the expert 18 out of 20 times.
- Appraiser 2 agreed with the expert 19 out of 20 times.
- Note that even if one sample of two agree with the attribute, it is counted as a disagreement.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab

Within Appraisers

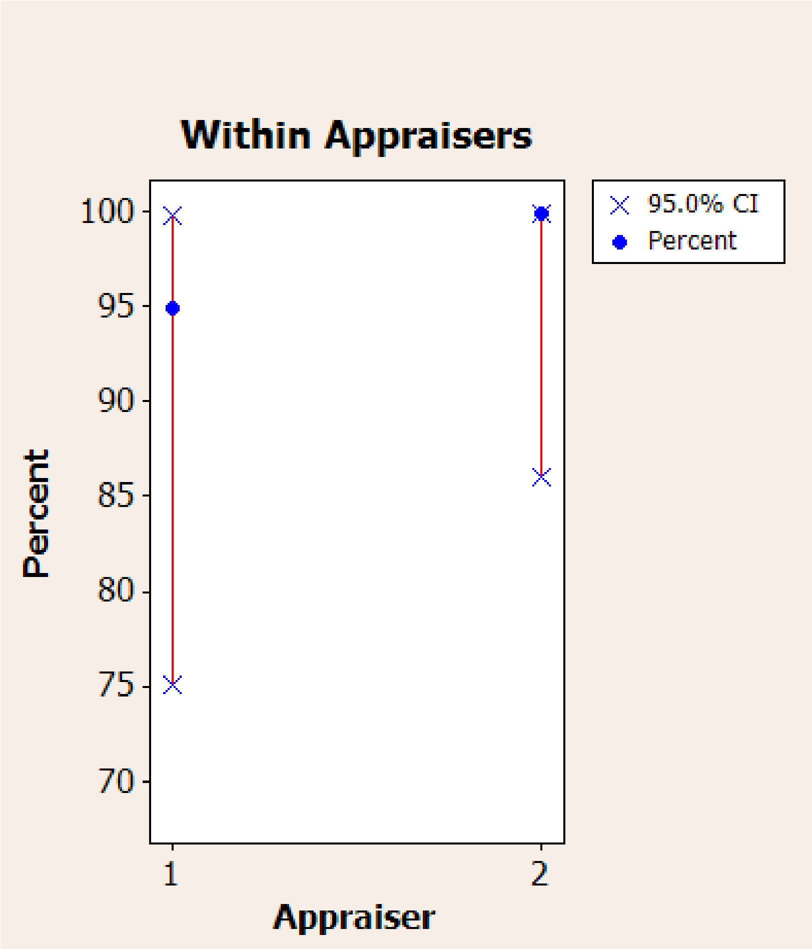
```
Assessment Agreement

Appraiser # Inspected # Matched Percent (%)       95.0% CI
1                  20        19        95.0  (75.13,  99.87)
2                  20        20       100.0  (86.09, 100.00)

# Matched: Appraiser agrees with him/herself across trials.
```

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab



- To the left, is Minitab's graphical output for the within appraisers evaluation.
- The graph summarizes the information shown in the session window.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab

Each Appraiser vs Standard

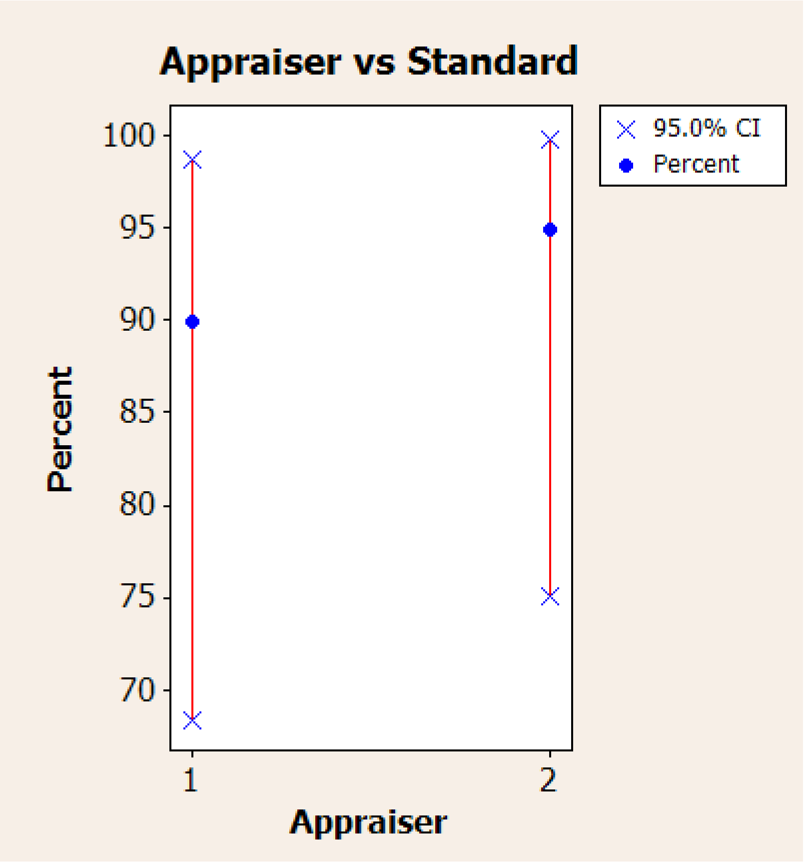
```
Assessment Agreement
Appraiser # Inspected # Matched Percent (%)       95.0% CI
1                   20         18         90.0 ( 68.30,  98.77)
2                   20         19         95.0 ( 75.13,  99.87)
# Matched: Appraiser's assessment across trials agrees with standard.
```

```
Assessment Disagreement
Appraiser   # NG/G Percent (%)   # G/NG Percent (%)  # Mixed Percent (%)
1                0       0.00         1      25.00         1       5.00
2                0       0.00         1      25.00         0       0.00
# NG/G: Assessments across trials = NG / standard = G.
# G/NG: Assessments across trials = G / standard = NG.
# Mixed: Assessments across trials are not identical.
```

- Above, is Minitab's session window output for each appraiser versus standard evaluation.
- Note that under "Assessment Disagreement," Minitab indicates the cases where appraisers have mixed results.
- The graphical output is shown on the next page.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab



- To the left, is Minitab's graphical output for each appraiser versus standard evaluation.
- The graph summarizes the information shown in the session window.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab

		Appraiser 1		Appraiser 2	
Sample	Attribute	First	Repeat	First	Repeat
1	G	G	G	G	G
2	G	G	G	G	G
3	G	G	G	G	G
4	G	G	G	G	G
5	G	G	G	G	G
6	G	**NG**	**G**	**G**	**G**
7	G	G	G	G	G
8	G	G	G	G	G
9	NG	**G**	**G**	**NG**	**NG**
10	NG	**NG**	**NG**	**G**	**G**
11	G	G	G	G	G
12	G	G	G	G	G
13	NG	NG	NG	NG	NG
14	G	G	G	G	G
15	G	G	G	G	G
16	G	G	G	G	G
17	NG	NG	NG	NG	NG
18	G	G	G	G	G
19	G	G	G	G	G
20	G	G	G	G	G

- Examining the data, there are three rows where the appraisers failed to agree with each other.
- Note that even if one sample of two agree with the other appraiser, it is counted as a disagreement.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab

Between Appraisers

```
Assessment Agreement

# Inspected # Matched Percent (%)       95.0% CI
         20        17       85.00 ( 62.11,  96.79)

# Matched: All appraisers' assessments agree with each other.
```

- Above, is Minitab's session window output for the between appraiser evaluation.
- There is no graphical output of this result.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab

		Appraiser 1		Appraiser 2	
Sample	Attribute	First	Repeat	First	Repeat
1	G	G	G	G	G
2	G	G	G	G	G
3	G	G	G	G	G
4	G	G	G	G	G
5	G	G	G	G	G
6	G	**NG**	**G**	**G**	**G**
7	G	G	G	G	G
8	G	G	G	G	G
9	NG	**G**	**G**	**NG**	**NG**
10	NG	**NG**	**NG**	**G**	**G**
11	G	G	G	G	G
12	G	G	G	G	G
13	NG	NG	NG	NG	NG
14	G	G	G	G	G
15	G	G	G	G	G
16	G	G	G	G	G
17	NG	NG	NG	NG	NG
18	G	G	G	G	G
19	G	G	G	G	G
20	G	G	G	G	G

- Examining the data, there are three rows where the appraisers failed to agree with the attribute set by the expert.
- Note that even if one sample of two agree with the other appraiser, it is counted as a disagreement.
- This is the overall measure of Attribute Gage R&R.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Agreement Analysis using Minitab

All Appraisers vs Standard

```
Assessment Agreement

# Inspected # Matched Percent (%)        95.0% CI
         20        17      85.00 ( 62.11,  96.79)

# Matched: All appraisers' assessments agree with standard
```

- Above, is Minitab's session window output for the all appraisers versus standard evaluation.
- This is the overall attribute gage R&R of this measurement system.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Interpreting the results

- **Within Appraiser** is the consistency within one person.
- **Each Appraiser vs Standard** is a measure of how well the operator's evaluation agrees with that of the "expert".
- **Between Appraisers** is a measure of how well the operators agree with each other.
- **All Appraisers vs Standard** is an overall measure of consistency between operators and agreement with the "expert".

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Key Points

- Before collecting new data, evaluate the gage using either variable or attribute gage R&R
- Before using existing data, try to estimate gage R&R so that you can estimate the trustworthiness of the data.
- Don’t delay a Six Sigma project due to a poor Gage R&R. Keep the project moving and improve the gage as the project goes forward.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Determine Process Capability

What is Process Capability?

- Process capability is a simple tool which helps us determine if a process, given its natural variation, is capable of meeting the customer requirements or specifications
- Helps to determine if there has been a change in the process
- Also enables the six sigma team to determine the percent of the product/service not meeting the customer requirement

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Measuring the Capability

- Sigma is a statistical unit of measure, used to denote the value of standard deviation in a set of variable data
- For a given process, Sigma Level is a metric that indicates how well a process is performing
- In Six Sigma, the capability of a process to meet customer specification is captured by the Process Sigma Level.
- Hence as value of Standard Deviation (σ) decreases the Process Sigma Level increases.

Let us look at different scenarios in a process

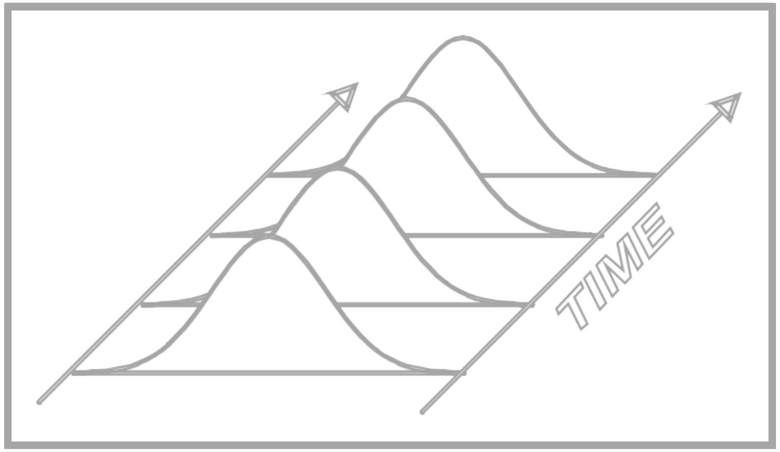
KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

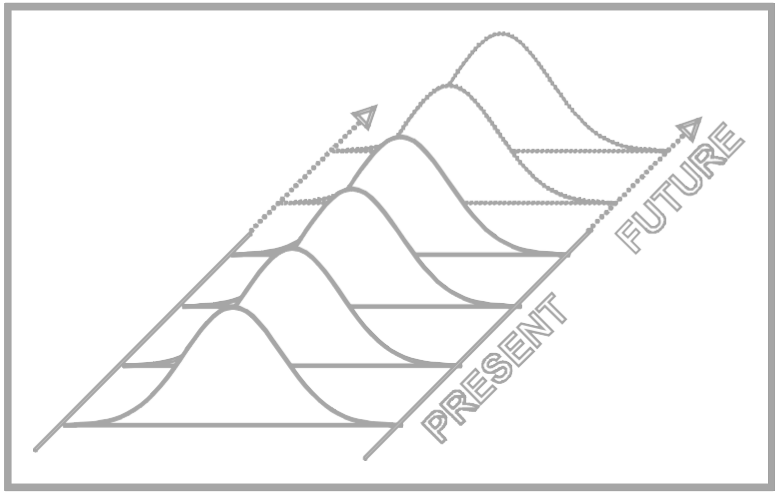
What is Process Control?

A Process is in statistical control when it is stable...

100% of the individual data values are within the spread of natural variation (+-3σ) of the process

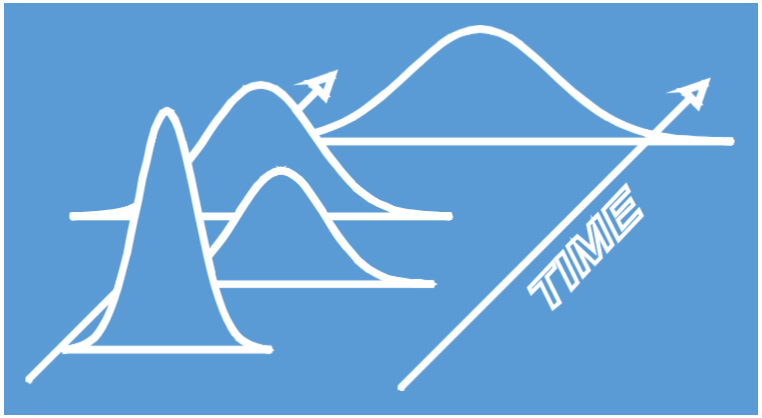


And, therefore, predictable...



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

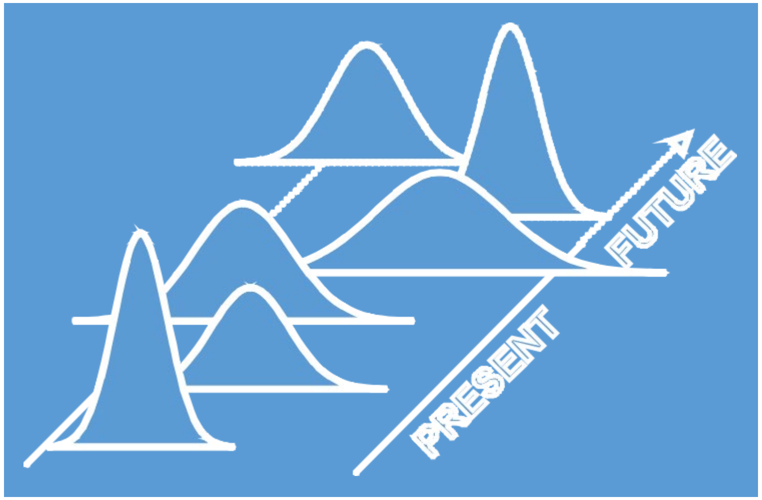
What is "Out of Control" process?



An unstable process...

100% of the individual data values are NOT within the spread of natural variation ($\pm 3\sigma$) of the process

Few data values fall beyond the $\pm 3\sigma$ limits and are referred to as "OUTLIERS"



...is unpredictable

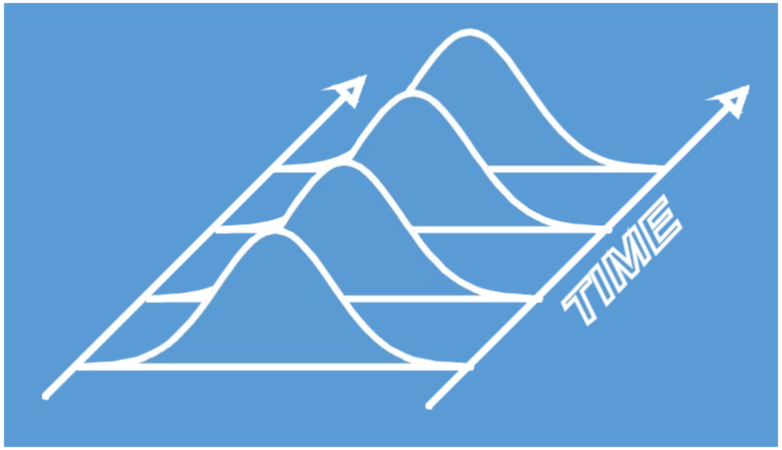
KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Common Cause vs. Special Cause

Type of variation	Characteristics
COMMON CAUSE	**Inherent to the process** • Expected • Predictable • Normal • Random
SPECIAL CAUSE	**Not always present** • Unexpected • Unpredictable • Not normal • Not random

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Common Cause vs. Special Cause



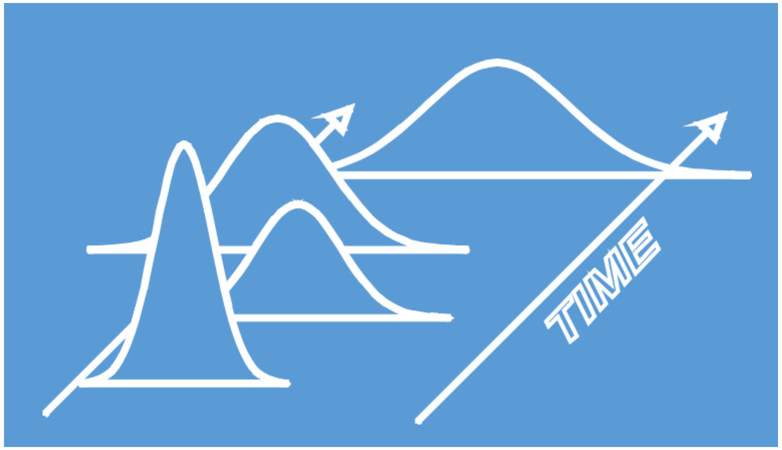
Only "COMMON" causes of variation are present in a Stable Process.

This variation is Random and Routine.

"SPECIAL" causes of variation are present in an Unstable Process.

This variation is Non-Random and Sporadic

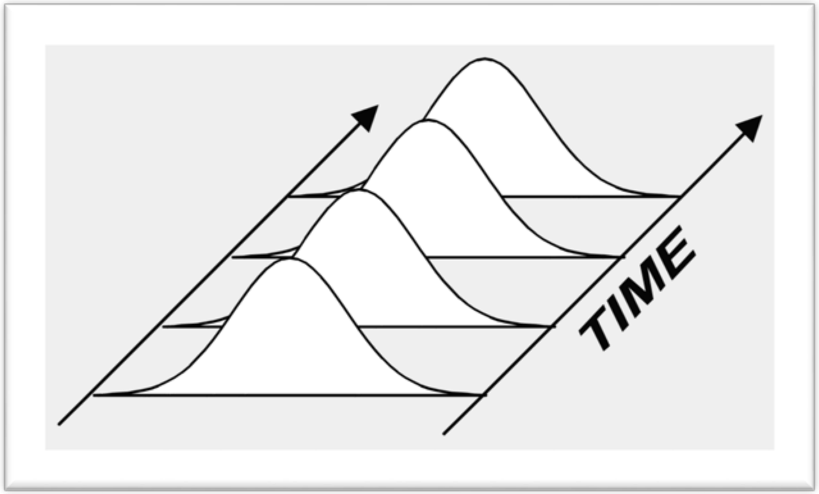
Some points in time are exceptional when compared to the rest.



KPMG

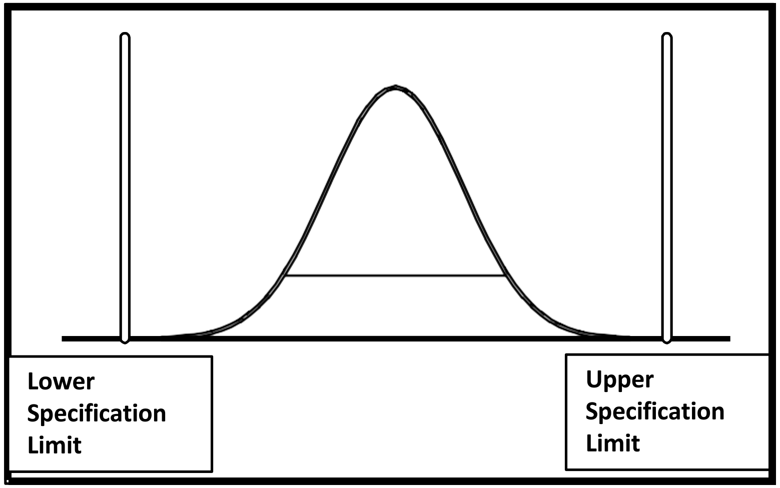
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Process Control vs. Process Capability



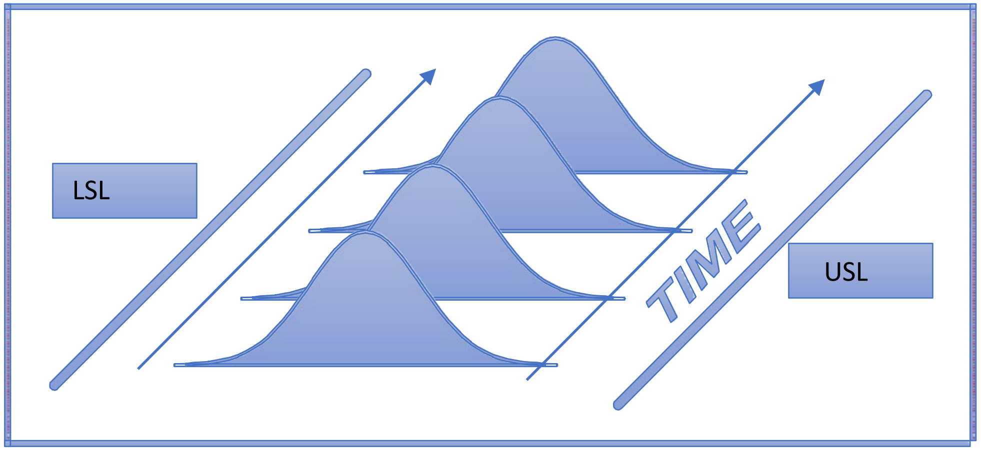
Process Control = Stability over time.

Process Capability = Ability of a stable process to meet specifications.



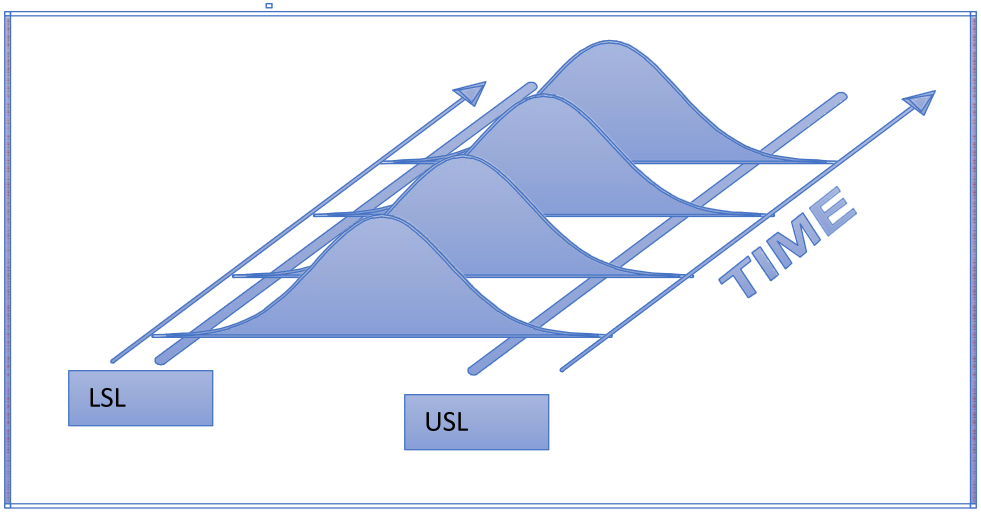
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Examples of Stable and Capable processes



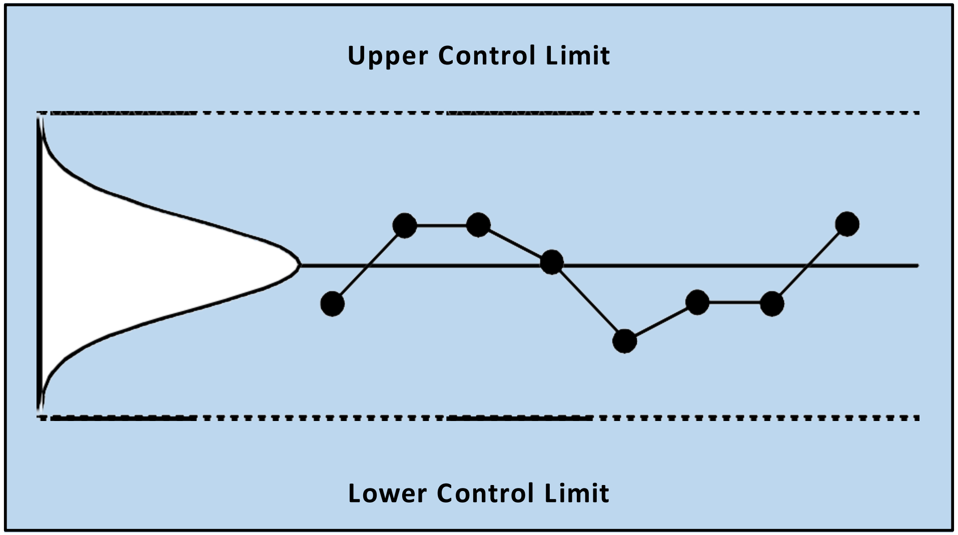
Process is Stable and Capable

Process is Stable but not Capable



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Control Limits vs. Specification Limits



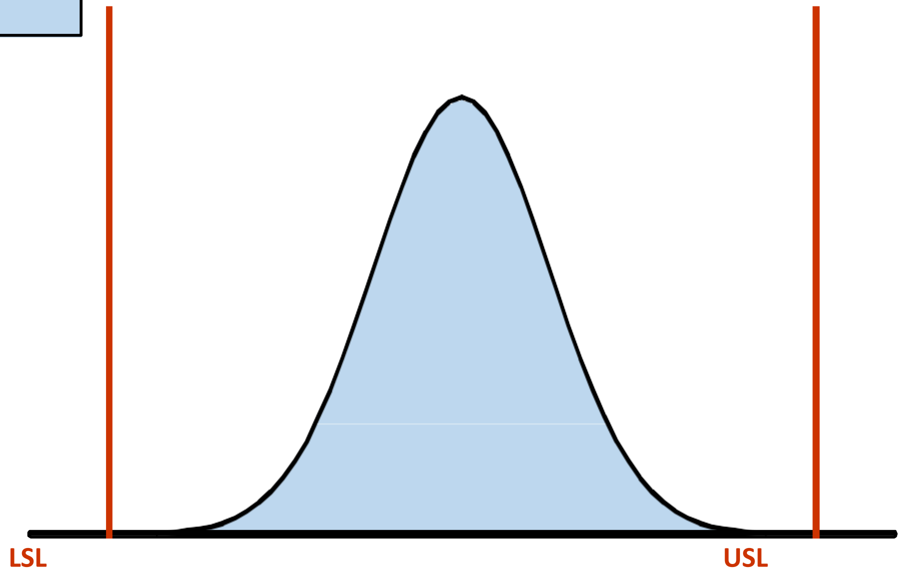
Control Limits are statistical bounds (the natural bounds of the data) used to determine process stability.

Statistically Control Limits are equivalent to $\pm 3\sigma$

Control limits are determined by the data (voice of the process).

Specification Limits are applied to individual measurements.

Specification limits are decided by people (voice of the process)

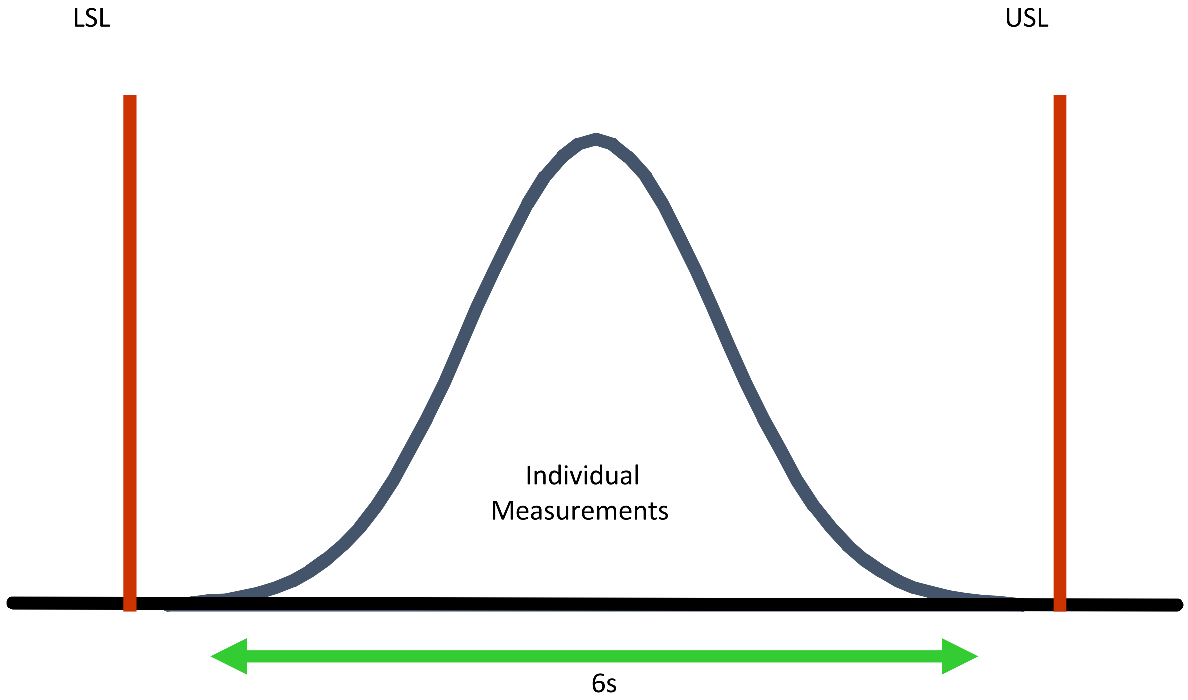


© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Determine Process Capability

Variable Data

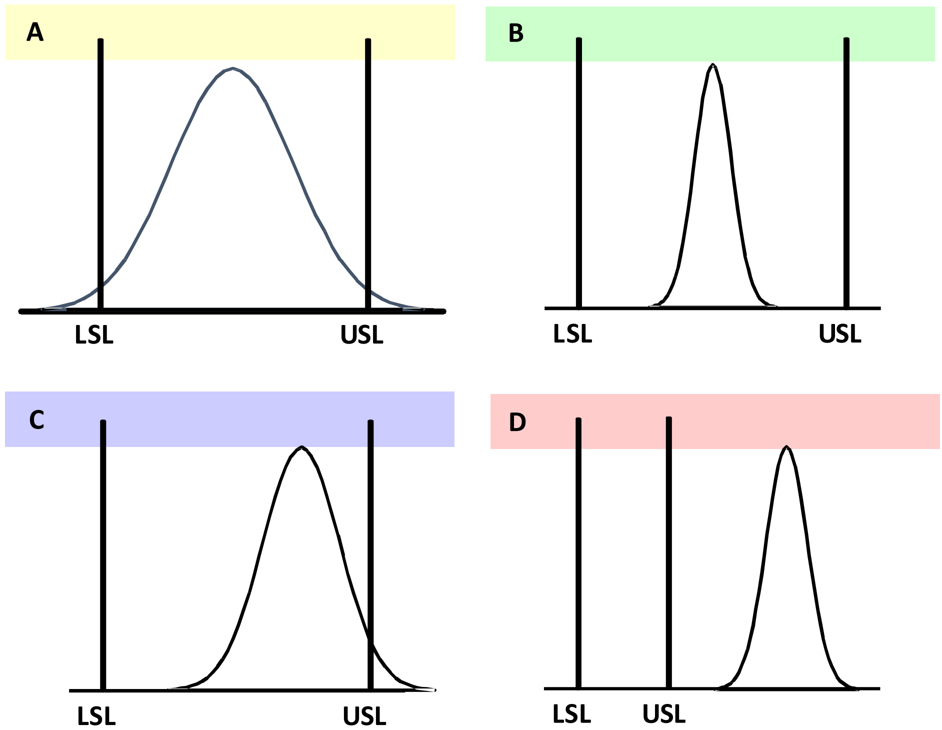
Stable and Capable Process



- A stable process provides the most reliable estimates of process capability
- A process is said to be capable when the ± 3σ points of the distribution of individual measurements are contained well within the specification limits.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

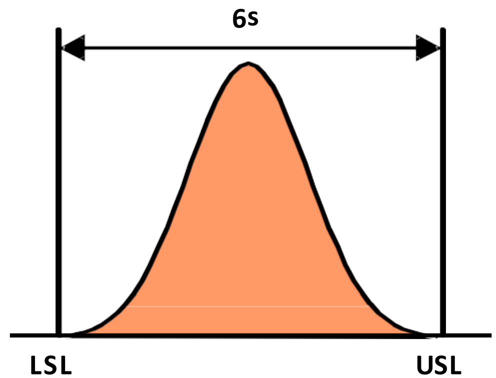
Stable and Capable Process



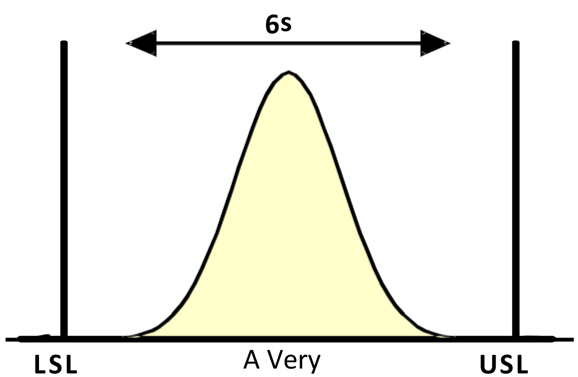
What can be said about the capability of these fourprocesses?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

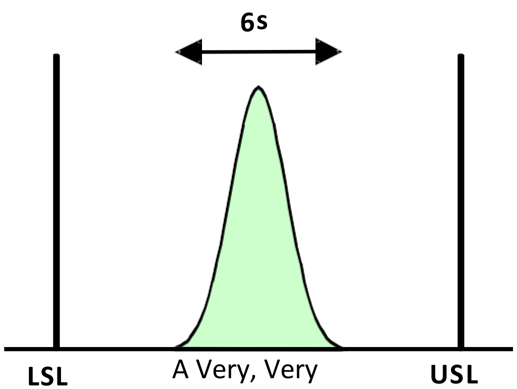
How capable is the process?



Capable Process



A Very
Capable Process



A Very, Very
Capable Process

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Steps for evaluating process capability for variable data

- Assure the data is normally distributed
- Estimate the average and standard deviation of the process
- Determine the process' potential capability
- Quantify process performance

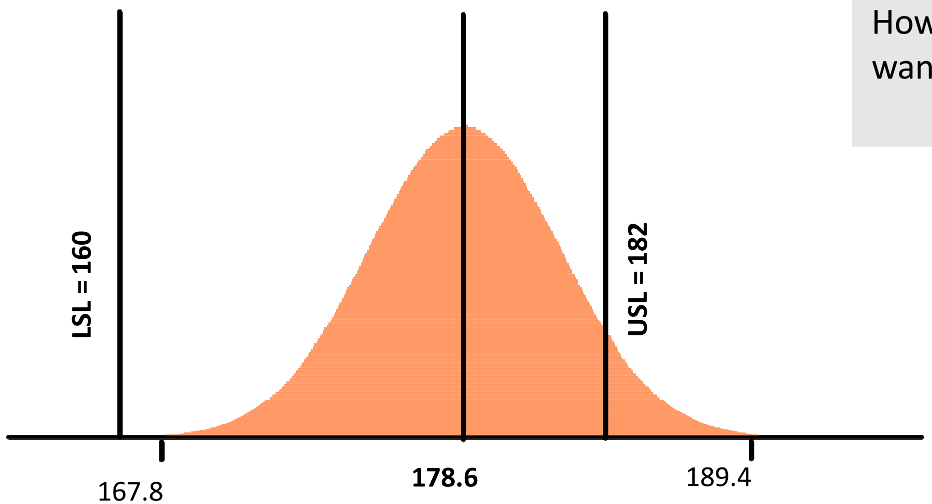
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Process variation vs Specification

Let us assume that for a normally distributed stable process, the average is 178.6 and the standard deviation is 3.6. The process target is 171, USL = 182, LSL = 160

Process Variation:
At $\pm$ 3σ that is 99.73% of the time, the process is producing products that falls between 167.8 and 189.4.

However, as per customer specification, we want all product to fall between 160 and 182

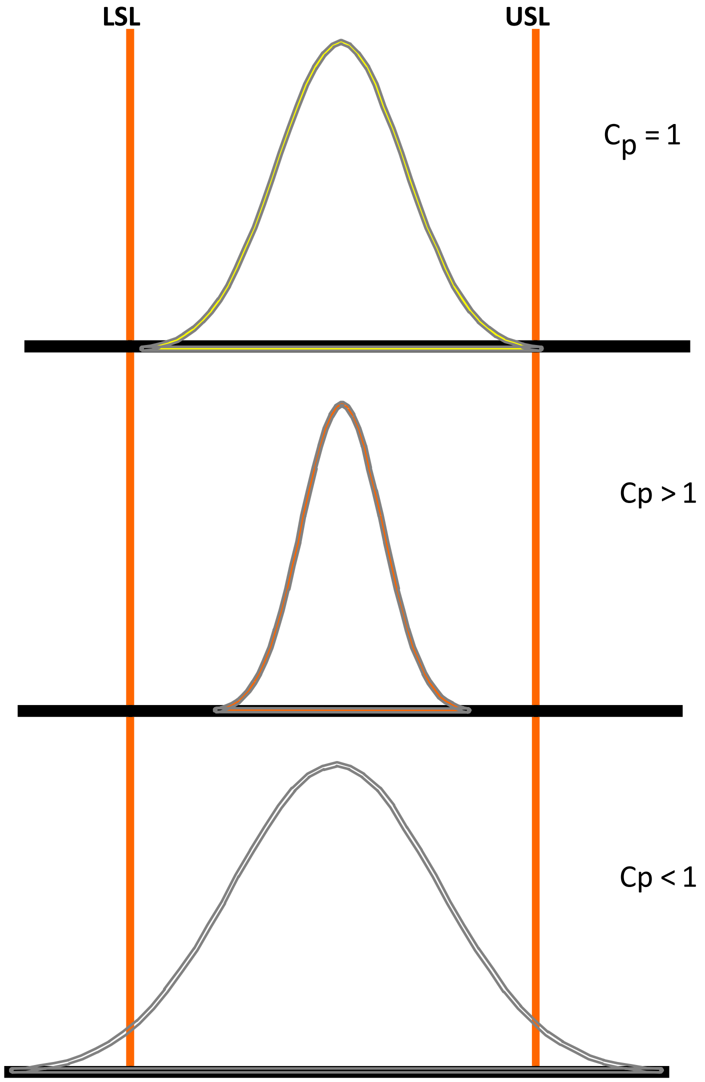


$\bar{\bar{x}}$

$$\bar{\bar{x}} - 3\sigma \qquad \bar{\bar{x}} + 3\sigma$$

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Determine the potential process capability



Process is just about meeting the specification but centering needs attention

Process variation well under control. Larger the Cp, better the process

Process variation exceeds specification in this case. Process is producing defects

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Determine the potential process capability

The Cp index reflects the potential of the process if the average were perfectly centered between the specification limits.

$$C_p = \frac{USL - LSL}{6\sigma}$$

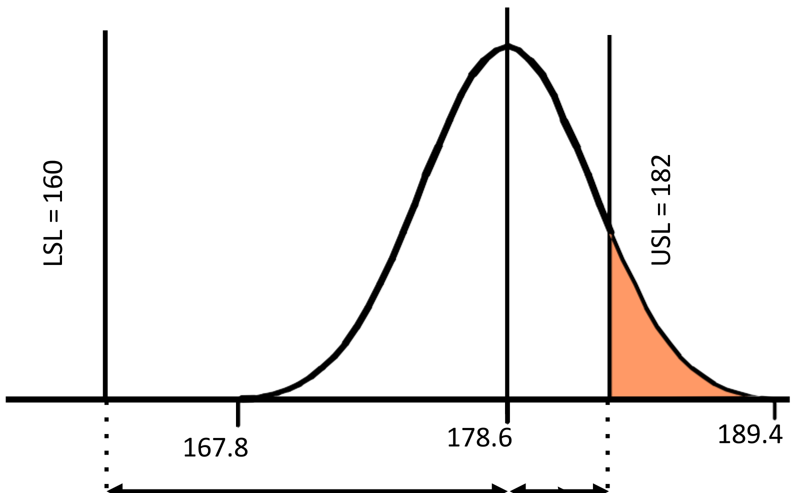
$$C_p = \frac{182 - 160}{6\sigma}$$

For the given example, the potential process capability Cp is 1.01

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quantify actual process performance (Cpk)

To estimate the percentage of product / process that falls outside the specification limits, we compute Cp (upper) and Cp (lower)



z_{lower} is the number of standard deviations between the Process Average and the Lower Specification Limit.

z_{upper} is the number of standard deviations between the Process Average and the Upper Specification Limit.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quantify actual process performance (Cpk)

The Cp index reflects the potential of the process if the average were perfectly centered between the specification limits.

$$Z_{(upper)} = \frac{USL - \bar{X}}{\sigma}$$

$$Z_{(lower)} = \frac{\bar{X} - LSL}{\sigma}$$

$$C_{PK} = \frac{\text{Minimum of } Z(\text{upper}) \ \& \ Z(\text{lower})}{3}$$

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quantify actual process performance (Cpk)

The Cp index reflects the potential of the process if the average were perfectly centered between the specification limits.

$$Z_{(upper)} = \frac{182.0 - 178.6}{3.6}$$

$$Z_{(lower)} = \frac{178.6 - 160}{3.6}$$

$Z_{(upper)} = 0.94$

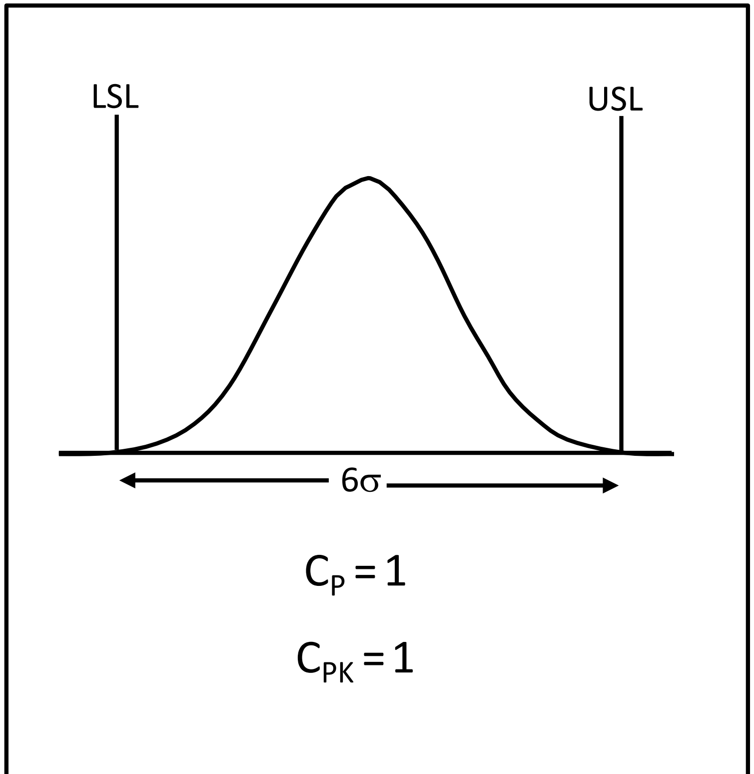
$Z_{(lower)} = 5.17$

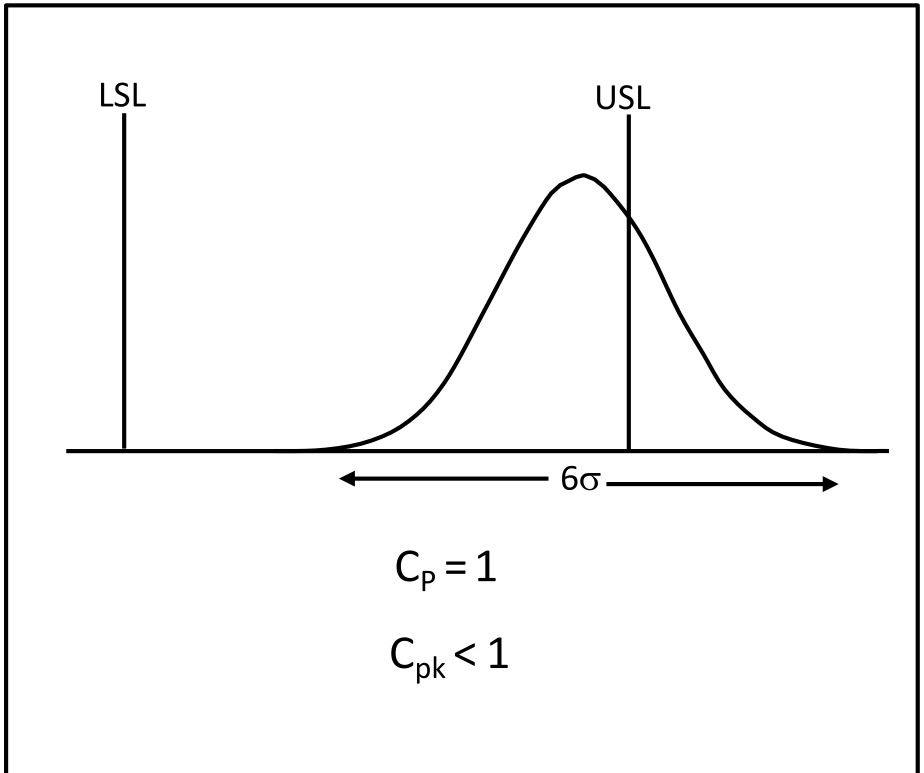
Cpk = 0.94 / 3 = 0.31

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Quantify actual process performance (Cpk)

Unlike the Cp, the Cpk index takes into account off-centering of the process. The larger the Cpk index, the better.

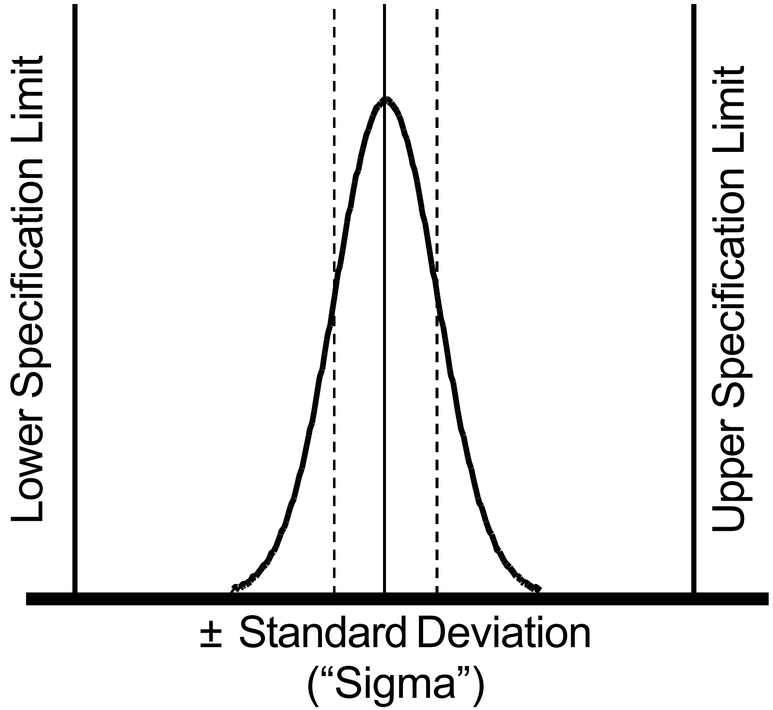




© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

What is Sigma?

“Sigma” (Standard deviation) is a measure of variation

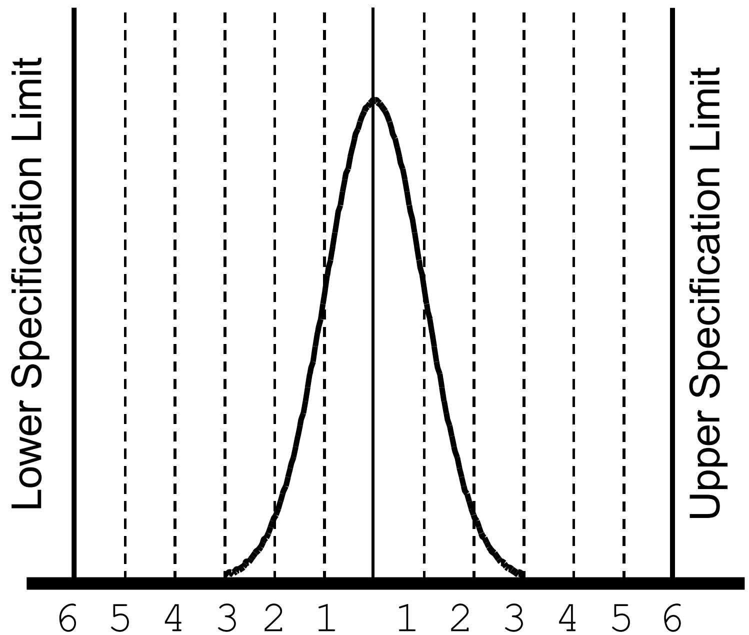


Plus or minus one standard deviation around the mean is about 68% of the total process output.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

What is a Six Sigma process?

If we can squeeze six standard deviations in between our process average and the customer's requirements…

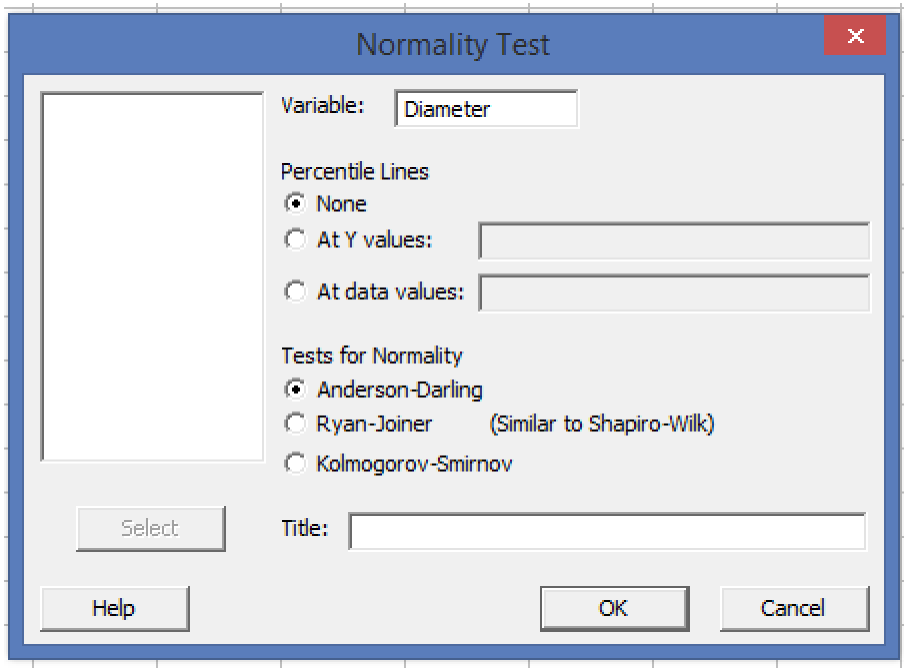


then
99.99966% of our "opportunities" meet customer requirements!

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

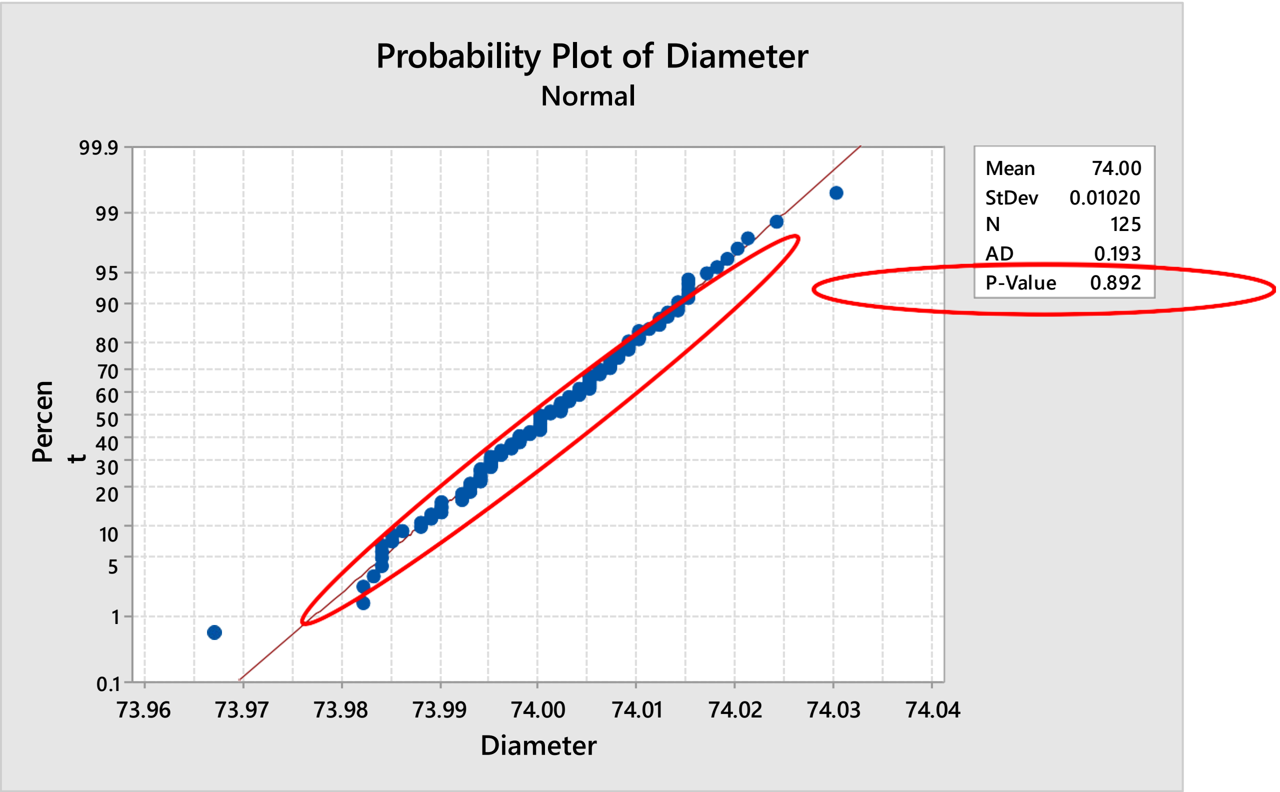
Checking stability for variable data in Minitab

- To identify the average and standard deviation of a process, we first need to check the distribution i.e. if it is normal distribution?
- Let us refer the data sheet – Piston.MTW in Minitab for the same We
- check the normality of the data
- **Choose Stat> Basic Statistics> Normality Test**



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Checking stability for variable data in Minitab

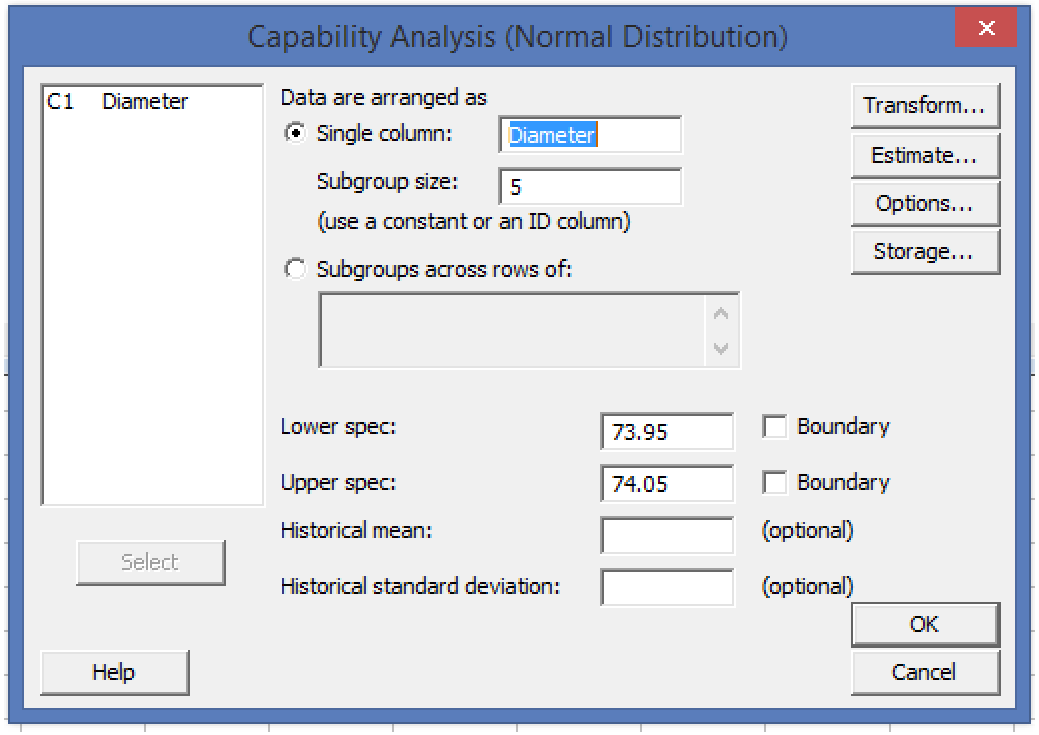


The Anderson-Darling test's p-value at 0.892 indicates that the data follows a normal distribution.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

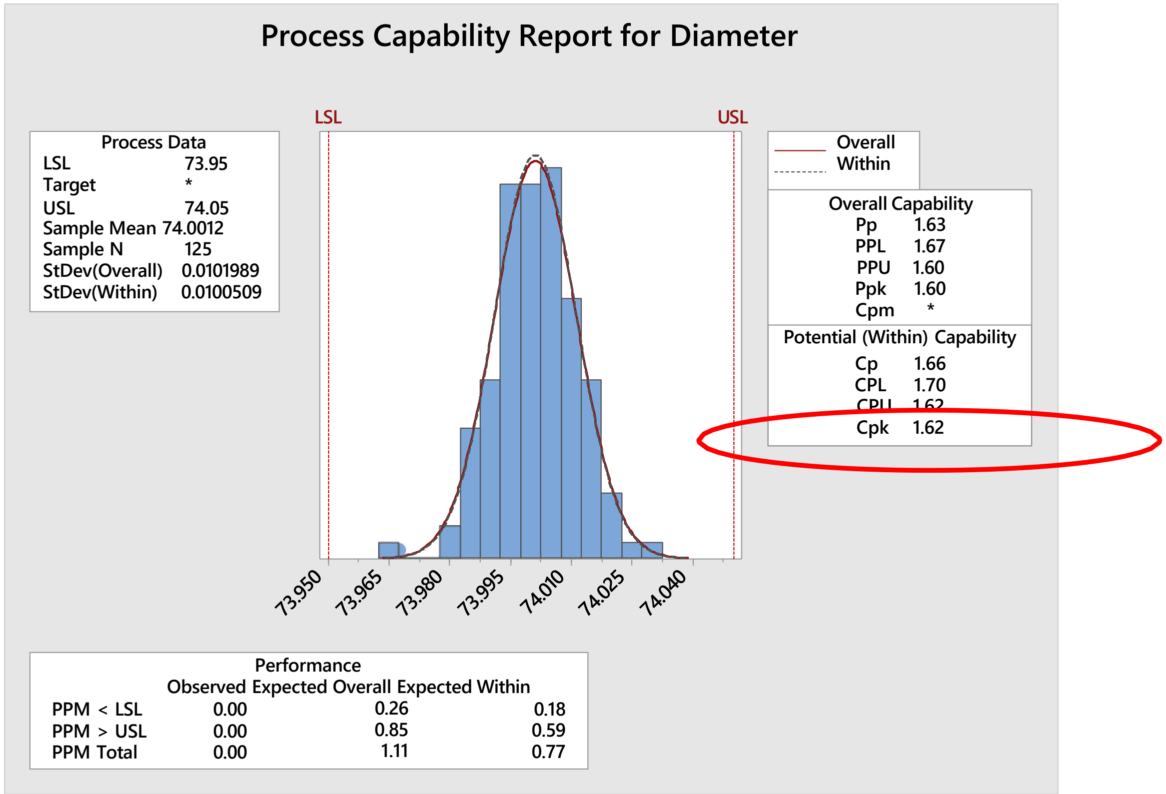
Cp and Cpk in Minitab

- Since the data follows a normal distribution, we can now check the capability of the data.
  Let us refer the same data sheet – Piston.MTW in Minitab Choose
- Stat> Quality Tools> Capability Analysis> Normal
-



KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Cp and Cpk in Minitab



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Determine Process Capability

Attribute Data - DPMO

Attribute Data - DPMO

Item	Definition	Example
Unit	The Item produced or Processed. It is the specific Product or Service used to evaluate whether or not Customer Requirements are met	Transaction processed/ Query addressed.
Defect	Any Event that does not meet the specifications of a CTQ	Incorrect information provided/ Incorrect address etc.
Defective	Any Unit that contains one or more Defects. Such a unit is called a "Defective Unit".	A Bill with two Defects.(Name & Amount)
Defect Opportunity	Any Event which can be Measured that provides a CHANCE of not meeting a Customer Requirement.	Name, Address & Amount on a Bill Each represent an opportunity for Defect

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Data - DPMO

DPU = Total Number of defects / Total number of units inspected

Department: Purchase
Defect: Incorrect entry
Total Defects: 56
Unit: Each MIS report (monthly)
No of units inspected: 5,000

DPU = 56 / 5000 = 0.0112
PPM = DPU * 10^6 = 11,200 = 3.8 Sigma

$$\text{DPO} = \frac{\text{Total Number of Defects}}{\text{Total number of units inspected * Opportunities per unit}}$$

Department: Purchase
Defect: Incorrect entry
Total Defects: 56
Unit: Each MIS report (monthly)
No of units inspected: 5,000
No of Opportunities per report: 6
DPO = 56 / 5,000 *6 = 0.001866
DPMO = DPO*10^6 = 1,866 = 4.4 Sigma

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Sigma and DPMO table

Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO
0.01	931888	0.26	892512	0.51	838913	0.76	770350	1.01	687933	1.26	594835	1.51	496011	1.76	397432
0.02	930563	0.27	890651	0.52	836457	0.77	767305	1.02	684386	1.27	590954	1.52	492022	1.77	393580
0.03	929219	0.28	888767	0.53	833977	0.78	764238	1.03	680822	1.28	587064	1.53	488033	1.78	389739
0.04	927855	0.29	886860	0.54	831472	0.79	761148	1.04	677242	1.29	583166	1.54	484047	1.79	385908
0.05	926471	0.30	884930	0.55	828944	0.80	758036	1.05	673645	1.30	579260	1.55	480061	1.80	382089
0.06	925066	0.31	882977	0.56	826391	0.81	754903	1.06	670031	1.31	575345	1.56	476078	1.81	378281
0.07	923641	0.32	881000	0.57	823814	0.82	751748	1.07	666402	1.32	571424	1.57	472097	1.82	374484
0.08	922196	0.33	878999	0.58	821214	0.83	748571	1.08	662757	1.33	567495	1.58	468119	1.83	370700
0.09	920730	0.34	876976	0.59	818589	0.84	745373	1.09	659097	1.34	563559	1.59	464144	1.84	366928
0.10	919243	0.35	874928	0.60	815940	0.85	742154	1.10	655422	1.35	559618	1.60	460172	1.85	363169
0.11	917736	0.36	872857	0.61	813267	0.86	738914	1.11	651732	1.36	555670	1.61	456205	1.86	359424
0.12	916207	0.37	870762	0.62	810570	0.87	735653	1.12	648027	1.37	551717	1.62	452242	1.87	355691
0.13	914656	0.38	868643	0.63	807850	0.88	732371	1.13	644309	1.38	547758	1.63	448283	1.88	351973
0.14	913085	0.39	866500	0.64	805106	0.89	729069	1.14	640576	1.39	543795	1.64	444330	1.89	348268
0.15	911492	0.40	864334	0.65	802338	0.90	725747	1.15	636831	1.40	539828	1.65	440382	1.90	344578
0.16	909877	0.41	862143	0.66	799546	0.91	722405	1.16	633072	1.41	535856	1.66	436441	1.91	340903
0.17	908241	0.42	859929	0.67	796731	0.92	719043	1.17	629300	1.42	531881	1.67	432505	1.92	337243
0.18	906582	0.43	857690	0.68	793892	0.93	715661	1.18	625516	1.43	527903	1.68	428576	1.93	333598
0.19	904902	0.44	855428	0.69	791030	0.94	712260	1.19	621719	1.44	523922	1.69	424655	1.94	329969
0.20	903199	0.45	853141	0.70	788145	0.95	708840	1.20	617911	1.45	519939	1.70	420740	1.95	326355
0.21	901475	0.46	850830	0.71	785236	0.96	705402	1.21	614092	1.46	515953	1.71	416834	1.96	322758
0.22	899727	0.47	848495	0.72	782305	0.97	701944	1.22	610261	1.47	511967	1.72	412936	1.97	319178
0.23	897958	0.48	846136	0.73	779350	0.98	698468	1.23	606420	1.48	507978	1.73	409046	1.98	315614
0.24	896165	0.49	843752	0.74	776373	0.99	694974	1.24	602568	1.49	503989	1.74	405165	1.99	312067
0.25	894350	0.50	841345	0.75	773373	1.00	691462	1.25	598706	1.50	500000	1.75	401294	2.00	308538

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Sigma and DPMO table

Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO
2.01	305026	2.26	223627	2.51	156248	2.76	103835	3.01	65522	3.26	39204	3.51	22216	3.76	11911
2.02	301532	2.27	220650	2.52	153864	2.77	102042	3.02	64255	3.27	38364	3.52	21692	3.77	11604
2.03	298056	2.28	217695	2.53	151505	2.78	100273	3.03	63008	3.28	37538	3.53	21178	3.78	11304
2.04	294599	2.29	214764	2.54	149170	2.79	98525	3.04	61780	3.29	36727	3.54	20675	3.79	11011
2.05	291160	2.30	211855	2.55	146859	2.80	96800	3.05	60571	3.30	35930	3.55	20182	3.80	10724
2.06	287740	2.31	208970	2.56	144572	2.81	95098	3.06	59380	3.31	35148	3.56	19699	3.81	10444
2.07	284339	2.32	206108	2.57	142310	2.82	93418	3.07	58208	3.32	34380	3.57	19226	3.82	10170
2.08	280957	2.33	203269	2.58	140071	2.83	91759	3.08	57053	3.33	33625	3.58	18763	3.83	9903
2.09	277595	2.34	200454	2.59	137857	2.84	90123	3.09	55917	3.34	32884	3.59	18309	3.84	9642
2.10	274253	2.35	197663	2.60	135666	2.85	88508	3.10	54799	3.35	32157	3.60	17864	3.85	9387
2.11	270931	2.36	194895	2.61	133500	2.86	86915	3.11	53699	3.36	31443	3.61	17429	3.86	9137
2.12	267629	2.37	192150	2.62	131357	2.87	85343	3.12	52616	3.37	30742	3.62	17003	3.87	8894
2.13	264347	2.38	189430	2.63	129238	2.88	83793	3.13	51551	3.38	30054	3.63	16586	3.88	8656
2.14	261086	2.39	186733	2.64	127143	2.89	82264	3.14	50503	3.39	29379	3.64	16177	3.89	8424
2.15	257846	2.40	184060	2.65	125072	2.90	80757	3.15	49471	3.40	28717	3.65	15778	3.90	8198
2.16	254627	2.41	181411	2.66	123024	2.91	79270	3.16	48457	3.41	28067	3.66	15386	3.91	7976
2.17	251429	2.42	178786	2.67	121000	2.92	77804	3.17	47460	3.42	27429	3.67	15003	3.92	7760
2.18	248252	2.43	176186	2.68	119000	2.93	76359	3.18	46479	3.43	26803	3.68	14629	3.93	7549
2.19	245097	2.44	173609	2.69	117023	2.94	74934	3.19	45514	3.44	26190	3.69	14262	3.94	7344
2.20	241964	2.45	171056	2.70	115070	2.95	73529	3.20	44565	3.45	25588	3.70	13903	3.95	7143
2.21	238852	2.46	168528	2.71	113139	2.96	72145	3.21	43633	3.46	24998	3.71	13553	3.96	6947
2.22	235762	2.47	166023	2.72	111232	2.97	70781	3.22	42716	3.47	24419	3.72	13209	3.97	6756
2.23	232695	2.48	163543	2.73	109349	2.98	69437	3.23	41815	3.48	23852	3.73	12874	3.98	6569
2.24	229650	2.49	161087	2.74	107488	2.99	68112	3.24	40930	3.49	23295	3.74	12545	3.99	6387
2.25	226627	2.50	158655	2.75	105650	3.00	66807	3.25	40059	3.50	22750	3.75	12224	4.00	6210

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Sigma and DPMO table

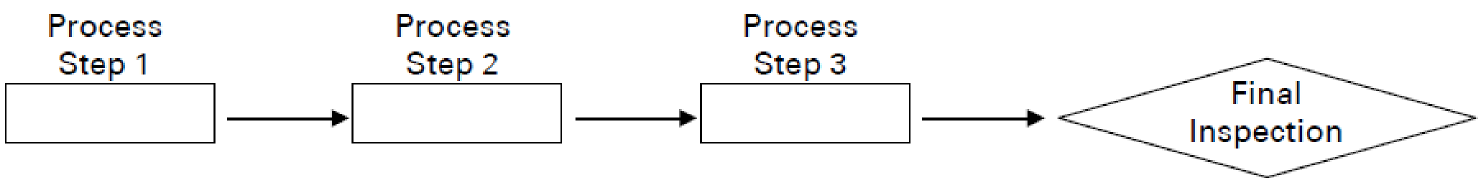
Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO	Sigma	DPMO
4.01	6037	4.26	2890	4.51	1306	4.76	557	5.01	224	5.26	85	5.51	30.4	5.76	10.2
4.02	5868	4.27	2803	4.52	1264	4.77	538	5.02	216	5.27	82	5.52	29.1	5.77	9.8
4.03	5703	4.28	2718	4.53	1223	4.78	519	5.03	208	5.28	78	5.53	27.9	5.78	9.4
4.04	5543	4.29	2635	4.54	1183	4.79	501	5.04	200	5.29	75	5.54	26.7	5.79	8.9
4.05	5386	4.30	2555	4.55	1144	4.80	483	5.05	193	5.30	72	5.55	25.6	5.80	8.5
4.06	5234	4.31	2477	4.56	1107	4.81	467	5.06	185	5.31	70	5.56	24.5	5.81	8.2
4.07	5085	4.32	2401	4.57	1070	4.82	450	5.07	179	5.32	67	5.57	23.5	5.82	7.8
4.08	4940	4.33	2327	4.58	1035	4.83	434	5.08	172	5.33	64	5.58	22.5	5.83	7.5
4.09	4799	4.34	2256	4.59	1001	4.84	419	5.09	165	5.34	62	5.59	21.6	5.84	7.1
4.10	4661	4.35	2186	4.60	968	4.85	404	5.10	159	5.35	59	5.60	20.7	5.85	6.8
4.11	4527	4.36	2118	4.61	936	4.86	390	5.11	153	5.36	57	5.61	19.8	5.86	6.5
4.12	4397	4.37	2052	4.62	904	4.87	376	5.12	147	5.37	54	5.62	19.0	5.87	6.2
4.13	4269	4.38	1988	4.63	874	4.88	362	5.13	142	5.38	52	5.63	18.1	5.88	5.9
4.14	4145	4.39	1926	4.64	845	4.89	350	5.14	136	5.39	50	5.64	17.4	5.89	5.7
4.15	4025	4.40	1866	4.65	816	4.90	337	5.15	131	5.40	48	5.65	16.6	5.90	5.4
4.16	3907	4.41	1807	4.66	789	4.91	325	5.16	126	5.41	46	5.66	15.9	5.91	5.2
4.17	3793	4.42	1750	4.67	762	4.92	313	5.17	121	5.42	44	5.67	15.2	5.92	4.9
4.18	3681	4.43	1695	4.68	736	4.93	302	5.18	117	5.43	42	5.68	14.6	5.93	4.7
4.19	3573	4.44	1641	4.69	711	4.94	291	5.19	112	5.44	41	5.69	14.0	5.94	4.5
4.20	3467	4.45	1589	4.70	687	4.95	280	5.20	108	5.45	39	5.70	13.4	5.95	4.3
4.21	3364	4.46	1538	4.71	664	4.96	270	5.21	104	5.46	37	5.71	12.8	5.96	4.1
4.22	3264	4.47	1489	4.72	641	4.97	260	5.22	100	5.47	36	5.72	12.2	5.97	3.9
4.23	3167	4.48	1441	4.73	619	4.98	251	5.23	96	5.48	34	5.73	11.7	5.98	3.7
4.24	3072	4.49	1395	4.74	598	4.99	242	5.24	92	5.49	33	5.74	11.2	5.99	3.6
4.25	2980	4.50	1350	4.75	577	5.00	233	5.25	88	5.50	32	5.75	10.7	6.00	3.4

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Rolled Throughput Yield

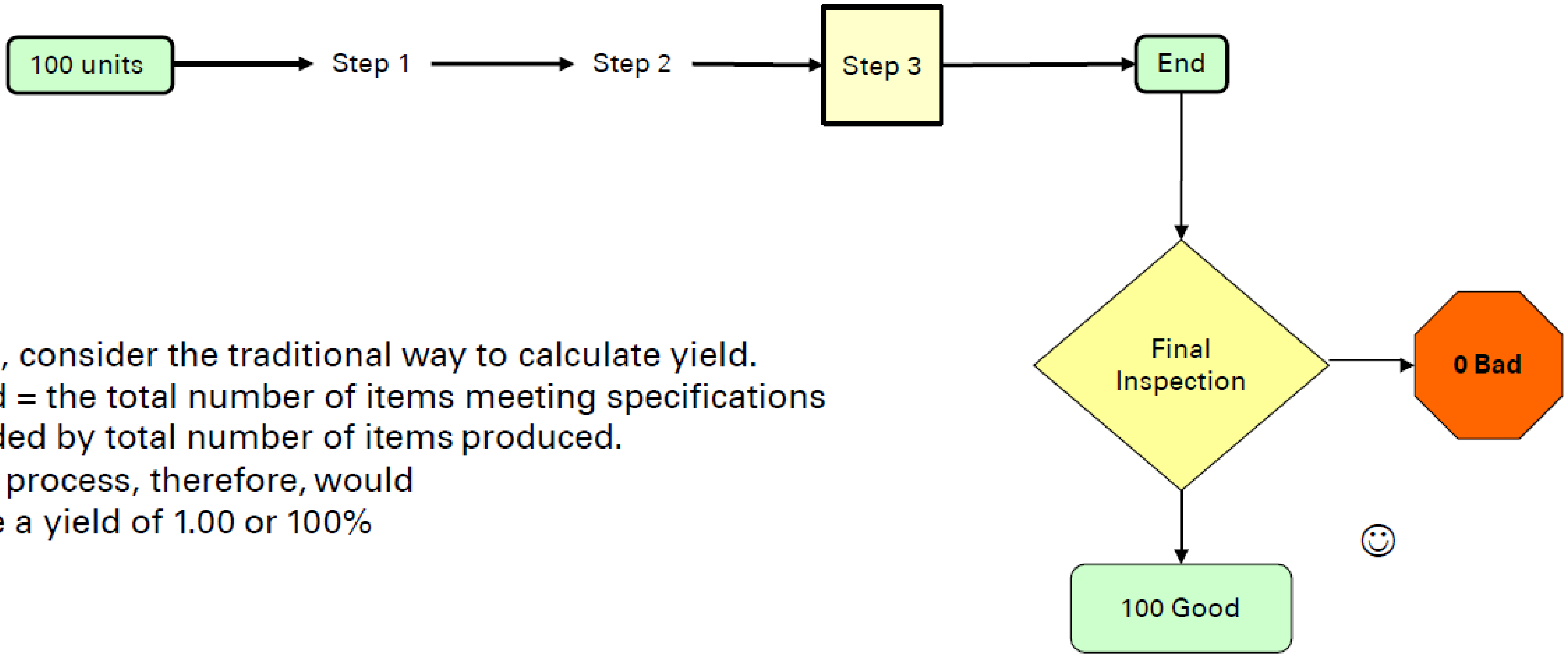
Attribute Data - Rolled Throughput Yield

- Another useful metric in Six Sigma is the Rolled Throughput Yield (RTY)
- RTY is the probability of obtaining a unit with zero defects.
- Let us consider a process as below



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

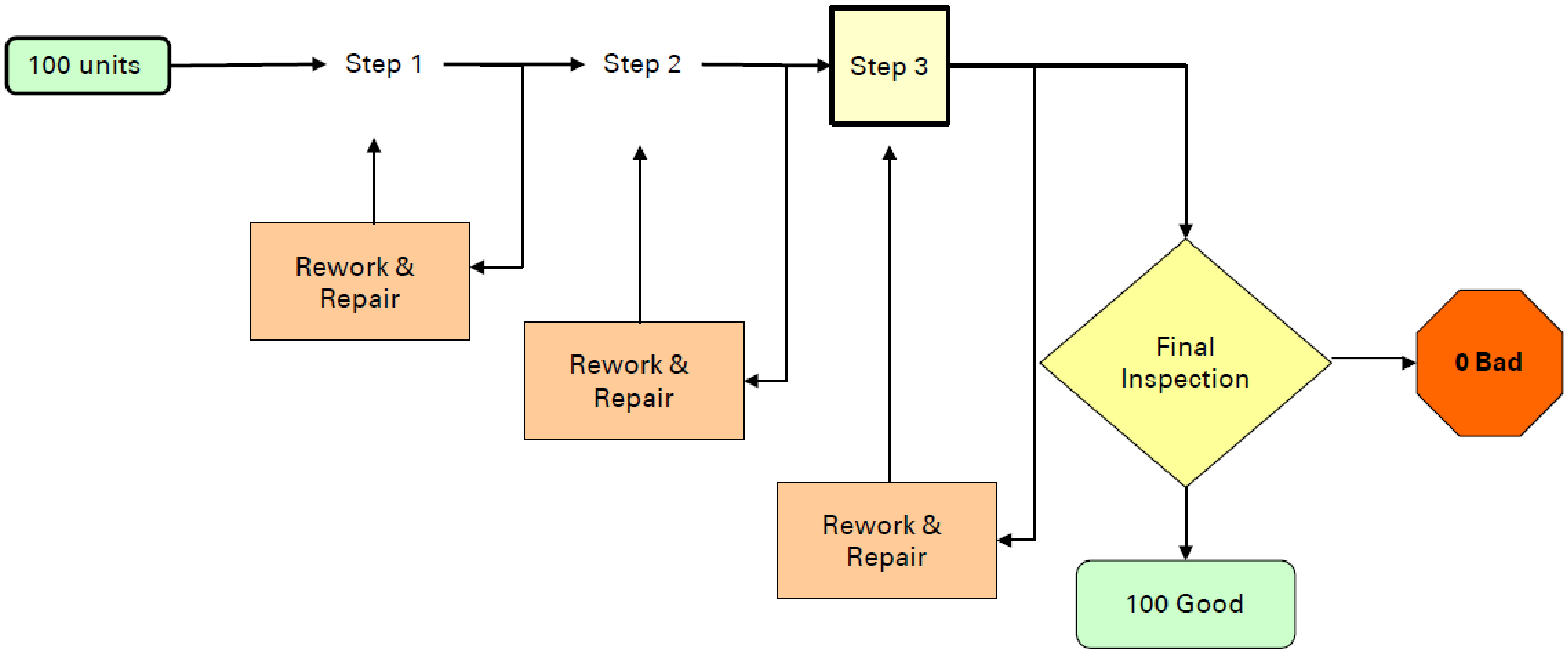
Attribute Data - Rolled Throughput Yield



First, consider the traditional way to calculate yield.
Yield = the total number of items meeting specifications divided by total number of items produced.
This process, therefore, would have a yield of 1.00 or 100%

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Data - Rolled Throughput Yield

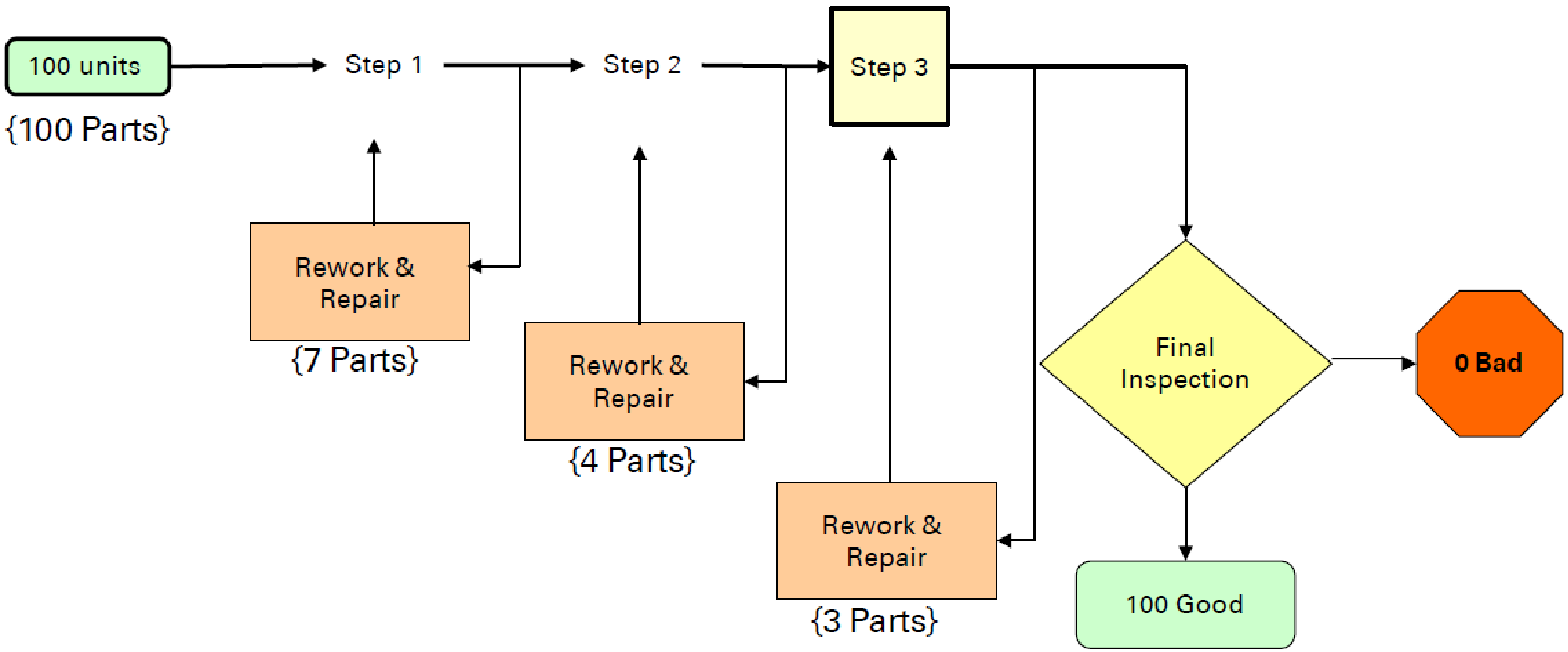


Upon closer inspection of the process, however, we begin to discover rework & repair at every station.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

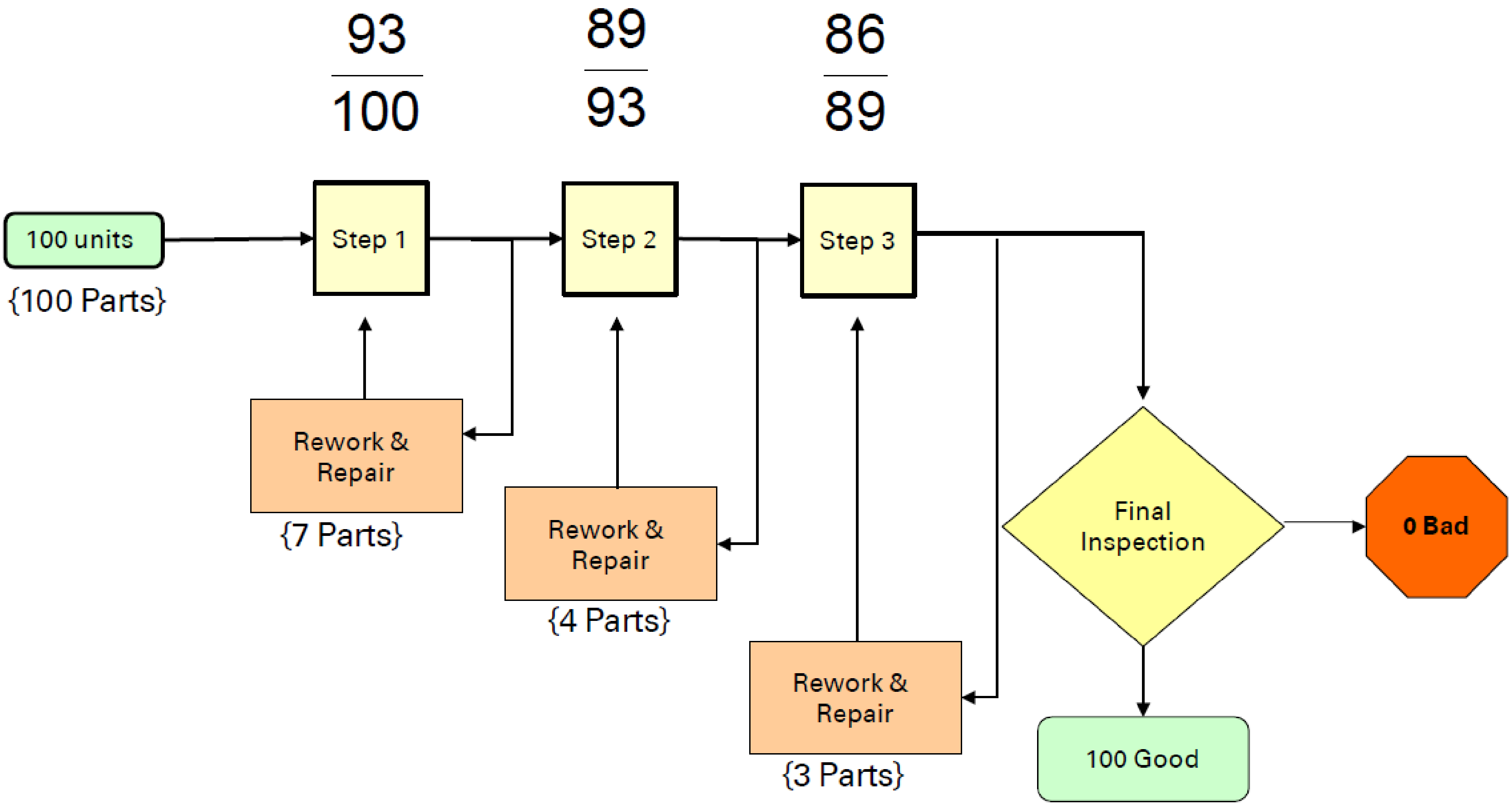
Attribute Data - Rolled Throughput Yield



If we start 100 parts through the process, how many are reworked or repaired before final inspection?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Data - Rolled Throughput Yield

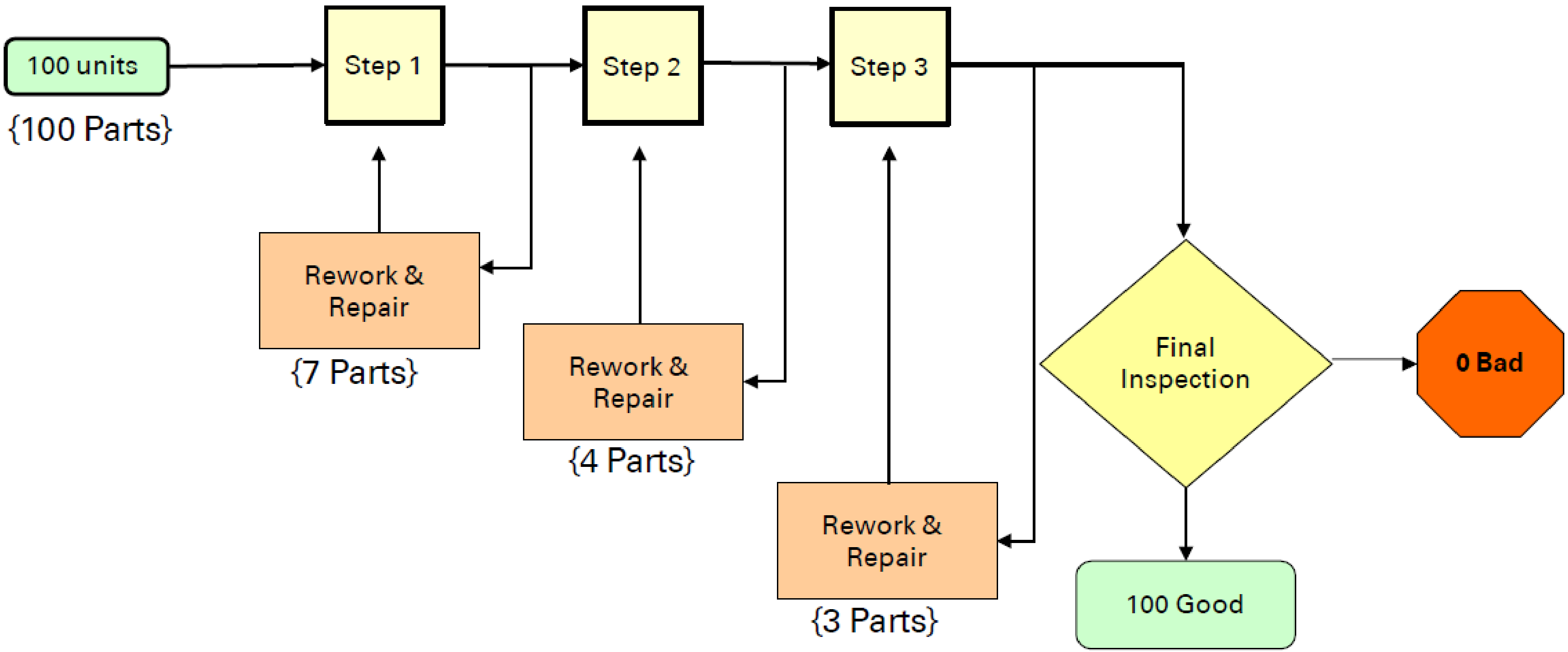


© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Data - Rolled Throughput Yield

RTY

$$\frac{93}{100} \times \frac{89}{93} \times \frac{86}{89} = 0.86 \text{ or } 86\%$$



KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Attribute Data - Rolled Throughput Yield

RTY

$$\frac{93}{100} \times \frac{89}{93} \times \frac{86}{89} = 0.86 \text{ or } 86\%$$

Start → Step 1 → Step 2 → Step 3 → End

This means that 14% of the effort expended in this process is due to defects which result in rework and repair. If there is a labor content of $430,000 per year, approximately 14% of that ($60,200) is waste.

That's the "hidden factory."

RTY can be used ...

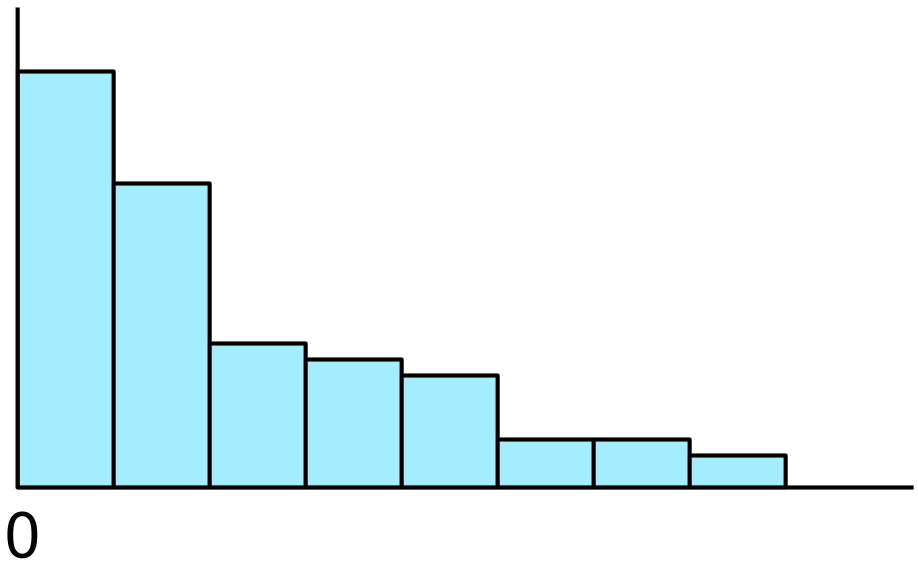
- To identify opportunities for improvement.
- To characterize the process steps, unveiling the hidden factory.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Introduction to Non-Normal Data & Data Transformation

Non-Normal Data

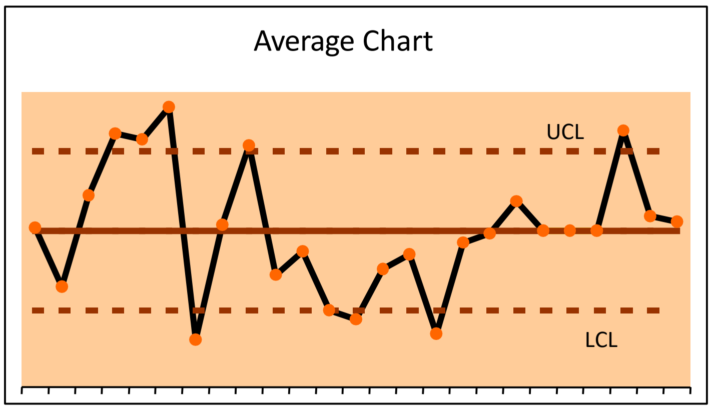
- Some data sets are not normally distributed. For example, some measures have zero or some other constant as a physical block.
- Examples: Time, concentricity, warpage, flatness

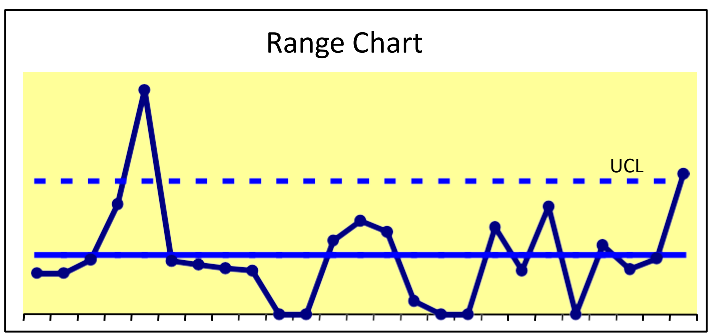


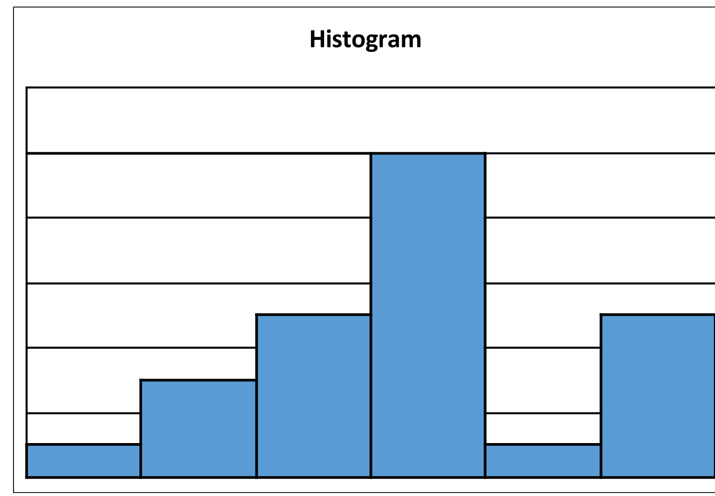
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Non-Normal Process

Sometimes data sets are not normal because the process is not yet in statistical control.



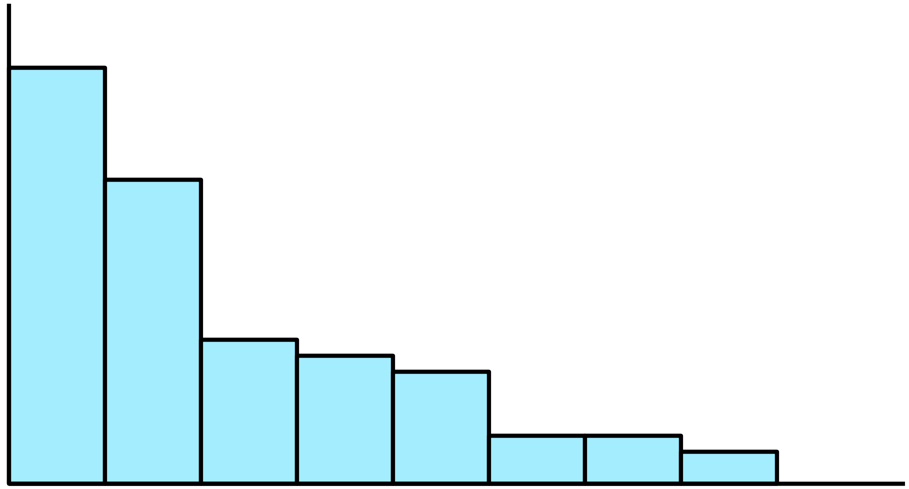




Once the process is stabilized, the data *may* form a Normal Distribution

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Handling Non-Normal Data



When a data set is naturally non-normal, we can still make predictions about the population.

In such a case, we must either…

Transform the data (for example, take the log of every data value) to achieve a Normal Distribution. Use some other type of distribution that approximates the actual shape of the data.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

What is data transformation?

Data Transformation is a technique whereby we transform a non-normal data by applying a function to the data that changes its values so that they more closely follow a normal distribution.

Why transform data?

- We transform the data to achieve a normal distribution
- However, if the data is not being transformed, one could use the Weibull Distribution to make predictions.

Types of Transformation Techniques

- There are various traditional approaches to data transformation. But the most commonly used are Box-Cox Transformation and Johnson Transformation

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Box-Cox Transformation

- Box Cox transformation is a popular power transformation method developed by George E. P. Box and David Cox.
- Data transforms are usually applied so that the data appear to more closely meet assumptions of a statistical inference model to be applied or to improve the interpret-ability or appearance of graphs.
- This transformation is easy to understand and provides both within-subgroup and overall capability statistics.
- Minitab finds the optimum power transformation, wherein it finds the best value for lambda (λ)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Box-Cox Transformation

- The formula for Box-Cox transformation is

$$\begin{cases} y = \dfrac{x^{\lambda} - 1}{\lambda} \text{ where } \lambda \neq 0 \\ y = \ln x \text{ where } \lambda = 0 \end{cases}$$

Where:

- y is the transformation result
- x is the variable under transformation
- λ is the transformation parameter.

- Common lambdas (λ) for data transformation:

Lambda (λ) value	Transformation
$\lambda = 2$	$Y' = Y^2$
$\lambda = 0.5$	$Y' = \sqrt{Y}$
$\lambda = 0$	$Y' = \ln Y$
$\lambda = -0.5$	$Y' = 1 / (\sqrt{Y})$
$\lambda = -1$	$Y' = 1 / Y$

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Johnson Transformation

- The Johnson transformation uses a different algorithm than the Box-Cox transformation.
- The Johnson transformation function is selected from three families of functions in the Johnson system as it covers wide variety of distributions by changing parameters. Minitab helps one to find this optimum parameter.
- Johnson Transformation is very powerful, and it can be used with data that include zero and negative values
- Use this when the Box-Cox transformation does not find a suitable transformation.

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Transformation using Minitab

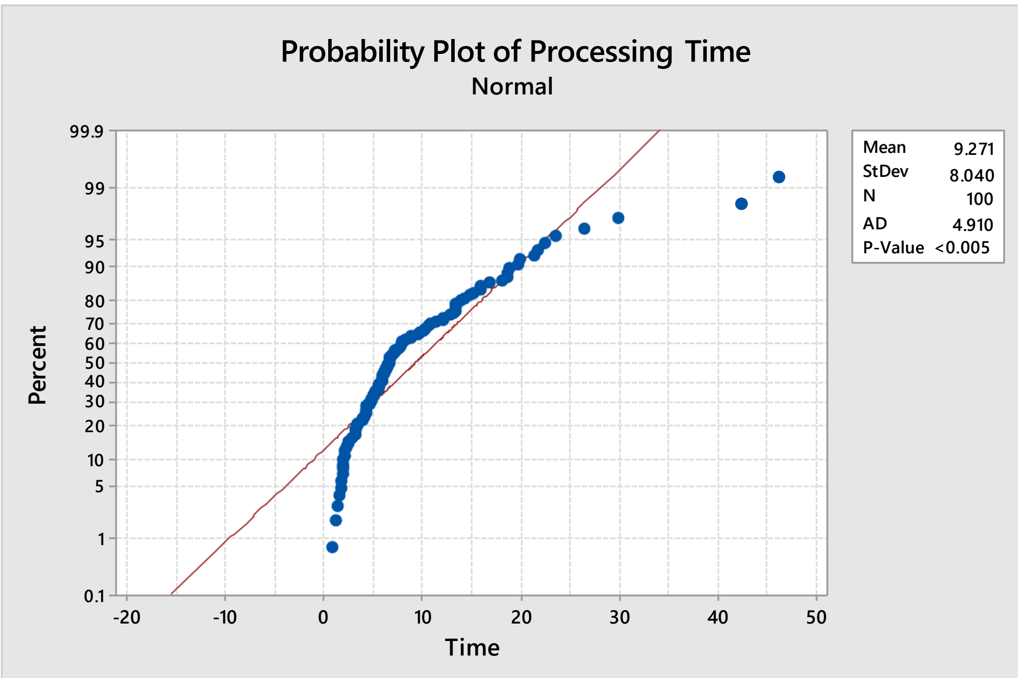
- Let us consider an example of ‘Processing Time’
- Test for Normality
- If P < 0.05, try to transform the data

Processing Time			
7.84	1.85	10.04	14.15
6.17	2.32	2.17	1.88
5.91	3.91	5.76	46.2
29.78	8.1	16.75	2
3.02	3.18	4.25	4.68
1.91	12.72	6.73	5.06
3.9	13.26	4.28	6.53
7.42	5.94	3.21	21.2
3.04	1.92	11.35	7.04
6.53	18.69	1.74	13.28
10.58	0.77	4.49	5.83
7.65	6.4	5.39	13.78
3.97	12.05	13.03	1.02
42.4	26.3	4.88	1.53
19.59	1.37	13.32	7.83
6.36	10.71	5.29	11.97
8.7	7	2.46	6.62
6.06	5.39	5.16	22.44
15.17	6.14	6.64	8.8
10.1	18.63	13.26	7.02
14.7	4.26	23.46	9.73
19.84	5.38	18.58	4.54
21.54	18.05	1.64	2.7
9.42	3.29	1.95	4.71
5.69	15.84	15.85	4.15

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Data Transformation using Minitab

The P value is < than 0.05, the data does not fall a normal distribution.

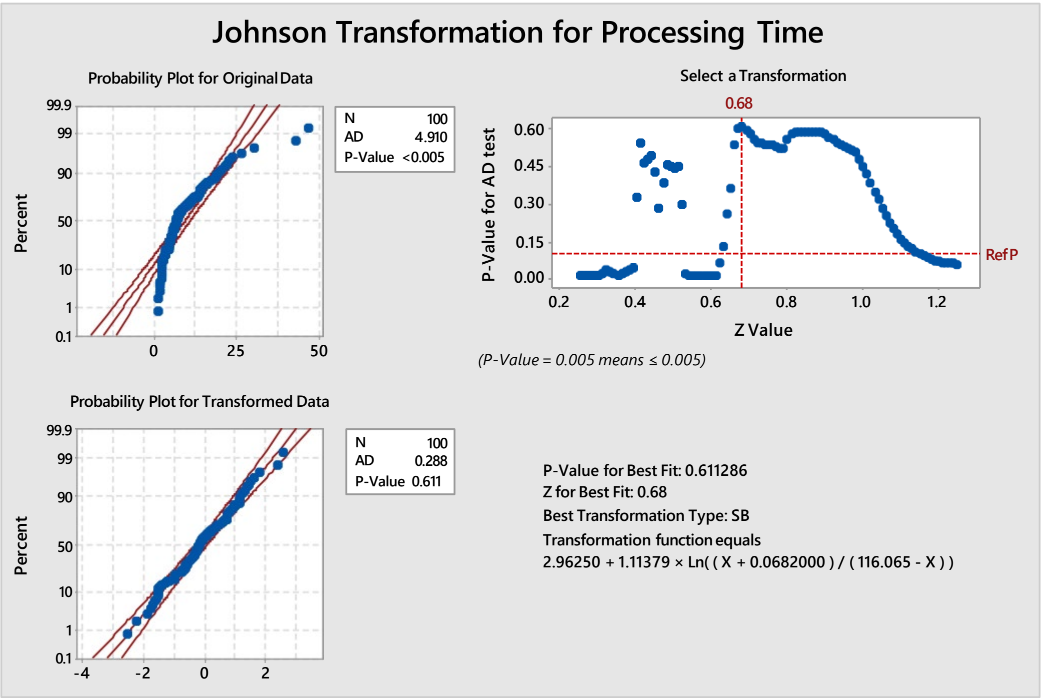


Processing Time			
7.84	1.85	10.04	14.15
6.17	2.32	2.17	1.88
5.91	3.91	5.76	46.2
29.78	8.1	16.75	2
3.02	3.18	4.25	4.68
1.91	12.72	6.73	5.06
3.9	13.26	4.28	6.53
7.42	5.94	3.21	21.2
3.04	1.92	11.35	7.04
6.53	18.69	1.74	13.28
10.58	0.77	4.49	5.83
7.65	6.4	5.39	13.78
3.97	12.05	13.03	1.02
42.4	26.3	4.88	1.53
19.59	1.37	13.32	7.83
6.36	10.71	5.29	11.97
8.7	7	2.46	6.62
6.06	5.39	5.16	22.44
15.17	6.14	6.64	8.8
10.1	18.63	13.26	7.02
14.7	4.26	23.46	9.73
19.84	5.38	18.58	4.54
21.54	18.05	1.64	2.7
9.42	3.29	1.95	4.71
5.69	15.84	15.85	4.15

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Transformation using Minitab

Stat > Quality Tools > Johnson Transformation



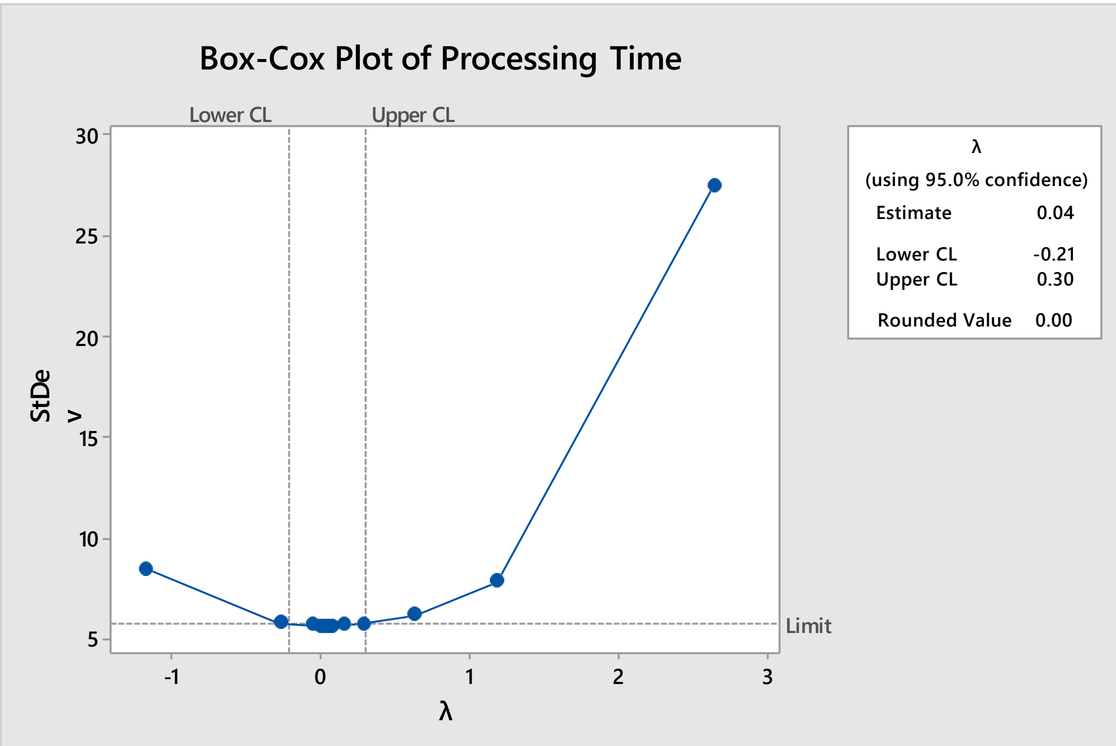
The P value for the transformed data is > than 0.05, the data now falls a normal distribution.

Processing Time			
7.84	1.85	10.04	14.15
6.17	2.32	2.17	1.88
5.91	3.91	5.76	46.2
29.78	8.1	16.75	2
3.02	3.18	4.25	4.68
1.91	12.72	6.73	5.06
3.9	13.26	4.28	6.53
7.42	5.94	3.21	21.2
3.04	1.92	11.35	7.04
6.53	18.69	1.74	13.28
10.58	0.77	4.49	5.83
7.65	6.4	5.39	13.78
3.97	12.05	13.03	1.02
42.4	26.3	4.88	1.53
19.59	1.37	13.32	7.83
6.36	10.71	5.29	11.97
8.7	7	2.46	6.62
6.06	5.39	5.16	22.44
15.17	6.14	6.64	8.8
10.1	18.63	13.26	7.02
14.7	4.26	23.46	9.73
19.84	5.38	18.58	4.54
21.54	18.05	1.64	2.7
9.42	3.29	1.95	4.71
5.69	15.84	15.85	4.15

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Transformation using Minitab

Stat > Control Charts > Box Cox Transformation



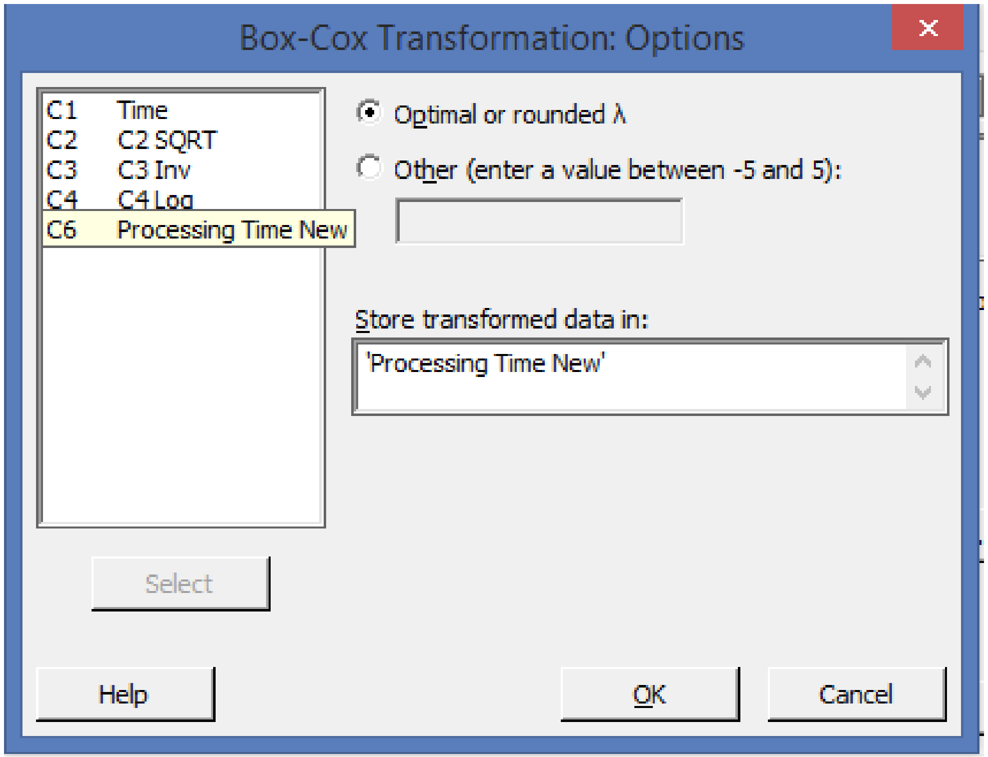
Minitab calculates the optimum value wherein the sum of squares of error is minimum. In this case the minimum value is 0.04

Processing Time			
7.84	1.85	10.04	14.15
6.17	2.32	2.17	1.88
5.91	3.91	5.76	46.2
29.78	8.1	16.75	2
3.02	3.18	4.25	4.68
1.91	12.72	6.73	5.06
3.9	13.26	4.28	6.53
7.42	5.94	3.21	21.2
3.04	1.92	11.35	7.04
6.53	18.69	1.74	13.28
10.58	0.77	4.49	5.83
7.65	6.4	5.39	13.78
3.97	12.05	13.03	1.02
42.4	26.3	4.88	1.53
19.59	1.37	13.32	7.83
6.36	10.71	5.29	11.97
8.7	7	2.46	6.62
6.06	5.39	5.16	22.44
15.17	6.14	6.64	8.8
10.1	18.63	13.26	7.02
14.7	4.26	23.46	9.73
19.84	5.38	18.58	4.54
21.54	18.05	1.64	2.7
9.42	3.29	1.95	4.71
5.69	15.84	15.85	4.15

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Transformation using Minitab

Click on options button in the Box Cox Transformation window



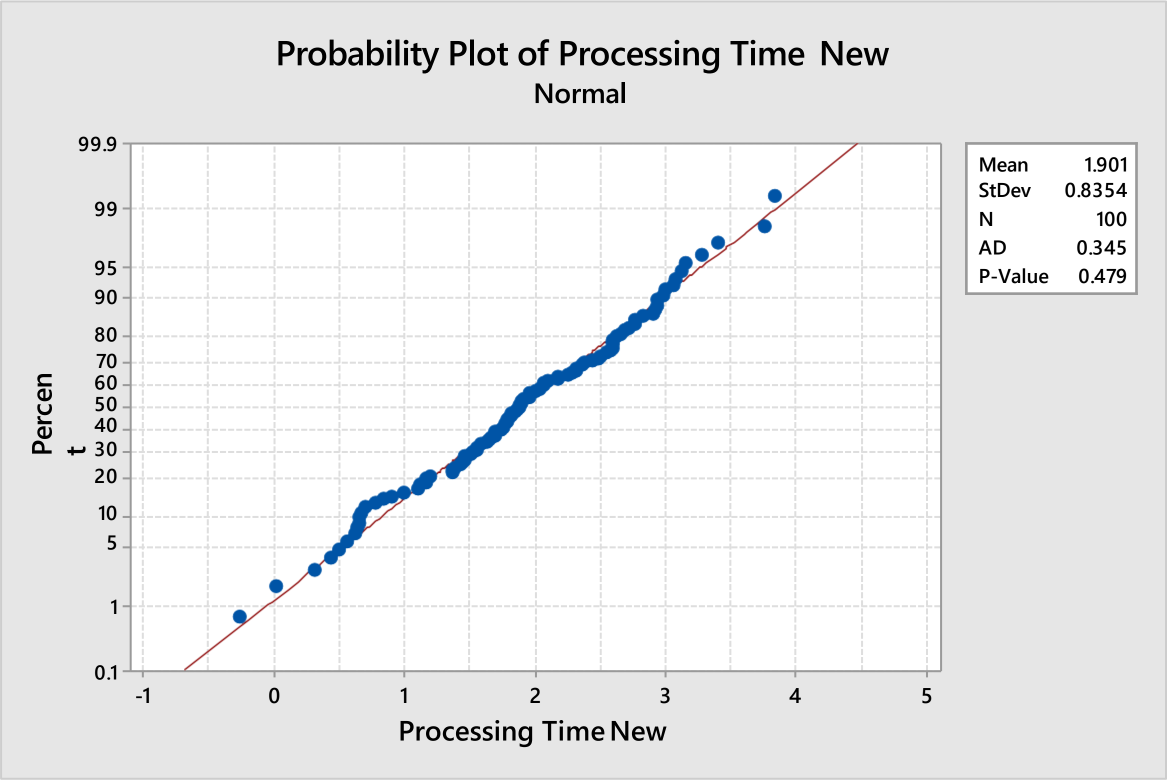
Create a new column in Minitab – Processing Time New

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Transformation using Minitab

Once you have the data in the new column, run the Test of Normality to check for the distribution

Stat > Basic Statistics > Normality Test



The P value for the transformed data is greater than 0.05, the data now falls a normal distribution.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

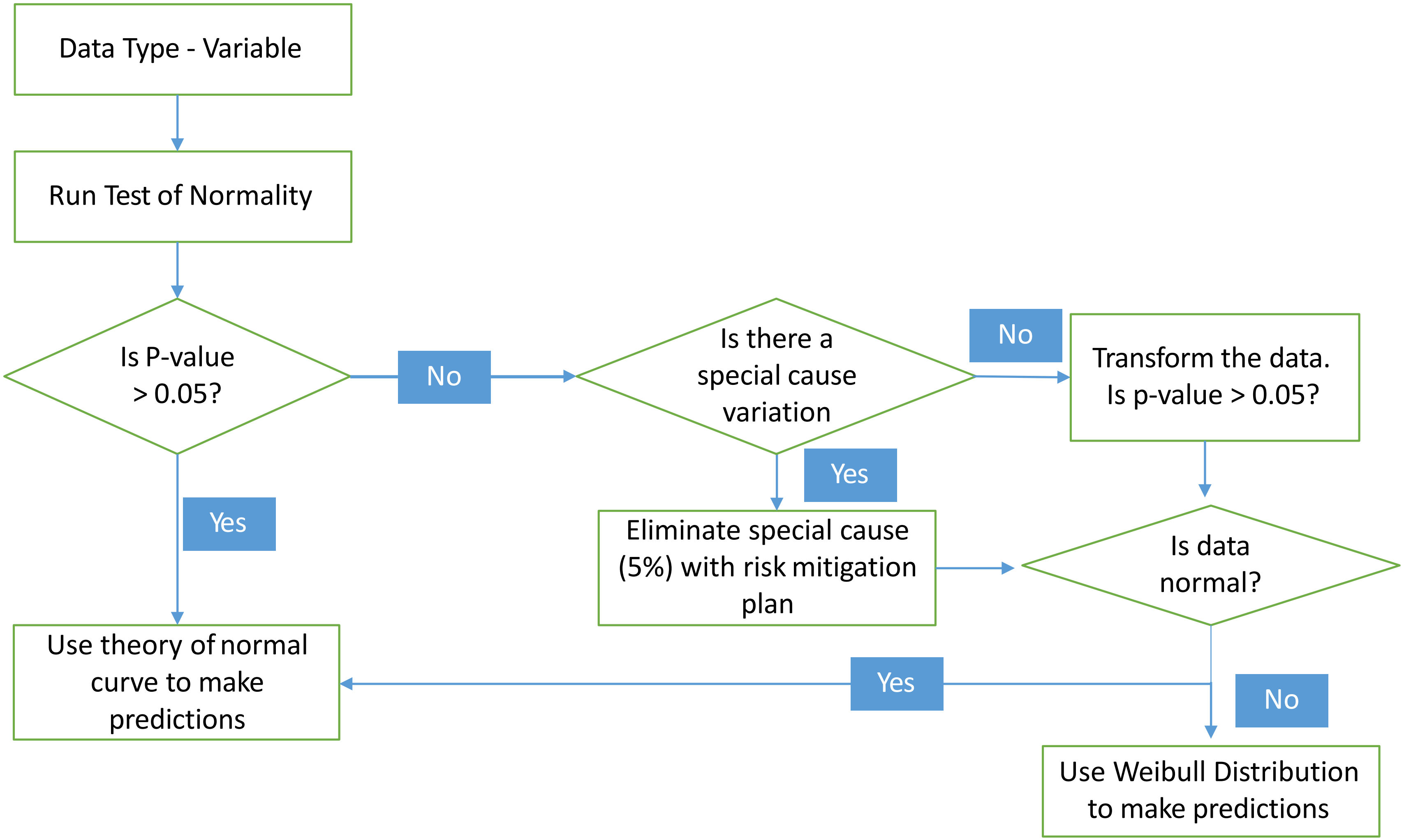
Non-Normal data with Weibull Distribution

- If we are unable to find a transformation that successfully converts non-normal data into a normal distribution, then can use the Weibull distribution, directly, to estimate the percentage of product beyond specifications.

- The Weibull is a family of distributions
- The Weibull has two parameters
  - β or the "shape" parameter
  - α or the "scale" parameter
- Minitab estimates these two parameters from the data set that is being analyzed.
- When $\beta \approx 1$, the Weibull distribution is exponential. When $\beta \approx 3.5$ the Weibull distribution approximates the Normal distribution.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Using Non-Normal Data



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Inference from Measure Phase

- The Defect (Y) and Possible Opportunities for the Defect to Occur
- Data Type – Attribute or Continuous
- Data Transformation techniques
- Types of distributions
- Types of Sampling
- Data Collection Check Sheets
- Operational Definitions and Measurement Method Statements
- Measurement System Analysis
- Current Process Capability (DPMO; RTY; Cp and Cpk)

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Analyze Phase

Analyze Phase - Roadmap

Identify Possible Causes → Process Door Analysis with Lean → Hypothesis Testing → Correlation and Regression Analysis

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Root Cause Analysis

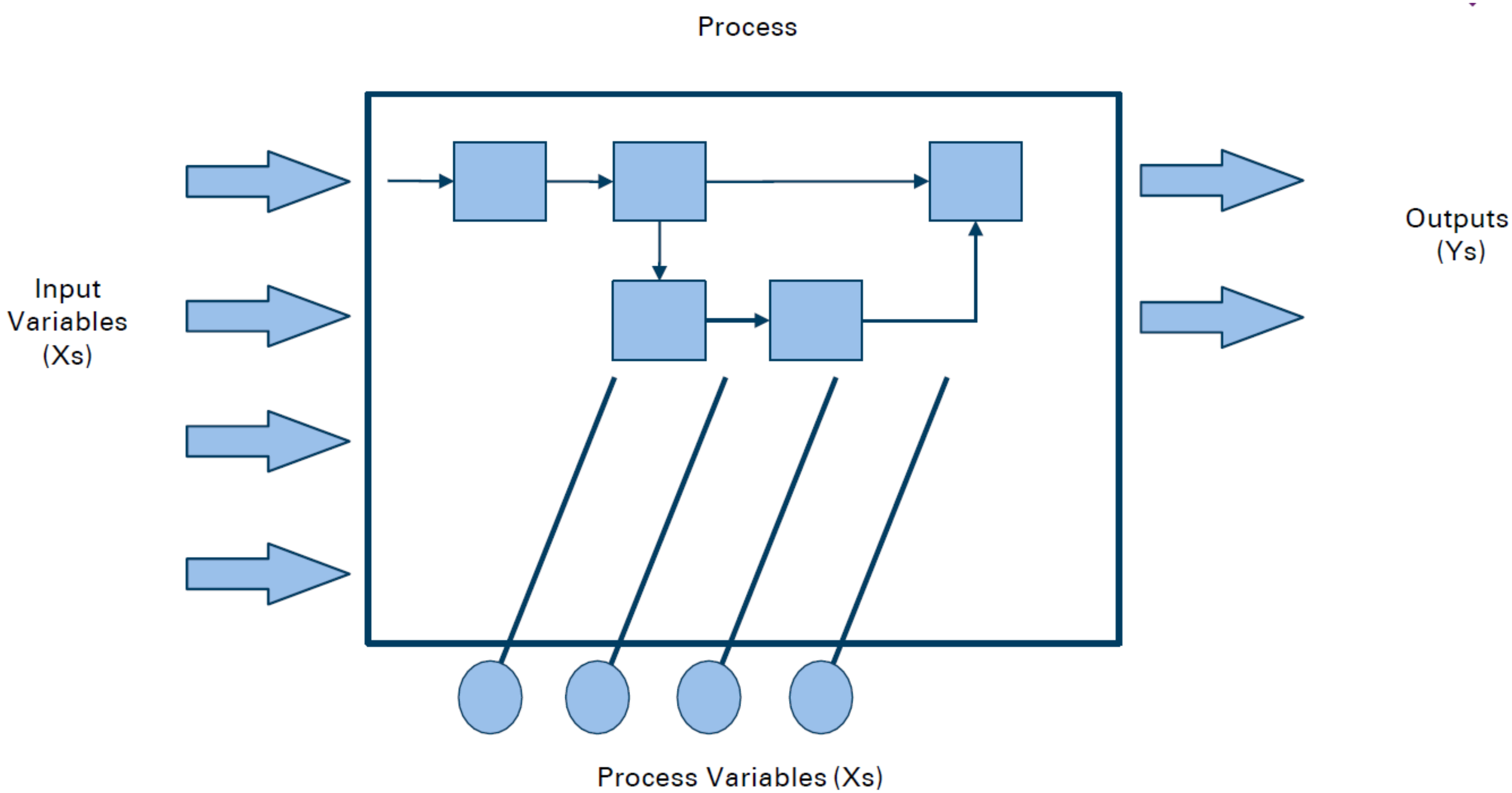
Root Cause Analysis

Why do a root cause analysis?

- Use the process data to understand the problem and identify the vital few root causes in order to reduce variation that the customer experiences
- Eliminate actions based on intuition and preconceived ideas
- Recalibrate project scope
- Establish performance goals for the process
- Allows teams to develop sustainable process improvements that will lead to long-term benefits
- Determine potential benefit of project

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Root Cause Analysis



What Are The Vital Few Process Inputs And Variables (X's) That Affect CTQ Performance Or Output Measures (Y's) ?

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Root Cause Analysis

Root Cause Analysis

Process Door Analysis

1. Identify the possible cause
2. Process diagnosis
3. Examine each process step
4. Justify

Data Door Analysis

1. Identify the possible cause
2. Minimize subjectivity
3. Question each assumption
4. Manage risk of decision errors

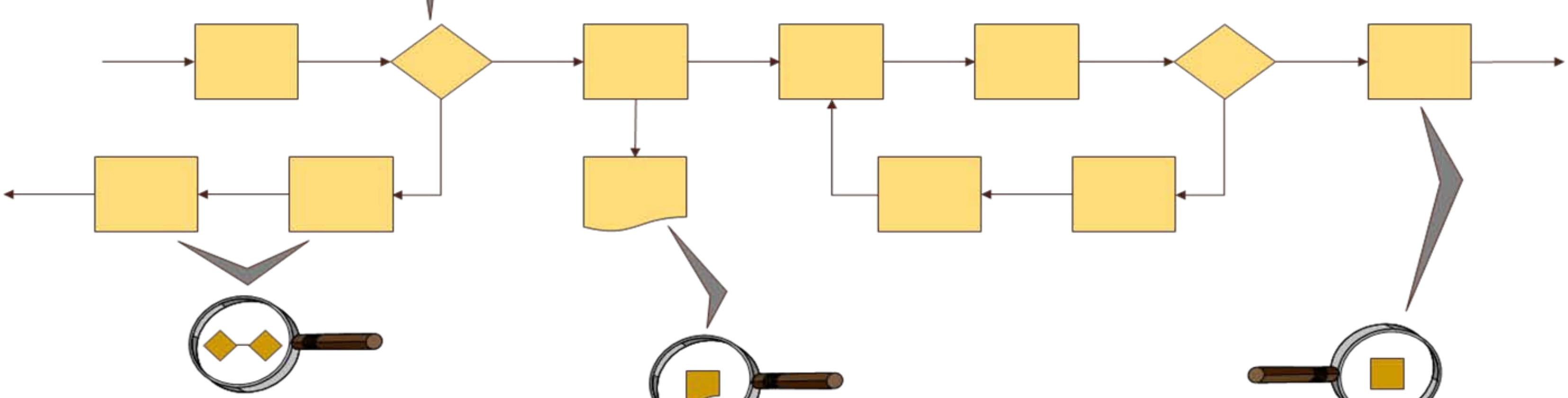
KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Process Door Analysis

Process Analysis

1. Examine Each Decision Symbol
   - Is this a checking activity?
   - Is this a complete check, or do some types of errors go undetected
   - Is this a redundant check?



2. Examine Each Rework Loop
   - Would we need to perform these activities if we had no failure?
   - How 'long' is this rework loop (steps, time lost, resources consumed, etc?)
   - Does this rework loop prevent the problem from reoccurring?

3. Examine Each Activity Symbol
   - Is this a redundant activity?
   - What is the value of this activity relative to its cost?
   - How have we prevented errors in this activity?

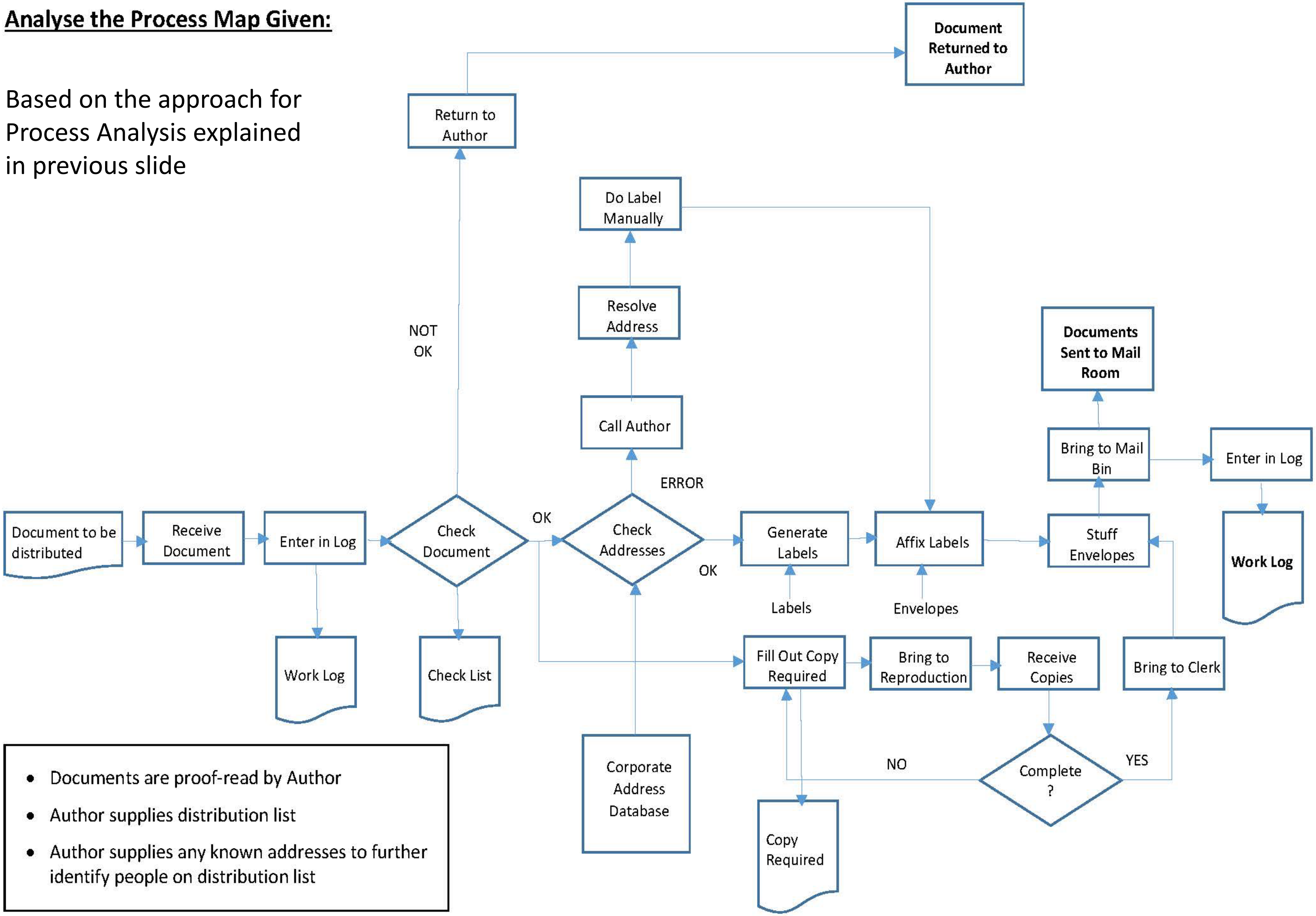
4. Examine Each Document or Database Symbol
   - Is this necessary?
   - How is this kept up to date?
   - Is there a single source for this information?
   - How can we use this information to monitor and improve the process?

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Analyse the Process Map Given:

Based on the approach for Process Analysis explained in previous slide



- Documents are proof-read by Author
- Author supplies distribution list
- Author supplies any known addresses to further identify people on distribution list

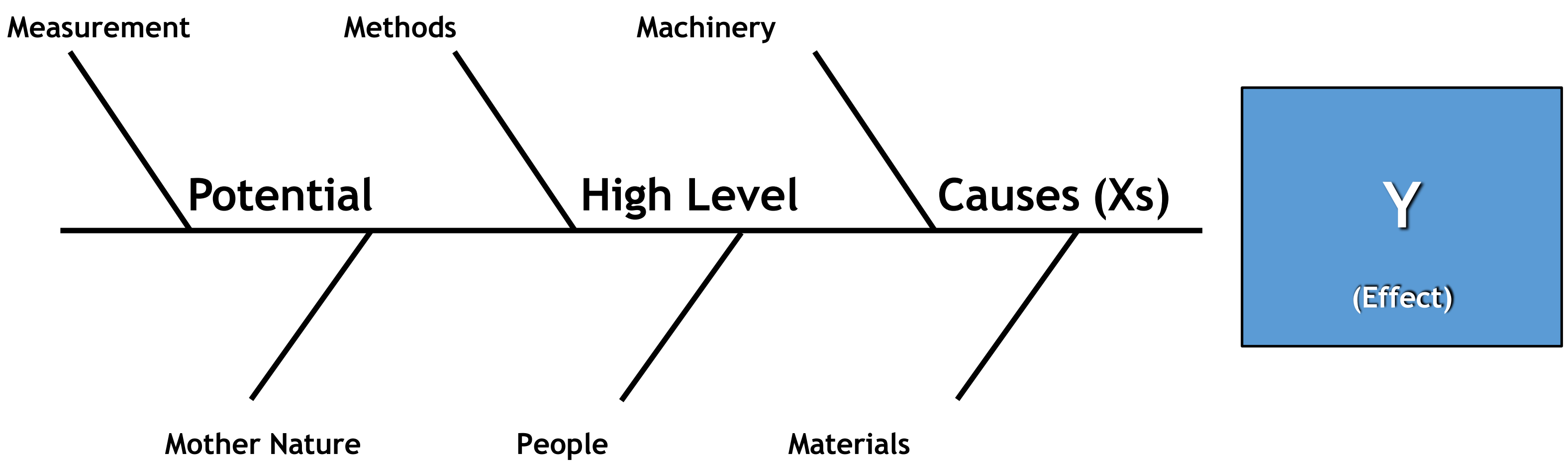
Data Door Analysis

Identify Possible Causes

Fishbone Diagram

Cause and Effect diagram

A visual tool used by an improvement team to **brainstorm** and **logically organize** possible causes for a specific problem or effect

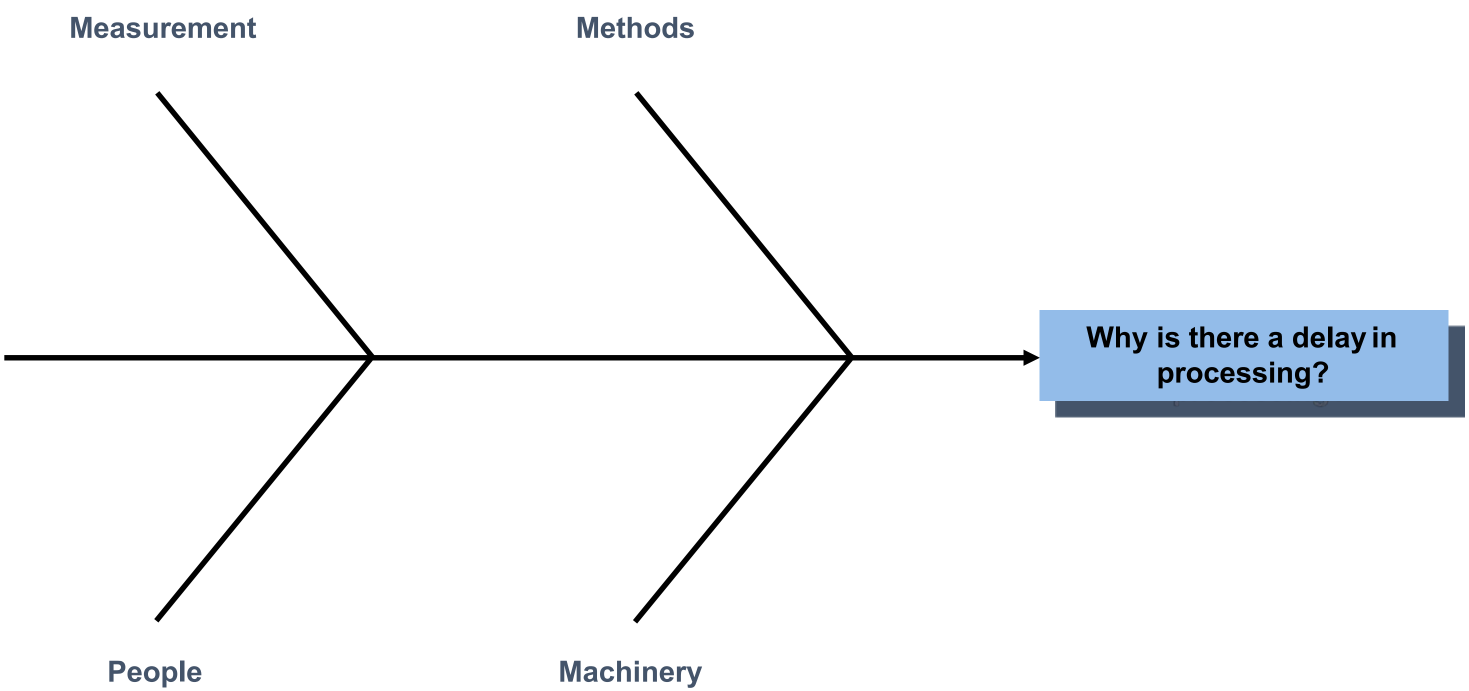


- Summarize potential high-level causes
- Provide visual display of potential causes
- Stimulate the identification of deeper potential causes

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

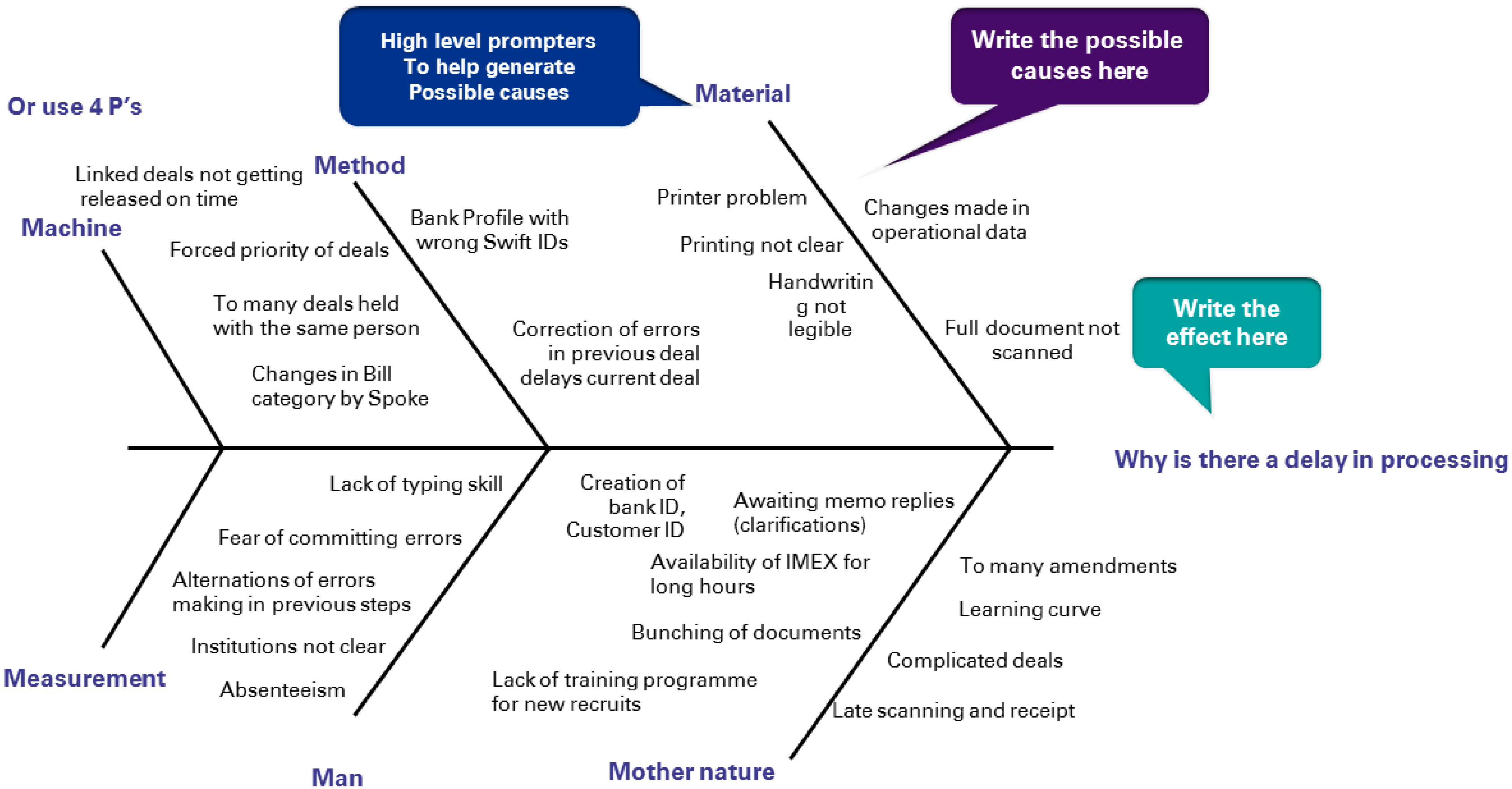
Why Cause and effect diagram

Brainstorm the "major" cause categories and connect to the centerline of the Cause & Effect diagram



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Cause and effect diagram - An Example



KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Sorting the Possible Causes

Non-Controllable Causes: These are causes that the team unanimously conclude are beyond the control of the present process boundaries or outside the physical location of the process execution.

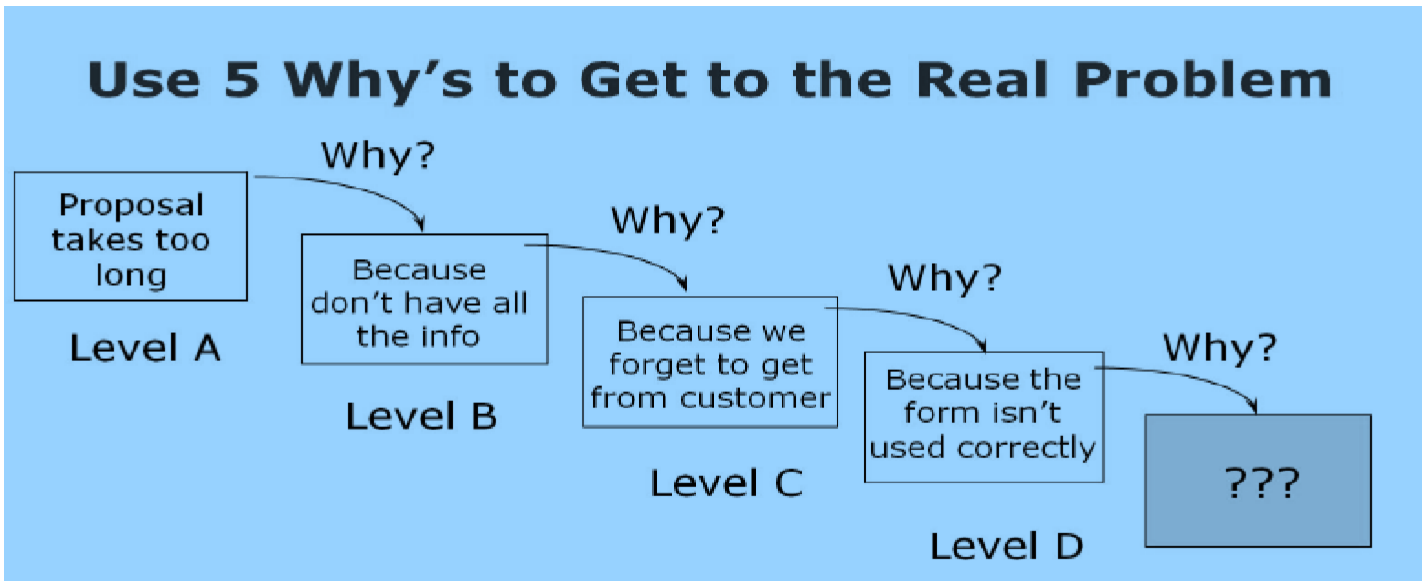
Lack of Solution OR Direct Improvements: These are causes that are actually solutions that can be implemented directly and need no further analysis. They are usually stated as lack of resources, equipment, tools or training.

Likely and Controllable Causes: These causes are the causes that have passed the above two filters and need further analysis.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Asking 5 Why's?

- The immediate next step after the segregation is to attack the likely and controllable causes and ask at least 3 – 5 why's?..... for each cause. This is called root cause drill down.
- Only after we have asked why 3 - 5 times to each of the likely causes, we will be able to arrive at the **possible root cause, also known as KPIV.**



Our project then becomes: Improving the percentage of RFQ's that the form is filled out correctly on.

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

5 Why's? - An Example

WHY do we have poor and declining participation in improvement programs?
1. Because people resist change .

Why do people resist change?
2. Because they fear making mistakes.

Why do people fear making mistakes?
3. Because they are criticized for mistakes.

Why are people criticized for mistakes? No ideas, let's move on.

Okay, then Why else do people fear making mistakes
4. Because they are penalized for mistakes.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Door Analysis

Pareto Analysis

Pareto Analysis

What is a Pareto Diagram?

- A Pareto chart is a bar chart that graphically ranks defects from largest to smallest, which can help prioritize quality problems and focus improvement efforts on areas where the largest gains can be made.
- Pareto analysis was conceptualized by an Italian economist – Vilfredo Pareto
- In his endeavor, Pareto tried to prove that distribution of wealth and income in societies is not random, but there is a consistent pattern.
- In his study – Principle of Unequal Distribution, he proved that 80% of wealth in Italy is controlled by 20% of elite
- The concept was formulated in six sigma by Dr Joseph Juran.
- Juran extended this principle of 80-20 to quality control stating that most defects in a production are a result of small percentage of the cause of the defects – which he described as “vital few from trivial many”
- Therefore, Pareto analysis is based on the principle that 80% of problems find their roots in 20% causes.
- In other words, it is a diagram that shows 20% of the inputs (X’s) cause 80% of the problems with dependent process outputs (Ys)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

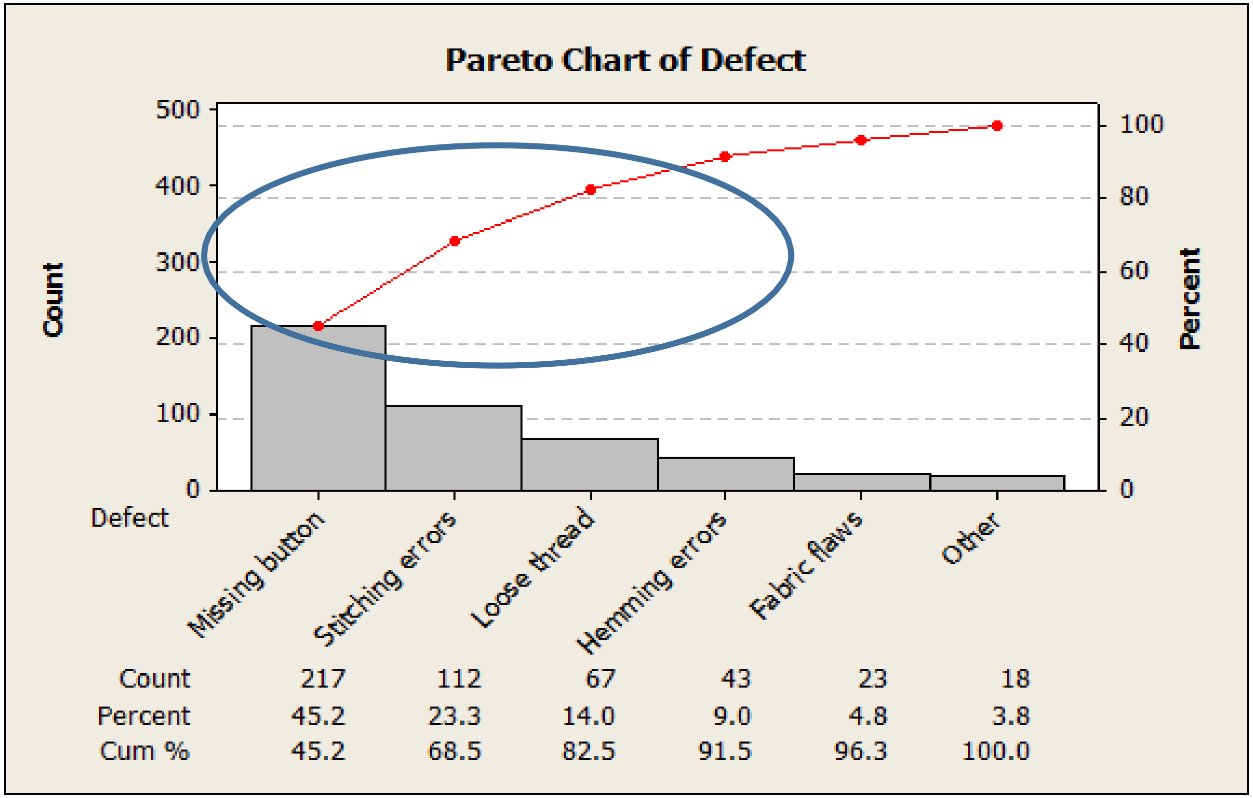
Pareto Analysis

Minitab path:
Refer the data: “Clothing Defect”

- Click on Stat → Quality Tools → Pareto Chart
- In Defects or attribute data in, enter *Defect*.
- In Frequencies in, enter *Count*.
- Select Combine remaining defects into one category after this percent, and enter 95.
- Click OK.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Pareto Analysis



In the above graph, 45.2% of the defects are missing buttons and 23.3% are stitching errors. The cumulative percentage for missing buttons and stitching errors is 68.5%. Thus, the largest improvement to the entire clothing process might be achieved by solving the missing button and stitching problems

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Tackling the other causes

- The non-controllable causes need to be factored into the improved process so that they do not affect the Process Capability.
- The team should launch initiatives to implement actions for the lack of solutions or direct improvements. This can happen in parallel as the project progresses.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Process Analysis with lean

Types of Wastes

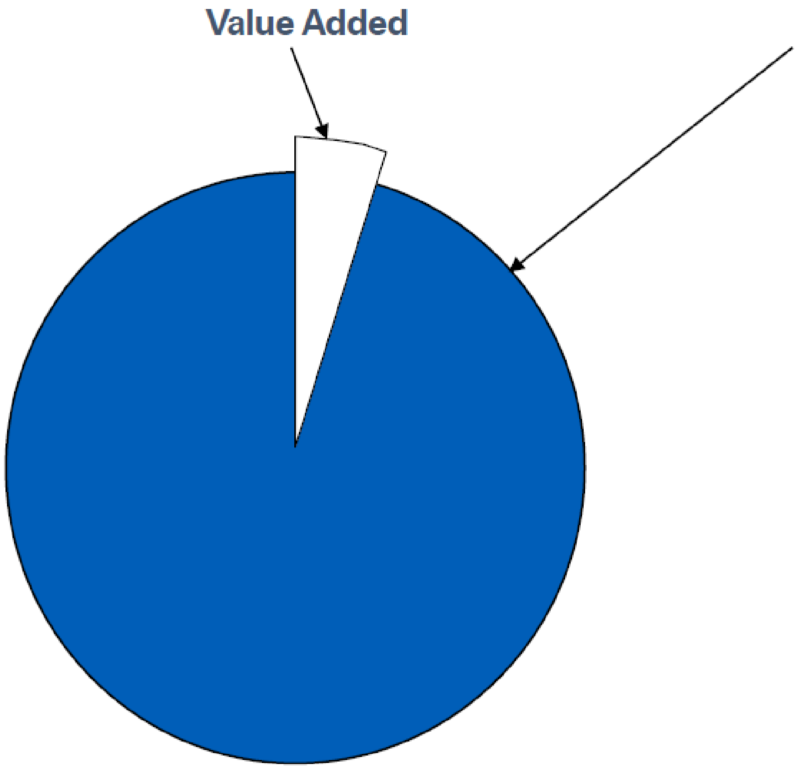
Types of Wastes

What is a Waste?

- Any element of the production, processing, service, delivery, or distribution that adds no value to the end user or customer or business or the final product is termed as waste.
- Waste adds cost, effort and time
- Waste is a symptom and not a root cause of the problem

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Identifying Wastes



- Transportation
- Inventory
- Motion
- Human Intellect
- Waiting
- Overproduction
- Over Processing
- Defects

Typically 95% of all lead time is non-value added

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Identifying Wastes

- Transportation – poor layout, poor flow
- Inventory – large batches, complexity to complete task
- Motion – poor organization, no standard work
- Human Intellect– old guard thinking, politics, high turnover, low investment in training
- Waiting – unbalanced work loads, slow system response, incomplete information, approvals
- Overproduction – releasing work before next process can work on them, unbalanced work loads
- Over-processing – excess communication, lack of communication, unnecessary approvals, customer requirements are not clearly understood
- Defects – incomplete or incorrect information

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Wastes Examples

Waste Type	Definition	Example
Transportation	Unnecessary movement of items resulting in wasted efforts and cost.	Conveyor Belts for transporting raw material from warehouse to plant
Inventory	Pile up of semi processed or finished goods that block working capital and hold up cash flow.	Loan Applications pied up on the managers table for approval.
Motion	Unnecessary movement of people to perform an activity.	Pharmacist moving around the whole shop in order to fetch medicines from various drawers
Human Intellect	Overutilization or underutilization of talent.	Engineer being used as operator for operating machine.
Waiting	Waiting for the next process step.	Waiting for your turn to meet the doctor at OPD in hospital
Over-Processing	Too much processing of information.	Five to six reviews / signatures on Purchase Orders before sending to Supplier and yet wrong material gets delivered.
Over-Production	Producing more than the requirement.	Producing goods for maintaining machine utilization in spite of no customer need or order. Over production will always lead to inventory buildup.
Defects	Any process or activity that results in rework.	Car doors rejected in paint shop due to dents or surface unevenness.

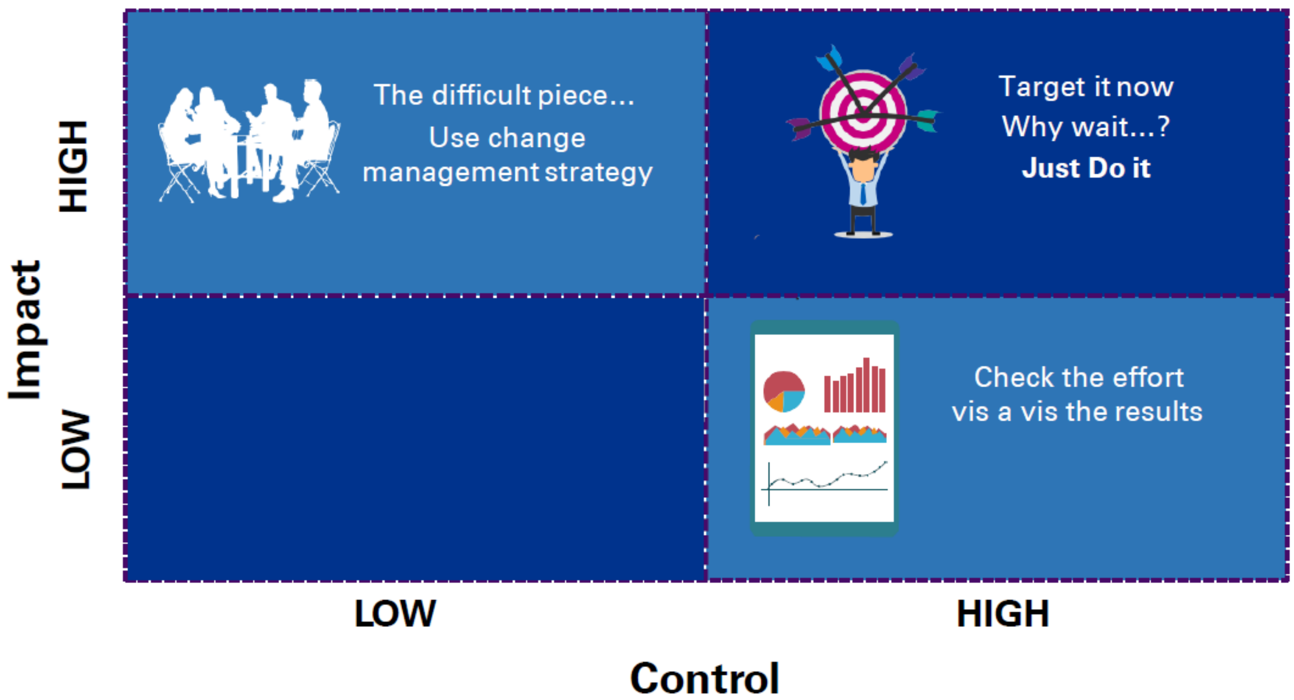
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Identify Possible causes

Control Impact Matrix

Control Impact Matrix

When we know the possible root causes...can we attack all?



We prioritize the root causes obtained in our root cause analysis using Control Impact Matrix

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Control Impact Matrix

- Classify all the causes that the group arrives at in the brainstorming session for the control and impact matrix
- Use your team's process knowledge and business experience to list possible Xs in a control/impact matrix, then use process data to verify or disprove placement of the Xs

Prioritization steps

- Using the control/impact matrix, examine each X in light of two questions:
  a) What is the impact of this X on our process?
  b) Is this X in our team's control or out of our team's control?
- With your team, place each X in the appropriate box on the matrix
- Use process data to verify or disprove your assertions
- The validated matrix is a guide used to addressing the Xs. Begin with the 'high impact/in our control' category
- 'Out of our control' Xs may require special solutions and CAP (change accelerating process) tools for successful sustainable solutions

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Data Door Analysis

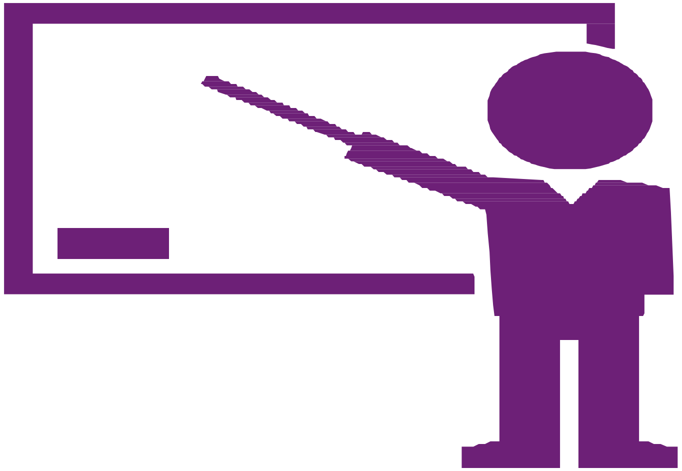
Hypothesis Testing

What is a Hypothesis?

A Hypothesis is a claim or statement about a property of a population (such as mean, variance, proportion).

The American Heritage College Dictionary defines Hypothesis as–

“a tentative explanation that accounts for a set of facts and can be tested by further investigation.”



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

What is Hypothesis Testing?

Measurements are organized into statistics to provide us insights by looking at the spread, shape, consistency and location of the process

Hypothesis Testing is simply a statistical method of comparing reality to an assumption and re-confirming – *did things actually change?*

In other words, Hypothesis Testing is a statistical way of checking whether observed differences between two or more samples are due to random chance or is there an actual difference in the sample.

We use hypothesis testing to test the assumptions established during problem analysis and investigation.

The practical meaning of Hypothesis Testing is that decisions taken based on the test results, with regards to implementing change will yield true and sustainable results.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Why Hypothesis Testing?

We use hypothesis testing in the Analyze phase of our DMAIC journey

- to verify that a suspected cause (X) truly impacts the CTQ (Y)

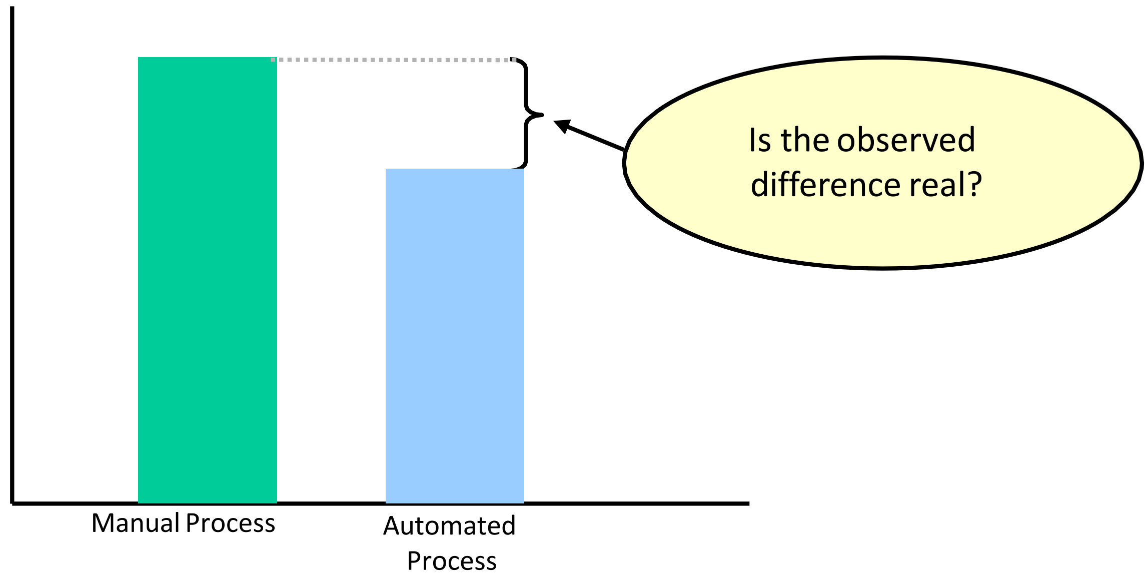
And post implementation of solutions

- to verify the improvement or change in CTQ (Y)

Let us look at an example here:

A team suspects that manual processing of documents (AS-IS process being followed) results in rework and delay in opening accounts for their customers.

Y = High TAT for account opening



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Key Terms in Hypothesis Testing

Null Hypothesis (Ho)

- There is “no evidence of difference”
- It is assumed to be true unless proven otherwise
- We never prove it, we only fail to reject it
- Can only contain the condition of ‘equality’
- In other words, the Null Hypothesis is the antithesis to our claim regarding the relationship of two or more data sets.

Alternate Hypothesis (Ha)

- The statement that we would like to show is true
- It is a claim that is being studied
- Also known as research hypothesis
- It usually defines the direction of desirable change, can be either < / > / not equal
- It is the hypothesis that we are attempting to prove or test
- Is the statement that must be true if Null Hypothesis is false

The Null and Alternate Hypothesis are mutually exclusive and together complete the entire set of probabilities.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Group Exercise

1. The average time taken to fill open positions for an analyst is 56 days. After implementing improvements, new data was collected. The average is now 28 days. Workers claim the process has changed.

2. Two methods (A and B) exist for generating orders. You wish to discover whether or not the methods are different in terms of the number of Purchase Orders generated with errors.

3. You are testing samples of Fabric A to make sure it has an average tensile strength greater than 75 lbs. per square inch.

4. Handling of door panels prior to the foam (foam a) curing has caused imprints resulting in rejections. To deal with this problem, we want to study the cure time of an alternative foam (call it foam b). Does foam b take less time, on average, to cure?

For the above, define the Null and Alternative Hypothesis

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Hypothesis Testing: P- Value

Our basis of taking a decision depends on the P-value

So what is a P-value?

The P-value is the probability of obtaining a particular sample if the null hypothesis (Ho) is true

The P-value is based on an actual or assumed reference distribution like Normal Distribution, F-Distribution, Chi Square Distribution, etc..

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Hypothesis Testing: Level of Significance

How do we take a decision now?

Generally, if P-Value < (less then) α, we reject the null hypothesis.

What is the level of significance (α)?

- The level of significance (α) is always set before the hypothesis test is done
- The α is most often set at 0.05 at 95% confidence

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Hypothesis Testing - Decisions

How do we infer the Hypothesis Tests?

As a result of the hypothesis test, we will either….

- Reject the Null Hypothesis, or
- Fail to Reject the Null Hypothesis

We always work with the Null Hypothesis; the test result will tell us if we can rejector fail to reject the Null Hypothesis

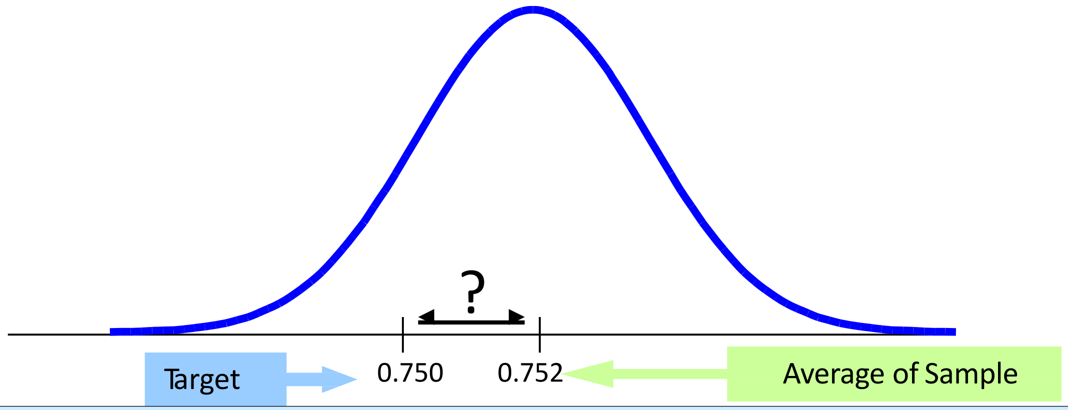
KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Hypothesis Testing: How to interpret the P-value

Let us consider a situation as below:

Ho: The process mean equals target mean
Ha: The process mean does not equal target mean



P- value = 0.04 ➡ We reject the Null Hypothesis!

From the P-value (Minitab Output) we can interpret that there is only a 4% chance of obtaining this sample if, indeed, the process mean equals the target. Therefore, we are led to believe the process mean does not equal the target. Therefore, we reject the Null Hypothesis.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Risk of Hypothesis

Whenever a hypothesis test is run, there is a risk associated with the decision that is made.

There are two types of errors (risks):

Type I error: It is the probability of rejecting the null hypothesis when it is true. This is also known as alpha risk, denoted by α.

Type II error: The probability of accepting the null hypothesis when it is false. This also known as beta risk, denoted by β

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Type I and Type II errors

Whenever a hypothesis test is run, there is a risk associated with the decision that is made. There are two types of errors (risks):

Decision \ Reality	Null Hypothesis True	Null Hypothesis False
Accept Null Hypothesis	Correct decision	Type II error
Reject Null Hypothesis	Type I error	Correct decision

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Steps for conducting Hypothesis Testing

- Define the Null and Alternative Hypotheses
- Determine the level of significance (α)
- Randomly select a representative sample of data
- Compute the P-value
- Compare the P-value to the level of significance (α)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Group Exercise

Define the Null and Alternate Hypothesis for the following situations:

- There is a difference between the average processing time from the two departments.
- A bolt with thread type A has a stronger torque, on average, than the bolt with thread type B
- There is a difference in the proportion of lost prototype seats between the two Business Units.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Steps for conducting Hypothesis Testing

Let us look at the steps with some illustration

Write Null Hypothesis	H_0 : x Sample A = x Sample B
Write Alternate Hypothesis	H_A : There is a difference between Samples A and B
Decide on the p value	p = 0.05 (typical for DMAIC projects at 95% confidence)
Choose hypothesis test	Choose the correct test, given the type of X and Y data
Gather evidence and test/conduct analysis	Collect data, run analysis, get p value
Reject H_0 /not reject H_0 and draw conclusion	If $p>0.05$ conclude H_0 is true If $p<0.05$ conclude H_0 is not true

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Types of Hypothesis Testing

- There are many types of hypothesis tests.
- The choice of test depends on the …
  - Type and distribution of the data and the
  - Kind of comparison that is being made.

Common Hypothesis Tests	Application
• One sample - t Test • Two sample – t Test • Paired t Test • One way ANOVA • z Test	Comparing Population **Means**
• One Proportion Test • Two Proportion Test • Chi Squared Test	Comparing Population **Proportions** or Percentages
• One Variance Test • Two Variance Test • Test of Equal Variances	Comparing Population **Variances**
• One Sign Test • Mann Whitney Test • Moods' Median Test	Comparing Population **Median**

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Sample Size - Larger vs Smaller samples

When performing hypothesis test with variable data…

- The sample size is considered large when $n > 30$.
- The sample size is considered small when $n \leq 30$.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Hypothesis Testing

Test of Means

1 sample z-Test: An Overview

The 1-sample Z test is used when….

- Testing the equality of a population mean to a specific value, and
- Sample size is large ($n > 30$)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

1 sample z-Test: Example

You are attempting to assess the speed of delivery when ordering a specific commodity via two different modes.

Delivery via mode A has been traditionally assumed to generate the best response; but we need to test that assumption against mode B.

Mode A:

Average delivery time = 6 days
Standard deviation = 2 days

A random sample of size 36 was collected from the Mode B, yielding:

Mode B:

Average delivery time = 4.7 days
Standard deviation = 2.0 days

Is there a difference in average delivery speed between Mode A and Mode B? Define the

Null and Alternate Hypothesis

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

1 sample z-Test: Example

- Define the Hypothesis

  Null Hypothesis: Average delivery time Mode A equals historical mean of 6 days Alternate

  Hypothesis: Average delivery time using Mode B does not equal 6 days.

- Randomly select a representative sample of data (36 data points were collected with average of 4.7 days and standard deviation of 2 days)

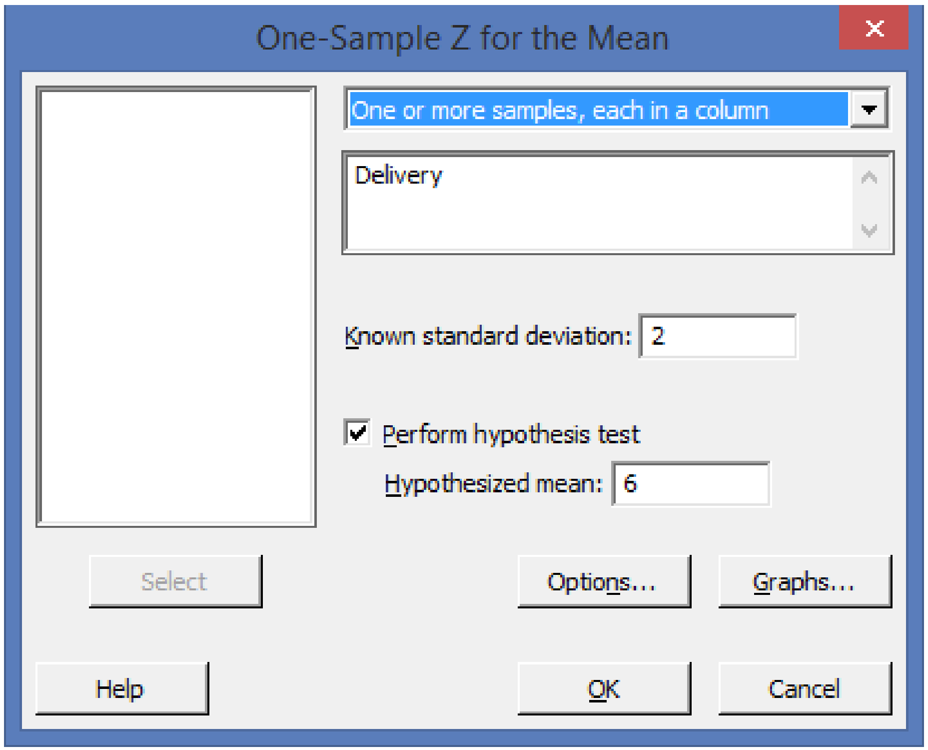
- Compute the P-value: the probability of obtaining the observed sample if the null hypothesis is true. Using Minitab…

  Select: Stat > Basic Statistics > 1-Sample Z

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

1 sample z-Test: Example

We are performing a 1-sample Z test because we are testing the equality of a population mean to a specificvalue (6 days), and we have a large sample (n ≥ 30).



Delivery Time	
6	6
7	3
3	2
6	2
7	4
8	3
5	4
3	6
2	4
5	7
3	4
9	4
5	1
5	4
4	7
6	3
4	4
5	9

Enter the data in Minitab. Standard Deviation 2, Hypothesized Mean6

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

1 sample z-Test : Example

One-Sample Z: Delivery

```
Test of mu = 6 vs mu not = 6

The assumed sigma = 2
```

Variable	N	Mean	StDev	SE Mean
Delivery	36	4.722	1.966	0.333

Variable	95.0% CI	Z	P
Delivery	(4.069, 5.376)	-3.83	0.000

Since the P-value of 0.000 is less than the level of significance 0.05, we reject the Null Hypothesis.

The data provides sufficient evidence that the average delivery time when using Mode A does not equal the historical value of 6 days.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One Sample t-Test: Overview

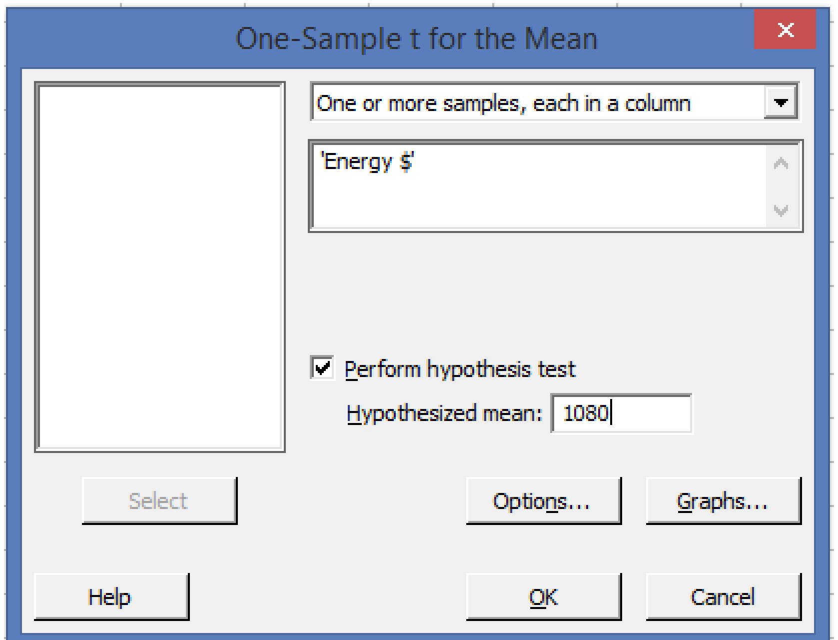
The 1-sample t- test is used when….

- Testing the equality of a population mean to a specific value, and
- Sample size is small (n $\leq$ 30)
- Refer worksheet ENERGY.MTW in Minitab
- **Select: Stat > Basic Statistics > 1-Sample t**

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One Sample t-Test: Example

- Refer worksheet ENERGY.MTW in Minitab
- Select: Stat > Basic Statistics > 1-Samplet



Energy $
1211
1572
1668
1250
1478
1307
1184
865
1162
1308
1188
1111
1747
1326
1142

Define the Null and Alternate Hypothesis

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One Sample t-Test : Example

- Define the Hypothesis

  Null Hypothesis: Mean expenditure =$1080

  Alternate Hypothesis: Mean expenditure does not equal $1080

- Compute the P-value: the probability of obtaining the observed sample if the null hypothesis is true.

One-Sample T: Energy $

Test of μ = 1080 vs ≠ 1080

Variable	N	Mean	StDev	SE Mean	95% CI	T	P
Energy $	15	1301.3	231.0	59.6	(1173.3, 1429.2)	3.71	0.002

Since the P-value of 0.002 is less than the level of significance 0.05, we reject the Null Hypothesis.

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Two Sample Test: Overview

In this section we discuss about the 'Two sample tests' viz:

- Two Sample z-test
- Two Sample t-test

The Two-sample z test is used when testing the equality of two population means and the two samples are large ($n > 30$).

The Two-sample t test is used when testing the equality of two population means and the two samples are small ($n \leq 30$).

Minitab does not offer a 2-Sample Z test. Therefore, we use the Two-Sample t test instead.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

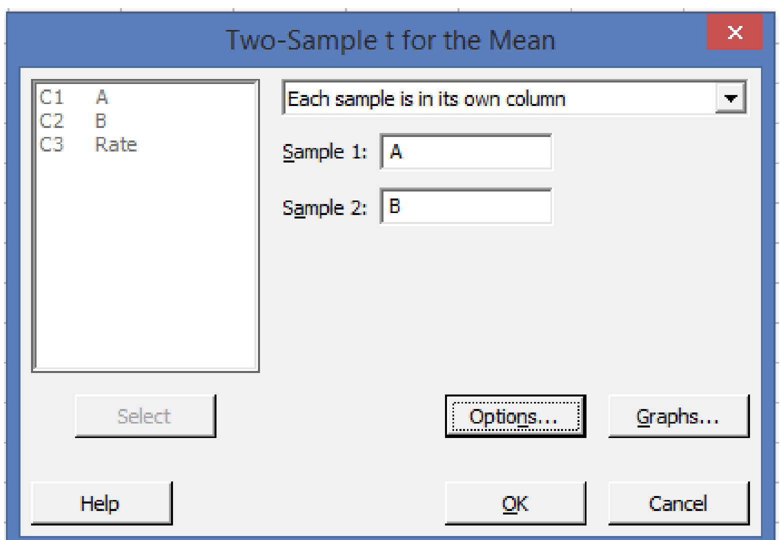
Two Sample Test: Overview

The Two-sample t test is used when the two samples are independent (not related, Unpaired data).

The conditions for using the student t distribution to conduct hypothesis tests with small samples are:

1. The population standard deviations are unknown.
2. The parent populations each have a distribution that is essentially normal.

- Refer worksheet HOSPITALRATINGS.MTW in Minitab
- **Select: Stat > Basic Statistics > 2-Sample t**



A	B
81	89
77	64
75	35
74	68
86	69
90	55
62	37
73	57
91	42
98	49
	59
	58
	65
	71
	67

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Two Sample t-Test : Example

- Define the Hypothesis

  Null Hypothesis: Mean ratings for Hospital A and Hospital B are the same. Alternate

  Hypothesis: Mean ratings for Hospital A and Hospital B are different.

- Compute the P-value: the probability of obtaining the observed sample if the null hypothesis is true.

Two-Sample T-Test and CI: A, B

```
Two-sample T for A vs B

    N  Mean  StDev  SE Mean
A  10  80.7   10.6      3.4
B  15  59.0   14.2      3.7

Difference = μ (A) - μ (B)
Estimate for difference:  21.70
95% CI for difference:  (11.38, 32.02)
T-Test of difference = 0 (vs ≠): T-Value = 4.36  P-Value = 0.000  DF = 22
```

Since the P-value of 0.000 is less than the level of significance 0.05, we reject the Null Hypothesis. This means that there is a difference in the mean rating for hospital A and hospital B.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Paired t-Test: Overview

- If one sample is related to another, the samples are <u>dependent (paired)</u>.
- For instance, if you made a claim about a drug designed to lower cholesterol, you would make cholesterol measurements on the same individuals before and after the use of the drug. You would need paired data.
- If you take a measurement on the same circuit boards before and after burn in, there is a relationship or dependency between the measurements.
- In these cases, you do not examine two different data sets. Instead, you look at the difference between the before and after data to see if the resulting differences are significant.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Paired t-Test : Example

A Black Belt conducted a study to determine if training was effective at reducing the time it takes to process orders. The process is not automated and is highly dependent on the knowledge and skill level of individual processors.

The orders used throughout the study are essentially identical in magnitude.

Processor	Before Training	After Training
1	23	21
2	17	11
3	8	6
4	9	10
5	7	5
6	25	22
7	16	11
8	11	16
9	12	10
10	9	5

- Define the Null and Alternate Hypothesis
- Use Minitab to test the equality of two dependent samples
- Select: Stat > Basic Statistics > Paired t
- Interpret the P-value

Data is processing time in hours.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One way ANOVA: Overview

When we need to test the equality of more than two population means we use One Way Analysis of Variance (ANOVA)

Assumptions:

- Independent random samples have been drawn from "r" normal populations with means:
  $\mu 1 = \mu 2 = \mu 3 = \mu 4 ,\ldots, = \mu r$
- Each population has the same variance
- The random samples from each population do not need to be of the same size

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One way ANOVA: Example

Suppose you are testing to see if there is a significant difference in average stiffness among three foam formulations. You collected the random samples:

Formulation A	Formulation B	Formulation C
338.2	340.4	344.4
331.0	328.2	338.5
323.0	338.5	348.9
317.9	333.7	326.6
327.9	332.8	342.6
328.1	323.4	355.4
308.3	335.3	337.0
312.8	326.2	339.2
324.0	344.0	331.3
342.4	310.3	336.3
326.2	317.5	342.3
328.1	322.0	338.6
337.6	321.2	323.9
320.3	321.4	345.7
311.0	342.6	333.1
328.1	334.7	331.3

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One way ANOVA: Example

- Define the Null and Alternate Hypothesis
  - Null Hypothesis: Formulation A=B=C
  - Alternate Hypothesis: At least one mean is different
- Random samples of 16 foam pads were tested for each formulation.
- Use Minitab to test the hypothesis
- Select: Stat > Basic Statistics > One-Way
- Interpret the P-value

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One way ANOVA: Example

One-way ANOVA: Formulation A, Formulation B, Formulation C

```
Source   DF      SS     MS     F      P
Factor    2  1428.8  714.4  8.36  0.001
Error    45  3846.6   85.5
Total    47  5275.4

S = 9.246   R-Sq = 27.08%   R-Sq(adj) = 23.84%


                                   Individual 95% CIs For Mean Based on
                                   Pooled StDev
Level           N    Mean  StDev  ------+---------+---------+---------+---
Formulation A  16  325.35   9.72  (-------*-------)
Formulation B  16  329.52   9.75         (-------*-------)
Formulation C  16  338.43   8.18                        (-------*-------)
                                  ------+---------+---------+---------+---
                                    324.0     330.0     336.0     342.0


Pooled StDev = 9.25
```

P-value = 0.001 < a = 0.05

Therefore, we reject the null hypothesis. The data provides sufficient evidence thatat least one formulation is different from the others.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Hypothesis Testing

Test of Proportion

One proportion test: Overview

The 1-proportion test is used when….

- Testing the equality of a population proportion to a specific value
- Calculate a range of values that is likely to include the population proportion

Example: Purchase Orders

When purchase orders are selected out of the accounting system and examined for whether or not they contain a project number, history has shown that 18% do not. A new mistake proofing is attempted and 40 recent purchase orders are audited. Only 2 P.O.s now have a missing project number.

At the 5% level of significance, has the process been improved?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One proportion test: Example

Define the Hypothesis

- Null Hypothesis: The new process yields 18% or greater of purchase orders without project numbers
- Alternate Hypothesis: The new process yields less than 18% of purchase orders without project numbers

Randomly select a representative sample of data (Sample of 40 were randomly selected, only 2 P.O's were found to be missing project numbers)

Compute the P-value: the probability of obtaining the observed sample if the null hypothesis is true using Minitab…

Select: Stat > Basic Statistics > 1-Proportion

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One proportion test: Example

Test and CI for One Proportion

Test of p = 0.18 vs p < 0.18

Sample	X	N	Sample p	95.0% Upper Bound	Exact P-Value
1	2	40	0.050000	0.149152	0.017

Since the P-value is less than level of significance (0.05), we reject the null hypothesis.

This proves that the sample provides sufficient evidence that the proportion of defective purchase orders has decreased.

The process has improved!

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Two proportion Test: Overview

The 2-proportion test is used to….

- Determine whether the proportions of two groups differ
- Calculate a range of values that is likely to include the difference between the population proportions

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Two proportion Test: Example

A Black Belt is comparing two methods of processing cell phone requests in standard paper vs. intranet forms to determine which method is more accurate

Method	No. Audited	Number with Errors	Proportion Defective
Standard Paper	80	12	0.150
Intranet Form	80	5	0.063

Define the Hypothesis

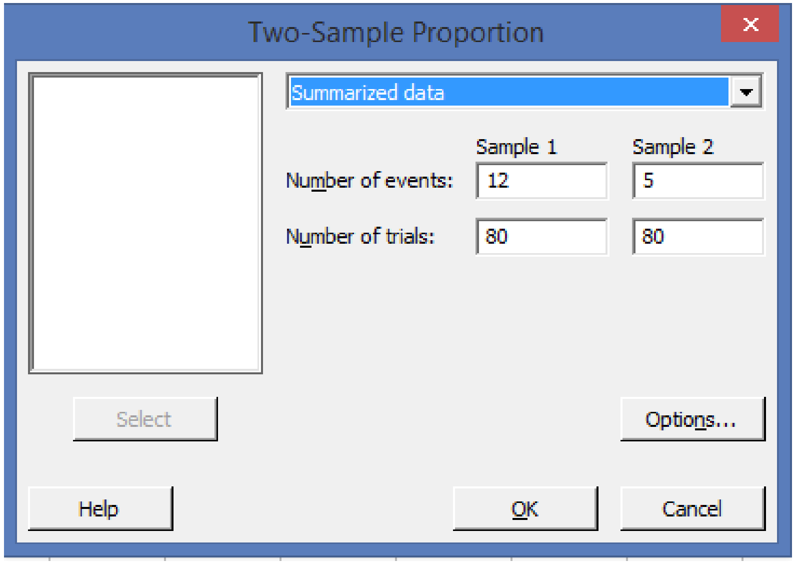
- Null Hypothesis: The methods of processing cell phone requests in standard paper and intranet form is the same
- Alternate Hypothesis: The methods of processing cell phone requests in standard paper and intranet form is significantly different

Compute the P-value: the probability of obtaining the observed sample if the null hypothesis is true. Using Minitab…

Select: Stat > Basic Statistics > 2-Proportions

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Two proportion Test: Example



Since the P-value and Fisher's P value is greater than level of significance (0.05), we fail to reject the null hypothesis.

Thus proving that the methods of processing cell phone requests in standard paper and intranet form is the same.

Test and CI for Two Proportions

```
Sample     X   N  Sample p
1         12  80  0.150000
2          5  80  0.062500

Difference = p (1) - p (2)
Estimate for difference:  0.0875
95% CI for difference:  (-0.00702985, 0.182030)
Test for difference = 0 (vs ≠ 0):  Z = 1.81  P-Value = 0.070

Fisher's exact test: P-Value = 0.122
```


© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Chi Square: Overview

The Chi Square test is used to….

- Test the equality of more than two population proportions

- A contingency table, using the Chi-Square test, can be used to determine if a difference exists among populations for proportion data.

- This test actually tests whether or not two variables are dependent on eachother.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Chi Square: Example

The Personnel Department wants to see if there is a link between age (old and young) and whether that person gets hired

	Hired	Not Hired	Total
Old	30	150	180
Young	45	230	275
Total	75	380	455

Define the Null and Alternate Hypothesis

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Chi Square : Example

Define the Hypothesis

- Null Hypothesis: Age and Hiring are not dependent (independent)
- Alternate Hypothesis: Age and Hiring are dependent

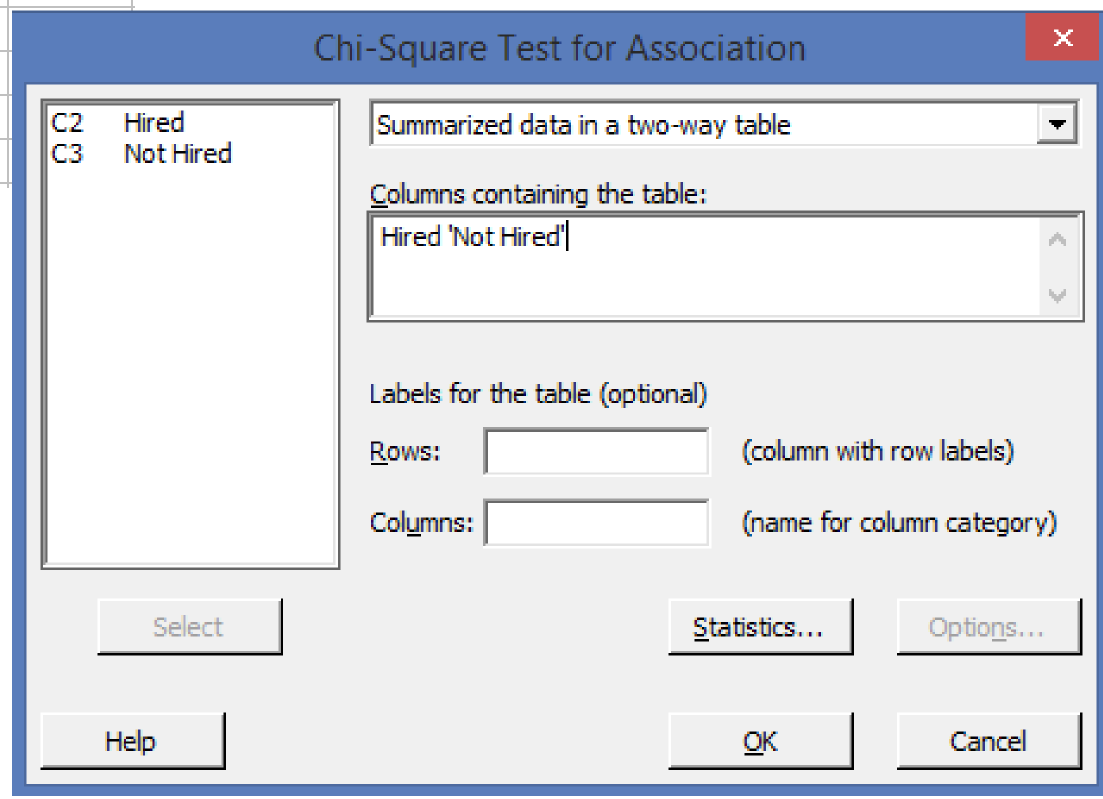
Compute the P-value: the probability of obtaining the observed is true. sample if the null hypothesis
Using Minitab…

Select: Stat > Tables > Chi Square Test for Association

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Chi Square : Example

↓	C1-T	C2	C3	C4	C5
		Hired	Not Hired		
1	Old	30	150		
2	Young	45	230		
3					
4					
5					
6					



KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Chi Square : Example

Chi-Square Test for Association

```
Rows: Worksheet rows   Columns: Worksheet columns

       Hired  Not Hired  All

1         30        150  180
       29.67     150.33

2         45        230  275
       45.33     229.67

All       75        380  455

Cell Contents:      Count
                    Expected count


Pearson Chi-Square = 0.007, DF = 1, P-Value = 0.932
Likelihood Ratio Chi-Square = 0.007, DF = 1, P-Value = 0.932
```

Since the P-value is greater than level of significance (0.05), we fail to reject thenull hypothesis.

Thus proving that Age and Hiring are independent

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Hypothesis Testing

Test of Variance

One variance test: Overview

The 1-variance test is used when….

- Testing the equality of a population variance to a specific value

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One variance test: Example

Example:

Refer worksheet “AIRPLANEPIN.mtw”

You are a quality control inspector at a factory that builds high precision parts for aircraft engines, including a metal pin that must measure 15 inches in length. Safety laws dictate that the variance of the pins' length must not exceed 0.001in. Previous analyses determined that pin length is normally distributed. You collect a sample of 100 pins and measure their length in order to conduct a hypothesis test for the population variance.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

One variance test: Example

Define the Hypothesis

Compute the P-value: the probability of obtaining the observed hypothesis is sample if the null true. Using Minitab…

Select: Stat > Basic Statistics > 1Variance

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One variance test: Example

Define the Hypothesis:

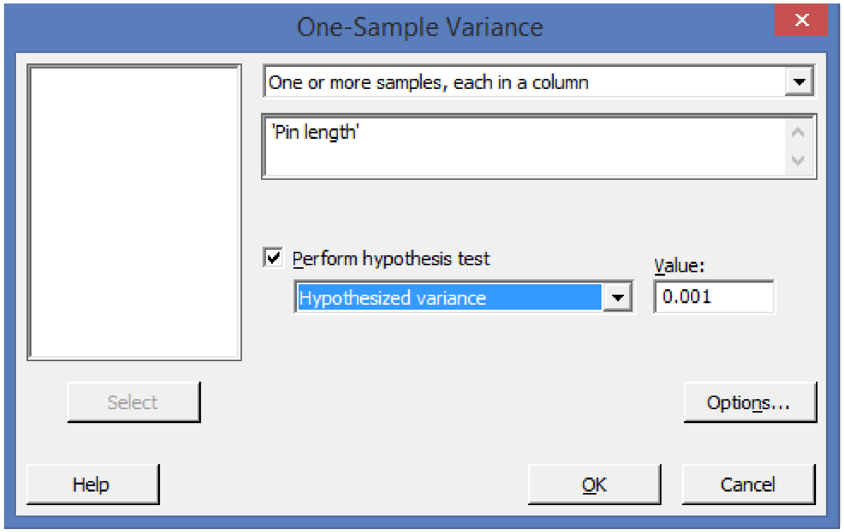
- Ho: Variance of metal pins equals 0.001in^2
- Ha: Variance of metal pins < 0.001in^2

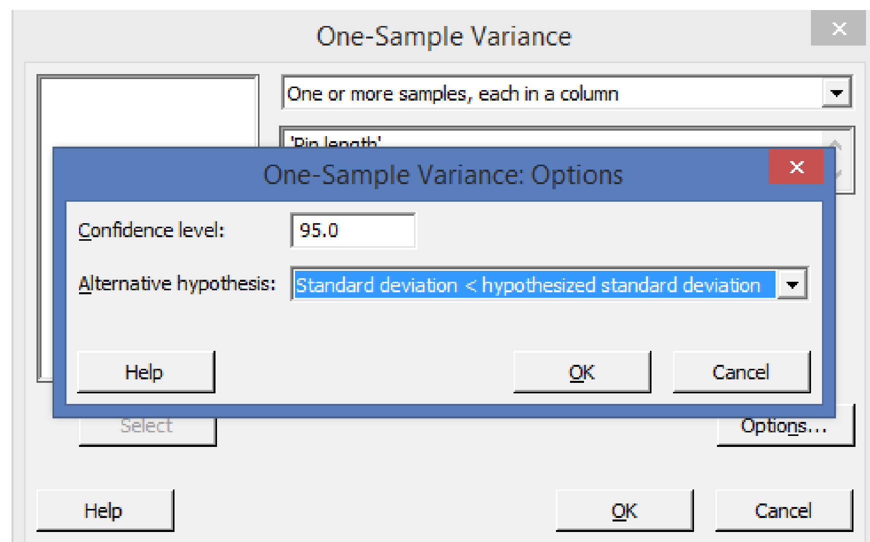
Compute the P-value: the probability of obtaining the observed hypothesis is sample if the null true. Using Minitab…

Select: Stat > Basic Statistics > 1Variance

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One variance test: Example





Set the hypothesis on Minitab

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One variance test: Example

Test and CI for One Variance: Pin length

```
Method

Null hypothesis          σ-squared = 0.001
Alternative hypothesis   σ-squared < 0.001

The chi-square method is only for the normal distribution.
The Bonett method is for any continuous distribution.


Statistics

Variable      N   StDev  Variance
Pin length  100  0.0267  0.000715


95% One-Sided Confidence Intervals

                          Upper Bound   Upper Bound
Variable    Method          for StDev  for Variance
Pin length  Chi-Square         0.0303      0.000919
            Bonett             0.0296      0.000878


Tests

                             Test
Variable    Method      Statistic  DF  P-Value
Pin length  Chi-Square      70.77  99    0.014
            Bonett              —   —    0.004
```

Since the P-value is lesser than level of significance (0.05), we reject the null hypothesis.

Thus proving that variance of metal pins < 0.001in²

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Two variances test: Overview

The 2-variances test is used when….

- Testing the equality of a two population variances and standard deviations

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Two variances test: Example

Example:

Refer worksheet “FURNACE.MTW”

A study was performed in order to evaluate the effectiveness of two devices for improving the efficiency of gas home-heating systems. Energy consumption in houses was measured after one of the two devices was installed. The two devices were an electric vent damper (Damper = 1) and a thermally activated vent damper (Damper = 2). You want to determine whether the variability in energy consumption is greater for one device than for the other.

2

The energy consumption data are stacked in one column (BTU.In). A separate column (Damper) identifies which sample each row belongs to

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Two variances test: Example

Define the Hypothesis

Compute the P-value: the probability of obtaining the observed hypothesis is   sample if the null true. Using Minitab…

Select: Stat > Basic Statistics > 2 Variances

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Two variances test: Example

Define the Hypothesis:

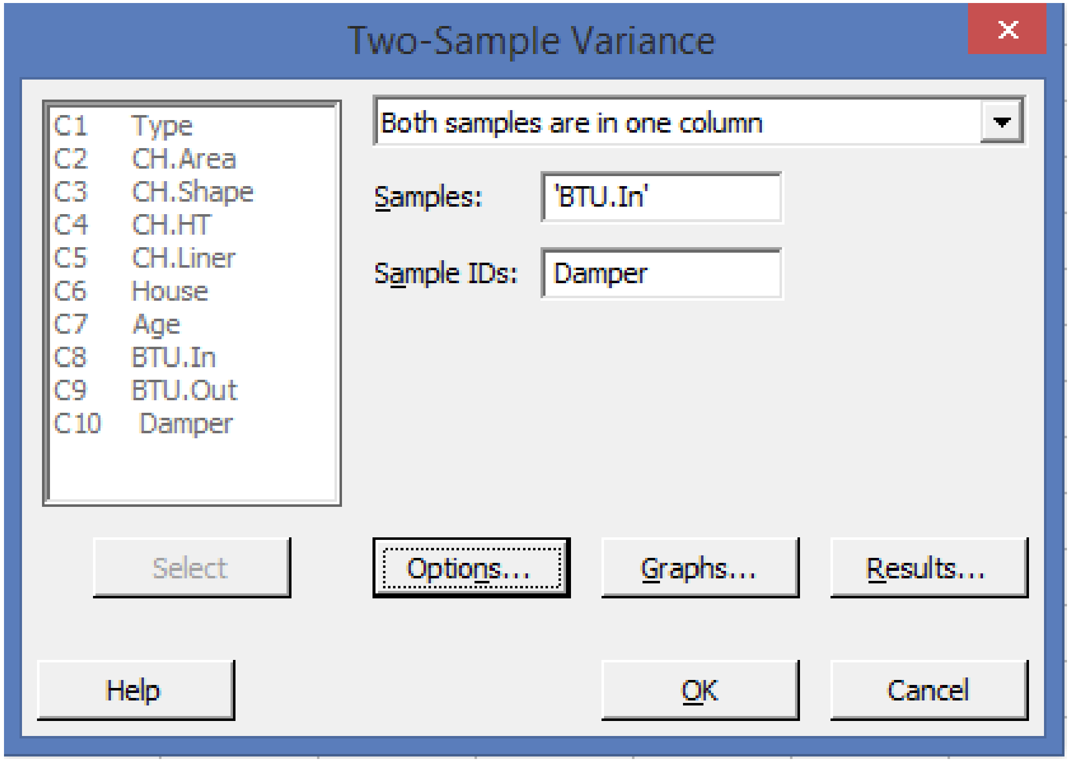
- Ho: Variability in energy consumption for both the devices is the same
- Ha: Variability in energy consumption is greater for one device than the other

Compute the P-value: the probability of obtaining the observed sample if the null hypothesis is true. Using Minitab…

Select: Stat > Basic Statistics > 2 Variances

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Two variances test: Example



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Two variances test: Example

Test and CI for Two Variances: BTU.In vs Damper

```
Method

Null hypothesis          σ(1) / σ(2) = 1
Alternative hypothesis   σ(1) / σ(2) ≠ 1
Significance level       α = 0.05

Statistics
                                        95% CI for
Damper    N  StDev   Variance             StDevs
1        40  3.020      9.120     (2.347, 4.085)
2        50  2.767      7.656     (2.335, 3.413)

Ratio of standard deviations = 1.091
Ratio of variances = 1.191

95% Confidence Intervals
                                  CI for
          CI for StDev          Variance
Method        Ratio               Ratio
Bonett   (0.763, 1.494)    (0.581, 2.232)
Levene   (0.697, 1.412)    (0.486, 1.992)

Tests
                          Test
Method   DF1   DF2   Statistic   P-Value
Bonett     —     —           —     0.586
Levene     1    88        0.00     0.996
```

Since the P-value is greater than level of significance (0.05), we fail to reject the null hypothesis.

Thus proving that variability in energy consumption for both the devices is the same

These data do not provide enough evidence to claim that the two populations have unequal standard deviations.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Two variances test: Bonett Vs. Levene

By default, Minitab's Two-Sample Variance command displays results for Bonett's method and Levene's method.

For most continuous distributions, both methods give you a type 1 error rate that is close to your specified significance level (also known as alpha or a).

Bonett's procedure is usually more powerful, so you should base your conclusions on Bonett's confidence interval (CI) and test, unless the your samples have less than 20 observations each.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Equal variances test: Overview

The equal variances test is used when….

- Testing the equality of more than two population variances and standard deviations

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Equal variances test: Example

Example:

Refer worksheet "FURNACE.MTW"

A study was performed in order to evaluate the effectiveness of two devices for improving the efficiency of gas home-heating systems. Energy consumption in houses was measured after one of the two devices was installed. The two devices were an electric vent damper (Damper = 1) and a thermally activated vent damper (Damper = 2). You want to determine whether the variability in energy consumption is greater for one device than for the other.

2

The energy consumption data are stacked in one column (BTU.In). A separate column (Damper) identifies which sample each row belongs to

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Equal variances test: Example

Define the Hypothesis

Compute the P-value: the probability of obtaining the observed hypothesis is   sample if the null true. Using Minitab…

Select: Stat > Basic Statistics > 2 Variances

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Equal variances test: Example

Define the Hypothesis:

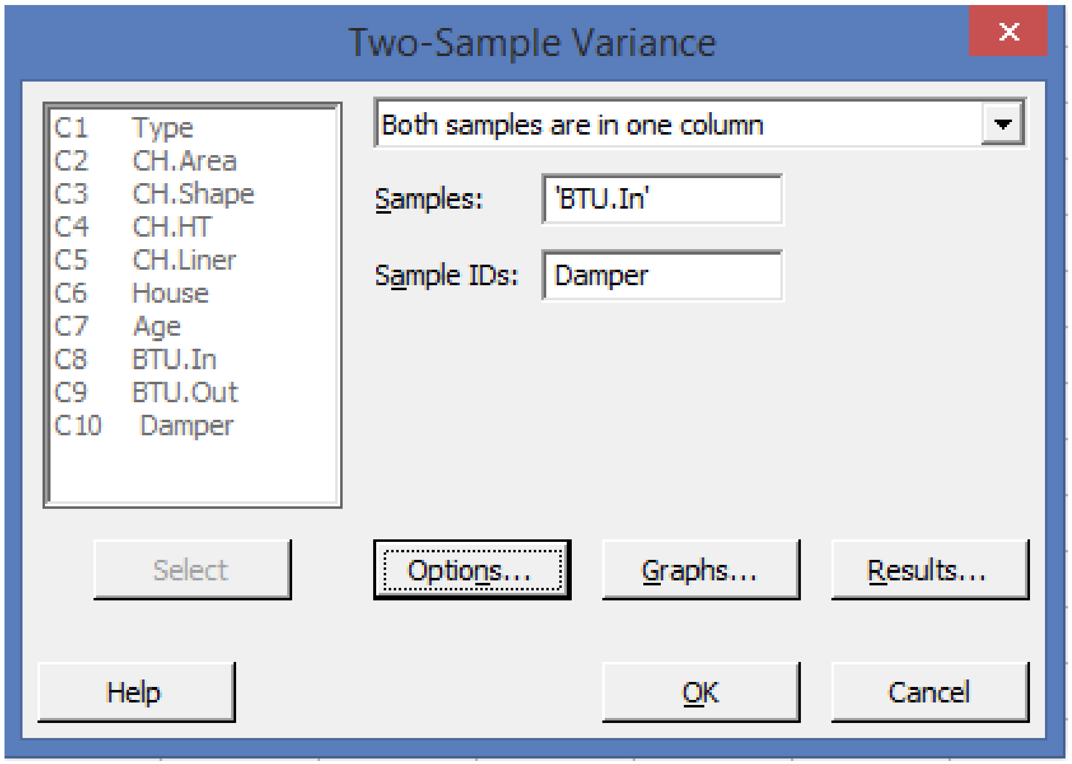
- Ho: Variability in energy consumption for both the devices is the same
- Ha: Variability in energy consumption is greater for one device than the other

Compute the P-value: the probability of obtaining the observed sample if the null hypothesis is true. Using Minitab…

Select: Stat > Basic Statistics > 2 Variances

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Equal variances test: Example



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Equal variances test: Example

Test and CI for Two Variances: BTU.In vs Damper

Method

Null hypothesis	$\sigma(1) / \sigma(2) = 1$
Alternative hypothesis	$\sigma(1) / \sigma(2) \neq 1$
Significance level	$\alpha = 0.05$

Statistics

Damper	N	StDev	Variance	95% CI for StDevs
1	40	3.020	9.120	(2.347, 4.085)
2	50	2.767	7.656	(2.335, 3.413)

Ratio of standard deviations = 1.091
Ratio of variances = 1.191

95% Confidence Intervals

Method	CI for StDev Ratio	CI for Variance Ratio
Bonett	(0.763, 1.494)	(0.581, 2.232)
Levene	(0.697, 1.412)	(0.486, 1.992)

Tests

Method	DF1	DF2	Test Statistic	P-Value
Bonett	—	—	—	0.586
Levene	1	88	0.00	0.996

Since the P-value is greater than level of significance (0.05), we fail to reject the null hypothesis.

Thus proving that variability in energy consumption for both the devices is the same

These data do not provide enough evidence to claim that the two populations have unequal standard deviations.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Equal variances test

By default, Minitab's Two-Sample Variance command displays results for Bonett's method and Levene's method.

For most continuous distributions, both methods give you a type 1 error rate that is close to your specified significance level (also known as alpha ora).

Bonett's procedure is usually more powerful, so you should base your conclusions on Bonett's confidence interval (CI) and test, unless the your samples have less than 20 observations each.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Hypothesis Testing

Test of Median (Non Parametrics)

One sign test: Overview

The 1-sign test of the median is used when….

- Testing the equality for the median of a single population

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One sign test: Example

Example:

Refer worksheet "EXH_STAT.MTW."

Price index values for 29 homes in a suburban area in the Northeast were determined. Real estate records indicate the population median for similar homes the previous year was 115. This test will determine if there is sufficient evidence for judging if the median price index for the homes was greater than 115

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One sign test: Example

Define the Hypothesis

Compute the P-value: the probability of obtaining the observed sample if the null hypothesis is true.
Using Minitab…

Select: Stat > Nonparametrics > 1-Sample Sign

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One sign test: Example

Define the Hypothesis:

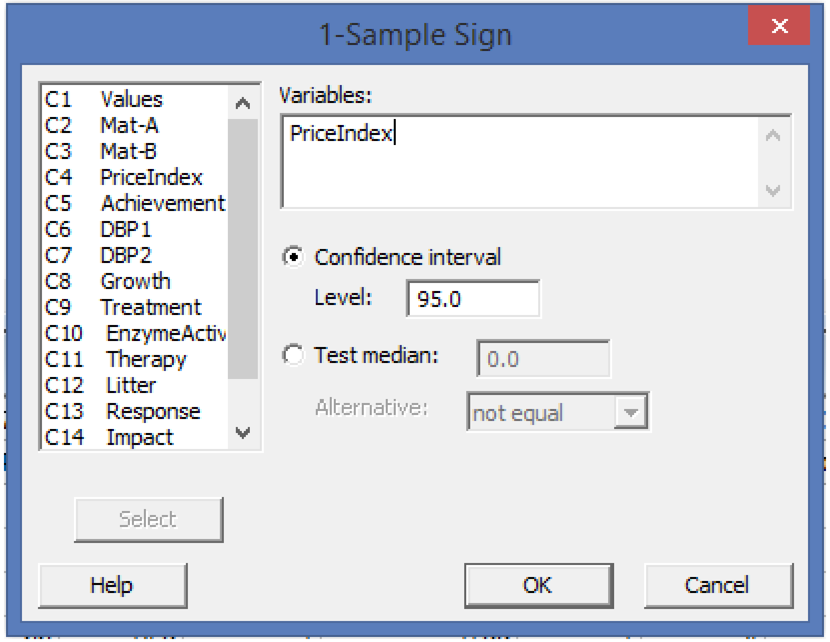
- Ho: Price index for homes $\leq$ to 115
- Ha: Price index for homes > than 115

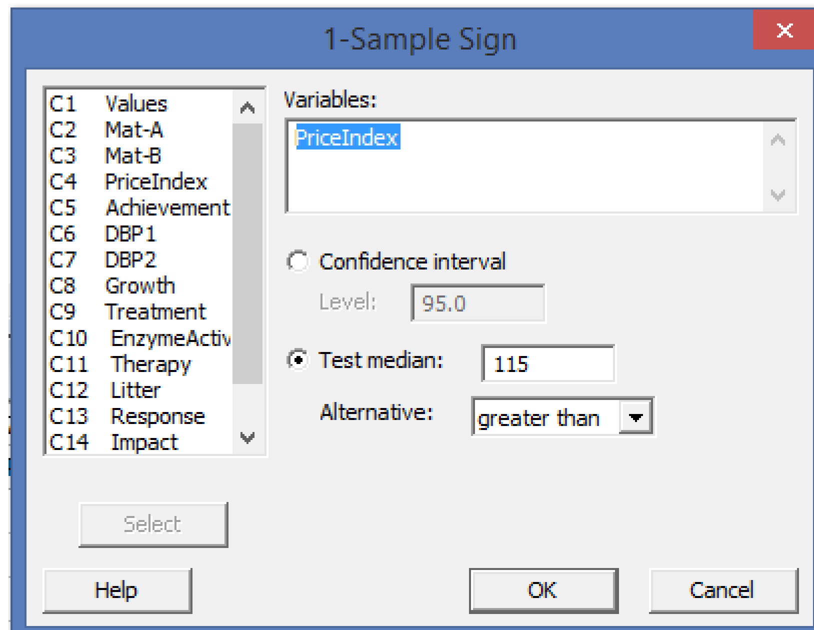
Compute the P-value: the probability of obtaining the observed hypothesis is sample if the null true. Using Minitab…

Select: Stat > Nonparametrics > 1-Sample Sign

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One sign test: Example





Set the hypothesis on Minitab

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

One sign test: Example

Sign Test for Median: PriceIndex

```
Sign test of median =  115.0 versus > 115.0

              N  Below  Equal  Above        P  Median
PriceIndex   29     12      0     17   0.2291   144.0
```

Since the P-value is greater than level of significance (0.05), we fail to reject the null hypothesis.

Therefore we fail to conclude that the price index for homes is greater than 115

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Mann Whitney test: Overview

The Mann Whitney test of the median is used to….

- determine whether two populations have the same population median

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Mann Whitney test : Example

Example:

Refer worksheet "EXH_STAT.MTW."

Samples were drawn from two populations and diastolic blood pressure was measured. You will want to determine if there is evidence of a difference in the population locations without assuming a parametric model for the distributions. Therefore, you choose to test the equality of population medians using the Mann-Whitney test with a = 0.05 rather than using a two- sample t-test, which tests the equality of population means.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Mann Whitney test : Example

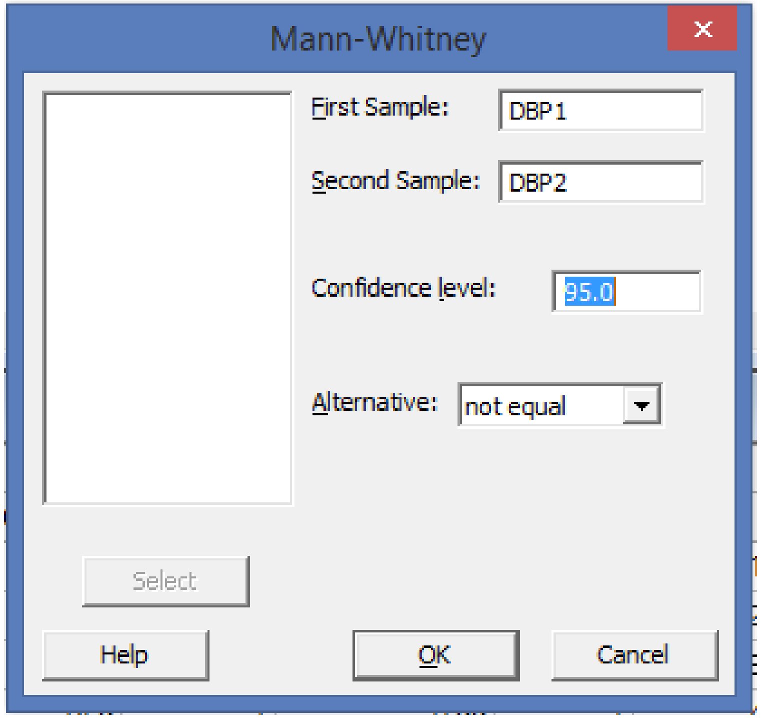
Define the Hypothesis

Compute the P-value: the probability of obtaining the observed hypothesis is sample if the null true. Using Minitab…

Select: Stat > Nonparametrics > Mann-Whitney

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Mann Whitney test : Example



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Mann Whitney test : Example

Mann-Whitney Test and CI: DBP1, DBP2

```
      N  Median
DBP1  8   69.50
DBP2  9   78.00


Point estimate for η1 - η2 is -7.50
95.1 Percent CI for η1 - η2 is (-18.00,4.00)
W = 60.0
Test of η1 = η2 vs η1 ≠ η2 is significant at 0.2685
The test is significant at 0.2679 (adjusted for
ties)
```

Since the P-value is greater than level of significance (0.05), we fail to reject the null hypothesis.

Therefore we conclude that there is no difference between population medians

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Mood's Median test: Overview

The Mood's Median test of the median is used when….

- Mood's median test can be used to test the equality of medians from two or more populations and, like the Kruskal-Wallis Test, provides an nonparametric alternative to the one-way analysis of variance.

- Mood's median test is sometimes called a median test or sign scores test.

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Mood's Median test : Example

Example:

Refer worksheet "CARTOON.MTW."

One hundred seventy-nine participants were given a lecture with cartoons to illustrate the subject matter. Subsequently, they were given the OTIS test, which measures general intellectual ability.

Participants were rated by educational level: 0 = preprofessional,
1 = professional,
2 = college student

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Mood's Median test : Example

Define the Hypothesis

Compute the P-value: the probability of obtaining the observed hypothesis is sample if the null true. Using Minitab…

Select: Stat > Nonparametrics > Mood's Median Test

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Mood's Median test : Example

Define the Hypothesis:

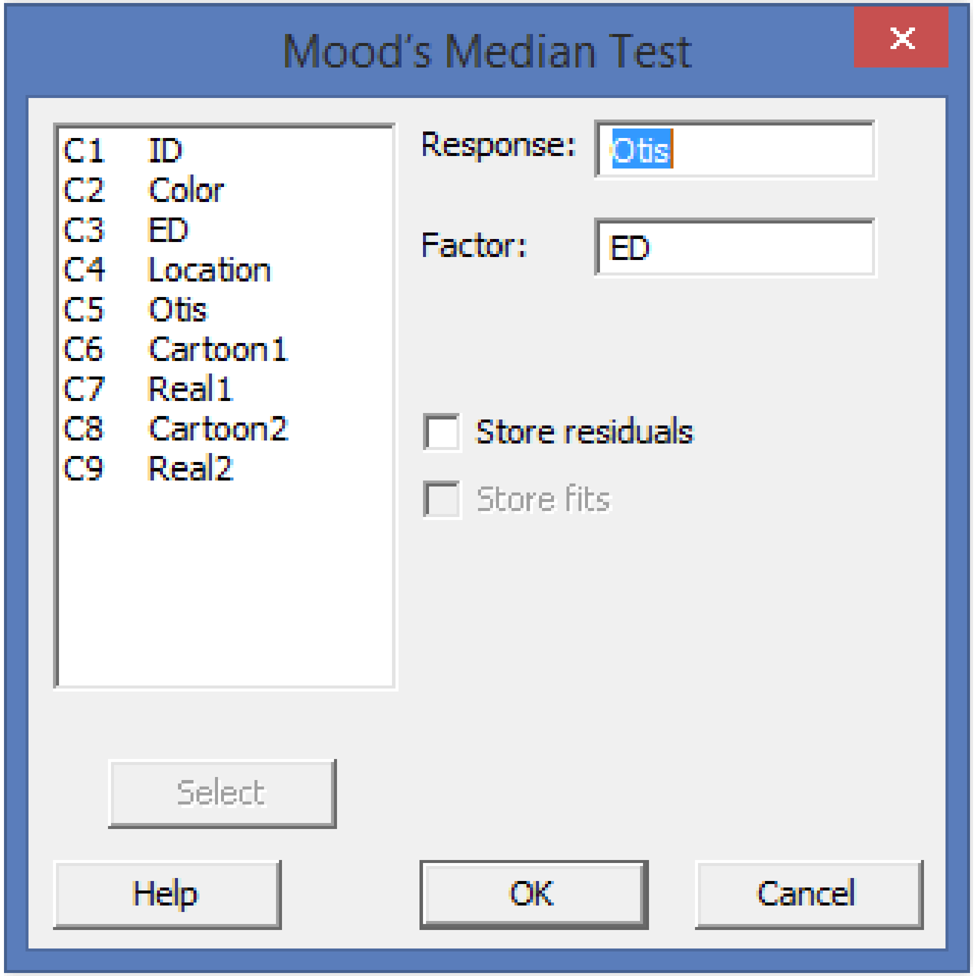
- Ho: All medians are equal
- Ha: At least one median is different

Compute the P-value: the probability of obtaining the observed sample if the null hypothesis is true. Using Minitab…

Select: Stat > Nonparametrics > Mood's Median Test

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Mood's Median test : Example



KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Mood's Median test : Example

Results for: Cartoon.MTW

Mood Median Test: Otis versus ED

```
Mood median test for Otis
Chi-Square = 49.08    DF = 2    P = 0.000

                                   Individual 95.0% CIs
ED  N≤  N>  Median  Q3-Q1  ----+---------+---------+---------+--
0   47   9    97.5   17.3  (-----*-----)
1   29  24   106.0   21.5          (------*------)
2   15  55   116.5   16.3                         (----*----)
                           ----+---------+---------+---------+--
                            96.0     104.0     112.0     120.0

Overall median = 107.0
```

Since the P-value is less than level of significance (0.05), we reject the null hypothesis.

Here, the participant scores are classified as being below or above the overall median.
The p-value < than 0.05 indicates that the medians are not equal.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Group Exercises

1. Sachin wants to test the assumption that there is no difference between the mean life of motors of two companies. For his study, he picks up a sample of 10 motors with mean life of 4,050 hours and standard deviation of 200 hours from the first company and a sample of 9 motors with mean life of 4,300 hours and standard deviation of
260 hours from the second company. You are requested to help Sachin test the hypothesis, that there is no difference in the mean life of both the brands of motor at 95 percent level of confidence.

2. A team wants to see if there is relationship between ambient temperature and the viscosity of a material.

3. Ajay is exploring a brand of canned rasogullas to be sold in the chain of grocery shops which he represents. For him to select a particular brand, the drained rasogullas should not weigh less than 260 grams (in a 500 grams can). From the 11 cans he sampled at the distributor, the mean weight of drained rasogullas was 247 grams with a standard deviation of 30 grams. You are requested to help Ajay take a decision regarding procurement of Rasogullas at 95 percent level of confidence.

What type of tool would you use ? ______________________________

Define: Ho and Ha

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Group Exercises

4. The Personnel Department wants to see if there is a link between age (old and young) and whether that person gets hired.

5. A brand marketing firm wants to understand the television viewing pattern during cricket matches across countries. During a league cricket match involving players of 4 nationalities, following viewing patterns were observed in a randomly selected sample of 50 viewers across these countries (marked A, B, C and D). Please help the team understand if there is a relationship between the nationalities and the viewing patterns at 5 percent level of significance.

	A	B	C	D
Viewed	12	10	6	14
Did Not View	38	40	44	36

What type of tool would you use? ______________________________

Define: Ho and Ha

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

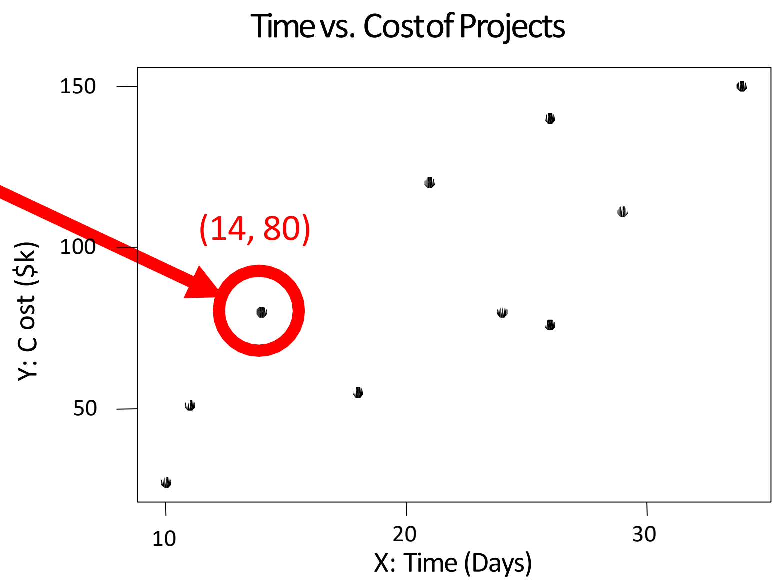
Correlation and Regression

Scatter Plot

What is a Scatter Plot?

A scatter diagram is the graphical representation of paired (x,y) data. This type of graph is appropriate when the values in one data set correspond to values in another data set, and you wish to understand the relationship between the two.

Project	Time (Days)	Cost ($k)
1.	14	80
2.	29	111
3.	26	76
4.	10	27
5.	18	55
6.	11	51
7.	34	150
8.	26	140
9.	24	80
10.	21	120

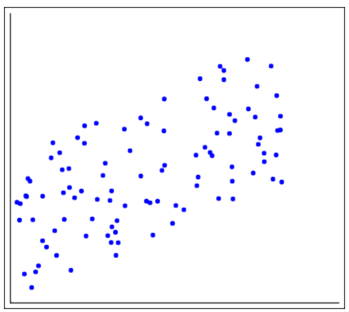


© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

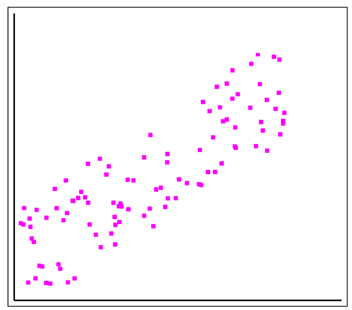
Interpreting a Scatter Plot

Let us assume for all charts below:

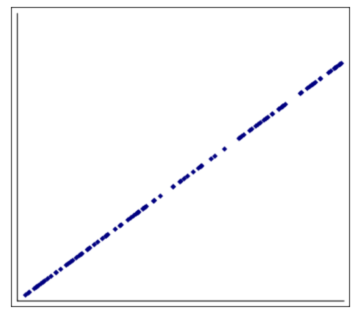
Y = Participant satisfaction (scale: 1 – worst to 100 – best) X = No of hours spent by the trainer



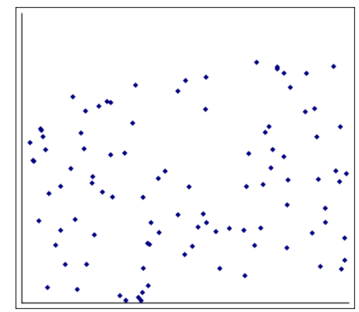
Positive Correlation



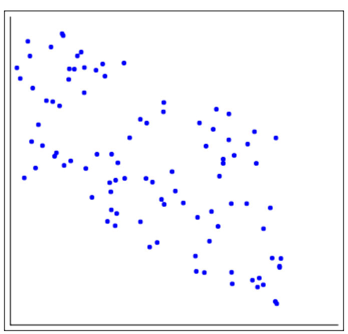
Strong Positive



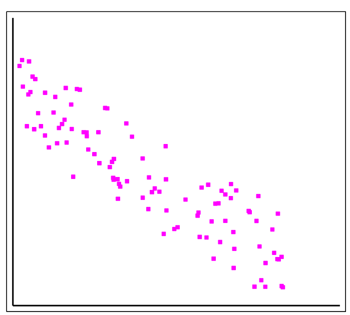
Perfect Positive



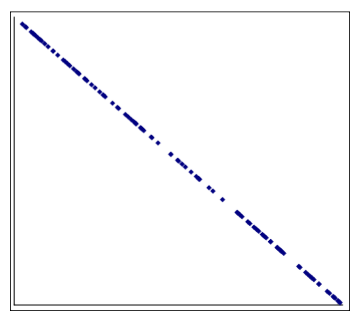
No Correlation



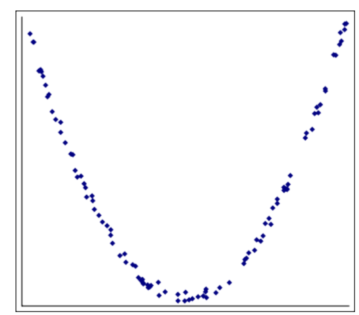
Negative Correlation



Strong Negative



Perfect Negative



Nonlinear Correlation

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Correlation and Regression

Correlation Analysis

What is Correlation?

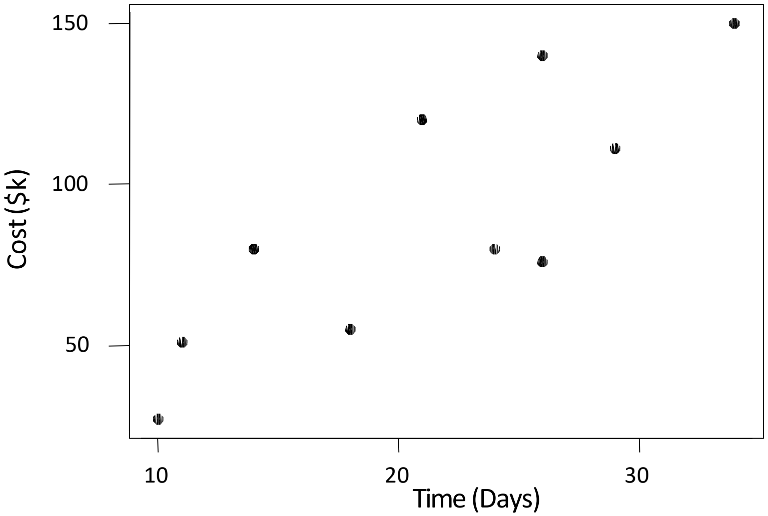
When two variables show a relationship on a scatter plot, they are said to be correlated, but this does not necessarily mean they have a cause-effect relationship.

If two variables X and Y, are related such that as Y increases / decreases with another variable X a correlation is said to exist between them. In other words, a correlation exists between two variables when they are related to one another in some way.

Let us consider this –

Project	Time (Days)	Cost ($k)
1.	14	80
2.	29	111
3.	26	76
4.	10	27
5.	18	55
6.	11	51
7.	34	150
8.	26	140
9.	24	80
10.	21	120

Time vs. Cost of Projects



From the above plot, we can infer that as the project time increases, so does the cost.

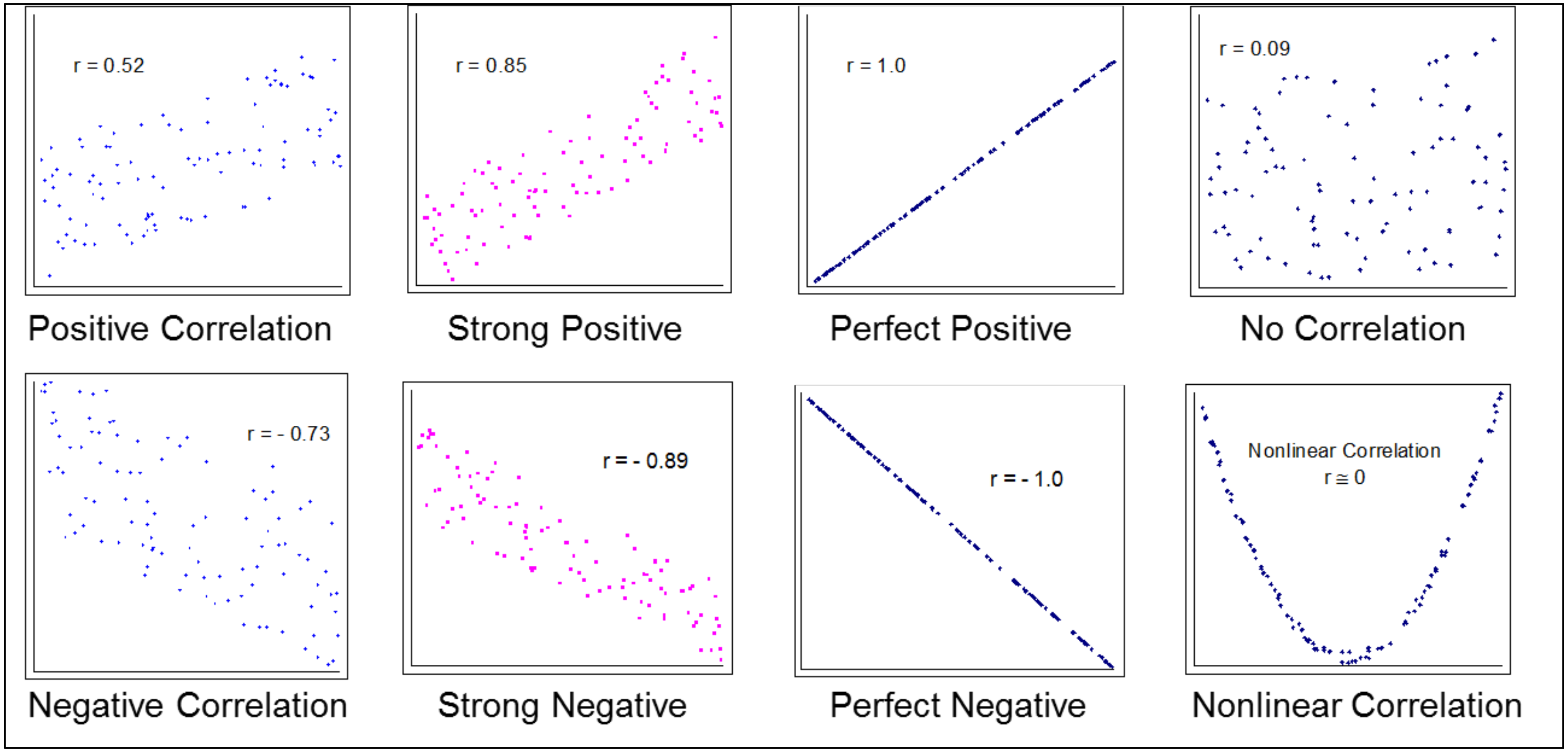
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Correlation Coefficient

- The correlation coefficient, r, is a statistical measure of the strength of the linear relationship between two variables.
- r is always between -1 and 1
- r is also known as Pearson's correlation coefficient
- When r is close to zero, no linear relationship is present.
- When a relationship exists, the variables are said to be correlated
  - Perfect negative relationship r = –1.0
  - No linear correlation r = 0
  - Perfect positive relationship r = +1.0

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Correlation Coefficient Examples



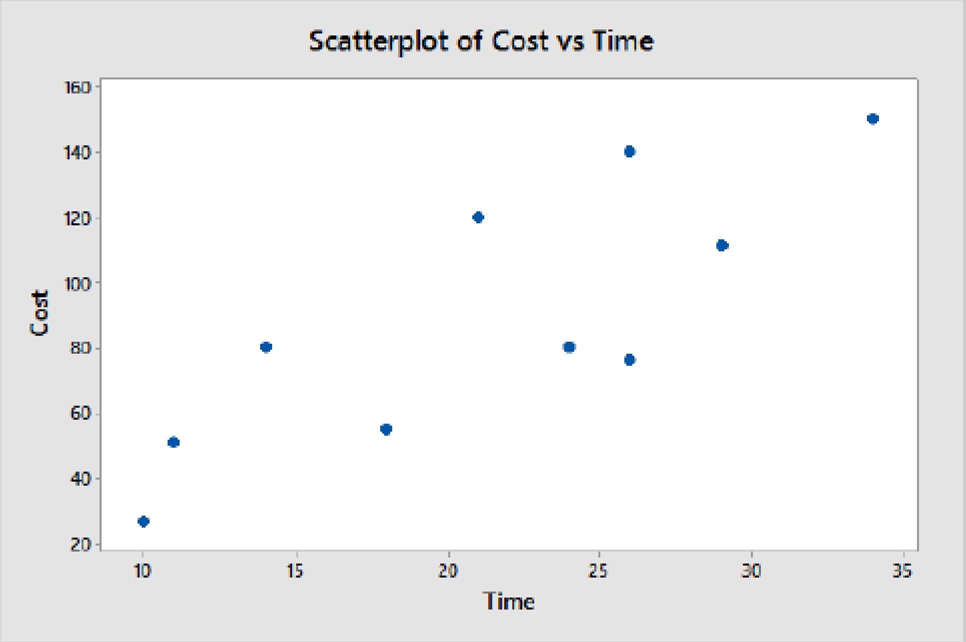
Note: r ≈ 0 means no linear relationship. The variables might be related, just not in a linear fashion.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Scatter Plot and Correlation with Minitab

Let us consider the following data set for our analysis in Minitab to plot a Scatter Diagram

Project	Time (Days)	Cost ($k)
1.	14	80
2.	29	111
3.	26	76
4.	10	27
5.	18	55
6.	11	51
7.	34	150
8.	26	140
9.	24	80
10.	21	120



- Scatter Plot
- Go to Minitab
- Select: Graph > Scatter Plot > Simple
- Double click C2 -> Cost is put into the Y box
- Double click C1 -> Time is put into the X box

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Scatter Plot and Correlation with Minitab

For the same set of data let us do a Correlation using Minitab

Correlation

- Go to Minitab
- Select: Stat >Basic Statistics > Correlation
- Double click C1, C2 -> C1, C2 are put into the variables box
- Click OK

Correlations: Time, Cost

Pearson correlation of Time and Cost (r) = 0.819 P-Value

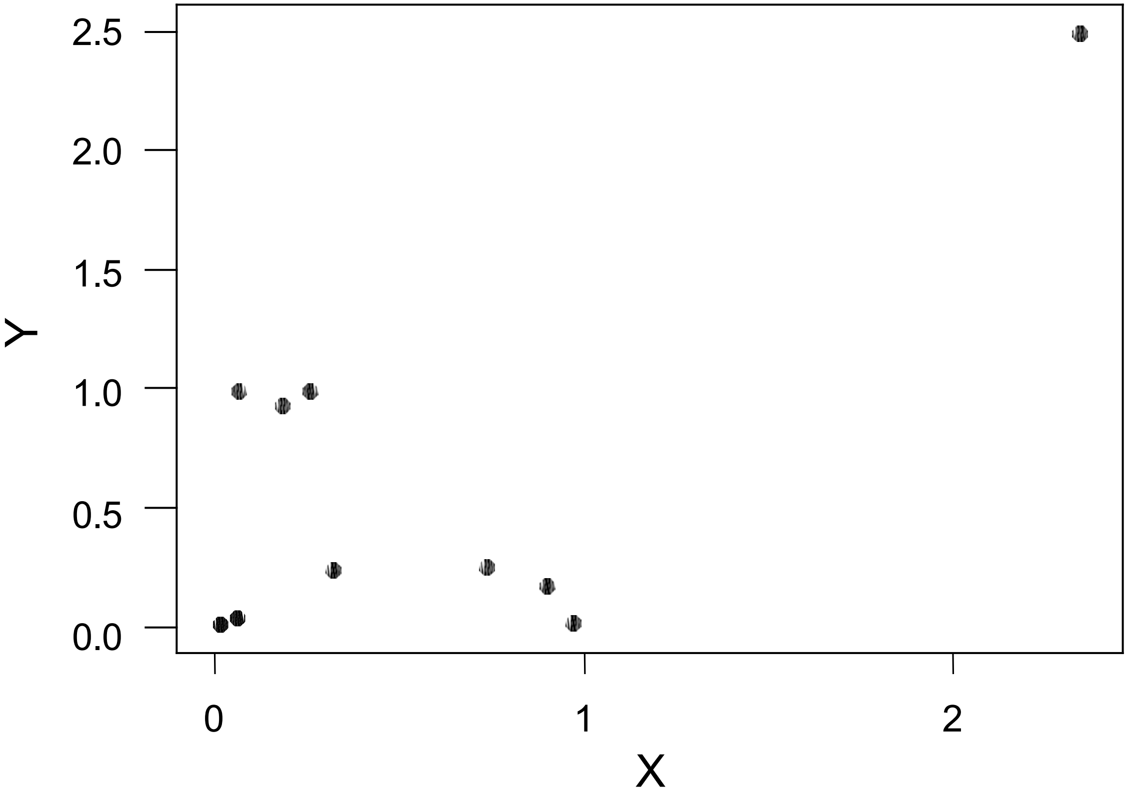
= 0.004

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Group Discussion

Y	X
0.991	0.261
0.255	0.736
0.015	0.973
0.175	0.900
0.012	0.015
2.490	2.345
0.991	0.067
0.239	0.323
0.036	0.062
0.926	0.186

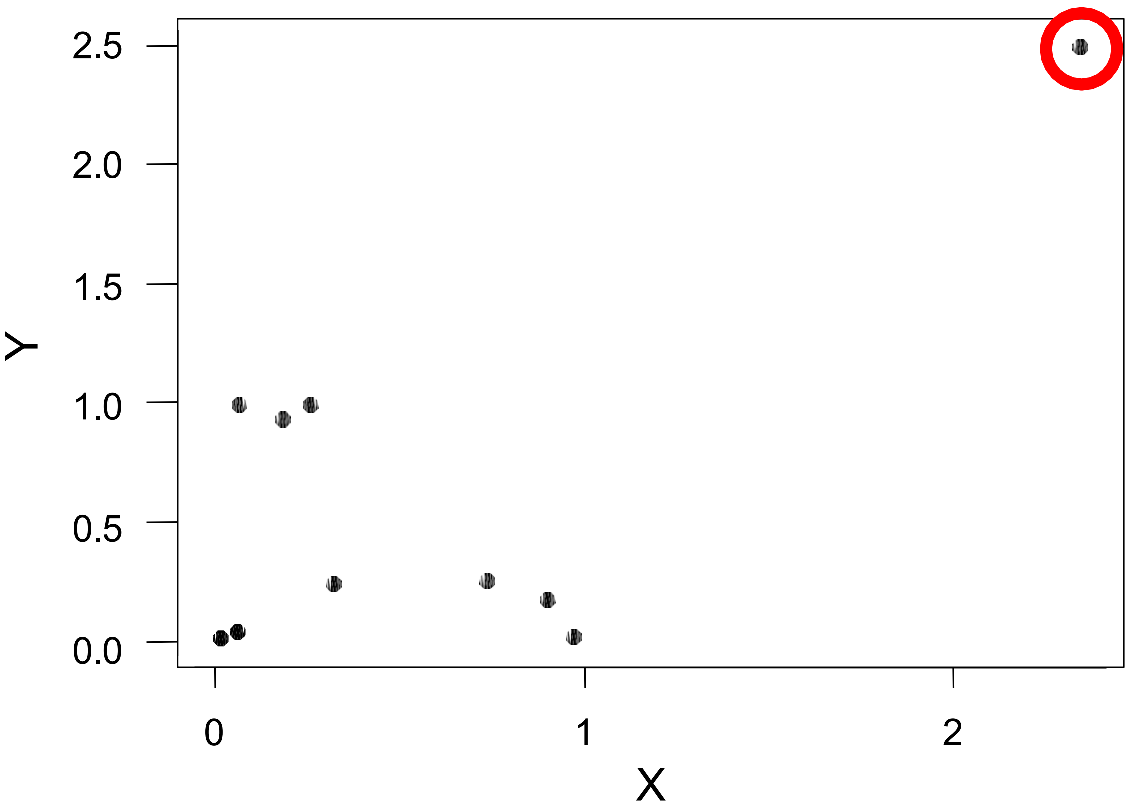


Is there a correlation between Y & X?
Why or Why not? Use Minitab to tabulate the results

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Group Discussion

Y	X
0.991	0.261
0.255	0.736
0.015	0.973
0.175	0.900
0.012	0.015
2.490	2.345
0.991	0.067
0.239	0.323
0.036	0.062
0.926	0.186



What is this point?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Group Discussion

Y	X
0.991	0.261
0.255	0.736
0.015	0.973
0.175	0.900
0.012	0.015
2.490	2.345
0.991	0.067
0.239	0.323
0.036	0.062
0.926	0.186

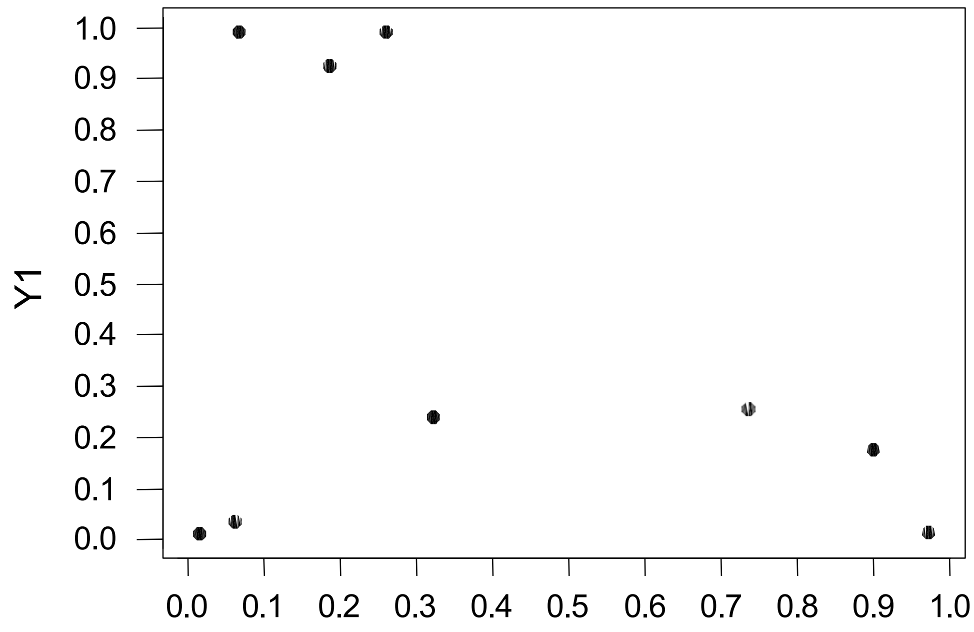
Can this point be removed? What do we term them as?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Group Discussion

- Below is the data, graph and correlation test with the one dominant x,y pair removed:

Y1	X1
0.991	0.261
0.255	0.736
0.015	0.973
0.175	0.900
0.012	0.015
0.991	0.067
0.239	0.323
0.036	0.062
0.926	0.186



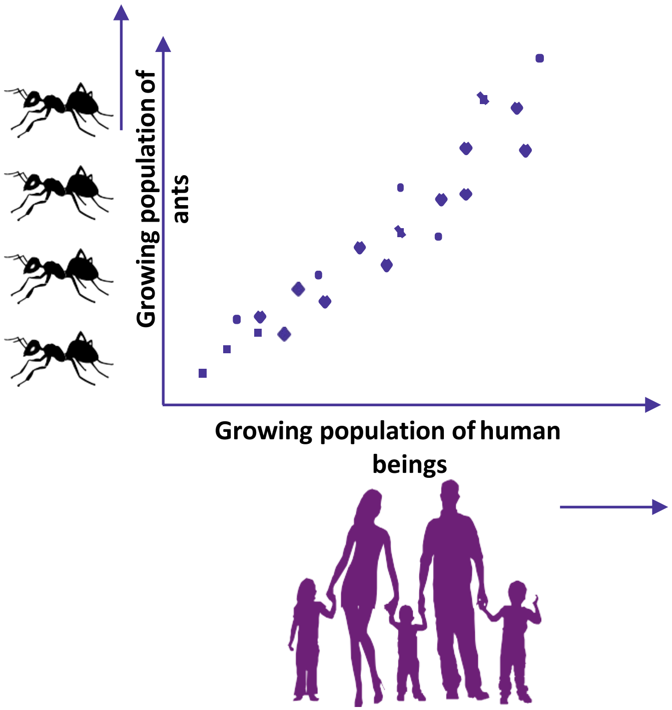
Is there a Correlation between them?

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Avoid Pitfalls

Avoid Pitfalls

- To avoid these problems:
- Always examine the x,y data.
- Always look at an x,y plot.
- If there seems to be an outlier, drop that data point and rerun the correlation test.
- Never jump too quickly to your conclusion.



REMEMBER: Correlation does not imply causation

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Correlation and Regression

Regression Analysis

What is Regression?

- Regression analysis is a statistical process for estimating the relationships among variables.
- Regression analysis is a form of predictive modelling technique which investigates the relationship between a dependent and independent variables.
- This technique is used for forecasting, time series modelling and finding the causal effect relationship between the variables.

For example, relationship between rash driving and number of road accidents by a driver is best studied through regression.

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

What is Regression?

There are multiple benefits of using regression analysis. They are asfollows:

- It indicates the significant relationships between dependent variable and independent variable.
- It indicates the strength of impact of multiple independent variables on a dependent variable.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

What is Regression?

- In statistical modeling, regression analysis is a statistical process for estimating the relationships among variables.

- Regression analysis is a form of predictive modelling technique which investigates the relationship between a dependent (target) and independent variable (s) (predictor).

- In simple linear regression, you obtain the graph and the equation of the straight line that best represent the relationship between two variables.

- Given a sample of paired data, the regression equation $y = b_0 +$

  b_1x

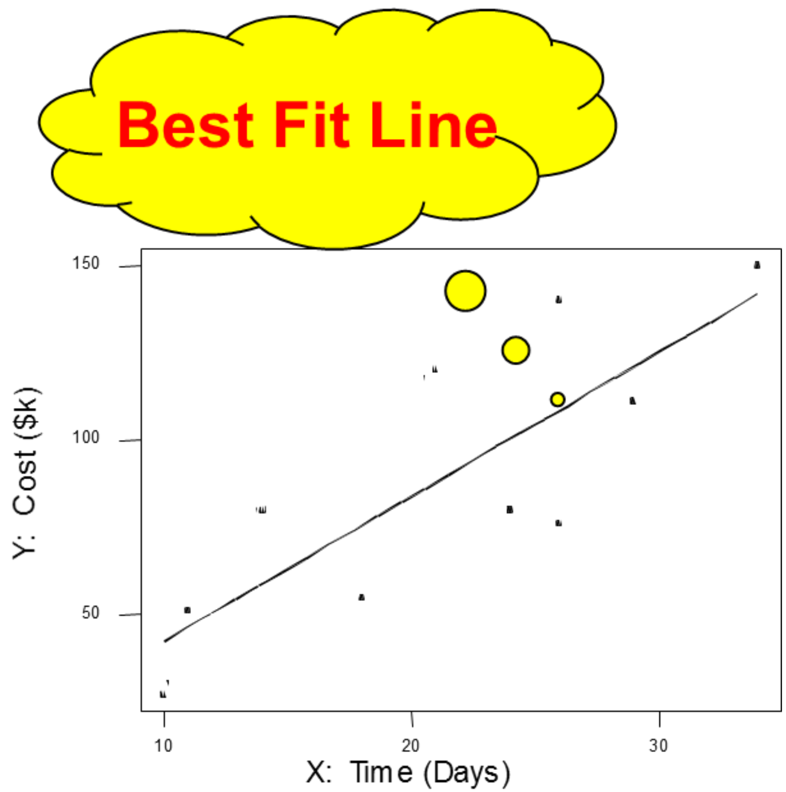
  describes the relationship between two variables.

- The graph of the regression equation is called the regression line (or best fit line).

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

What is Regression?

- The graph of the regression equation is called the regression line (or best fitline).



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Regression Equation

The regression equation:

Dependent Variable

$$y = b_0 + b_1 x$$

y-intercept

slope

Independent Variable

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Simple Linear Regression: Example

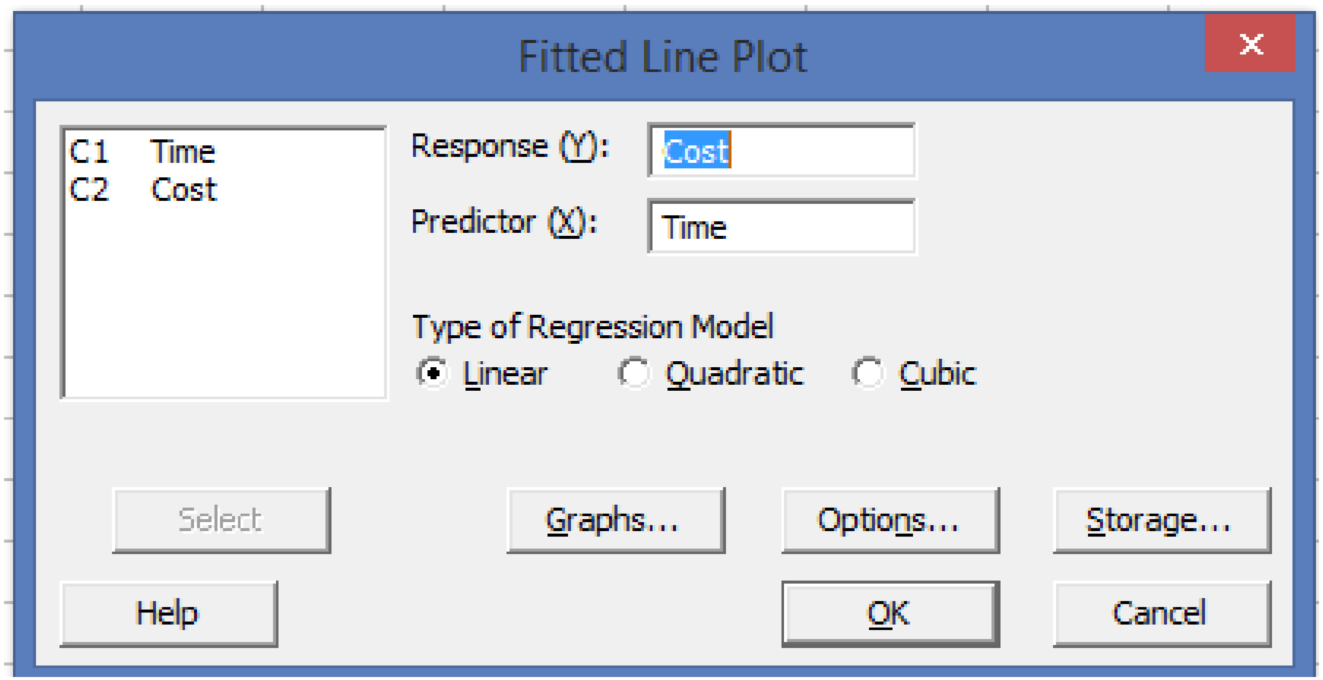
Consider the below example of a relationship between Cost and Time

Time	Cost
14	80
29	111
26	76
10	27
18	55
11	51
34	150
26	140
24	80
21	120

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Fitting a Regression Model: Using Minitab

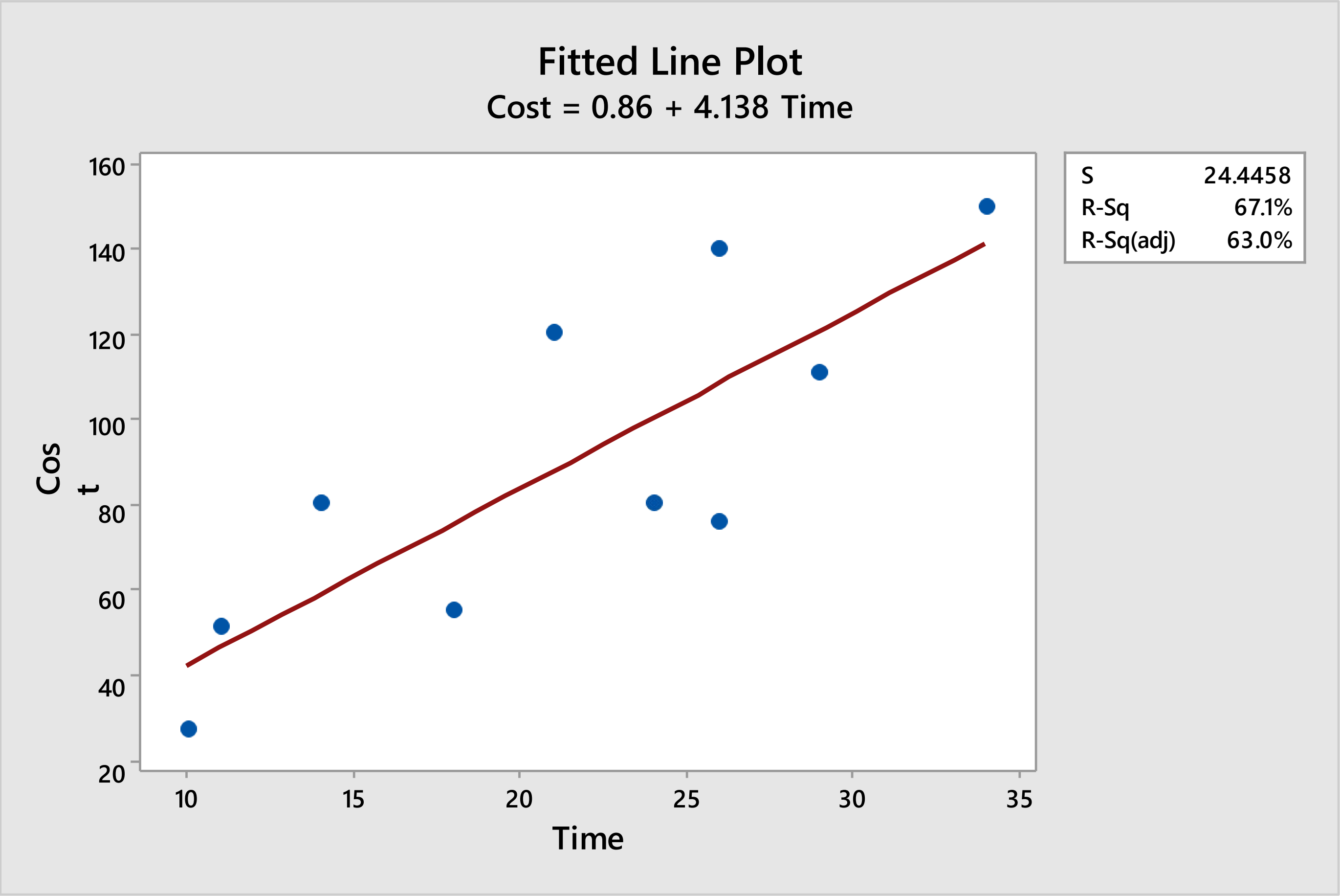
Select: Stat> Regression> Fitted Line Plot



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Fitting a Regression Model: Using Minitab

Select: Stat> Regression> Fitted Line Plot

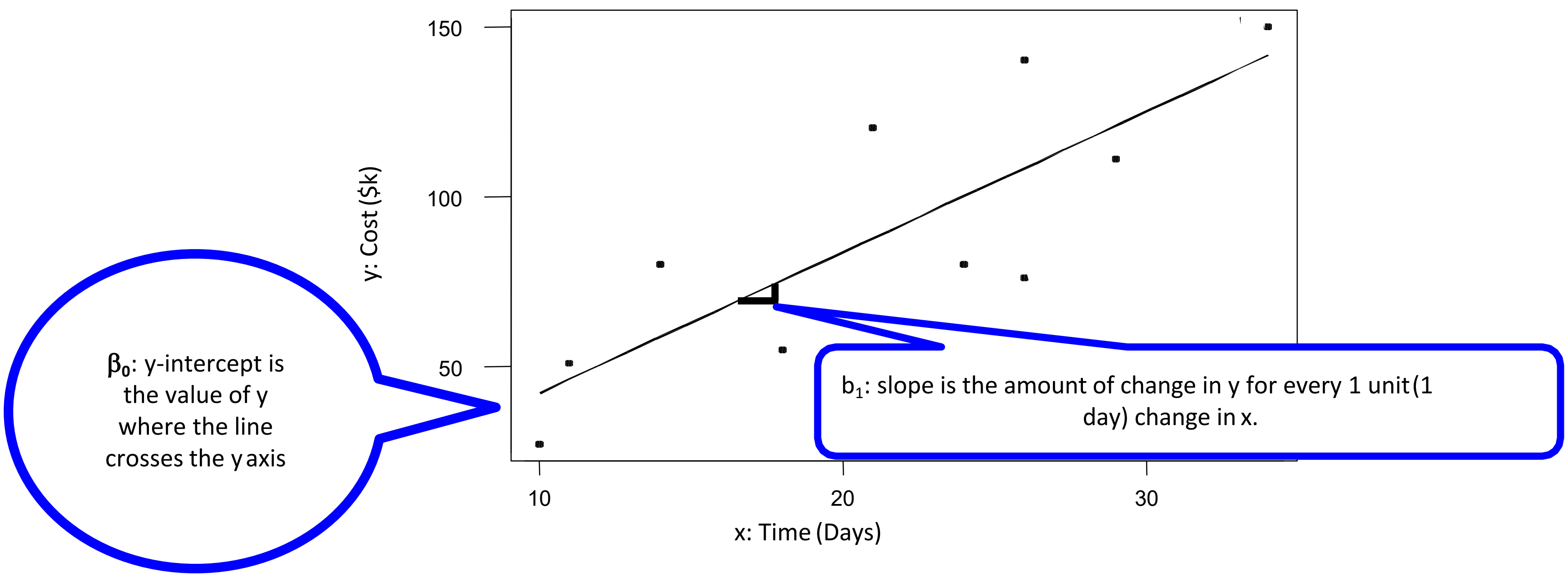


© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Simple Linear Regression: Using Minitab

Consider the below example of a relationship between Cost and Time in Minitab

Select: Stat> Regression> Regression> Fit Regression Model



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Simple Linear Regression: Using Minitab

Regression Analysis: Cost versus Time

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	9761	9761.2	16.33	0.004
Time	1	9761	9761.2	16.33	0.004
Error	8	4781	597.6		
Lack-of-Fit	7	2733	390.4	0.19	0.944
Pure Error	1	2048	2048.0		
Total	9	14542			

P - value = .004

Rejecting Null Hypothesis means we have convincing evidence of a linear relationship between Time and Cost.

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
24.4458	67.12%	63.01%	54.08%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	0.9	23.1	0.04	0.971	
Time	4.14	1.02	4.04	0.004	1.00

Regression Equation

Cost = 0.9 + 4.14 Time

The best fitting line is y = .9 + 4.14 x

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

R-Squared (R-Sq)

What is R-Sq?

The coefficient of determination, R-sq is the amount of the variation in y that is explained by the regression line.

R-Sq = Explained Variation *100 Total

Variation

In the given illustration,

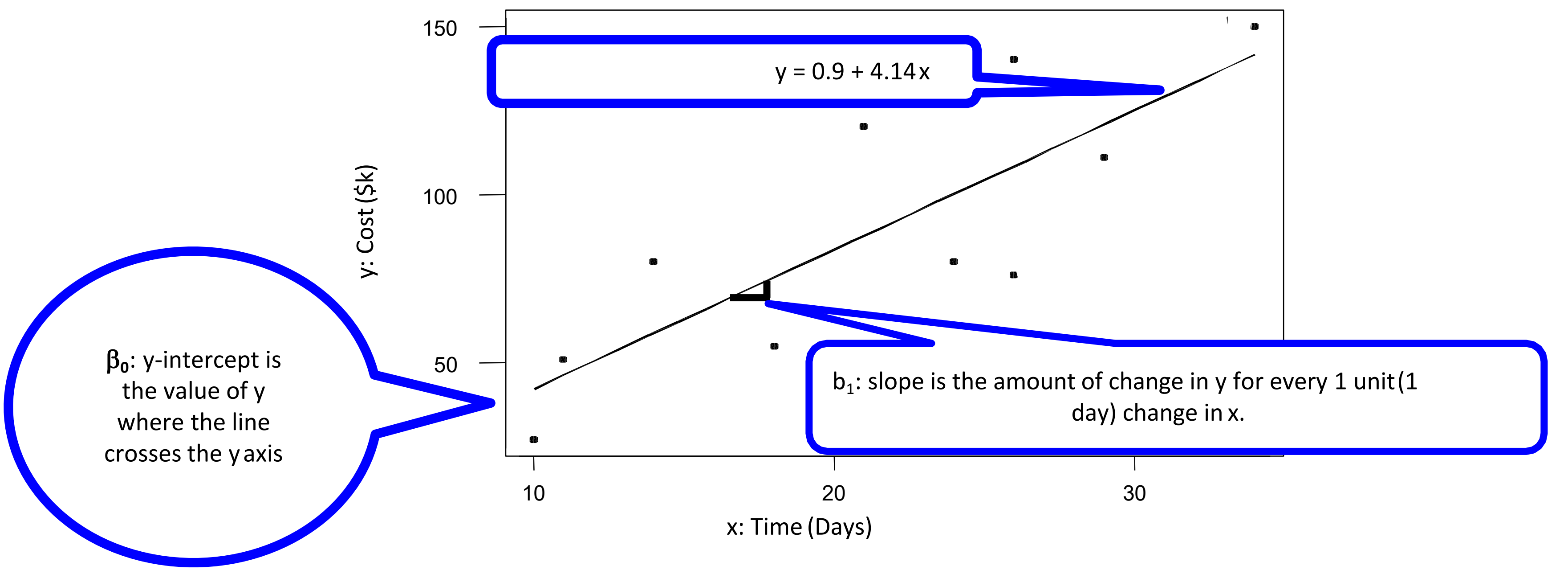
R-Sq = 9761/ 14542*100 = 67.1%

It means, 67.1% of the variability in cost can be explained by the relationship to time.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Regression Equation

Consider the below example of a relationship between Cost and Time



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Correlation and Regression

Simple Linear Regression

Simple Linear Regression: Example

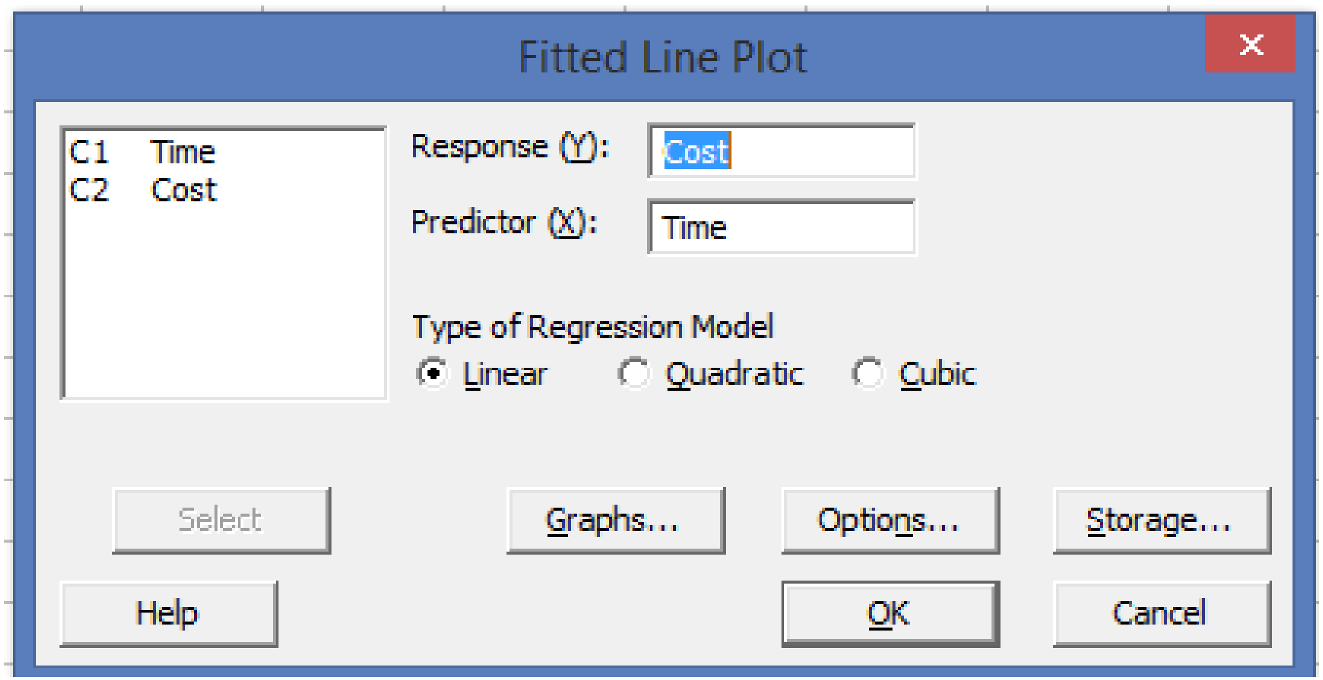
Consider the below example of a relationship between Cost and Time

Time	Cost
14	80
29	111
26	76
10	27
18	55
11	51
34	150
26	140
24	80
21	120

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Fitting a Regression Model: Using Minitab

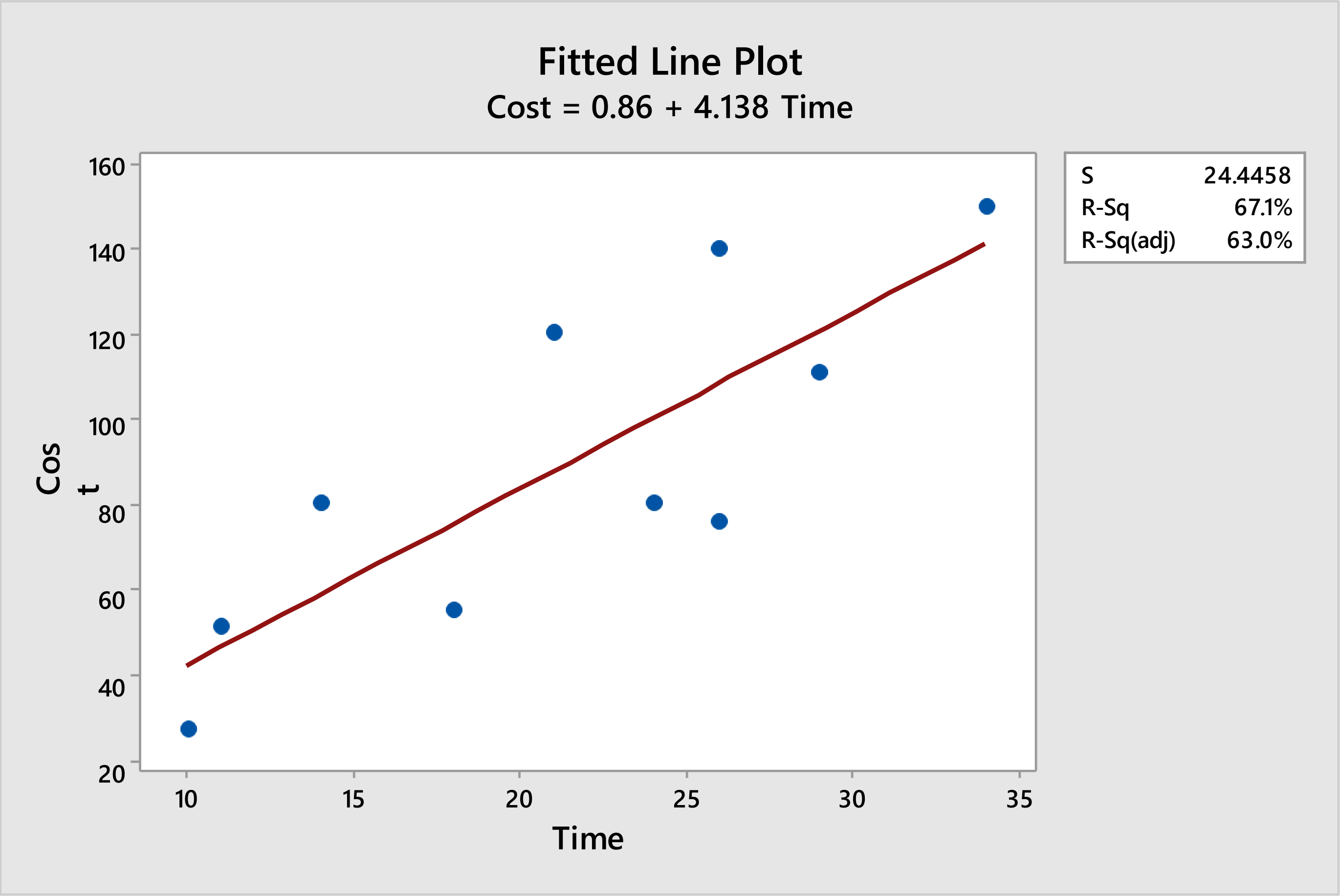
Select: Stat> Regression> Fitted Line Plot



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Fitting a Regression Model: Using Minitab

Select: Stat> Regression> Fitted Line Plot

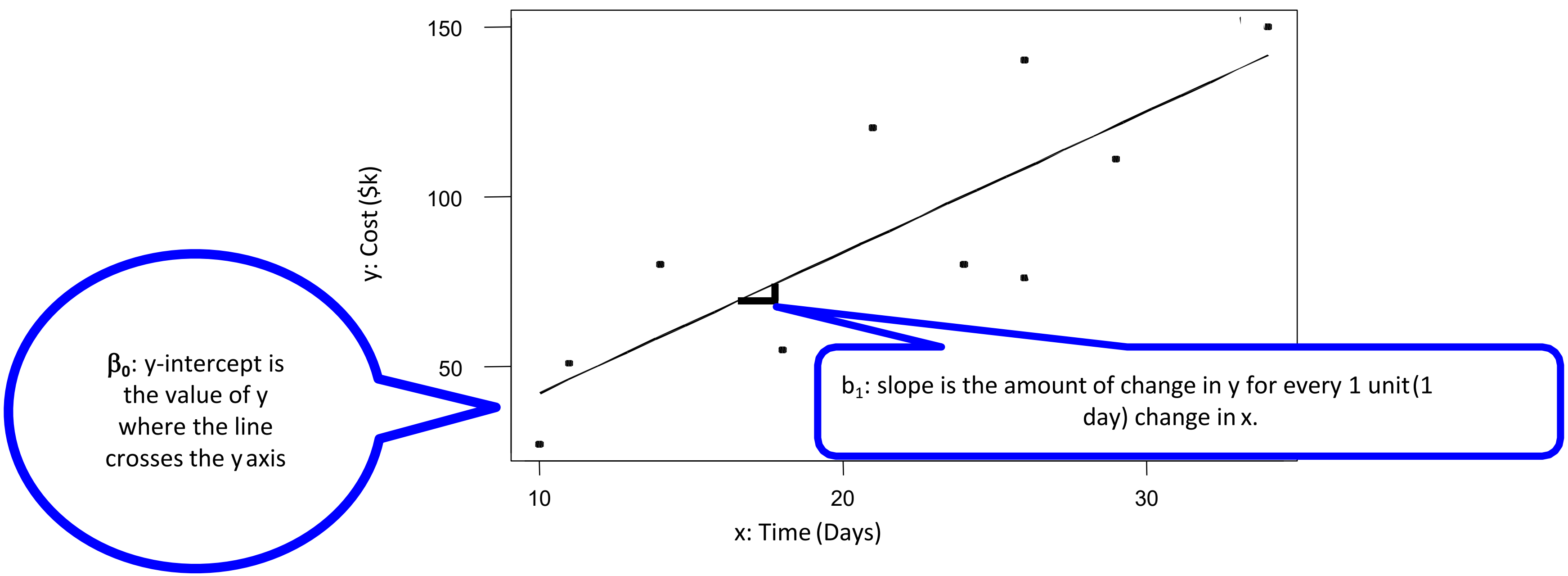


© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Simple Linear Regression: Using Minitab

Consider the below example of a relationship between Cost and Time in Minitab

Select: Stat> Regression> Regression> Fit Regression Model



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Simple Linear Regression: Using Minitab

Regression Analysis: Cost versus Time

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	9761	9761.2	16.33	0.004
Time	1	9761	9761.2	16.33	0.004
Error	8	4781	597.6		
Lack-of-Fit	7	2733	390.4	0.19	0.944
Pure Error	1	2048	2048.0		
Total	9	14542			

P - value = .004

Rejecting Null Hypothesis means we have convincing evidence of a linear relationship between Time and Cost.

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
24.4458	67.12%	63.01%	54.08%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	0.9	23.1	0.04	0.971	
Time	4.14	1.02	4.04	0.004	1.00

Regression Equation

Cost = 0.9 + 4.14 Time

The best fitting line is y = .9 + 4.14x

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

R-Squared (R-Sq)

What is R-Sq?

The coefficient of determination, R-sq is the amount of the variation in y that is explained by the regression line.

R-Sq = Explained Variation *100 Total

Variation

In the given illustration,

R-Sq = 9761/ 14542*100 = 67.1%

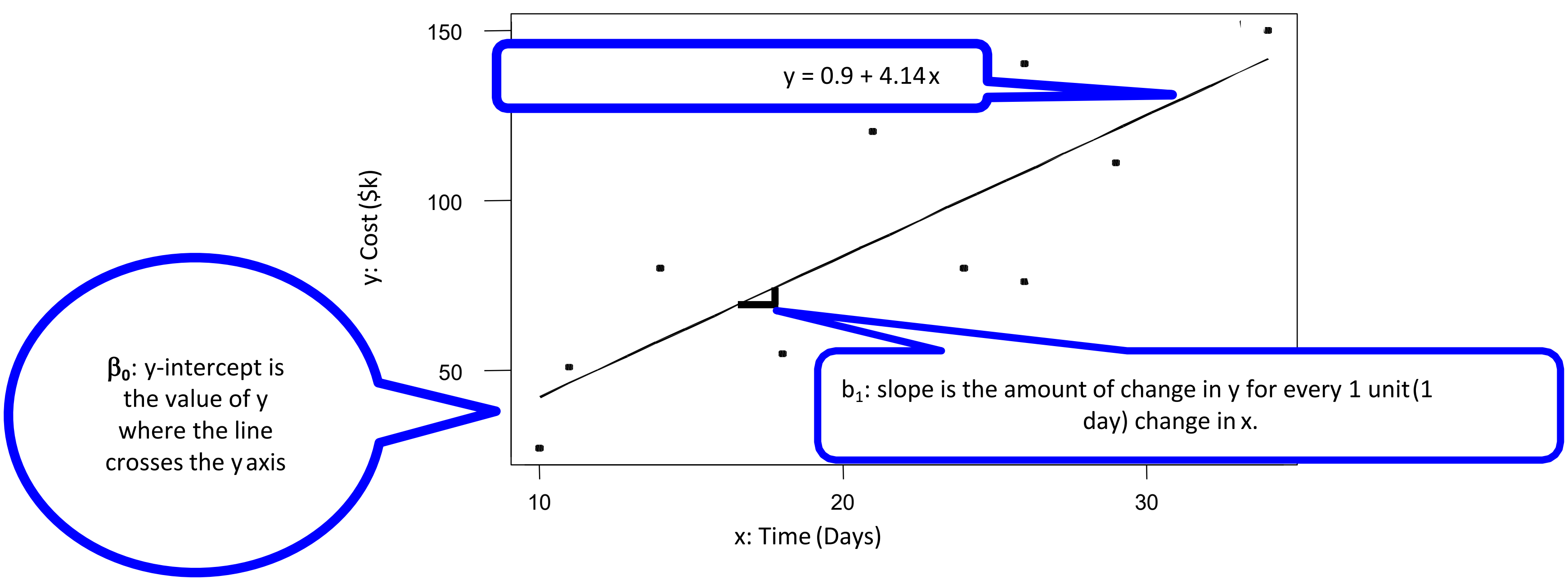
It means, 67.1% of the variability in cost can be explained by the relationship to time.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Regression Equation

Consider the below example of a relationship between Cost and Time



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Residuals

In regression analysis, the difference between the observed value of the dependent variable (y) and the predicted value (denoted by $\hat{y}$) is called the residual (e). Each data point has one residual.

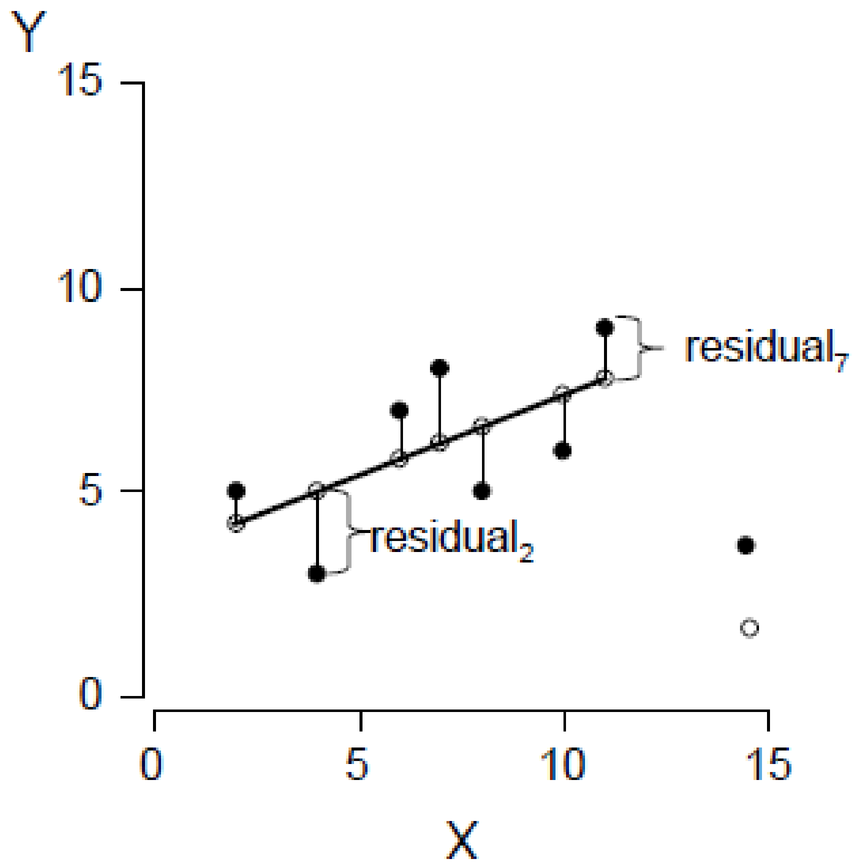
Regression Assumptions:

Key regression assumptions are based on properties of the residuals (not the original data). We assume residuals are:

- Not related to the X's
- Do not show any pattern (random)
- Stable and independent, do not change over time
- Constant, do not increase as predicted Ys increase (homogeneous)
- Residuals are normally distributed with mean of zero

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Residuals



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Residuals

To use the results of regression, assumptions about residuals must be satisfied

What are the assumptions about residuals?

- Residuals are normally distributed with mean of zero
- Residuals show no pattern (random)
- Residuals have constant variance (homogeneous variance or no heteroscedasticity)
- Residuals are independent of the values of regressor (x) variables
- Residuals are independent of each other

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Correlation and Regression

Multiple Linear Regression

Multiple Regression

Multiple Regression: Overview

- Multiple regression is to compare relationship between a continuous dependent variable (Y) and several continuous independent variables (X1,X2…)

Use of Multiple Regression

- When process input variables and the output is continuous, multiple regression can be used to investigate the relationship between the inputs and the outputs

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Multiple Regression

Let's take a quick look at an example

Some researchers observed the following data on 20 individuals with high blood pressure:

- blood pressure (y = *BP*, in mm Hg)
- age (x_1 = *Age*, in years)
- weight (x_2 = *Weight*, in kg)
- body surface area (x_3 = *BSA*, in sq m)
- duration of hypertension (x_4 = *Dur*, in years)
- basal pulse (x_5 = *Pulse*, in beats per minute)
- stress index (x_6 = *Stress*)

The researchers were interested in determining if a relationship exists between blood pressure and age, weight, body surface area, duration, pulse rate and/or stress level.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

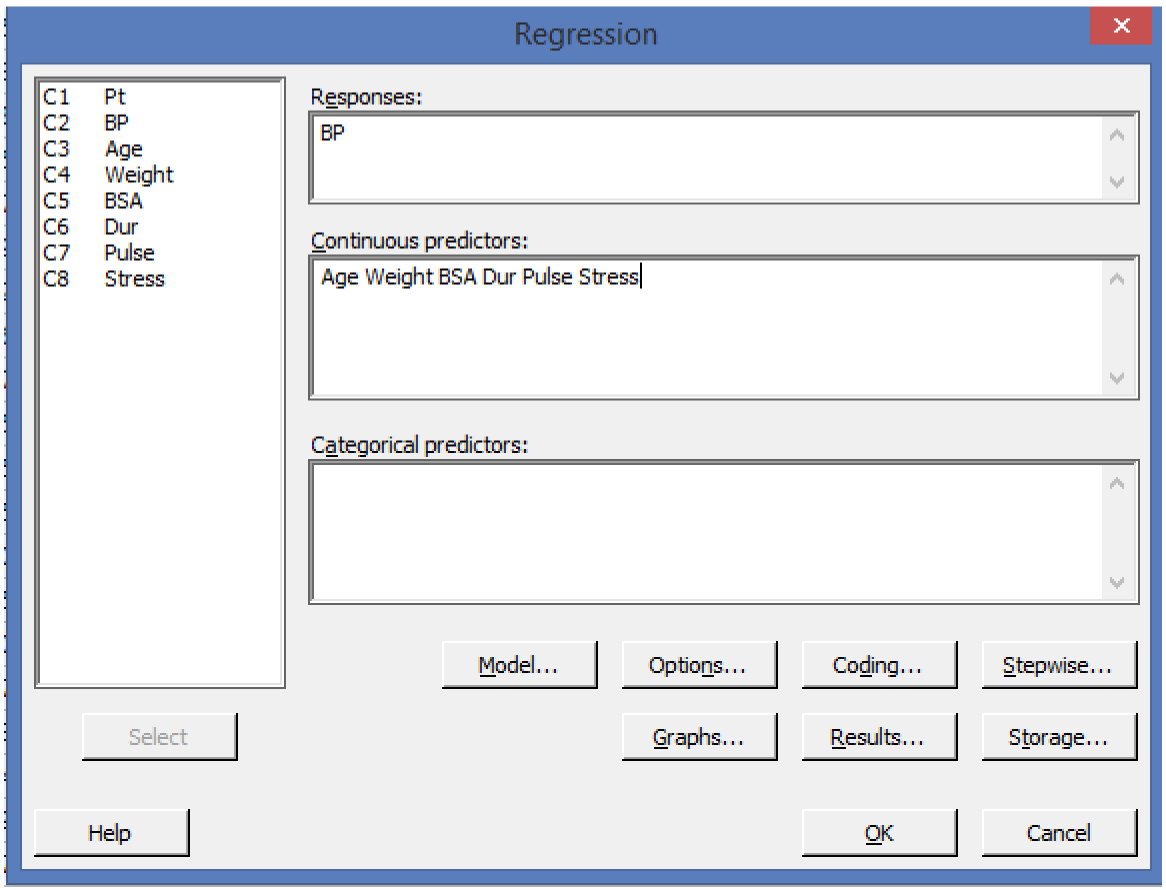
Multiple Regression

Pt	BP	Age	Weight	BSA	Dur	Pulse	Stress
1	105	47	85.4	1.75	5.1	63	33
2	115	49	94.2	2.1	3.8	70	14
3	116	49	95.3	1.98	8.2	72	10
4	117	50	94.7	2.01	5.8	73	99
5	112	51	89.4	1.89	7	72	95
6	121	48	99.5	2.25	9.3	71	10
7	121	49	99.8	2.25	2.5	69	42
8	110	47	90.9	1.9	6.2	66	8
9	110	49	89.2	1.83	7.1	69	62
10	114	48	92.7	2.07	5.6	64	35
11	114	47	94.4	2.07	5.3	74	90
12	115	49	94.1	1.98	5.6	71	21
13	114	50	91.6	2.05	10.2	68	47
14	106	45	87.1	1.92	5.6	67	80
15	125	52	101.3	2.19	10	76	98
16	114	46	94.5	1.98	7.4	69	95
17	106	46	87	1.87	3.6	62	18
18	113	46	94.5	1.9	4.3	70	12
19	110	48	90.5	1.88	9	71	99
20	122	56	95.7	2.09	7	75	99

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Multiple Regression

Select: Stat> Regression> Regression> Fit Regression Model



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Multiple Regression

Regression Analysis: BP versus Age, Weight, BSA, Dur, Pulse, Stress

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	6	557.844	92.9740	560.64	0.000
Age	1	33.331	33.3306	200.99	0.000
Weight	1	39.172	39.1718	236.21	0.000
BSA	1	0.947	0.9472	5.71	0.033
Dur	1	0.330	0.3305	1.99	0.182
Pulse	1	0.444	0.4444	2.68	0.126
Stress	1	0.442	0.4421	2.67	0.126
Error	13	2.156	0.1658		
Total	19	560.000			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.407229	99.62%	99.44%	99.08%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-12.87	2.56	-5.03	0.000	
Age	0.7033	0.0496	14.18	0.000	1.76
Weight	0.9699	0.0631	15.37	0.000	8.42
BSA	3.78	1.58	2.39	0.033	5.33
Dur	0.0684	0.0484	1.41	0.182	1.24
Pulse	-0.0845	0.0516	-1.64	0.126	4.41
Stress	0.00557	0.00341	1.63	0.126	1.83

The R-sq value indicates that the predictors explain 99.62% of the variation in BP.
Thus proving that the model fits the data well.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Multiple Regression

Regression Analysis: BP versus Age, Weight, BSA, Dur, Pulse, Stress

```
Regression Equation

BP = -12.87 + 0.7033 Age + 0.9699 Weight + 3.78 BSA + 0.0684 Dur - 0.0845
Pulse
     + 0.00557 Stress


Fits and Diagnostics for Unusual Observations
Obs       BP      Fit   Resid  Std Resid
 19  110.000  110.932  -0.932      -2.64  R

R  Large residual
```

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Multiple Regression

Now we can go to the Regression routines in Minitab and look at how well we can predict BP using Stepwise Regression.

- Stepwise: This procedure helps sift through all the inputs to produce a best guess "best" model.

Once the best model is selected, Regression is used to get more detailed diagnostics on that model.

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Stepwise Regression

- In a stepwise regression we have multiple predictor or independent variables (X's) and a response variable or dependent variable (Y)

- It begins by selecting the single independent variable (entire set of predictors) that is the 'best' predictor which maximizes R^2. Then it adds (eliminates) variables in sequential manner, in order of importance and at each step it increases R^2.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Stepwise Regression

Select: Stat> Regression> Regression> Fit Regression Model>

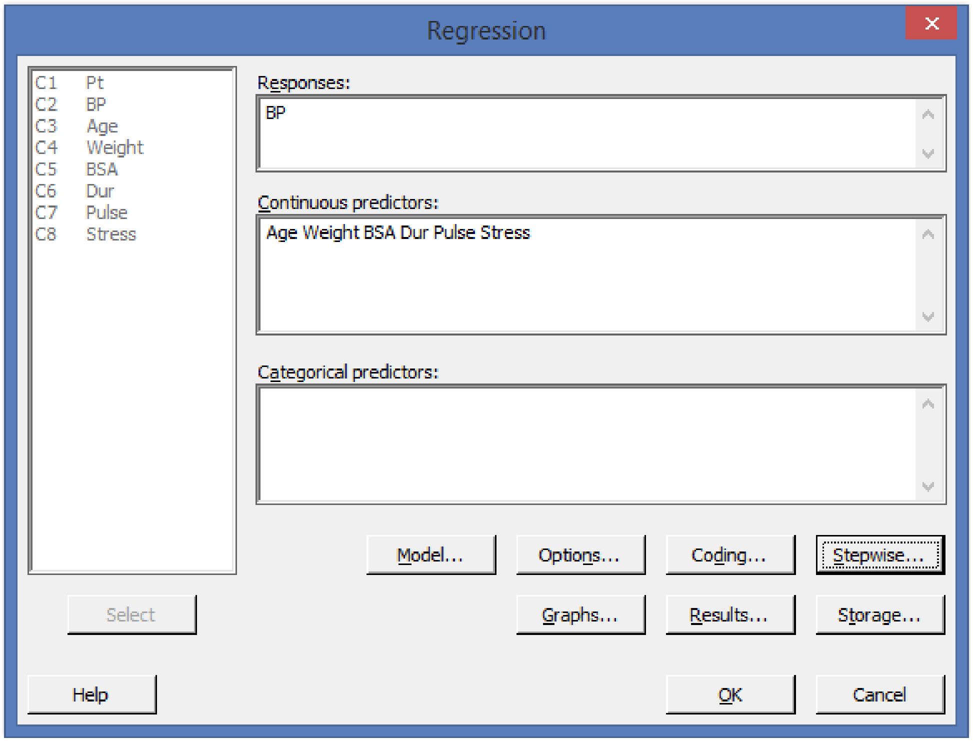
By default, this procedure starts with an empty model and then adds or removes a term for each step. You can specify terms to include in the initial model or to force into every model.

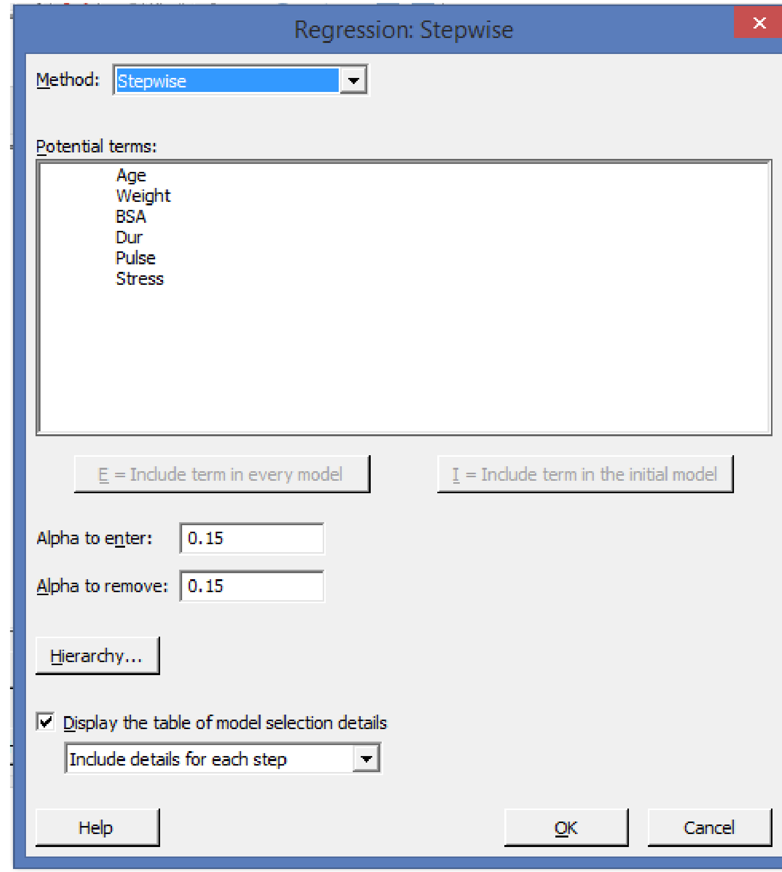
Select - Stepwise>Method>Select 'Stepwise'>Select 'Include details for each step'

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Stepwise Regression





© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Stepwise Regression

Regression Analysis: BP versus Age, Weight, BSA, Dur, Pulse, Stress

Stepwise Selection of Terms

Candidate terms: Age, Weight, BSA, Dur, Pulse, Stress

	-----Step 1----		-----Step 2-----		-----Step 3-----	
	Coef	P	Coef	P	Coef	P
Constant	2.21		-16.58		-13.67	
Weight	1.2009	0.000	1.0330	0.000	0.9058	0.000
Age			0.7083	0.000	0.7016	0.000
BSA					4.63	0.008
S		1.74050		0.532692		0.437046
R-sq		90.26%		99.14%		99.45%
R-sq(adj)		89.72%		99.04%		99.35%
R-sq(pred)		88.53%		98.89%		99.22%
Mallows' Cp		312.81		15.09		6.43

α to enter = 0.15, α to remove = 0.15

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	556.944	185.648	971.93	0.000
Age	1	48.658	48.658	254.74	0.000
Weight	1	65.303	65.303	341.89	0.000
BSA	1	1.768	1.768	9.25	0.008
Error	16	3.056	0.191		
Total	19	560.000			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.437046	99.45%	99.35%	99.22%

Here in this example, the Stepwise Regression with the best three variable model. Thus indicating that the other variables do not add any predictive value.

The stepwise model selection output is designed to present a concise summary of a number of fitted models. This table is followed by the full regression output for the procedure's final model.

For this example, the full regression output corresponds to the model in step 3.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Stepwise Regression

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-13.67	2.65	-5.16	0.000	
Age	0.7016	0.0440	15.96	0.000	1.20
Weight	0.9058	0.0490	18.49	0.000	4.40
BSA	4.63	1.52	3.04	0.008	4.29

Regression Equation

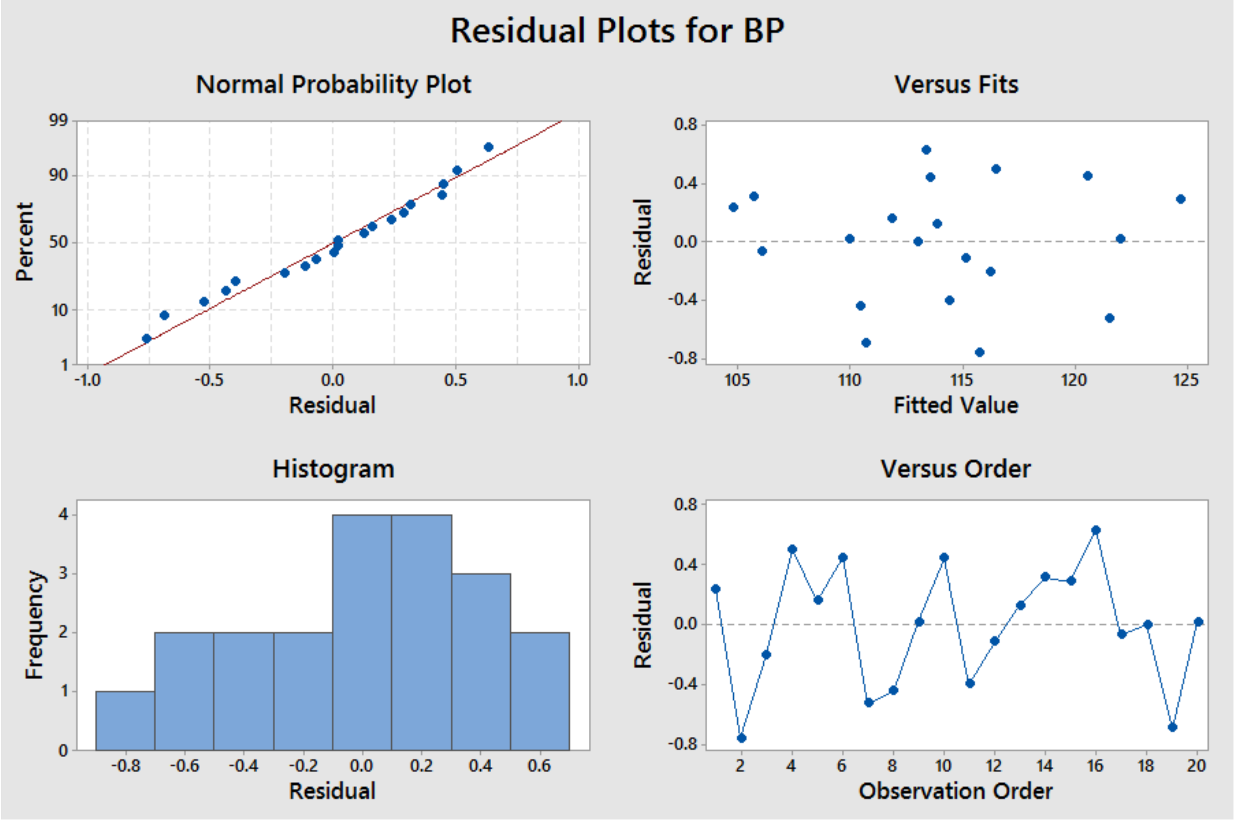
BP = -13.67 + 0.7016 Age + 0.9058 Weight + 4.63 BSA

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Model Diagnostics

Select: Stat> Regression> Regression> Fit Regression Model> Graphs> Four in One



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Model Diagnostics

Minitab displays four graphs:

- Histogram of residuals
- Normal Plot of residuals
- Plot of residuals versus fits
- Plot of residuals versus order

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Multi - Collinearity

- One of the assumption of model accuracy is that X's are not correlated
- Multi-Collinearity is a condition wherein high correlation exists between two independent variables
- This means the two variables contribute redundant information to the multiple regression model
- Can lead to unstable coefficients (large standard error)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Problems with Multi - Collinearity

Multicollinearity can cause severe problems:

- calculations of coefficients and standard errors are affected (unstable, inflated variances)
- difficulty in assessing any particular variable's effect
- opposite signs (from what is expected) in the estimated parameters
- if two input variables x1and x2are highly correlated, then p-value for both might be high

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Detecting Multi - Collinearity

Variance Inflationary Factor (VIF) is used to measure collinearity

$$VIF = \frac{1}{1 - R^2}$$

Where;
R^2 = Coefficient of determination

If VIF > 5, X is correlated with the other explanatory variables

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Checking Multi - Collinearity

Select: Stat> Basic Statistics> Correlation

Correlation: Age, Weight, BSA, Dur, Pulse, Stress, BP

	Age	Weight	BSA	Dur	Pulse	Stress
Weight	0.407					
BSA	0.378	0.875				
Dur	0.344	0.201	0.131			
Pulse	0.619	0.659	0.465	0.402		
Stress	0.368	0.034	0.018	0.312	0.506	
BP	0.659	0.950	0.866	0.293	0.721	0.164

Cell Contents: Pearson correlation

- High values of pairwise correlation (generally > 0.8) provide warnings of potential multicollinearity problems
- We see that Weight and Body Surface Area are most highly related to Blood Pressure.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Checking Multi - Collinearity - Relook at the output

Regression Analysis: BP versus Age, Weight, BSA, Dur, Pulse, Stress

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	6	557.844	92.9740	560.64	0.000
Age	1	33.331	33.3306	200.99	0.000
Weight	1	39.172	39.1718	236.21	0.000
BSA	1	0.947	0.9472	5.71	0.033
Dur	1	0.330	0.3305	1.99	0.182
Pulse	1	0.444	0.4444	2.68	0.126
Stress	1	0.442	0.4421	2.67	0.126
Error	13	2.156	0.1658		
Total	19	560.000			

Since the P-value for Dur, Pulse and Stress is higher, we can say that these factors do not impact BP

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
0.407229	99.62%	99.44%	99.08%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	-12.87	2.56	-5.03	0.000	
Age	0.7033	0.0496	14.18	0.000	1.76
Weight	0.9699	0.0631	15.37	0.000	8.42
BSA	3.78	1.58	2.39	0.033	5.33
Dur	0.0684	0.0484	1.41	0.182	1.24
Pulse	-0.0845	0.0516	-1.64	0.126	4.41
Stress	0.00557	0.00341	1.63	0.126	1.83

VIF values greater than 5 suggest that the regression coefficients are poorly estimated due to severe multicollinearity. We can see higher VIFs for Weight and BSA

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

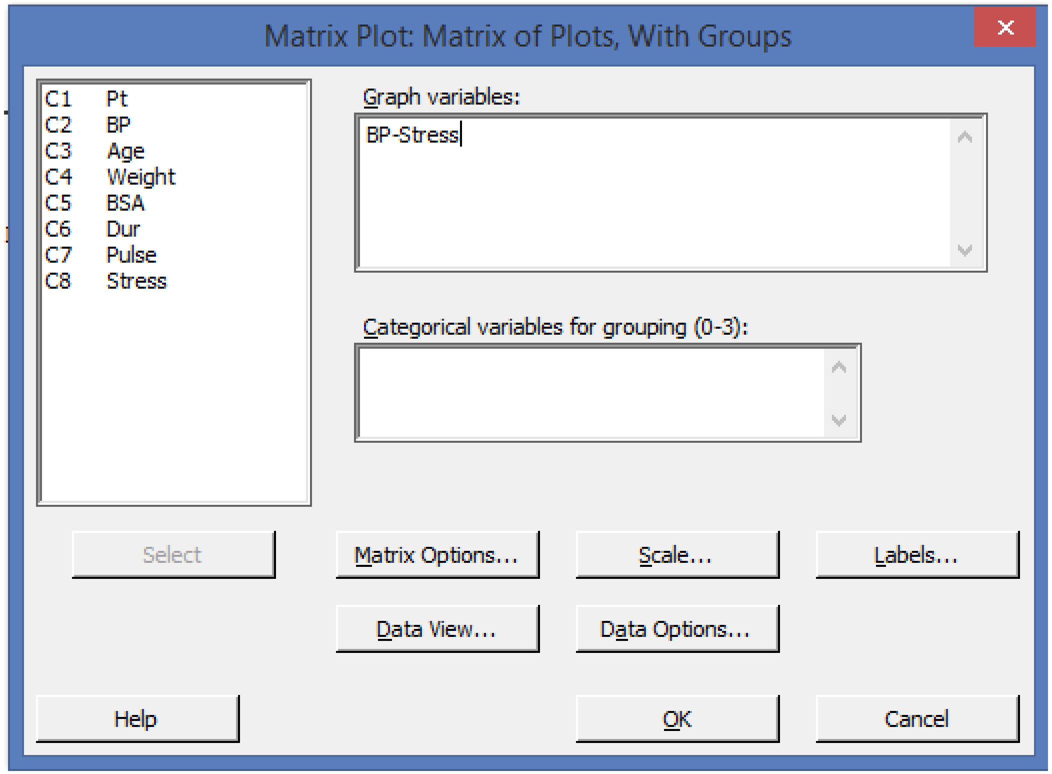
Checking Multi - Collinearity with Matrix Plot

- A matrix plot is a two-dimensional matrix of individual plots.
- Matrix plots are good for, among other things, seeing the two variable relationships among a number of variables all at once
- Matrix Plots help to identify the meaningful relationships with a single graph.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

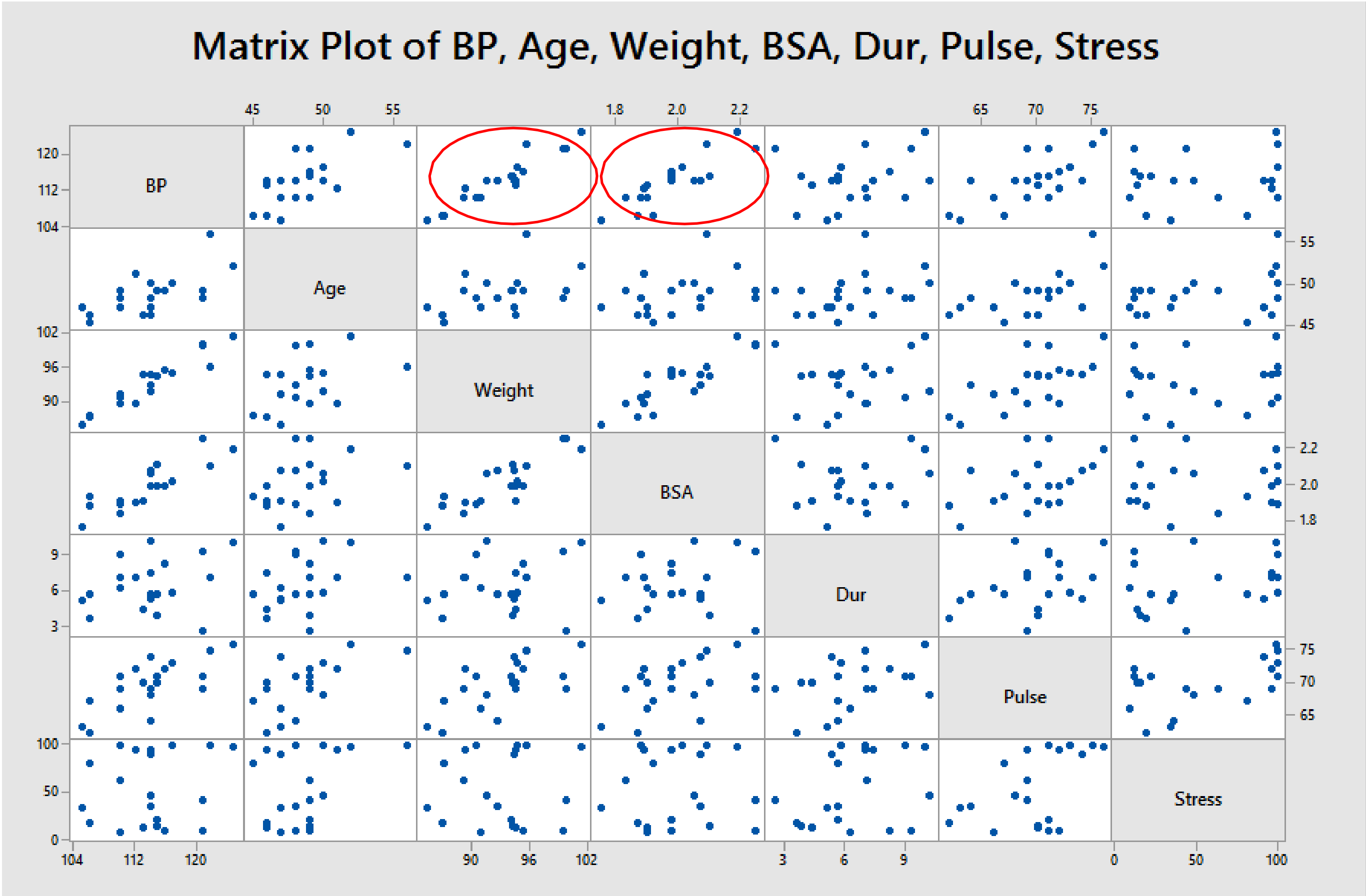
Checking Multi - Collinearity with Matrix Plot

Select: Graph> Matrix Plot> With Groups



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Checking Multi - Collinearity with Matrix Plot



Multi-collinearity exist between Weight and Body Surface Area (BSA). The correlation coefficient between the two variables can be estimated through correlation

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Checking Multi - Collinearity with Correlation

Select: Stat> Basic Statistics> Correlation

Correlation: Age, Weight, BSA, Dur, Pulse, Stress, BP

	Age	Weight	BSA	Dur	Pulse	Stress
Weight	0.407					
BSA	0.378	0.875				
Dur	0.344	0.201	0.131			
Pulse	0.619	0.659	0.465	0.402		
Stress	0.368	0.034	0.018	0.312	0.506	
BP	0.659	0.950	0.866	0.293	0.721	0.164

Cell Contents: Pearson correlation

We see that Weight and Body Surface Area are most highly related to Blood Pressure.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Correlation and Regression

Logistic Regression

Logistics Regression

- Logistic Regression also creates a prediction equation, but for a discrete dependent variable (Y)
- Logistic Regression can be applied to output data that are Binary (yes or no), or Ordinal (high, medium, low), or Nominal (North, South, East, West)
- As part of this course, we only focus on Binary Logistic Regression
- While the result of logistic regression is similar to least squares regression (a prediction equation), the math to get there is much different as it is based on natural logarithm and odds ratio

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Logistics Regression

Usually a transactional projects often utilize non-continuous data like:

- Was the service provided in an appropriate manner?
- Was the order filled?
- Was the product available for shipping?
- Degree of agreement scales, satisfaction ratings
- Defect severity ratings (A is better than B is better than C)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Logistics Regression

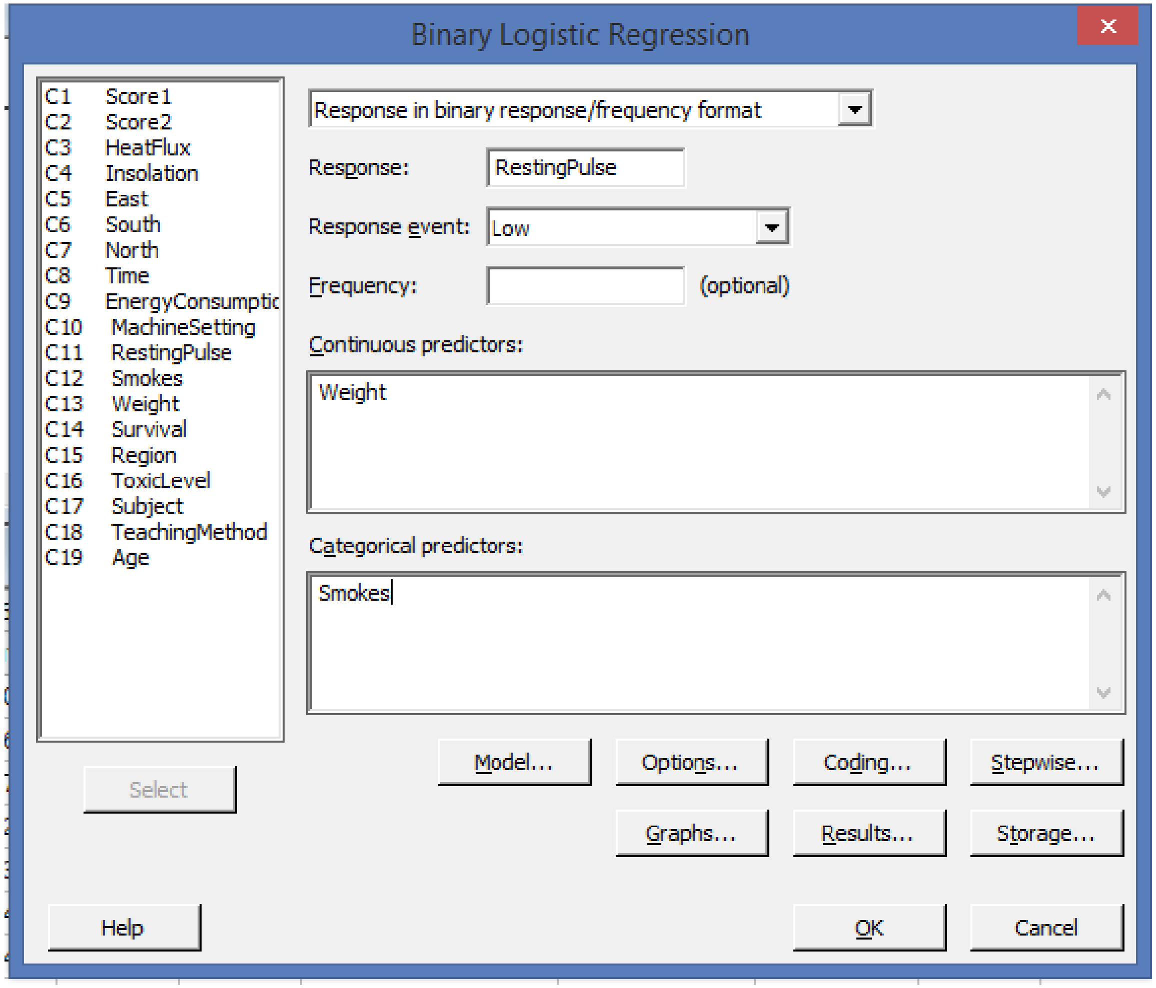
Refer the worksheet EXH_REGR.MTW.

Select: Stat> Regression> Binary Logistic Regression> Fit Binary LogisticModel

Example: You are a researcher who is interested in understanding the effect of smoking and weight upon resting pulse rate. The response-pulse rate have been categorized into low and high. Investigate the effects of smoking and weight upon pulserate.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

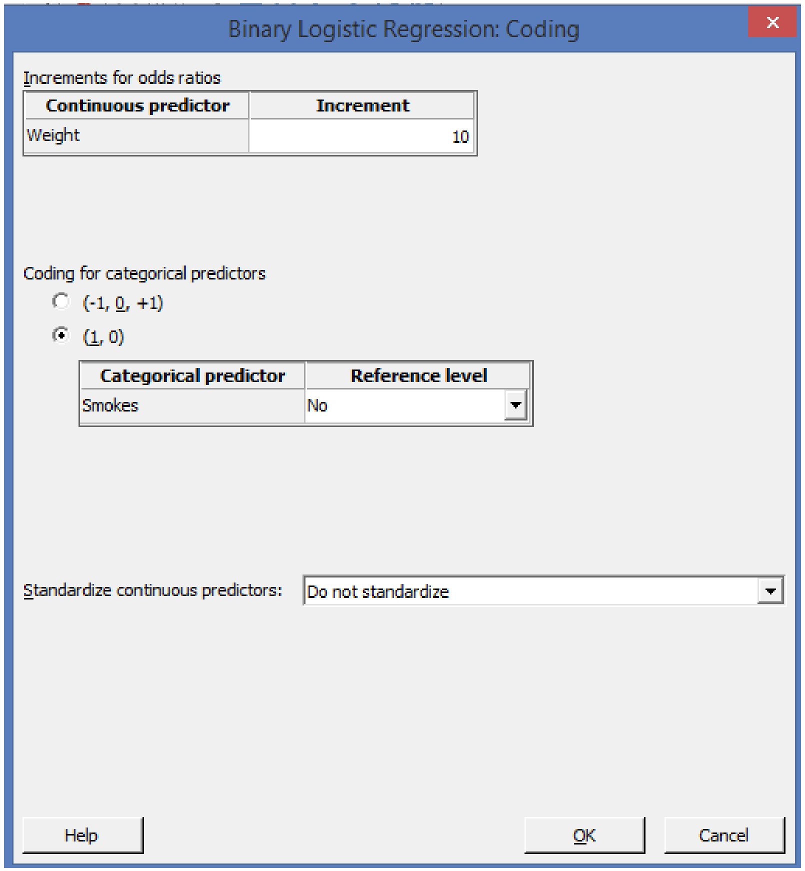
Logistics Regression



KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Logistics Regression



Increments for odds ratios:

Change the units for calculating the odds ratio from the units in the data for a continuous predictor.

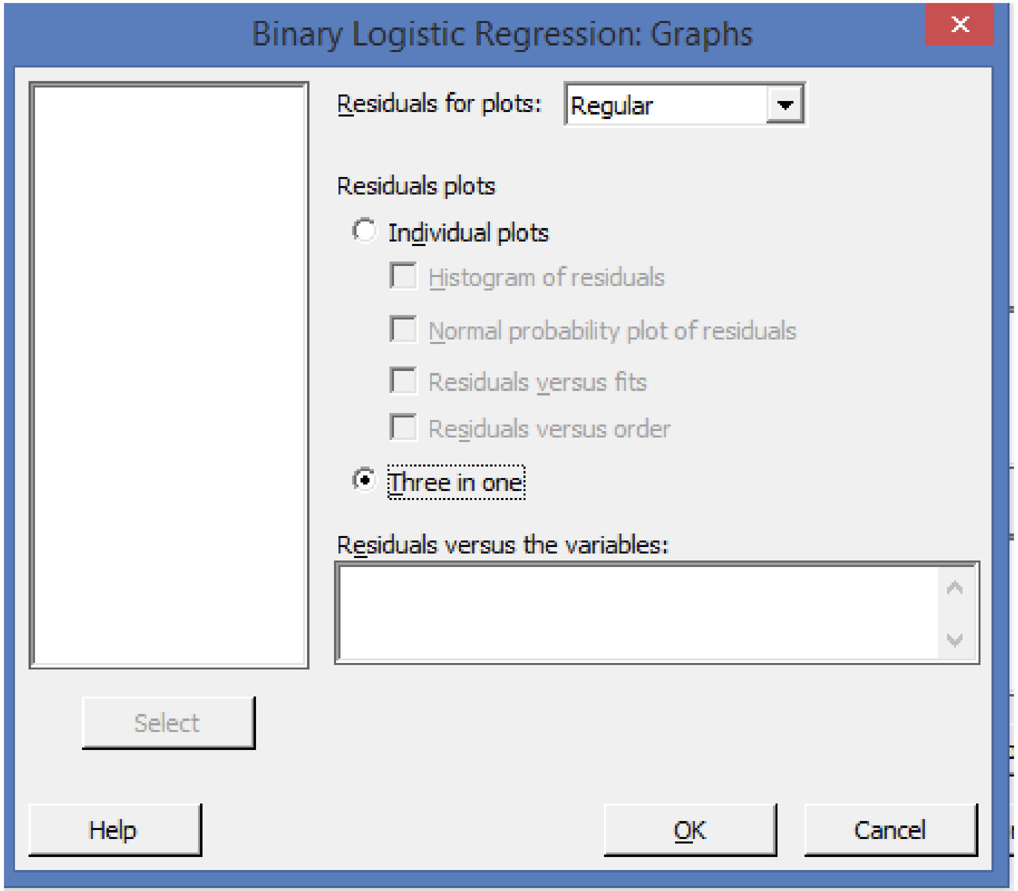
The coefficient for a variable when the model uses the logit link function represents the change in the log odds for an increase of one unit in the predictor.

In some cases, a change of one unit in the data is too small to be meaningful.

For example, if the predictor is mass in grams, a change of 1 gram can be too small to matter. Enter 1,000 to see the change in the odds ratio for a kilogram

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Logistics Regression



Select graphs – 3 in one

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Logistics Regression

Binary Logistic Regression: RestingPulse versus Weight, Smokes

Method

Link function	Logit
Categorical predictor coding	(1, 0)
Rows used	92

Response Information

Variable	Value	Count	
RestingPulse	Low	70	(Event)
	High	22	
	Total	92	

Deviance Table

Source	DF	Adj Dev	Adj Mean	Chi-Square	P-Value
Regression	2	7.574	3.787	7.57	0.023
Weight	1	4.629	4.629	4.63	0.031
Smokes	1	4.737	4.737	4.74	0.030
Error	89	93.640	1.052		
Total	91	101.214			

Since the P-value for Weight and Smoke is lesser than the level of significance, there is enough evidence to indicate that both has a significant impact on
Resting Pulse

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Logistics Regression

Binary Logistic Regression: RestingPulse versus Weight, Smokes

Model Summary

Deviance R-Sq	Deviance R-Sq(adj)	AIC
7.48%	5.51%	99.64

Coefficients

Term	Coef	SE Coef	VIF
Constant	-1.99	1.68	
Weight	0.0250	0.0123	1.12
Smokes			
Yes	-1.193	0.553	1.12

Odds Ratios for Continuous Predictors

	Unit of Change	Odds Ratio	95% CI
Weight	10	1.2843	(1.0101, 1.6330)

Odds Ratios for Categorical Predictors

Level A	Level B	Odds Ratio	95% CI
Smokes			
Yes	No	0.3033	(0.1026, 0.8966)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Logistics Regression

Binary Logistic Regression: RestingPulse versus Weight, Smokes

```
Odds ratio for level A relative to level B
Regression Equation

P(Low)  =  exp(Y')/(1 + exp(Y'))

Smokes
No       Y' = -1.987 + 0.02502 Weight

Yes      Y' = -3.180 + 0.02502 Weight

Goodness-of-Fit  Test
                   s
Test                  Chi-Square  P-Value
                 DF
Deviance         89        93.64    0.348
Pearson          89        88.63    0.491
Hosmer-Lemeshow   8         4.75    0.784
Fits and Diagnostics for Unusual Observations

        Observed
Obs  Probability    Fit     Resid  Std Resid
 56       0.0000  0.8689  -2.0159      -2.04  R
 86       1.0000  0.3828   1.3858       1.46     X

R  Large residual
X  Unusual X
```

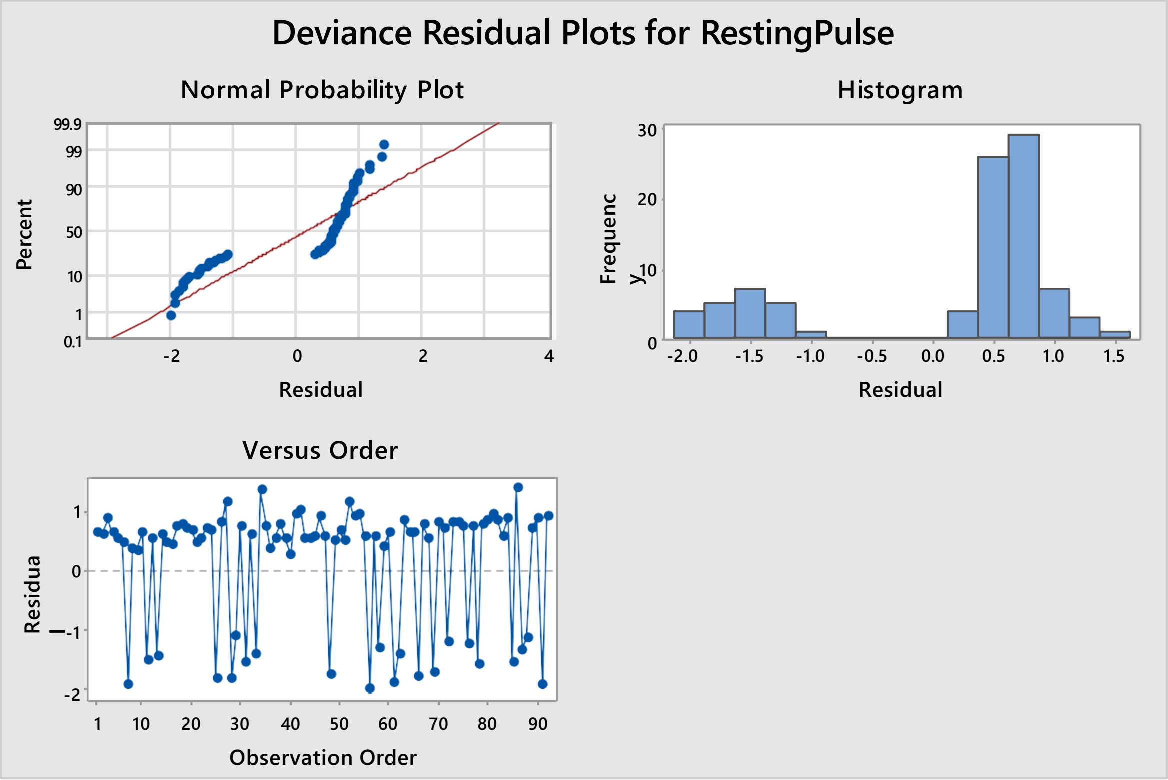
There are two equations, one for each level of the Smokes variable.

The constant term is more negative for the people who smoke, so these people have a lower probability of having a lower resting pulse.

Because there is no interaction with weight in the model, the coefficient is the same in both equations.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Logistics Regression



The Normality Plot does not follow a line & Histogram shows a bimodal pattern which means the data is not enough for normal approximation theory to apply.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

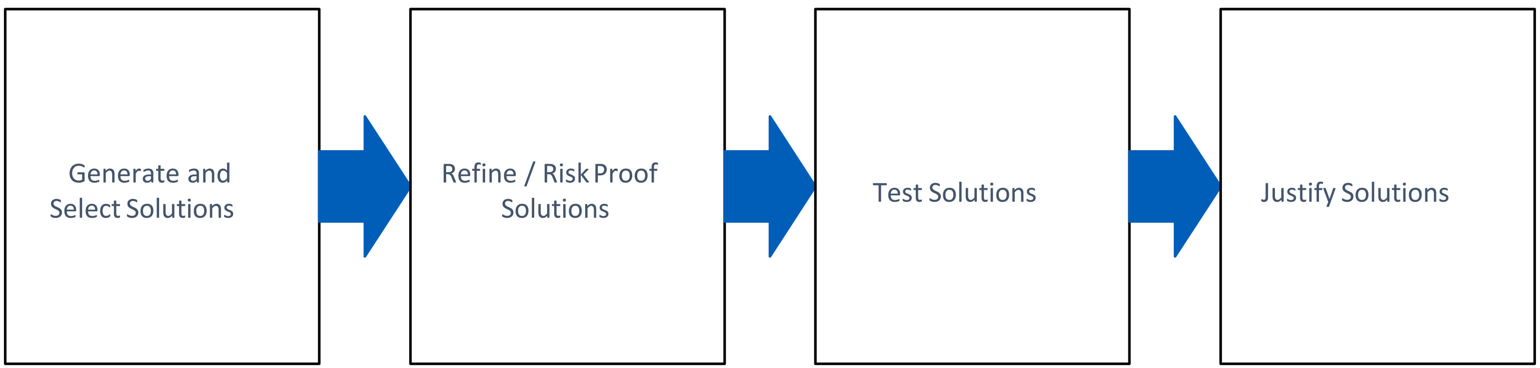
Inference from Analyze Phase

- Infer whether the causes were real i.e. were the causes statistically significant.
- Interpret scatter diagrams and the correlation coefficient, r, to determine if two variables are correlated.
- Perform Simple Linear Regression, Multiple Regression and Logistics Regression Analysis and interpret a regression equation.
- Interpret R^2 and P- values to determine the adequacy of a math model (regression equation).
- Make predictions using the math model.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

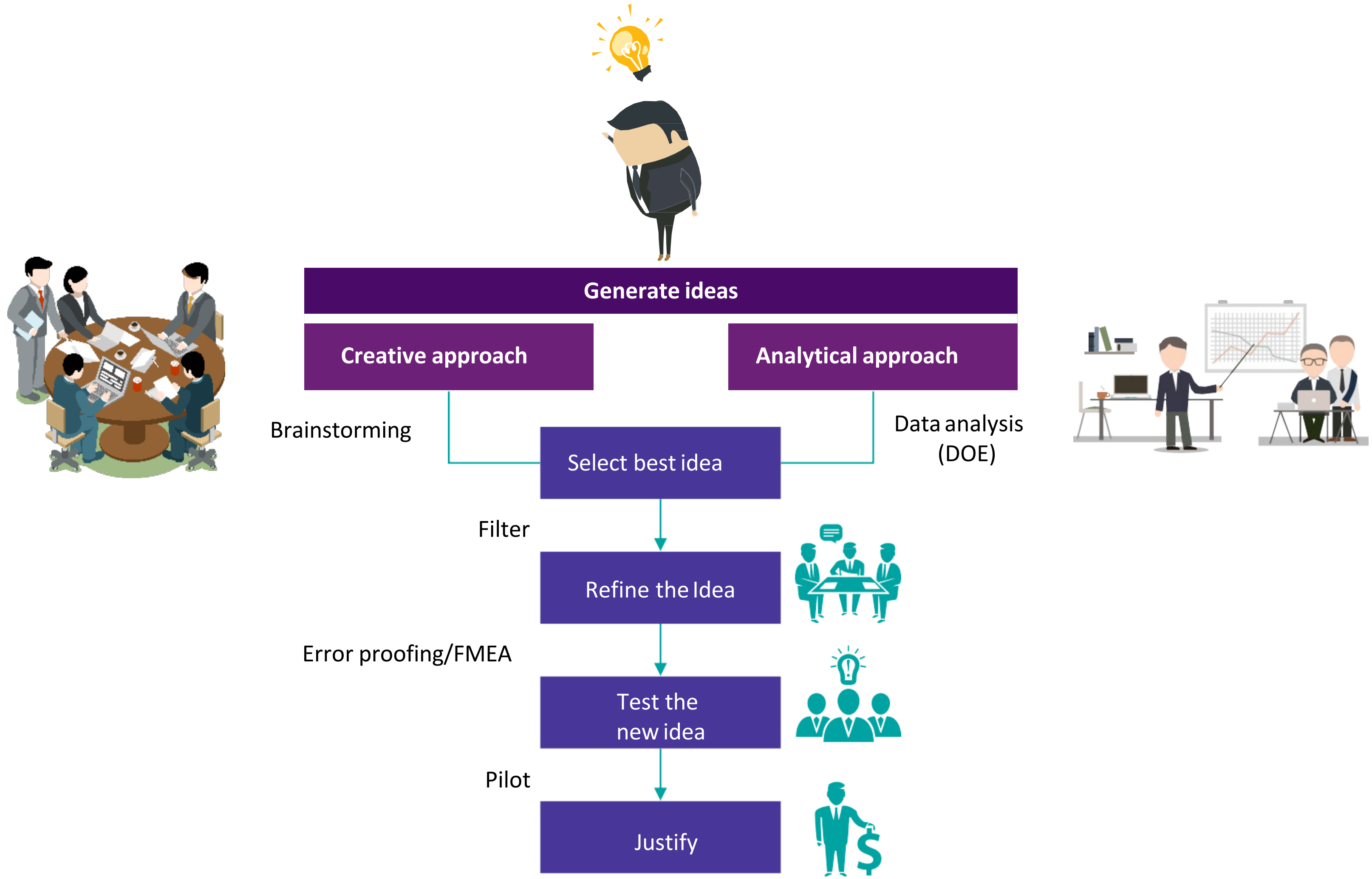
Improve Phase

Improve Phase - Roadmap



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Improve: Steps



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design Of Experiments

Design of Experiments

What is Design of Experiments (DOE)?

Design of experiments (DOE) is a valuable tool to optimize product and process designs to:

- to accelerate the development cycle
- to reduce development costs
- to improve the transition of products from research and development to manufacturing
- to effectively trouble shoot manufacturing problems

Today, Design of Experiments is viewed as a quality technology to achieve product excellence at lowest possible overall cost.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design of Experiments

When to use Design of Experiments (DOE)?

- For instance, supplier of a certain input for a given process can be an important factorfor shift in the performance (shift in the average).

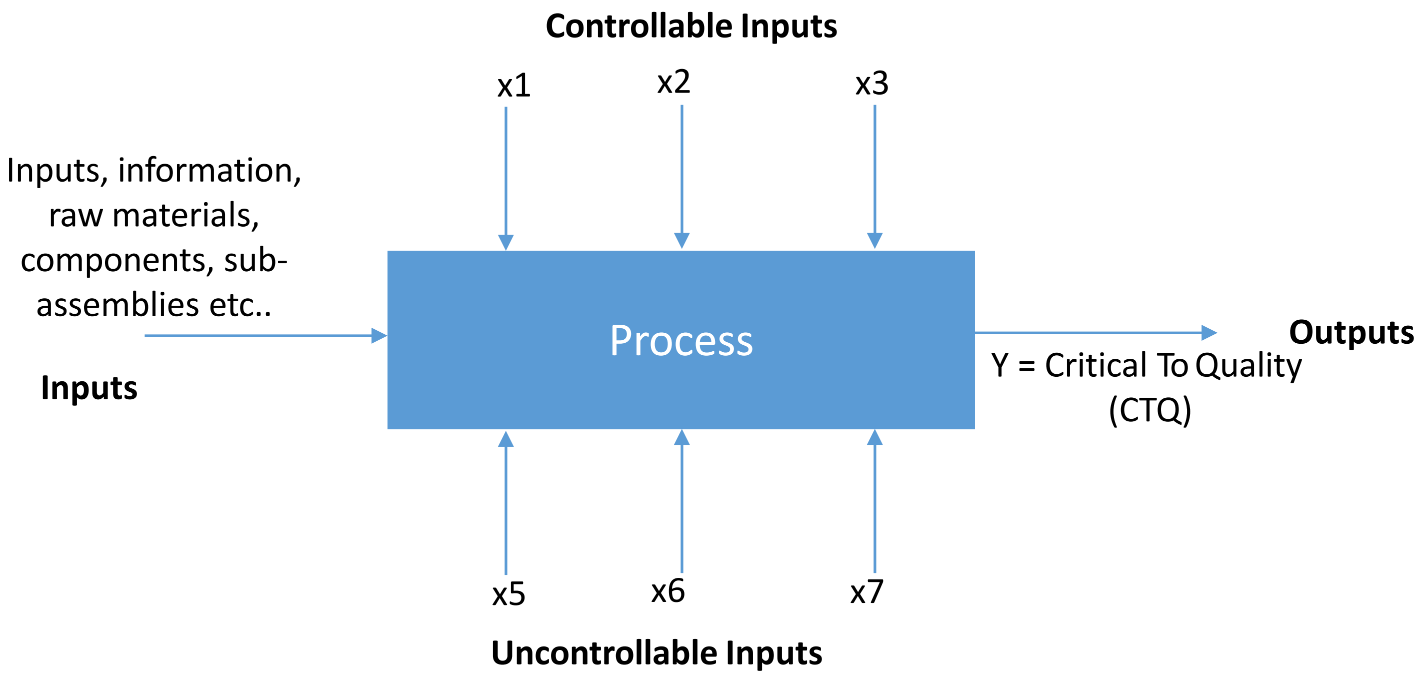
  Here you know that the change in the mean can be attributed to the change in the supplier and that this has been done with some intent and knowledge.

  When you know that a certain critical X has changed levelsand you need to find the optimum setting of the process for each level of X, we use DOE.

- If you need to find out for a given process at what level each of the X has to be set to give a favorable output, we rely on DOE.

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design of Experiments



In other words, DOE is a deliberate and systematic changes of input variables (factors) to observe the corresponding change in the output (response)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design of Experiments

Objectives of conducting an experiment

- Determining which input variable (X) are having most influence on the response variable (Y)
- Determining where to set the influential X's such that Y is optimum
- Determining where to set X's so that the variability in Y is minimum
- Determining where to set the influential X's such that the effects of uncontrollable input are minimized

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

DOE Terminologies

- **Response:** A measurable outcome of interest. For Example: yield, output, strength etc..
- **Factors:** These are process input variables that are set at various levels to determine their individual and joint effects on the output. For Example: time, temperature etc.
- **Levels:** Levels refer to the values of factors for which the data is gathered. For Example:

  Level 1 – Mold Temperature: 600° 700°

  Level 2 – Material Type: A B
- **Interaction:** The degree to which factors depend on one another. For Example: If you want to find the interaction between Time * Temperature i.e. the best level for time depends on what temperature is set.
- **Main Effect:** The change in the average response produced by a change in the level of the factor.
- **Trials or Runs:** The specific test combinations of factors and levels that are run during the experiment
- **Noise**: Variables that affect product or process performance and whose values cannot be controlled.
- **Replication**: Replication is a systematic duplication of series of experimental runs. It provides the means of measuring precision by calculating experimental error.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Examples

- **Example 1: In a machine process**
  - **RESPONSE:** Surface Finish (Y)
  - **FACTORS:** Speed of Machine (X1), Depth of Cut (X2)
  - **LEVELS:** High and Low

- **Example 2: In a popcorn making process**
  - **RESPONSE:** Yield of Popcorn (Y)
  - **FACTORS:** Grade of corn (X1), Type of Popper (X2)
  - **LEVELS:** Budget, Regular and Luxury; Air and Oil

- **Example 3: In a credit card receivables process**
  - **RESPONSE:** Delinquency% (Y)
  - **FACTORS:** Employment Type, Qualification
  - **LEVELS:** Full Time or Part Time; Graduate or Post Graduate

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Major Approaches to DOE

- One Factor At A Time (OFAT) Experiments
- Factorial Experiments
- Taguchi Method
- Response Surface Design

As part of the curriculum, we only cover Factorial Design Experiments

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Factorial Experiments

Factorial Experiments are the most effective of experiments in dealing with several factors involved in a study. Primarily, the factors are varied together instead of one at a time.

The three basic principles of experimental design are:

- Replication
- Randomization
- Blocking

Replication: Replication means repeating all the experimental conditions or running combinations two or more times. Replicates are done to obtain an estimate of experimental error which provides measurement for determining if the observed differences in the data are statistically significant.

Randomization: Randomization assigns the order in which the experimental trails will be run in no specific order i.e. in a random order. Randomization is done to minimize the effects of lurking variables over all the other factors in the experiment. (Lurking variables: It is that unknown factor in a process that has an important effect on the input factors)

Blocking: An experiment is arranging the runs of the experiment in groups (blocks) so that runs within each group (block) has a minor variation in common. In other words, a block is a carefully chosen subset of the total runs in the experiment that are tested under similar conditions.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial

- 2^k Factorial Designs are experiments where all factors have only 2 levels

- The number of combinations (Runs) for Full Factorial Design is denoted as $n = 2^k$ (where k is the number of factors)

Factors

$$2^k$$

Levels

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Example

Consider the manufacture of product for use in the making of paint in a batch process. Fixed amounts of raw materials are heated under pressure in rector 1 for a fixed period of time and the product is then recovered. Currently, the process is operating at 225 ℃ and pressure 4.5 bar. As part of Six Sigma improvement, you are aimed at increasing the product yield. A 2^k factorial experiment with two replications was planned.

The expected yield was around 90 kg. The temperature was set at two levels 200 ℃ and 250 ℃. The level for pressure was set at 4.0 bar and 5.0 bar

Response: Product Yield (Y)

Factors: Temperature (X1) and Pressure (X2)

Levels: 200 ℃ and 250 ℃; 4.0 bar and 5.0 bar

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

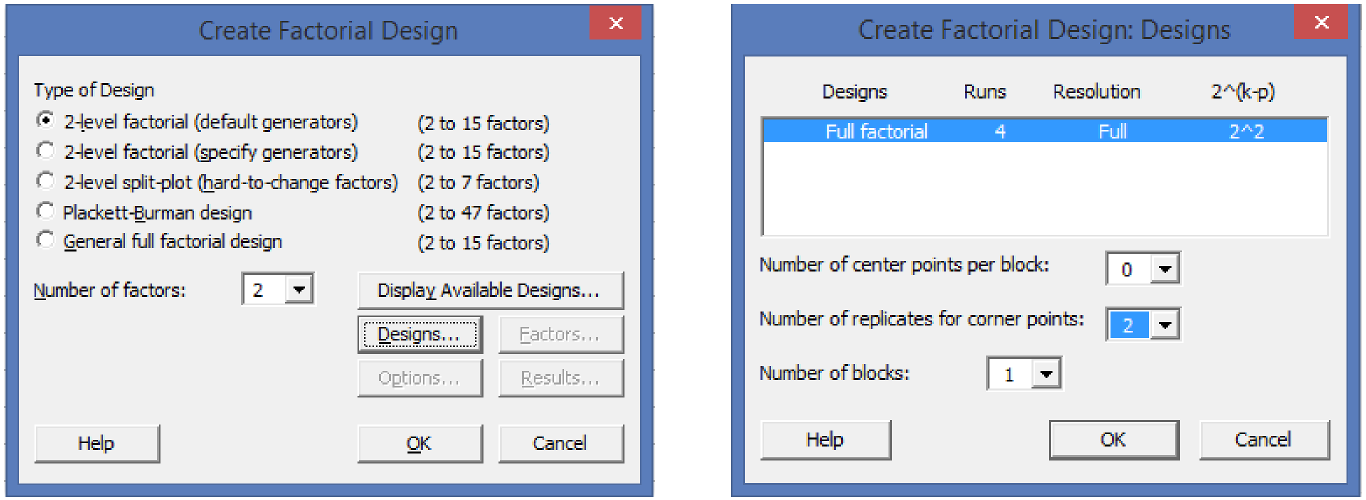
Steps in designing an experiment

1. Define the Problem
2. State the Hypothesis
3. Identify the Dependent (Y) and Independent (X) Variables
4. Determine the Test Levels for Each Independent Variable
5. Calculate the Number of Trials to Test All Combinations of Test Levels
6. Construct the Experimental Trials Table
7. Run the Experiment
8. Summarize the Data
9. Conclude and Recommend

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Minitab

Step 1: Create Factorial Design (Stat> DOE> Factorial> Create Factorial Design) Step 2: Select the number of factors (from the drop-down choose 2)



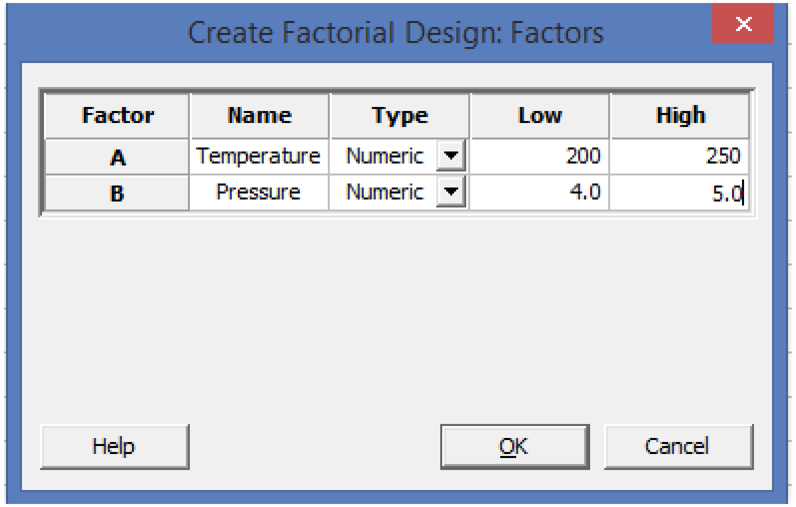
Step 3: Select Designs. Choose Full Factorial. From the drop-down choose the number of replicates as 2 (refer screen shot above)

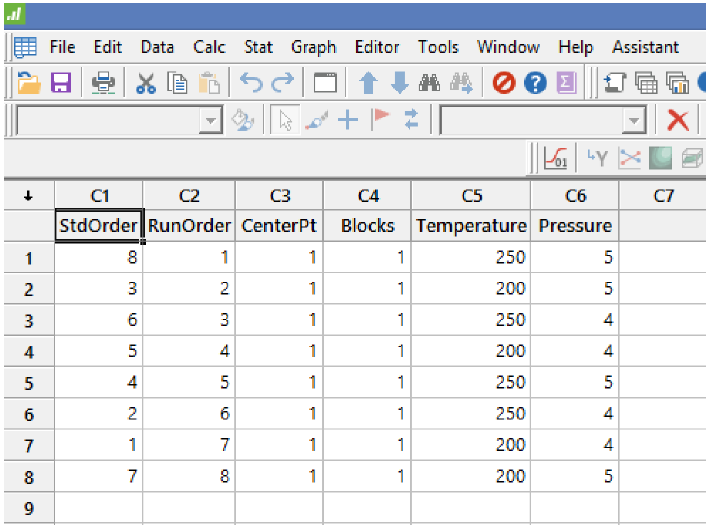
KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Minitab

Step 4: Select the Factors

Step 5: Mention the factor names and the corresponding high / low values





	C1	C2	C3	C4	C5	C6	C7
	StdOrder	RunOrder	CenterPt	Blocks	Temperature	Pressure	
1	8	1	1	1	250	5	
2	3	2	1	1	200	5	
3	6	3	1	1	250	4	
4	5	4	1	1	200	4	
5	4	5	1	1	250	5	
6	2	6	1	1	250	4	
7	1	7	1	1	200	4	
8	7	8	1	1	200	5	
9							

Step 6: Refer the RunOrder to carry out the experiments (refer above column C2)

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^K Factorial - Minitab

Step 7: Update the output of the experiment (refer column C7)

↓	C1	C2	C3	C4	C5	C6	C7	C8
	StdOrder	RunOrder	CenterPt	Blocks	Temperature	Pressure	Yield	
1	5	1	1	1	200	4	85	
2	6	2	1	1	250	4	90	
3	2	3	1	1	250	4	97	
4	4	4	1	1	250	5	90	
5	8	5	1	1	250	5	85	
6	3	6	1	1	200	5	101	
7	7	7	1	1	200	5	92	
8	1	8	1	1	200	4	96	
9								

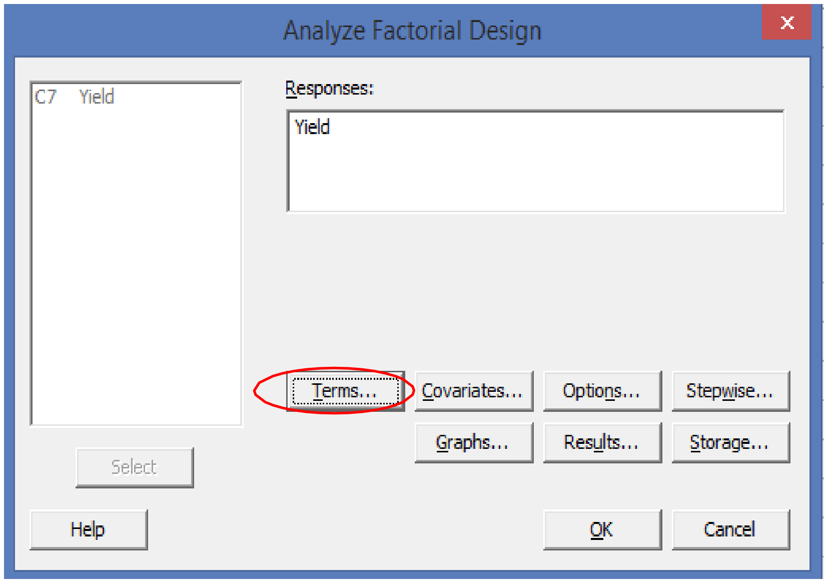
Step 8: Analyze the Factorial Design (Stat> DOE> Factorial> Analyze Factorial Design)

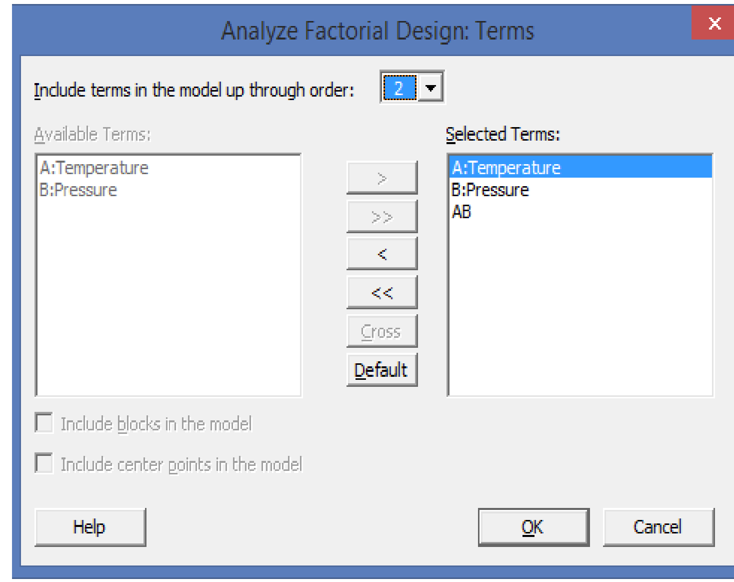
KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Minitab

Step 9: Select Responses – Yield





Step 10: Select "Terms" and ok

Step 11: Interpret the output, P-Value and the Pareto Diagram

Step 12: Look for graphical plots (Stat> DOE> Factorial> Factorial Plots) to check for interaction and impact

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Minitab

Factorial Regression: Yield versus Temperature, Pressure

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	3	200.000	66.667	9.52	0.027
Linear	2	72.000	36.000	5.14	0.078
Temperature	1	0.000	0.000	0.00	1.000
Pressure	1	72.000	72.000	10.29	0.033
2-Way Interactions	1	128.000	128.000	18.29	0.013
Temperature*Pressure	1	128.000	128.000	18.29	0.013
Error	4	28.000	7.000		
Total	7	228.000			

Since the P-value indicates that Pressure & Temp*Pressure interaction has a significant impact.

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
2.64575	87.72%	78.51%	50.88%

Coded Coefficients

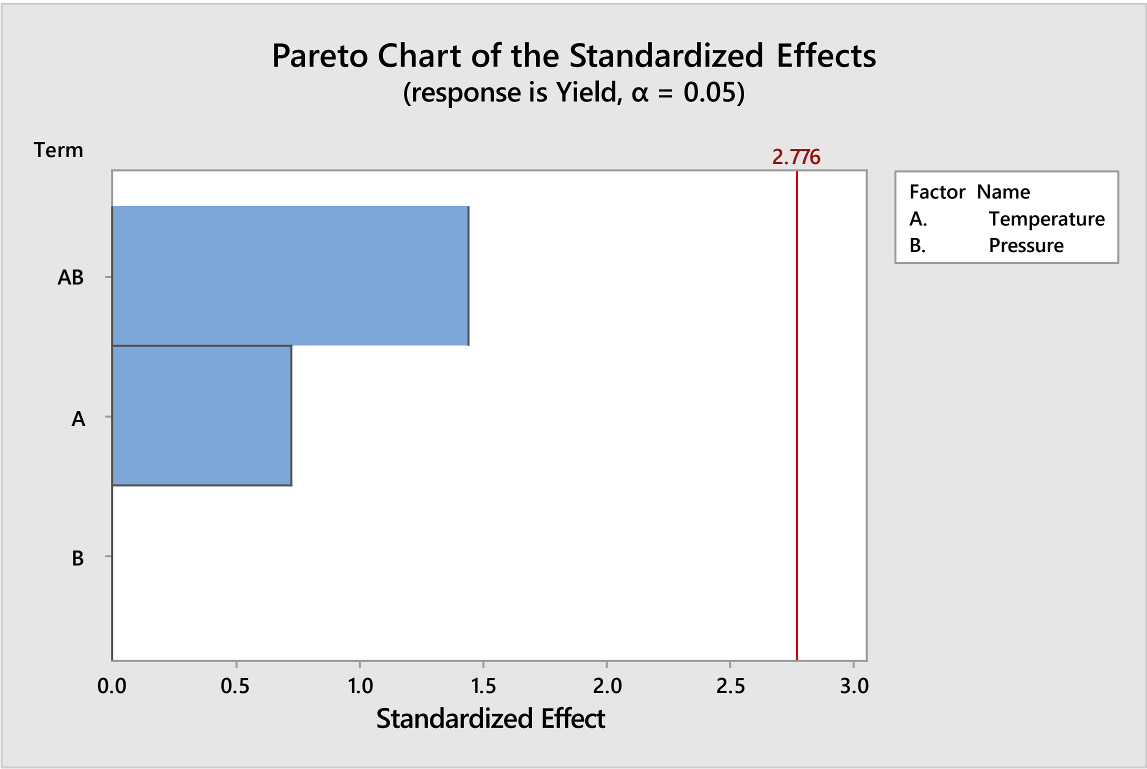
Term	Effect	Coef	SE Coef	T-Value	P-Value	VIF
Constant		92.000	0.935	98.35	0.000	
Temperature	-0.000	-0.000	0.935	-0.00	1.000	1.00
Pressure	-6.000	-3.000	0.935	-3.21	0.033	1.00
Temperature*Pressure	-8.000	-4.000	0.935	-4.28	0.013	1.00

Regression Equation in Uncoded Units

Yield = -205.0 + 1.440 Temperature + 66.0 Pressure - 0.3200 Temperature*Pressure

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^K Factorial - Minitab

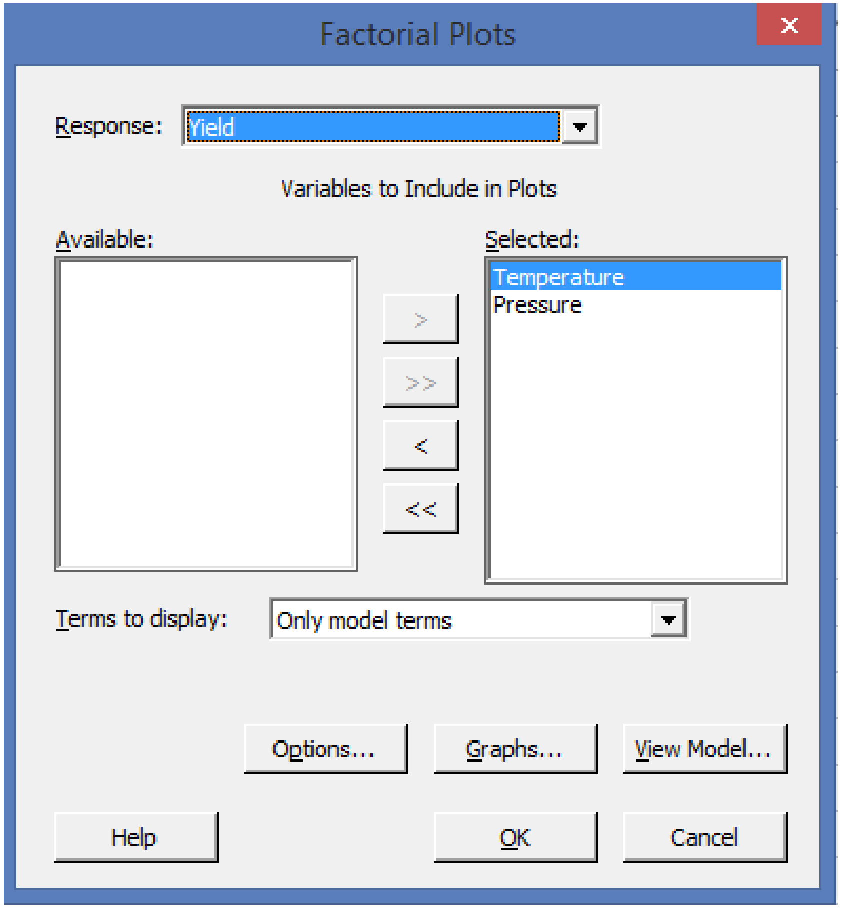


There is no significant impact of Pressure and Temperature on the yield

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Minitab

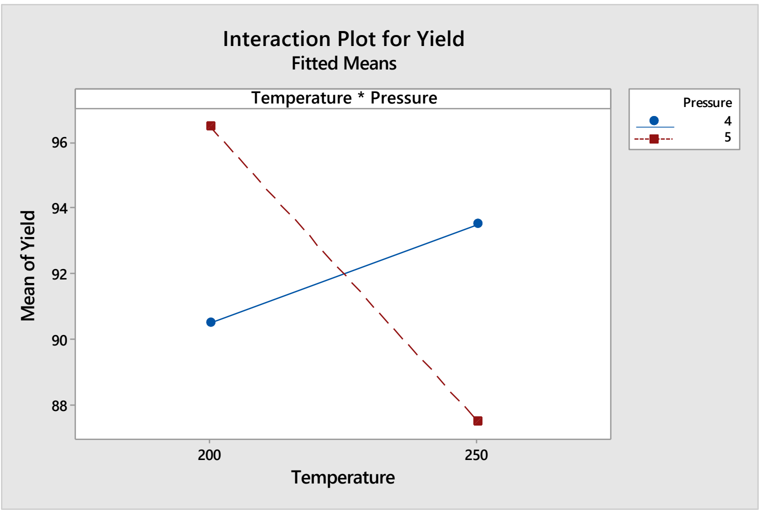
Select Temperature and Pressure

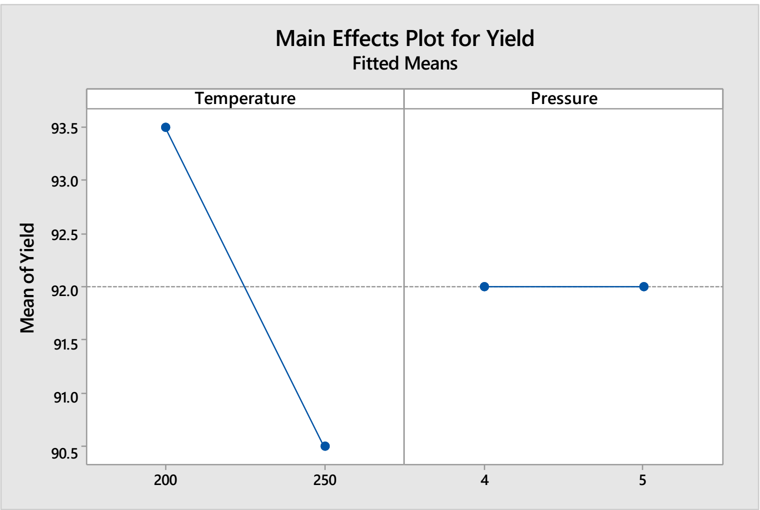


© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2 K Factorial - Minitab

Select Temperature and Pressure





Interaction Plot: The intersection graph displays interaction between Temperature and Pressure on Yield, thereby resulting in an average yield of 92

Main Effects Plot:
Increase in Temperature to 250 degrees C can result in an yield of 93.5
Pressure – High or Low has no impact on Yield

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Example

Refer the worksheet from the sample data – Insulation.MTW

A building materials manufacturer is developing a new insulation product. They would like to study four factors that they believe to impact the strength of the insulation. The four factors are: Material type (Material)

- Injection temperature (InjTemp)
- Injection pressure (InjPress)
- Cooling temperature (CoolTemp)

The manufacturer believes that temperature may also impact strength, but, because temperature cannot be controlled during the experiment, it is recorded as a covariate (MeasTemp).

Factors = 4; Levels = 2

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^K Factorial - Example

Create Factorial Design: Factors

Factor	Name	Type	Low	High
A	Material	Text	Formula1	Formula2
B	InjPress	Numeric	75	150
C	InjTemp	Numeric	85	100
D	CoolTemp	Numeric	25	45

Help OK Cancel

Full Factorial Design

```
Factors:    4   Base Design:        4, 16
Runs:      16   Replicates:             1
Blocks:     1   Center pts (total):     0


All terms are free from aliasing.
```

The output in Minitab gives you a basic approach for conducting the experiment

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Example

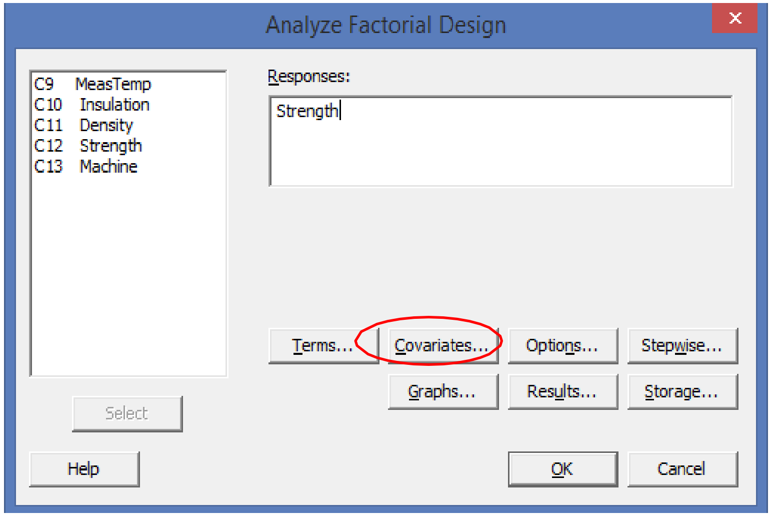
The example already has the response variable mentioned for each input variable as a result of an experiment in Column C12 - Strength

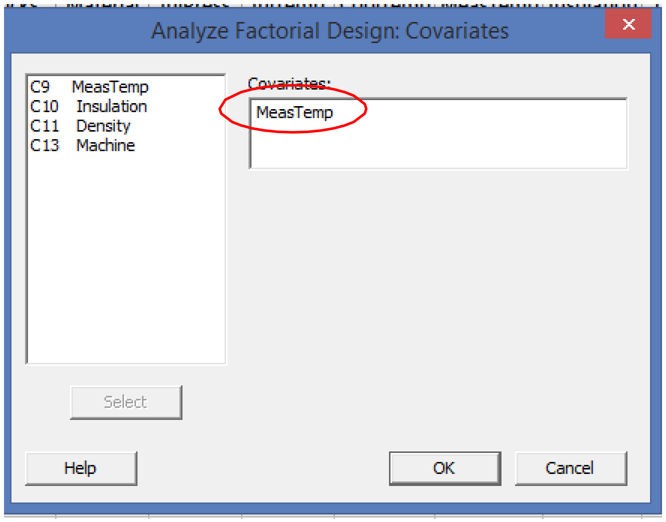
C1	C2	C3	C4	C5-T	C6	C7	C8	C9	C10	C11	C12	C13	C14
StdOrder	RunOrder	CenterPt	Blocks	Material	InjPress	InjTemp	CoolTemp	MeasTemp	Insulation	Density	Strength	Machine	
1	7	1	1	Formula1	75	85	25	21.0	13.2905	1.03359	22.3431	1	
2	6	1	1	Formula2	75	85	25	20.4	19.4389	0.73305	32.8514	1	
3	9	1	1	Formula1	150	85	25	22.9	17.0451	1.60314	30.0842	2	
4	14	1	1	Formula2	150	85	25	21.0	22.6308	1.09832	36.1254	2	
5	15	1	1	Formula1	75	100	25	23.1	14.0952	0.94079	25.9530	3	
6	2	1	1	Formula2	75	100	25	19.9	19.7534	0.68425	34.8078	3	
7	11	1	1	Formula1	150	100	25	21.9	16.7295	1.38776	32.3947	4	
8	5	1	1	Formula2	150	100	25	20.2	23.6961	1.08967	38.1821	4	
9	13	1	1	Formula1	75	85	45	21.2	15.3403	1.15338	19.2189	1	
10	10	1	1	Formula2	75	85	45	21.3	20.3478	0.43510	24.9666	1	
11	12	1	1	Formula1	150	85	45	22.1	19.0723	1.51854	23.9604	2	
12	16	1	1	Formula2	150	85	45	21.8	23.1256	0.98695	30.3459	2	
13	8	1	1	Formula1	75	100	45	21.7	19.7843	0.80379	26.3614	3	
14	3	1	1	Formula2	75	100	45	22.2	24.5312	0.52803	29.3391	3	
15	1	1	1	Formula1	150	100	45	22.6	24.3460	1.46323	29.8344	4	
16	4	1	1	Formula2	150	100	45	20.5	27.7156	1.07611	37.3687	4	

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Example

Let us analyze the output. Stat> DOE> Factorial> Analyze Factorial Design





In the given example we can see that the temperature can have an impact on the output but is difficult to control. Such continuous predictor variables are called Covariates. We include them in the experiment to reduce error variance. Here we have considered “MeasTemp”

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Example

Factorial Regression: Strength versus MeasTemp, Material, InjPress, InjTemp, CoolTemp

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	11	451.357	41.032	17.99	0.007
Covariates	1	3.591	3.591	1.58	0.278
MeasTemp	1	3.591	3.591	1.58	0.278
Linear	4	304.587	76.147	33.39	0.002
Material	1	35.053	35.053	15.37	0.017
InjPress	1	113.068	113.068	49.59	0.002
InjTemp	1	75.533	75.533	33.12	0.005
CoolTemp	1	38.666	38.666	16.96	0.015
2-Way Interactions	6	20.309	3.385	1.48	0.366
Material*InjPress	1	1.732	1.732	0.76	0.433
Material*InjTemp	1	3.045	3.045	1.34	0.312
Material*CoolTemp	1	0.095	0.095	0.04	0.848
InjPress*InjTemp	1	1.538	1.538	0.67	0.458
InjPress*CoolTemp	1	0.012	0.012	0.01	0.947
InjTemp*CoolTemp	1	14.694	14.694	6.44	0.064
Error	4	9.121	2.280		
Total	15	460.478			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
1.51005	98.02%	92.57%	70.86%

•The P-value shows significance for the main effects (0.002) is less than 0.05; Also material, injpress, injtemp and cooltemp has a significant impact on the Strength (Y)

•The p-value of **0.366** for the set of two-way interactions is not less than 0.05. Therefore, the group of two-way interactions is not significant.

•This model explains 98% variation in the data

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2 K Factorial - Example

Look at the effects and coefficients to determine the relative strength of the factors

Coded Coefficients

Term	Effect	Coef	SE Coef	T-Value	P-Value	VIF
Constant		56.0	21.0	2.66	0.056	
MeasTemp		-1.229	0.979	-1.25	0.278	5.87
Material	5.316	2.658	0.678	3.92	0.017	3.23
InjPress	5.645	2.822	0.401	7.04	0.002	1.13
InjTemp	4.355	2.177	0.378	5.76	0.005	1.00
CoolTemp	-3.457	-1.729	0.420	-4.12	0.015	1.24
Material*InjPress	-0.723	-0.361	0.415	-0.87	0.433	1.21
Material*InjTemp	-1.025	-0.512	0.443	-1.16	0.312	1.38
Material*CoolTemp	-0.208	-0.104	0.510	-0.20	0.848	1.82
InjPress*InjTemp	-0.837	-0.419	0.510	-0.82	0.458	1.82
InjPress*CoolTemp	-0.055	-0.027	0.382	-0.07	0.947	1.03
InjTemp*CoolTemp	1.933	0.966	0.381	2.54	0.064	1.02

Regression Equation in Uncoded Units

```
Strength = 52.7 - 1.229 MeasTemp + 10.43 Material + 0.216 InjPress + 0.007 InjTemp
           - 1.357 CoolTemp - 0.0096 Material*InjPress - 0.0683 Material*InjTemp
           - 0.0104 Material*CoolTemp - 0.00149 InjPress*InjTemp - 0.00007 InjPress*CoolTemp
           + 0.01288 InjTemp*CoolTemp
```

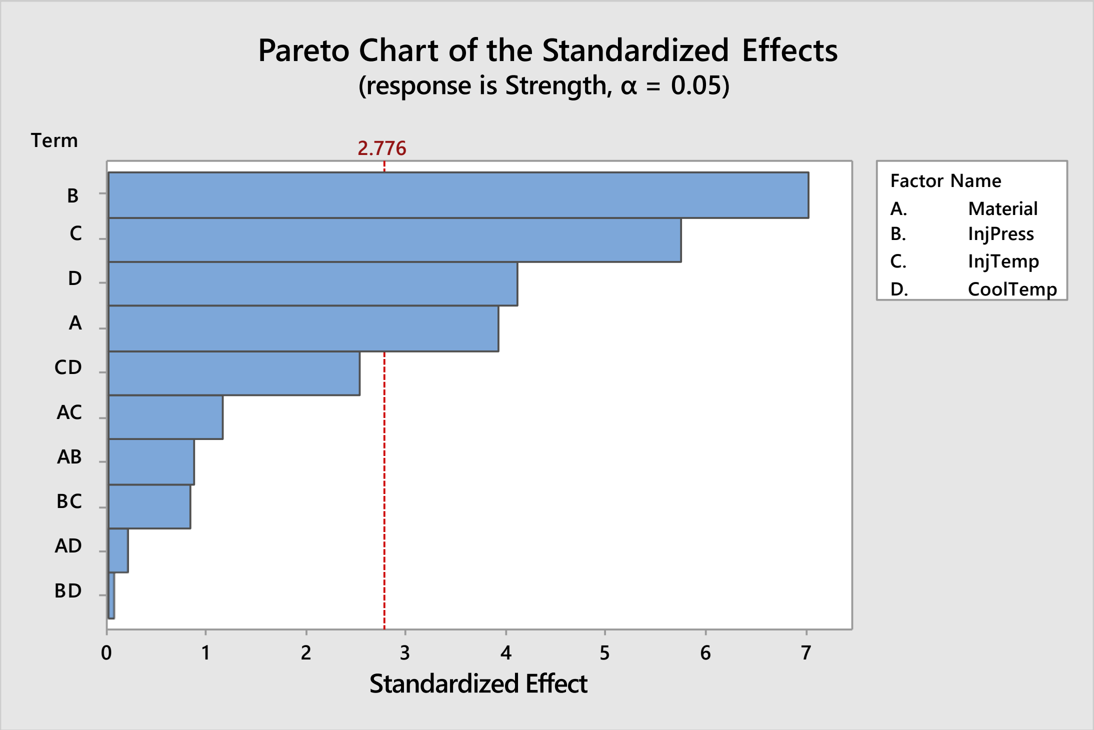
- InjPress has the greatest effect (**5.645**) on insulation strength
- Material has the second greatest effect (**5.316**) on insulation strength
- InjTemp has the third greatest effect (**4.355**) on insulation strength
- CoolTemp has the smallest effect (**-3.457**) on insulation strength

We can also observe VIF for the covariate (MeasTemp) is > 5, this means that the temperature is moderately correlated with the material

We can interpret each slope value for each predictor as the change in strength when the predictor increases by 1.
For example, when injection pressure increases by one unit, strength increases by 0.216.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG netw

2^K Factorial - Example

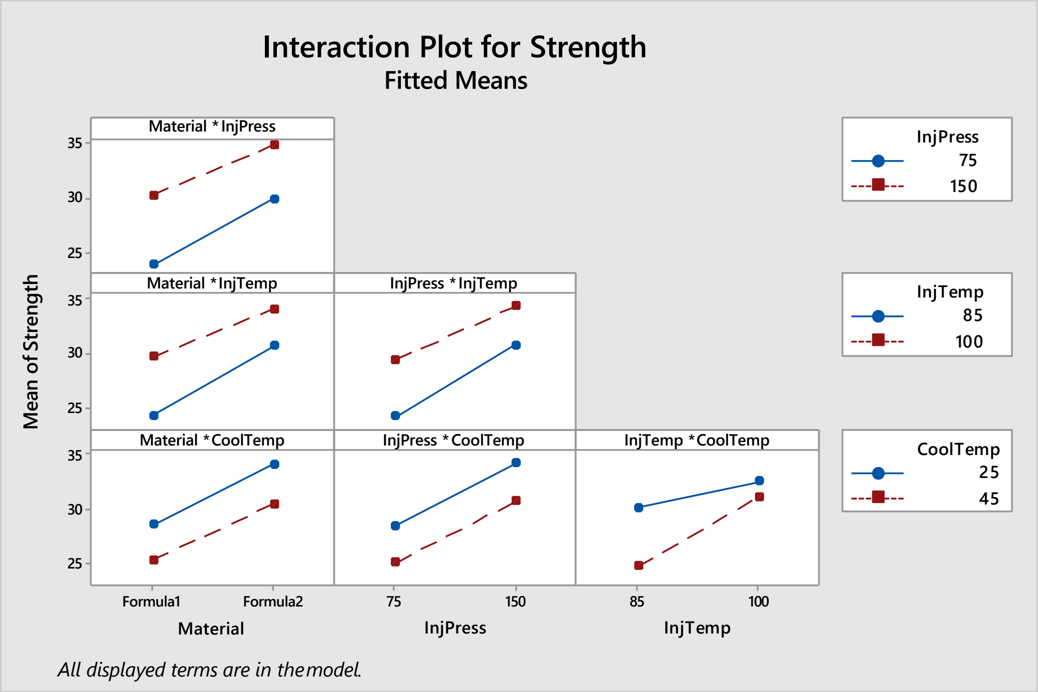


We can see that the largest effect is injection pressure (B) because it extends the farthest.
The effect for the injection pressure by cooling temperature interaction (BD) is the smallest because it extends the least.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

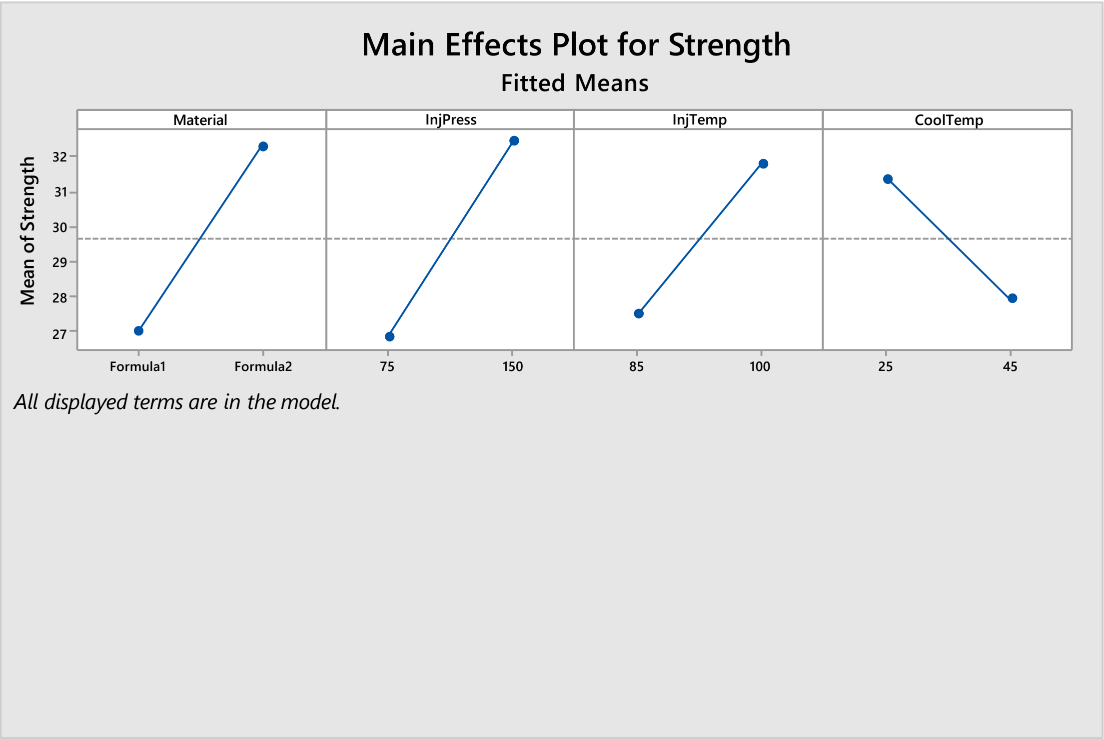
2^k Factorial - Example



The plot indicates that the increase in insulation strength when injection temperature changes from 85 to 100 depends on the cooling temperature. When cooling temperature is low, the change in strength is less than when cooling temperature is high

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Example



- Material: Formula 2 produced stronger insulation than Formula 1.
- InjPress: High injection pressure produced stronger insulation than low injection pressure.
- InjTemp: High injection temperature produced stronger insulation than low injection temperature.
- CoolTemp: Low cooling temperature produced stronger insulation than high cooling temperature.
- The overall mean (about 29.6) is plotted across each panel.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^K Factorial - Example

A plastic manufacturing company had formed a quality circle team consisting of engineers from various department. The objective was to improve yield of a coating process. After a series of brainstorming session, the team zeroed down on following factors and levels:

Response: Yield of coating process (Y)

Factors: Temperature (X1), Catalyst Concentration(X2), and Processing Time (X3)

Replicates: 3

Levels:

X1 – 400 $℉$ and 450 $℉$

X2 – 10% and 20%

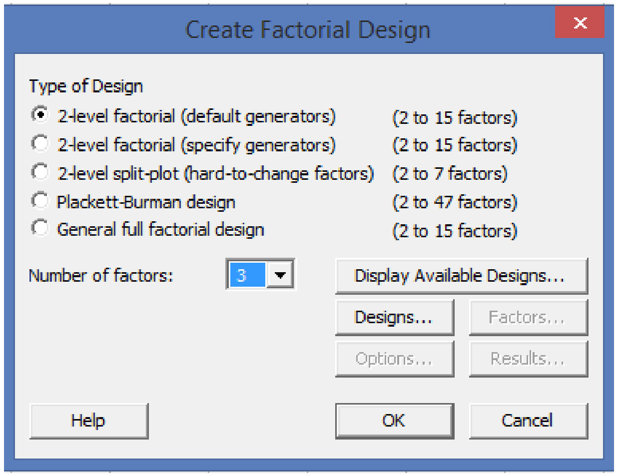
X3 – 45 seconds and 90 seconds

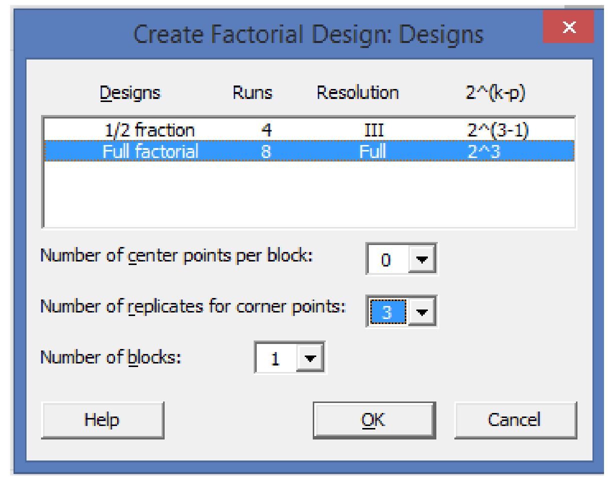
The design is a 3 factorial design, therefore it will be 2^3 = 2*2*2 = 8 replicated 3 times = 8*3 = 24 experiments

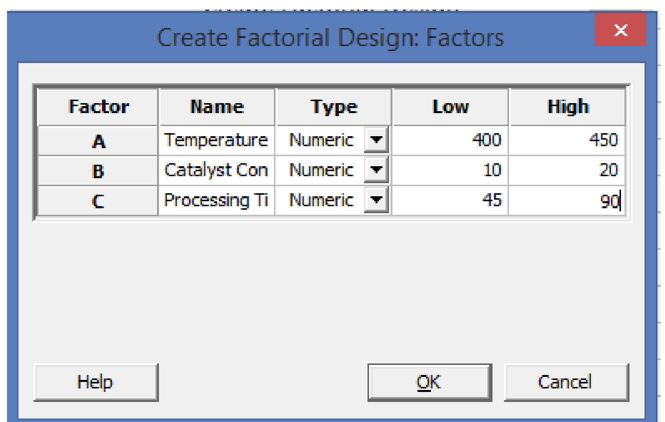
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Example

(Stat> DOE> Factorial> Create Factorial Design)







© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Example

Data with the results in column C8

↓	C1	C2	C3	C4	C5	C6	C7	C8	C9
	StdOrder	RunOrder	CenterPt	Blocks	Temperature	Catalyst Con	Processing Time	Results	
6	6	6	1	1	450	10	90	70.33	
7	1	7	1	1	400	10	45	63.93	
8	24	8	1	1	450	20	90	62.01	
9	9	9	1	1	400	10	45	67.73	
10	12	10	1	1	450	20	45	70.16	
11	15	11	1	1	400	20	90	72.80	
12	23	12	1	1	400	20	90	73.12	
13	5	13	1	1	400	10	90	54.72	
14	4	14	1	1	450	20	45	50.25	
15	8	15	1	1	450	20	90	60.31	
16	18	16	1	1	450	10	45	74.48	
17	10	17	1	1	450	10	45	56.46	
18	3	18	1	1	400	20	45	56.05	
19	11	19	1	1	400	20	45	57.85	
20	21	20	1	1	400	10	90	66.63	
21	20	21	1	1	450	20	45	66.91	
22	22	22	1	1	450	10	90	59.95	
23	19	23	1	1	400	20	45	67.54	
24	16	24	1	1	450	20	90	74.67	
25									

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^K Factorial - Example

Factorial Regression: Results versus Temperature, Catalyst Con, Processing Time

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	7	186.34	26.619	0.41	0.880
Linear	3	24.83	8.277	0.13	0.942
Temperature	1	5.95	5.950	0.09	0.765
Catalyst Con	1	16.52	16.517	0.26	0.619
Processing Time	1	2.36	2.363	0.04	0.850
2-Way Interactions	3	138.79	46.263	0.72	0.555
Temperature*Catalyst Con	1	3.99	3.994	0.06	0.806
Temperature*Processing Time	1	0.23	0.226	0.00	0.953
Catalyst Con*Processing Time	1	134.57	134.569	2.09	0.167
3-Way Interactions	1	22.72	22.718	0.35	0.561
Temperature*Catalyst Con*Processing Time	1	22.72	22.718	0.35	0.561
Error	16	1029.72	64.358		
Total	23	1216.06			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
8.02232	15.32%	0.00%	0.00%

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^k Factorial - Example

Coded Coefficients

Term	Effect	Coef	SE Coef	T-Value	P-Value	VIF
Constant		64.97	1.64	39.68	0.000	
Temperature	-1.00	-0.50	1.64	-0.30	0.765	1.00
Catalyst Con	-1.66	-0.83	1.64	-0.51	0.619	1.00
Processing Time	0.63	0.31	1.64	0.19	0.850	1.00
Temperature*Catalyst Con	0.82	0.41	1.64	0.25	0.806	1.00
Temperature*Processing Time	-0.19	-0.10	1.64	-0.06	0.953	1.00
Catalyst Con*Processing Time	4.74	2.37	1.64	1.45	0.167	1.00
Temperature*Catalyst Con*Processing Time	-1.95	-0.97	1.64	-0.59	0.561	1.00

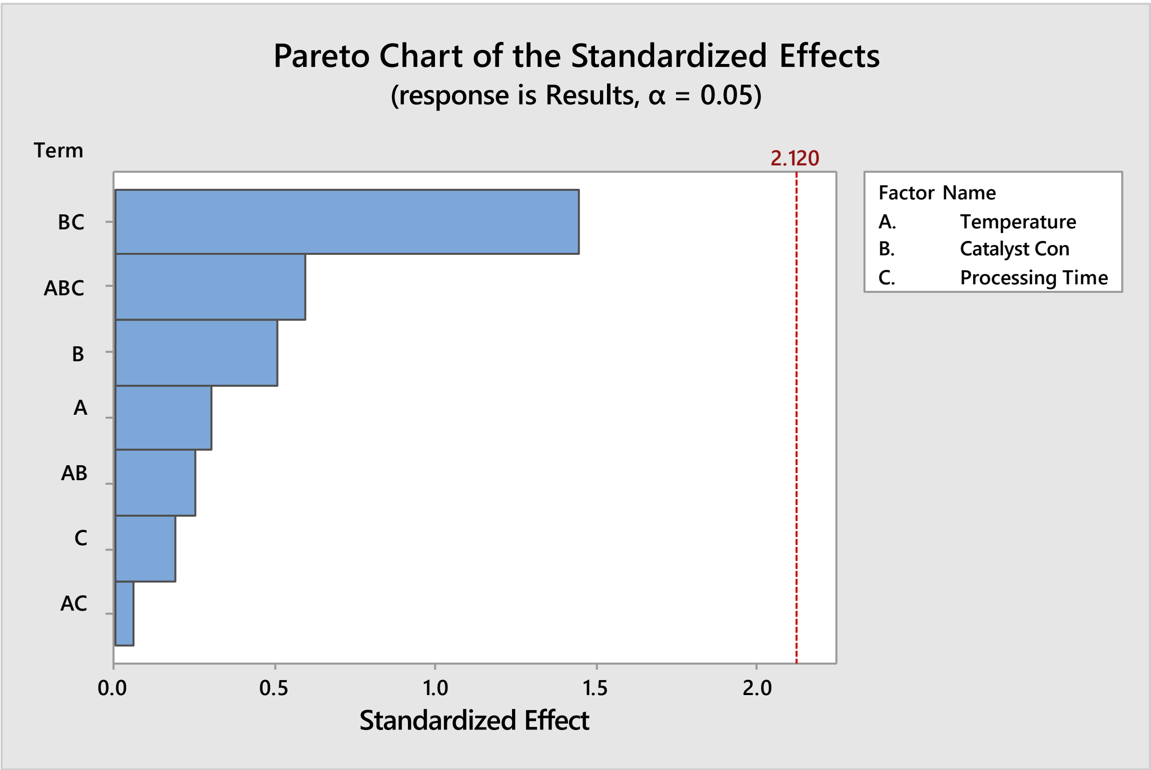
Regression Equation in Uncoded Units

```
Results = 261 - 0.407 Temperature - 12.9 Catalyst Con - 2.43 Processing Time
          + 0.0266 Temperature*Catalyst Con + 0.00502 Temperature*Processing Time
          + 0.168 Catalyst Con*Processing Time
          - 0.000346 Temperature*Catalyst Con*Processing Time
```

What do you conclude?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

2^{K} Factorial - Example



What do you conclude?

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

DOE Benefits

- Can be used to identify “vital few” sources of variation
- Defines the relationship between the inputs and outputs
- Allows you to measure the influence of the “vital few” variables on a response variable
- Is more effective and efficient than testing one factor at a time
- Minimizes the number of test runs you have to make to draw valid conclusions about X and Y linkages

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Generate and Select Solutions

Jugaad

Jugaad

- Jugaad is a colloquial Hindi word that roughly translates as ‘‘an innovative fix;
- It is an improvised solution born from ingenuity and cleverness.’’
- Jugaad also refers to an innovative fix or a simple work-around
- A solution that bends the rules, or a resource that can be used in such a way.
- It is also often used to signify creativity—to make existing things work, or to create new things with meager resources.
- Jugaad is increasingly accepted as a management technique and is recognized all over the world as an acceptable form of frugal (economical) engineering at peak in India.
- Companies in India are adopting Jugaad as a practice to reduce research and development costs.
- Jugaad also applies to any kind of creative and out-of-the-box thinking or life hacking that maximizes resources for a company and its stakeholders.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Jugaad - Principles

The principles of Jugaad anchor the six practices of highly effective innovators in complex settings like emerging economies

- Seek opportunity in diversity
- Do more with less
- Think and act flexibly
- Keep it simple
- Include the margin
- Follow your heart

Collectively, these six principles of Jugaad help drive resilience, frugality, adaptability, simplicity, inclusivity, empathy, and passion, all of which are essential to compete and win in a complex world.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Jugaad principles

Principle 1: Seek opportunity in diversity

Kanak Das, who lives in a remote village in northeast India, grew tired of riding his bicycle on roads full of potholes and bumps. Rather than complain, he
turned this constraint to his advantage by retrofitting his bicycle with a makeshift device that converts the shocks it receives into acceleration energy—allowing his bicycle to run faster on bumpy roads.

Principle 2: Do more with less

The Indian engineers at Siemens—a leading European industrial conglomerate—and their German colleagues co-developed Fetal Heart Monitor, a device that monitors the heartbeat of fetuses in the womb. This device uses simple but ingenious microphone technology instead of costly ultrasound technology. As such, the device delivers more value at less cost.

Principle 3: Think and Act Flexibly

Yuri Malina, flexible-minded entrepreneur who co-founded Design for America (DFA). Yuri's team, along with his DFA advisors, collaborated with a local hospital in Chicago to design a portable ''roll-on'' hand-sanitizer called SwipeSense for busy physicians and nurses so they can clean their hands on the go. SwipeSense is intuitive to use and easy to carry—it doesn't require clinicians to go back and forth to remote hand sanitizing stations—and it is very cheap (costing only $1.50 per unit). This is a frugal, flexible, and yet highly practical innovation since about two million people in the US get infected during hospital visits each year (and nearly a hundred thousand die prematurely from this infection.)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Jugaad principles

Principle 4: Keep it Simple

An open-source software provider called Ushahidi has created an elegantly simple solution—the Ushahidi Platform—that relies on mobile SMS (text messaging) to coordinate bottom-up responses to devastating events like hurricanes, earthquakes, or epidemic outbreaks. After being rolled out successfully in Africa, the Ushahidi Platform is now being widely deployed worldwide—including in the US—as a simple yet highly effective crisis management tool that works faster, better, and cheaper than traditional top-down, hit-or-miss relief management approach.

Principle 5: Include marginal groups

There are various examples of this as below:

NeuSoft (telemedicine in rural China), Zone V (cellphones for the blind), Yes Bank (for underserved Indians), KidZania (theme parks for underserved kids), Zica (salons for low-income Brasilian women), PayNearMe (for underbanked Americans), Renault (Logan, Darcia low-cost cars), Eli Lilly (YourEncore.com community of retirees), Johnson&Johnson (Text4Baby SMS service for mothers)

Principle 6: Follow your heart

The global design and innovation consultancy frog has launched ‘‘centers of passion’’ across its global network of design studios where employees freely discuss their left-field ideas and engage with colleagues, customers, or external partners who share similar passions.

Big Bazaar - design like a bazaar, not Western supermarket.

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Jugaad - Examples

- The Mitticool, an idea born out of adverse circumstances, shows how a resilient mindset can transform scarcity into opportunity. Combining limited resources and a never-say-die attitude, Prajapati tapped into his empathy and passion for his fellow community members to conjure up an ingenious solution that improved lives in Gujarat and beyond. Not only did he produce a cheap and effective cooling device, but he also created jobs for dozens of undereducated women.

- Millions of cellphone users in India rely on ‘‘missed calls’’ to communicate messages to each other using a prearranged protocol between the caller and receiver - think of it as *free textless* text messaging. For example, your carpooling partner may give you a ‘‘missed call’’ in the morning indicating he just left his house and is on his way to pick you up

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Generate and Select Solutions

Design Thinking

Design Thinking

- Design thinking is a human-centered, iterative approach to problem-solving
- Helps to understand in depth the unmet needs and wants of the customers
- Encourages creative consideration of a wide array of innovative solutions

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design Thinking is NOT..

- Only for creative people or product designers
- A narrow equation to aesthetics and crafts
- Just a brainstorming session
- One day process where problems can be solved in 24 hours
- Process to replace analytical problem solving
- A solution for all types of problems

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Traditional Thinking vs. Design Thinking

Traditional Thinking	Design Thinking
Flawless Planning	Enlightened Trial and Error
Avoid Failure	Room for failure
Rigorous Analysis	Rigorous Testing
Presentations	Lightweight Experiments
Research (arm’s length understanding of the customer requirements)	Deep customer immersion
Periodic	Continuous
Thinking	Doing

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Left Brain vs. Right Brain approach

Left Brain	Right Brain
Analytical	Creative
Rational	Holistic
Objective	Subjective
Historical and Present	Present & Future
Facts driven	Feelings
Planned	Spontaneous
Order / Pattern	Spatial

Design Thinking combines both analytical and creative approaches to generate solutions.

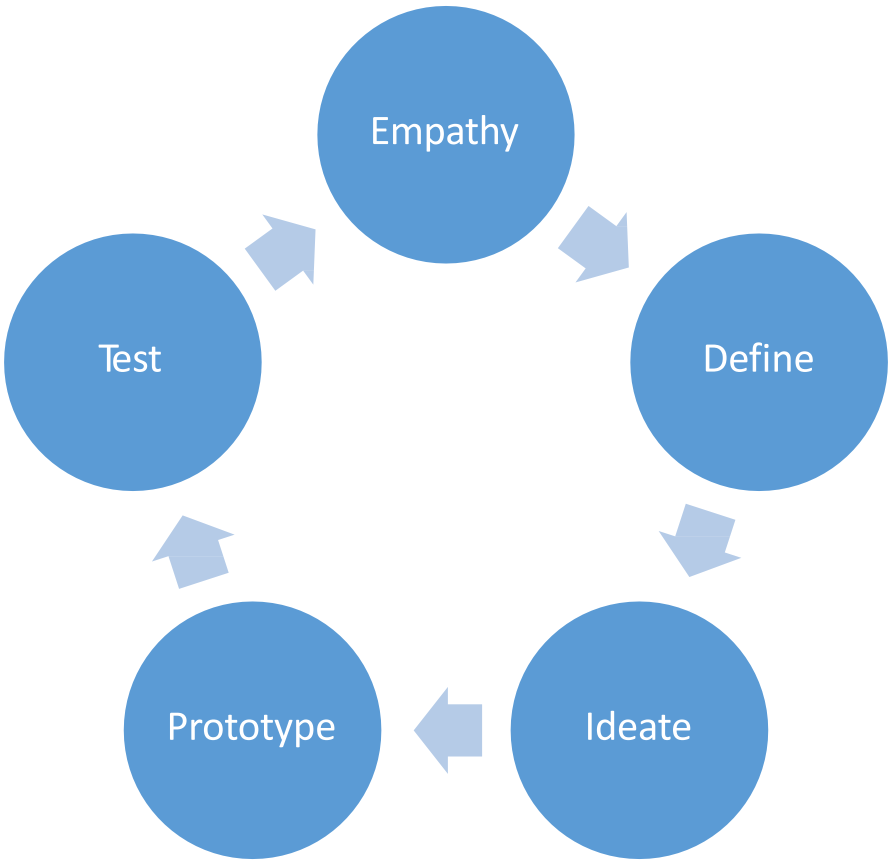
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design Thinking Mindset

- Think users first
- Ask the right questions
- Believe you can draw
- Commit to explore
- Prototype to test

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design Thinking Principles



………but Design Thinking is a Process….

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design Thinking Principles

Five action phases of Design Thinking

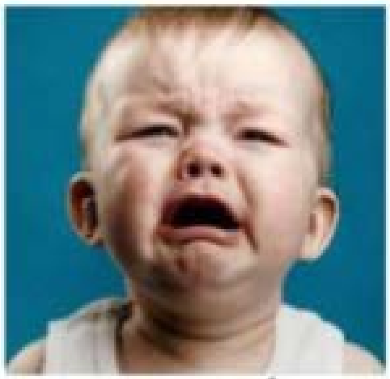
- Empathy – Learn about the customers for whom you are designing
- Define – Create a point of view based on user needs and insights
- Ideate – Brainstorm to explore with as many creative ideas and solutions
- Prototype - Build representation of one or more of ideas for visualization
- Test - Review and decide

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design Thinking Principles

Example: MRI for children

SOLUTION



MRI scans require a person not to move, but little kids cry and move around.

By immersing in the experience of a kid they learned that …

… for a kid an MRI room must be very stressful and a frightening experience.

Kid-friendly MRI. Simple commands to get the scan done accurately become part of an adventure.

Source: GE

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design Thinking Principles

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design Thinking Principles

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Design Thinking Benefits

- Complements other problem solving techniques
- Enables organizations to become more innovative
- Enables to solve human problems
- Increase customer satisfaction
- Gain competitive edge

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Generate and Select Solutions

Generate Solutions

TRIZ - Theory of inventive problem solving

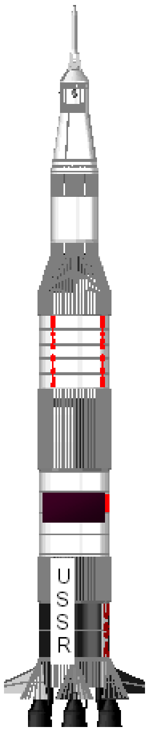
- A theory of problem solving based on the recognition of the power of psychological inertia and the importance of having a process to move problem solving activities beyond this inertia by identifying contradictions (barriers to an ideal solution) and providing specific tools and strategies for eliminating them

- Genrikh Altshuller (developer of TRIZ) screened thousands of patents looking for inventive problems and how they were solved. Only 40,000 had inventive solutions; the rest were straightforward improvements.

- Altshuller categorised these patents by identifying the problem-solving process that had been used repeatedly. He found that the problems had been solved using one of forty inventive principles.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

TRIZ - Key concepts



Overcoming psychological inertia

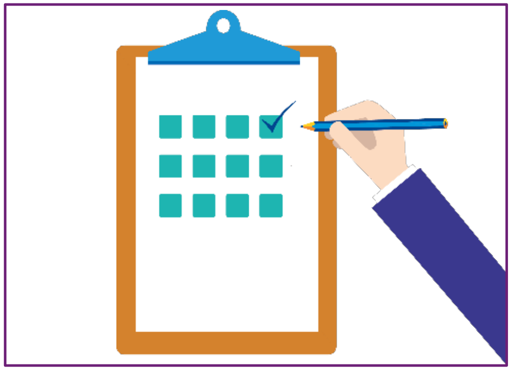
Psychological inertia is the sum total of an individual's personal bias, previous experience and technological knowledge. When challenged with a problem, the solution path will be constrained by the individual's psychological inertia. Psychological inertia predisposes not only the solution path, but the types of solutions as well. Psychological inertia defines the boundaries of the solution space; if the ideal solution for a problem demands 'out-of-the-box' thinking, psychological inertia confines the thought process to the conventional box.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

TRIZ - Key concepts

Inventive principles are rules for system transformations. Each principle is intended to eliminate a single contradiction or several contradictions. Also, these principles can be used simply to stimulate, or initiate, the thought process to create the ideal solution to a problem. They can be used for inspiration in conceptual design, directly utilized in the case- based design or used in various forms of analogical reasoning.

Two of the forty principles are listed in next pagewith accompanying service examples.



Apply selected inventive principles to your contradiction/problem to develop innovative solutions

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Brainstorming Techniques

- **Structured brainstorming:** This is a process in which the ideas are generated in a systematic method by moving from one person to another and obtaining an idea. The person who does not wish to provide an idea, passes. The process is continued until each participant passes and the facilitator determines no fresh ideas are being generated. As with all forms of brainstorming, the ideas are then taken up for discussion. At the end discard any ideas that are similar and debate on each idea to come up with the finalised list of possible solutions

- **Unstructured brainstorming**: In this form of brainstorming, ideas are generated randomly and there is no need to pass. The solutions are discussed freely and can be proposed at any time. This type of brainstorming process is effective when the representation of each of the diverse groups involved in the discussion is large. However, the facilitator should be careful that even in this randomized discussion with large representations, ideas may not be captured completely

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Brainstorming Techniques

Structured brainstorming

- Smooth flow of ideas
- Everyone has an opportunity to express opinions
- Efficient process in terms of idea generation
- Less creative in approach

Unstructured brainstorming

- Smooth
- Random flow of ideas
- All opinions may not be captured
- Lesser number of ideas generated
- Effective process for creative ideas

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

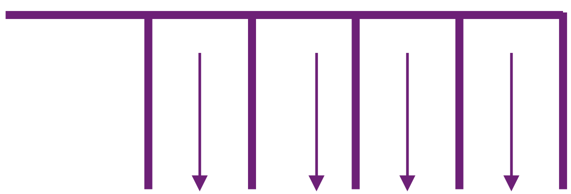
Brainstorming principles

Must be" for all brainstorming forms

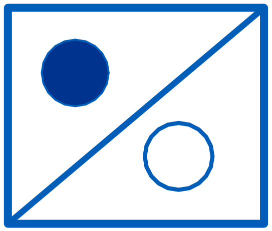
- Allow free flow of ideas
- Prevent biases (group/function/subject)
- Create heterogeneous groups
- Manage time
- Focus on critical Xs and CTQs
- Scribe all details
- Avoid large groups (6 - 10 optimum)
- Use voting techniques

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Prioritizing solutions



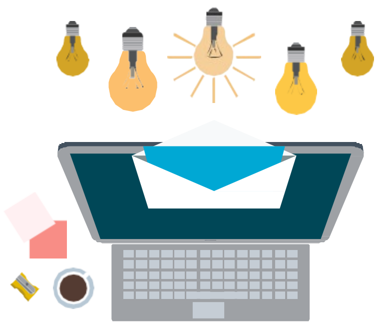
Structured approach to brainstorm on specific category of solutions available to the team



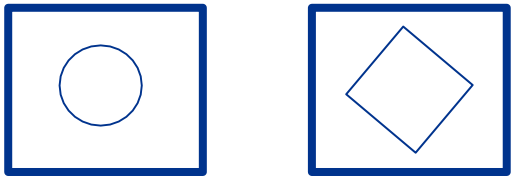
Structured approach to brainstorm on the opposite of the objectives of the discussion

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Prioritizing solutions

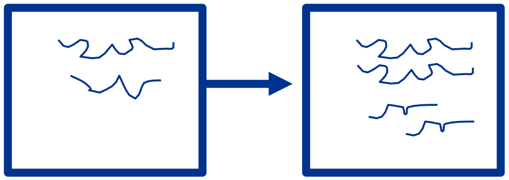


Analogy



Brainstorm on ideas on related topic to unblock the thought process

Brain writing

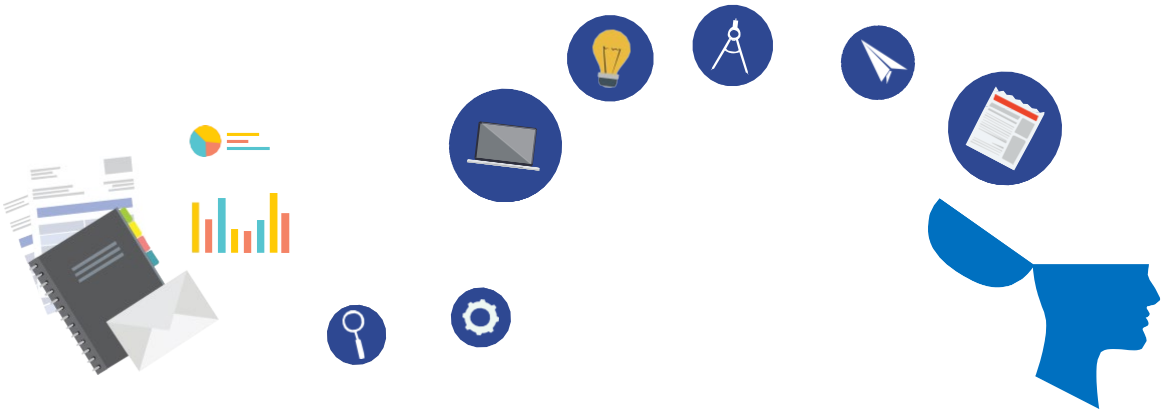


Build on ideas written on a sheet of paper randomly distributed amidst the group

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Prioritizing solutions



Channeling: This form of brainstorming entails channeling of thoughts of the participants similar to the discussion in fishbone diagram. The discussion revolves on specific stream/section of solution. E.g while discussion on methods to improve the sales of soaps, potential solutions can be channeled into: Improvements related to the packaging, fragrance, cutting of costs or distribution system for the product

Antisolution: Here the objective of the team is reversed to exactly the opposite of what is to be determined. E.g. In an effort to improve the training programme, the team can debate about the ways in which the training programme can be worsened. The solutions are opposite of what is determined during the discussion

Analogy: Analogy form of brainstorming encompasses a discussion on a process/product that is similar to the one which is being improved. E.g. Instead of discussing on 'what are ways to improve the productivity of call center associates for an insurance customer service call center, the team can discuss opinions on productivity of associates in the customer service call center for automobile leasing'. Make associations at the end of the discussion

Brain writing: Written ideas are exchanged in this process. Each person who receives a written idea tries to expand on it or adds a totally new idea to it. The pieces of paper are rotated and a collections of ideas or solutions are then debated and prioritised. This is used when the team members are unknown to each other

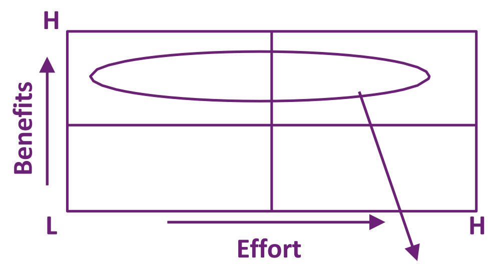
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Generate and Select Solutions

Select Solutions

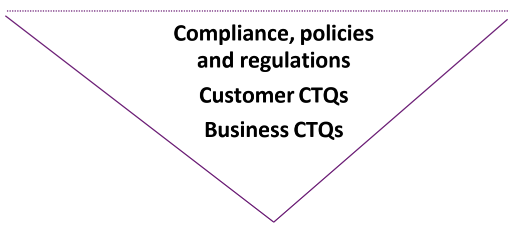
Solution selection

Pay off matrix

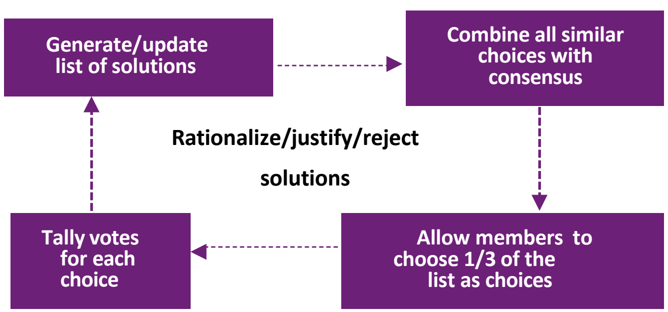


Solutions for further discussions

Screening against 'Must Be'



N/3 voting



Criteria based matrix

Establish Criteria	Assign Wt.	# of Votes	Soln. 1	Soln. 2
• Ease of Use				
• Intersite availability				
• Preparation of Charts				
• Information Real time				
• Auto escalation of unresolved issues				
• Filters and regulated access				
	Total score:			

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Solution selection

Pay-off matrix: After the possible solutions are generated, they can be assigned to different quadrants of the pay off matrix. The ideas which are low on benefits, are rejected. The discussion is then focused on the high effort/high benefits and these are rationalized for implementation. Some of the solutions in this quadrant are so prohibitively expensive that they can be eliminated without further analysis. Caution needs to be maintained when eliminating potential solutions from this quadrant

Screening against 'Must be' criteria: Company policies, local laws and social considerations are extremely important before proceeding further on any solution. Finally, the extent to which the customer CTQs of the project are satisfied is a key consideration for selection (use KANO model to evaluate expected extent of satisfaction generated)

N/3 voting: The key consideration in this form is that all rejected solutions should be justified

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Criteria Based Matrix (CBM)

Where to use CBM

- More than a few criteria for solution selection
- Solutions scoring differently under each criteria
- The weight for each criteria differs

How to use CBM

- Identify criteria and assign weightage
- The team members to give votes to each idea
- The votes are then multiplied with the weights of the criteria established for selection
- The total scores obtained at the bottom of the solution are used to select the same

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

N G T: Nominal Group Technique

What is the best method to reduce cost per transaction?

Solutions	Member 1	Member 2	Member 3	Totals
• Simplify Application process	6	4	6	16
• Automation	1	1	1	3
• Employ Six Sigma	3	3	2	8
• Consolidate sites	5	6	4	15
• Implement Web Services	4	5	3	12
• Reduce Workforce	2	2	5	9

Simplify Application process & consolidateSites

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Solutions with Lean Approach

Kaizen

Kaizen

- “Kai” means “change”
- “Zen” means “good (for the better)”
- Gradual, orderly, and continuous improvement
- Ongoing improvement involving everyone

What is a Kaizen?

- Kaizen is a Japanese word for the philosophy that defines management’s role in continuously encouraging and implementing small improvements involving everyone.
- It is the process of continuous improvement in small increments that make the process more efficient, effective, under control, and adaptable.
- It focuses on simplification by breaking down complex processes into their sub processes and then improving them.

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Why use Kaizen?

When to use Kaizen?

- To solve problems (without already knowing the solution)
- To eliminate waste (Muda)
- Transportation, Inventory, Motion, Waiting, Over-production, Over-processing, Defects
- Create ownership and empowerment
- Support lean thinking

What is a Kaizen Blitz?

- Total focus on a defined process to create radical improvement in a short period of time
- Dramatic improvements in productivity, quality, delivery, lead-time, set-up time, space utilization, work in process, workplace organization

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Kaizen Themes

Kaizen Themes:

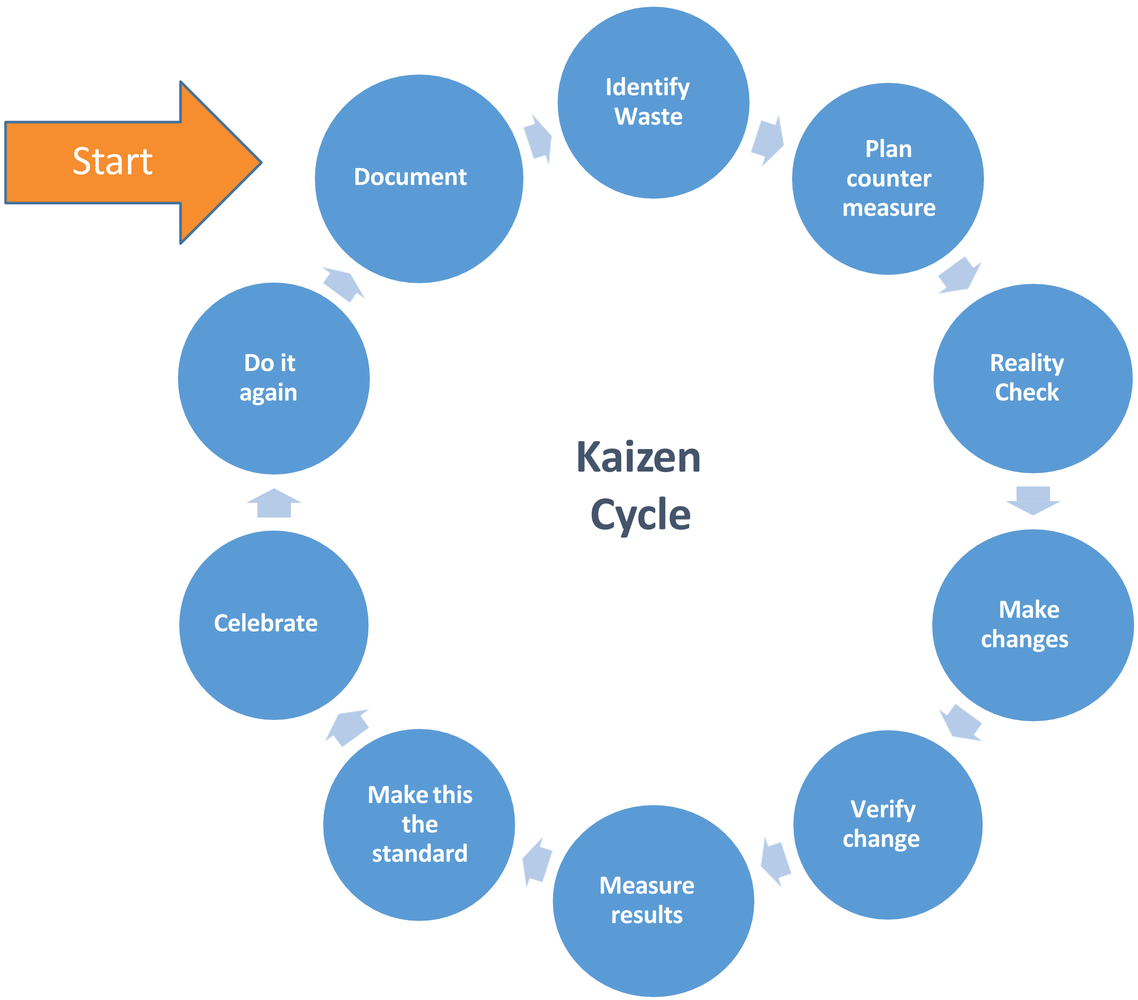
- Red Tagging (getting rid of clutter)
- Visual Control (instructions in the workplace)
- Better (any small improvement)
- Benchmark (adopt other industry service)
- Clarity (communication without confusion)
- Pit Stop (streamlining critical activity)
- Service Supreme (using our best experience as our standard)

Kaizen Advantages:

- Do Right Things (effectiveness)
- Do Things Right (efficiency)
- Do Things Better (improve)
- Do Away With Things (cut)
- Do Things Others Do Well (adapt)
- Do What No Other Is Doing (unbeatable)
- Do What Can’t Be Done (incredible)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Kaizen Cycle



KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Solutions with Lean Approach

Kanban

Kanban

- Kanban literally means “visual card,” “signboard,” or “billboard.”
- Toyota originally used Kanban cards to limit the amount of inventory tied up in “work in progress” on a manufacturing floor
- Not only is excess inventory waste, time spent producing it is time that could be expended elsewhere.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Why Kanban?

- Kanban is a “pull” system that involves cascading or signaling production and delivery instructions from downstream to upstream activities in which nothing is produced by the upstream supplier until the downstream customer signals a need.
- All production is based on consumer demand, with nothing "pushed" downstream.
- Simply said, nothing is produced without a signal from the next station in the line.
- Kanban acts as the means of signaling used for material & information movement
- Kanban is usually in a piece of paper in a vinyl envelope or container; outline marking on the floor

KPMG

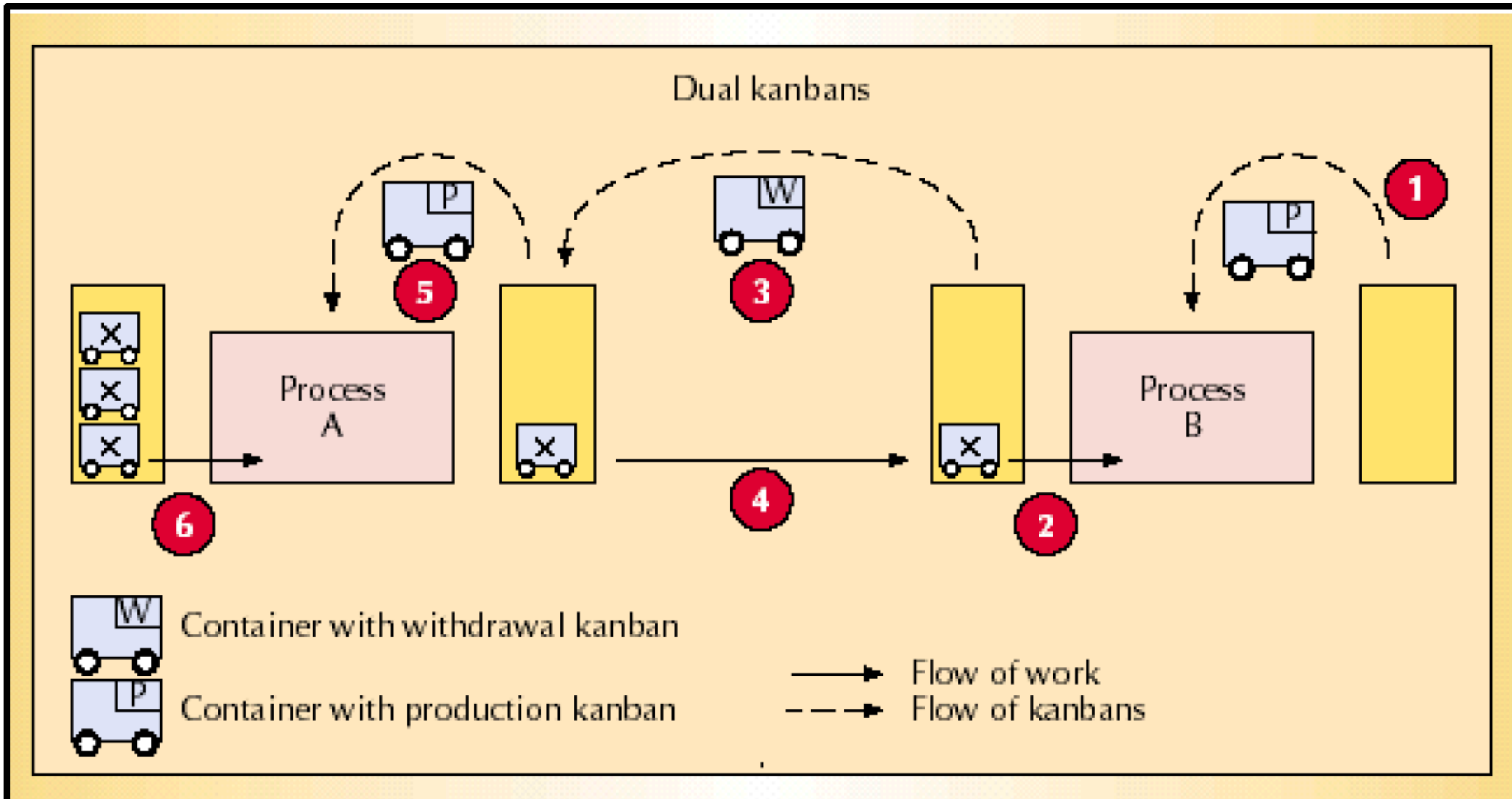
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Types of Kanban

- **Production Kanban:** authorizes production of goods
- **Withdrawal Kanban:** authorizes movement of goods
- **Kanban square:** a marked area designated to hold items
- **Signal Kanban:** a triangular Kanban used to signal production at the previous workstation
- **Material Kanban:** used to order material in advance of a process
- **Supplier Kanban:** rotates between the factory and suppliers

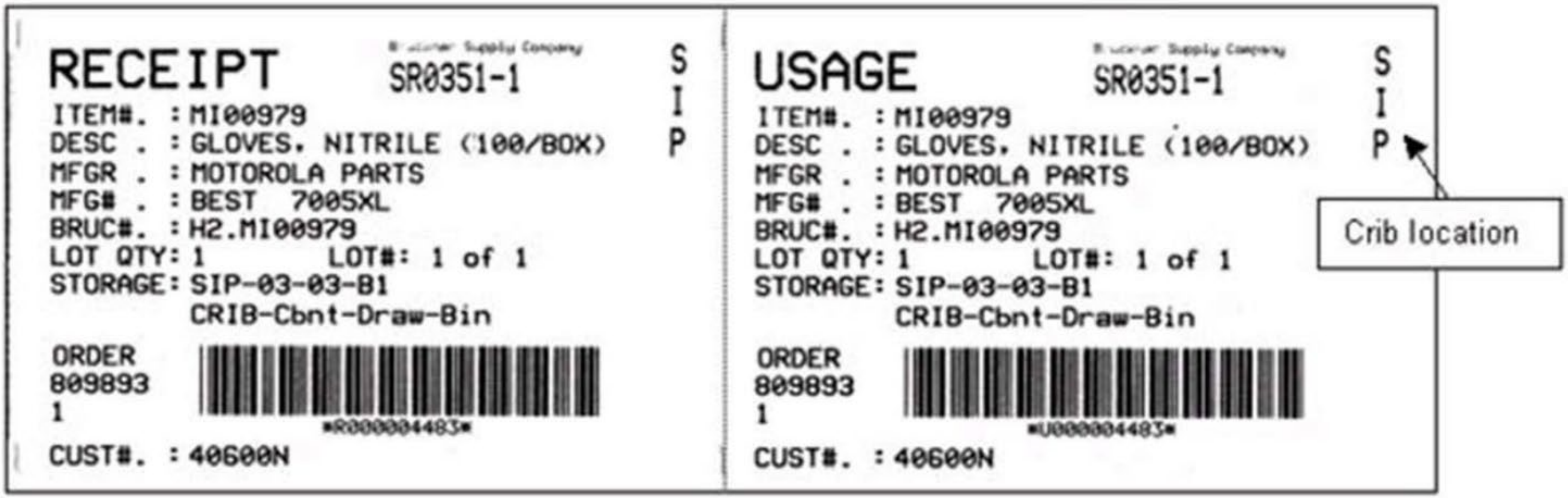
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Diagrammatic Representation of Kanban



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Kanban Card



Source: Slideshare.net

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Process Analysis with Lean

Five S (5's)

5S

What is 5S?

A systematic approach to organize and standardize the workplace. 5S was originally developed within the Toyota Production

Why 5S?

- To improve efficiency and productivity
- To maintain safety and cleanliness
- To maintain good control over the processes
- To maintain the good product quality

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

5S

- **SEIRI** → **Sort / Segregate**
- **SEITON** → **Self Arrange**
- **SEISO** → **Spic and Span**
- **SEIKETSU** → **Standardize**
- **SHITSUKE** → **Self Discipline**

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Objectives of 5S

Objectives of 5S

- Promote Safety
- Improve Work Flow
- Better Product Quality
- Reduce Inventory Waste
- Give People Control of Their Workplace

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Step 1: Sort (Seiri)

- Ensuring each item in a workplace is in its proper place or identified as unnecessary and removed.
- Sort items by frequency of use
- Get rid of unnecessary stuff

Can tasks be simplified?

Do we label items, and dispose of waste frequently?

While Sorting (Seiri) keep in mind:

- How often things are used.
- What is the life of the material.
- Cost of the material
- Be sure to throw the things, otherwise you may repent.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Step 1: Sort (Seiri)

Consequences of not Sorting (Seiri):

- The wanted is hard to find when required
- More space is demanded
- Unwanted items cause misidentification
- Misidentification causes errors in operation
- Maintenance cost of the equipment increases

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Step 2: Set in Order (Seiton)

Identifying places to arrange the things and placing them in proper order for prompt usage.

"A place for every thing and everything in its place."

While arranging or setting things in order (Seiton) keep in mind:

- The right location where the things will be used
- FIFO (First in First out) arrangement
- Labeling of the area and the equipment is very important
- Keep proper gaps between two things to avoid confusion
- Good SEITON includes use of labels signs, indications, display, cautions
- Use of labels signs, indications, display, cautions highlights difference between normality and abnormality.
- Non - users of the equipment also become aware of its use and precautions

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Step 2: Set in Order (Seiton)

Consequences of not arranging things in order (Seiton):

- Things are seldom available when needed
- Items get lost
- Items get mixed up
- Visual control not possible
- Failure to achieve targets

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Step 3: Shine (Seiso)

- Sweep your workplace thoroughly so that there is no dust/dirt/scrap anywhere
- The area should say “ Who I’m” and its neatness should give you a natural welcome

While arranging or setting things spic and span (Seiso) keep in mind:

- Cleaning should be done regularly
- Use the best cleaning agent
- All the nooks and corners should be cleaned
- Keep all the labels intact
- All the labels should correct, visible and legible to all

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Step 3: Shine (Seiso)

Consequences of not being spic and span (Seiso):

- Performance of machines deteriorates
- The quality / aesthetic quality deteriorates
- Dirty place is unpleasant and hazardous to health
- Sends uncaring and irresponsible message to the team members and society at large
- People working at dirty areas are generally found to have low desire to excel and their motivation level is low

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Step 4: Standardize (Seiketsu)

- Always aim at maintaining the standard level of cleanliness, hygiene and visual control
- Keep all the 4 M's (Man., Machine, Material and Method) intact, a lapse in any one of them will make you loose the rest of the three

While standardizing (Seiketsu) keep in mind:

- The standards should be arrived at unanimously
- Always keep the standards flexible to changes and improvements
- Standards should be known to all and displayed

Essence of standardizing (Seiketsu):

- It is the proof that 3-S (SEIRI, SEITON, SEISO) are being religiously carried out
- It is the barometer which indicates the control level based on the 5-S of all the workers

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Step 4: Standardize (Seiketsu)

Consequences of not standardizing (Seiketsu):

- Dual standards yield multiple results
- Multiple results lead to conflicts and confusions
- Rework increases
- Rework increases the basic cost of the finished product without any value addition

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Step 5: Sustain (Shitsuke)

If you are disciplined. :

- Rules will always be followed
- Laid down targets will be achieved
- Improvements will be promoted
- The no. of defects will be reduced
- The cost will not increase

To ensure success:

- Train all team members on 4-S
- Correct wrong practices on the spot
- Punctuality is the backbone of 5S
- Follow work instructions

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Advantages of 5S

Advantages:

- Assemble a 5S Lead team
- Define the work area 5S boundaries (list them)
- Assign work group members to their 5S areas
- Determine 5S targets, activities, and schedule
- Review/finalize plans with work group and site leadership

How to plan for 5S?

- Provides basis for being a world-class competitor
- Starts the foundation for a more systematic approach to the workplace
- Improves productivity and morale
- Empowers employees
- Increases profit

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

5S Implementation

- Obtain existing standards for color-coding and signage
- Decide on 5S color-coding and signage standards
- Communicate, communicate, communicate! (e-mail, signs, one on one)
- Install 5S communication board
- Set acceptable time table for completion

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

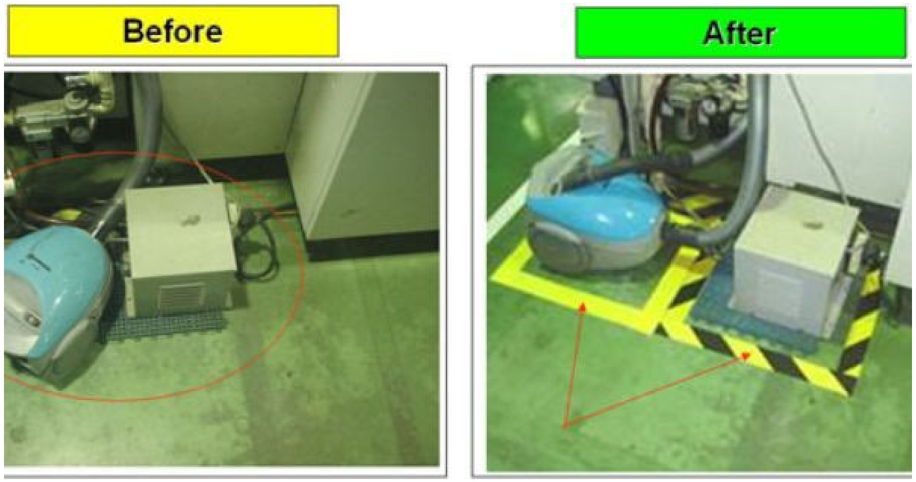
Roles for 5S implementation

Role	Responsibilities
TOP MANAGEMENT	• Play the role of mentor • Initiate the 5S programme • Provide resources • Appreciate the efforts
MIDDLE / LINE MANAGEMENT	• Play the role of facilitator • Take initiative in his area of work • Train the people in 5S • Give the feedback
EMPLOYEES	• Participate actively. • Give suggestions. • Respect the opinion of others. • Be a good team player, and • Maintain discipline

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Examples with 5S



Source: Internet

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Examples with 5S

Source: Internet

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Refine / Risk Proof Solutions

Poka-Yoke

Poka-Yoke

- Poka means "Mistake" or "Error" and Yoke means "Proofing" or "Avoid".

- In other words, Poka-Yoke means *Error Proofing* or *Mistake Proofing* or *Avoidance of Error*.

- Poka-Yoke is a Japanese improvement strategy for mistake-proofing to prevent defects (or nonconformities) from arising during production processes.

- The Poka-Yoke concept was created in the mid-1980s by Shigeo Shingo, a Japanese manufacturing engineer.

- Mistake Proofing is a method for avoiding errors in a process.

- The simplest definition of 'Mistake Proofing' is that is a technique for eliminating errors by making it impossible to make mistakes in the process.

- It is often considered the best approach to process control

- Poka-Yoke device is any mechanism that either prevents a mistake from being made or makes the mistake obvious at a glance.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Why Poka-Yoke?

- Error free designs or processes
- Eliminate the possibility of setting the X's beyond the limits
- Warns operators before the X's move to the outside limit so that the preventive action can be taken
- Can also be used in conjunction with the risk management or SPC
- The opportunity for error needs to be minimized or eliminated

Get it right during process design…!

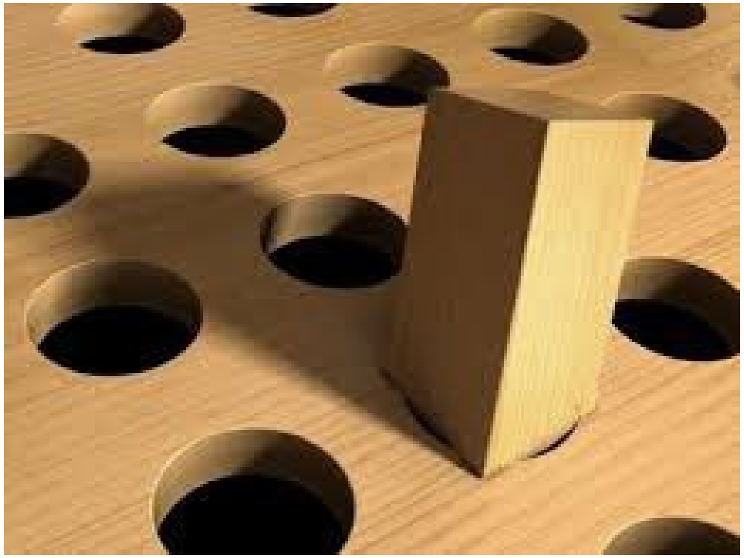
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Poka-Yoke - Examples

Source: Internet

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Poka-Yoke - Examples



Source: Internet

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Refine / Risk Proof Solutions

Failure Mode and Effects Analysis

Failure Mode Effects Analysis (FMEA)

A Failure Mode and Effects Analysis is a systemized team activity intended to:

- recognize and evaluate potential failure and its effects
- identify actions which will reduce or eliminate the chance of failure
- document analysis findings

A team activity with the subject matter experts…!

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

FMEA

Objective of FMEA

- Identify the high priority failure modes and causes of defects in an operational or transactional process.
- Identify high priority input variables (Xs) that impact important output variables (Ys).
- Evolve a consensus on the recommended corrective actions and procedures to follow.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

FMEA

When to use FMEA

- FMEA is designed to prevent failures from occurring or from getting to internal and external customers.
- FMEA is essential for situations where failures might occur and the effects of those failures occurring are potentially serious.
- FMEA can be used on all improvement projects.
- FMEA serves as an overall control document for any given process.

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

FMEA Steps

- FMEA is carried out on a new process/product or redesigned process/product
- For each process step, list requirements for each process step.
- For each requirement, list the failure mode for each requirement.
- For each failure mode, list the effect of failure for each failure mode.
- For each effect of failure, estimate the severity.
- For each failure mode, list causes.
- For each cause of failure, estimate the likelihood of occurrence.
- For each cause of failure, list the current process controls.
- For each process control, estimate the detection.
- For each cause of failure, calculate the Risk Priority Number by multiplying the scores associated with severity, occurrence and detection
- For high priority causes of failure and/or failure modes, develop recommended actions.
- For each recommended action, assign responsibility and completion dates.
- For each recommended action, implement the action and note its effect.
- For each implemented action, re-estimate the severity, occurrence and detection rankings and recalculate the RPN

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

FMEA Worksheet

Process Step/Part Number	Potential Failure Mode	Potential Failure Effects	S E V	Potential Causes	O C C	Current Controls	D E T	R P N	Actions Recommended	Resp.	Actions Taken	S E V	O C C	D E T	R P N
								0							
								0							
								0							
								0							

RPN = Severity X Occurrence X Detection

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

F M E A - Example

Process Step	Potential Failure Mode	Potential Failure Effect	SEV	Potential Causes	OCC	Current Controls	DET	Risk Priority Number (RPN)
Cash Dispense Process at ATM	ATM does not dispense cash	• Customer Dissatisfaction • Incorrect entry in customer account • Discrepancy in customer's cash balance	8	ATM out of cash	5	Cash Alert	5	200
				Machine jammed	3	Monthly audits	10	240
				Power Failure	2	None	10	160
	ATM dispenses too much cash	• Bank loses money • Discrepancy in the cash balance	7	Denomination in wrong tray	3	Visual Verification	7	147
				Cash Stuck	2	Loading Procedure	4	56
	ATM takes too long to dispense cash	• Customer dissatisfaction • Customer annoyed	4	Network traffic	7	None	10	280
				Power Failure	2	None	10	80

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Severity Scale

RPN = Severity X Occurrence X Detection

High

Low

More the severity,

higher is the rating

Severity scale	
Rating	**Criteria – A failure could:**
10	Injure a customer or employee
9	Be illegal
8	Render product or service unfit for use
7	Cause extreme customer dissatisfaction
6	Result in partial malfunction
5	Cause a loss of performance which is likely to result in a complaint
4	Cause minor performance loss
3	Cause a minor nuisance, but be overcome with no performance loss
2	Be unnoticed and have only minor effect on performance
1	Be unnoticed and not affect the performance

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Occurrence Scale

RPN = Severity X Occurrence X Detection

High

Low

More often it occurs, higher is the rating

Occurrence scale	
Rating	**Criteria – A failure could:**
10	More than once per day
9	Once every 3-4 days
8	Once per week
7	Once per month
6	Once every 3 months
5	Once every 6 months
4	Once per year
3	Once every 1-3 years
2	Once every 3-6 years
1	Once every 6-100 years

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Detection Scale

RPN = Severity X Occurrence X Detection

Low

High

Lower the ability to detect, higher is the rating

Detection Scale	
Rating	**Definition**
10	Defect caused by failure is not detectable
9	Occasional units are checked for defect
8	Units are systematically sampled and inspected
7	All units are manually inspected
6	Units are manually inspected with mistake-proofing modifications
5	Process is monitored (SPC) and manually inspected
4	SPC is used with an immediate reaction to out-of-control conditions
3	SPC as above with 100% inspection surrounding out - of-control conditions
2	All units are automatically inspected
1	Defect is obvious and can be kept from affecting the customer

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Risk Priority Number (RPN)

- The RPN number is calculated from the team's estimates of Severity, Occurrence and Detection.

- RPN = S x O x D

- If you are using a 1 - 10 scale for Severity, Occurrence and Detection, the worst RPN = 1000 (10 x 10 x 10), while the best would be RPN = 1 (1 x 1 x 1).

- Use RPN numbers to prioritize failure modes and/or causes of failures in order to work on the highest priority issues.

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

F M E A - Useful Tips

Suggestions for completion of the activity in time:

- Similar to a process map, FMEA is also a “live” document used throughout the DMAIC journey
- Make it a “team effort.”
- Analyze new processes to avoid problems before they happen.
- Address concerns from a process perspective and not business contingency perspective
- Analyze existing processes to find and fix problems.
- Analyze existing processes to discover the high priority (“key”) process input variables.

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Test Solutions

Testing for solutions

Why test solutions

- Confirm to potential solution
- Obtain feedback and buy-in for proposal
- Opportunity for refining solution

How to test solutions

- Small scale experimentation-piloting
- Modeling-physical models
- Simulation- computer models

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Testing Essentials

Create A data collection plan and Monitor Activities For data Consistency and stability

Verify that the process has improved-process capability, hypothesis testing for evaluating statistical significance of the old and the improved process

Why collect data? → **What to collect?** → **How to collect?** → **Ensure consistency & stability**

Help ensure robust measurement plan for testing the solution

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Justify Solutions

Justify solution

Tangible benefits

- Savings
- Incremental revenue
- Interest cost and income
- Cash flow
- Reduced workforce

Intangible benefits

- Customer retention
- Improved VOB/VOC
- Ease of work
- Better governance

Prepare net financial gains and supplement with intangible gains and obtain business and finance leader approvals before proceeding

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Cost Benefit Analysis (C B A)

Hard Gains		
Cost Heads	**Vendor**	**In House Training**
Trainer Cost	250,000	0
T and L	30,000	0
Q Team Support	4,444	11,111
Facilities	20,000	20,000
Equipment	1,000	1,000
Stationary and Manuals	15,000	10,000
GB Examination	50,000	2,222
Total Cost	370,444	44,333
Estimated annual cost		
	Cost with Vendor	Cost with Quality Team
Per Training	370,444	44,333
12 Training per year	4,445,333	532,000
Cost of creating the training manual		
Man Days Spent	Average per day cost	Total Cost incurred
30	2,222	66,667
Savings per year		
3,846,667		
Assumptions		
1. Hypothetical Costs assumed		
2. Recurring cost of upgrading the content is INR66,667		
* All Figures in INR		

Soft gains

Trainings customised to the need of the employees

Calendar and duration of the training according to organizations need

Benefits of training in-house Vs. Employing a vendor

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

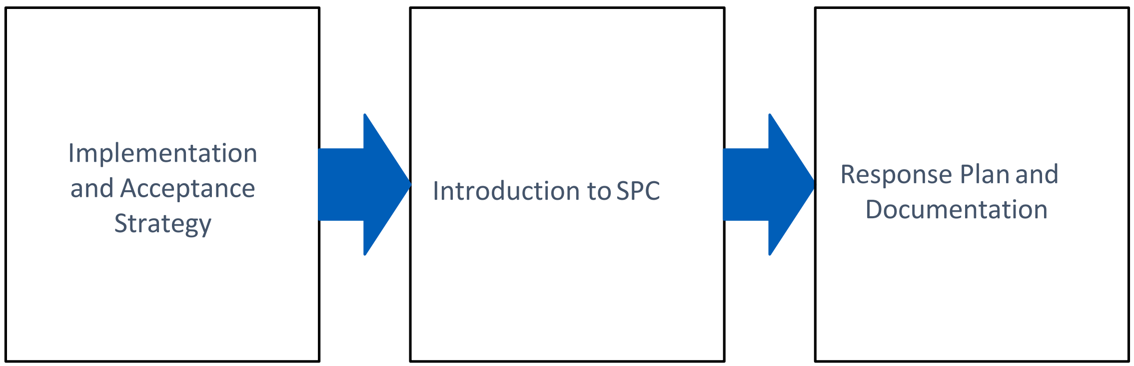
Inference from Improve Phase

- Shortlist the appropriate solution
- Risk proof the solution using FMEA
- Validate the improvement by testing the solutions
- Justify the benefits of the project to the management

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

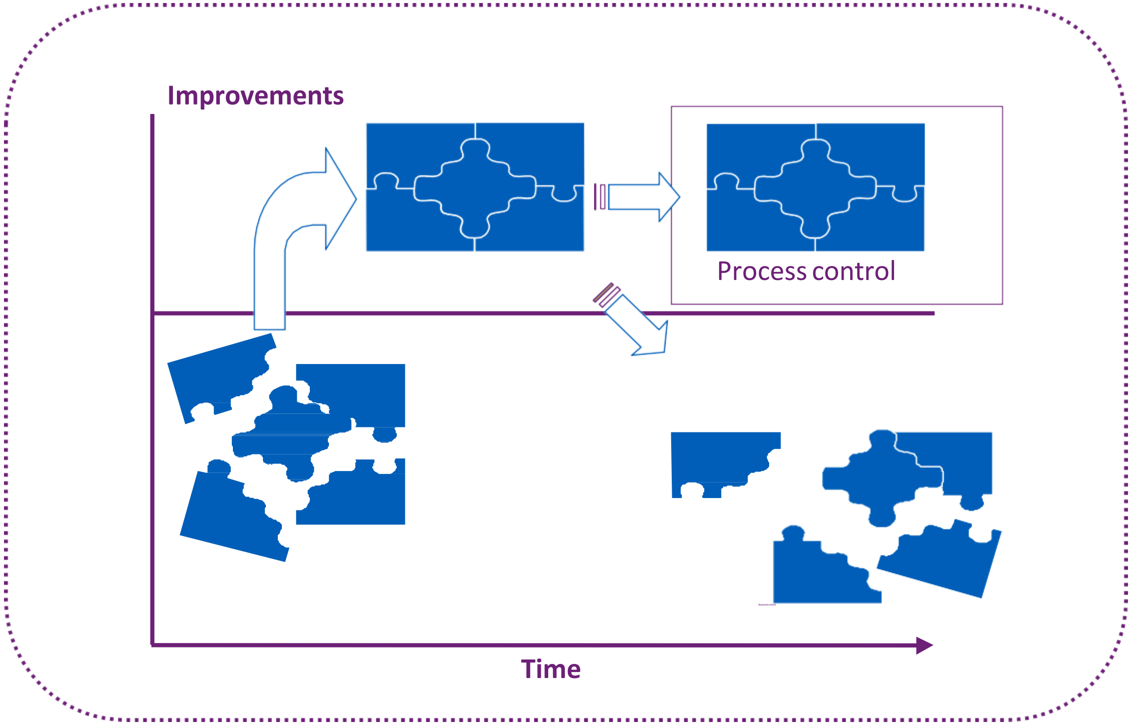
Control Phase

Control Phase - Roadmap



© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Why control?



To hold the gains of the DMAIC project there by help ensuring that the efforts of the project team are not written off

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Why control?

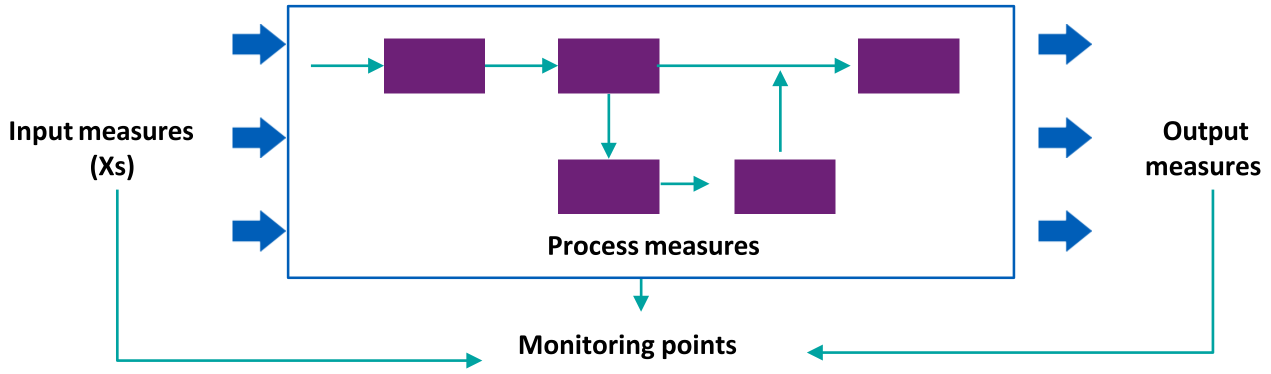
Process control is essential to prevent the loss of gains over a period of time

Detect the out of control state of processes and determine the appropriate actions

Control system incorporates risk management, SPC, measurement and audit plans, response plans, documentation and ownership

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

What to control?



- At the barest minimum, measure CTQs
- Strive to measure the critical Xs to provide chances to make corrections
- Install the appropriate data collection plan
- Audit appropriately
- Early warning system prevent $$ impact (rework reduction)

KPMG
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Implementation and Acceptance Strategy

Implementation

Elements of good implementation planning:

- Objectives
- Incorporation of pilot learning
- Resources and time frame
- Influence strategy
- Control plan
- Documentation

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Implementation

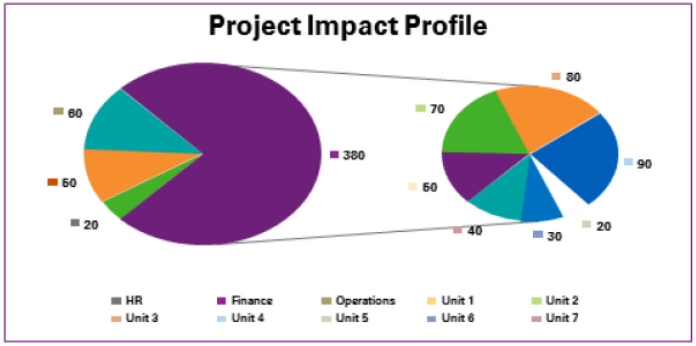
Elements of good implementation planning:

- Objectives
- Incorporation of pilot learning
- Resources and time frame
- Influence strategy
- Control plan
- Documentation

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

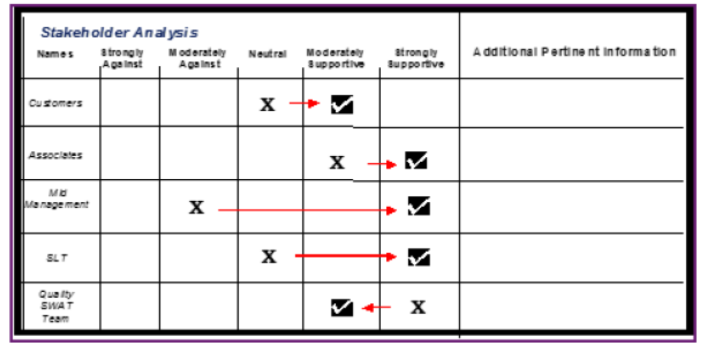
Acceptance Strategy

Key constituents map



Identify stake holders to be addressed first

Stakeholders analysis



Stakeholder Analysis

Names	Strongly Against	Moderately Against	Neutral	Moderately Supportive	Strongly Supportive	Additional Pertinent Information
Customers			X →	☑		
Associates				X →	☑	
Mid Management		X →			☑	
SLT			X →		☑	
Quality SWAT Team				☑	← X	

Identify stakeholder support profile

Influence strategy

	Issue/ Concern	Identify Wins	Strategy
Stakeholder 1			Data
Stakeholder 2			Demonstrate
Stakeholder 3			Demand

Determine influence strategy

TPC analysis

	Cause of resistance	Examples	Ranking
Technical			
Political			
Cultural			

Identify causes for stakeholder resistance

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Introduction to SPC

Statistical Process Control (SPC): An Overview

Control charts

- Statistical Process Control or SPC was conceptualized by Walter A Shewhart to determine if a business process (back then manufacturing process) is in a state of control.
- SPC primarily uses Control Charts
- Control charts are also known as Shewhart charts, or process behavior charts

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Statistical Process Control (SPC): An overview

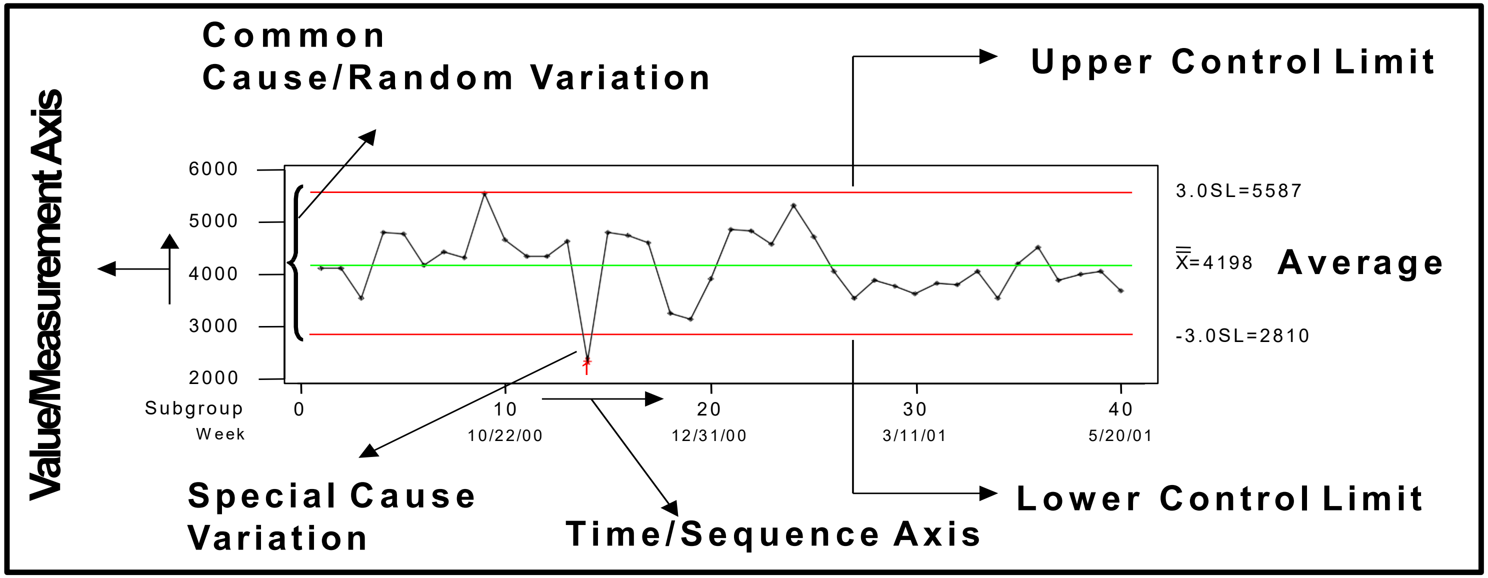
Fundamentals of SPC

- All processes display variation
- There are two types of variation i.e. Common Cause Variation and Special Cause Variation
- Common Cause Variation is inherent, steady, common, undiscoverable
- Special Cause Variation is intermittent, special, discoverable, removable
- The Control Charts are set at $\pm$ 3σ from the mean

KPMG

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Control Limits



- Select process, measures to be charted
- Establish data collection and sampling plan
- Calculate the statistics
- Plot control charts

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Special Cause vs. Common Cause Variation

True variation type…		Common causes	Special causes
	Common causes	Focus on systematic process change	**Mistake 1** **Tampering** (increases variation)
	Special causes	**Mistake 2** **Under-reacting** (missed prevention)	Investigate special causes for possible quick-fixes
		Low	**High**

In Our Control

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative (“KPMG International”), a Swiss entity. All rights reserved.

Statistical Process Control (SPC): Why and When Control Charts?

Control charts

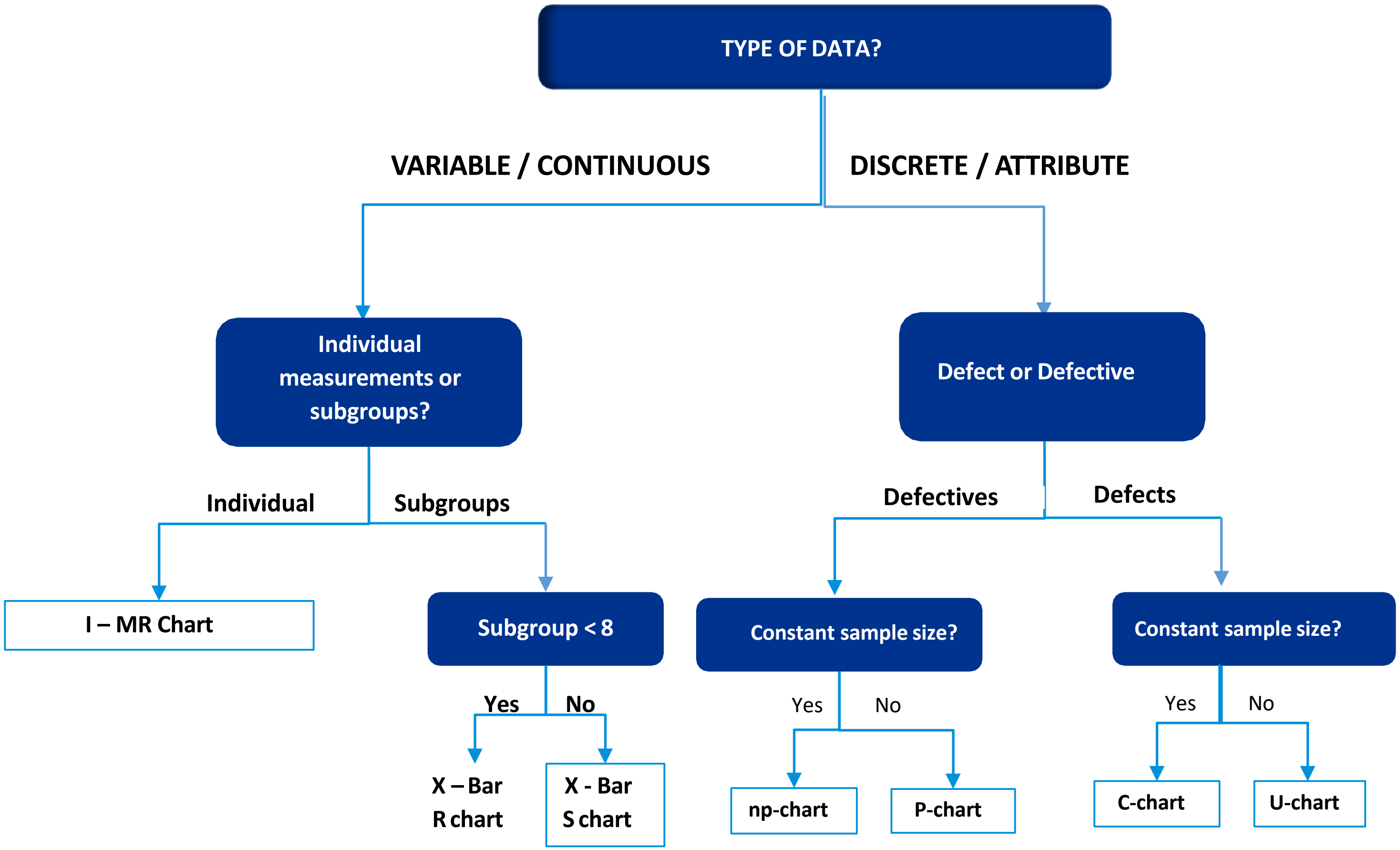
- Control Charts monitor changes in the Xs and detect changes that are due to special causes
- Used when the Xs cannot be mistake proofed or need to be controlled for inherent variations

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Introduction to SPC

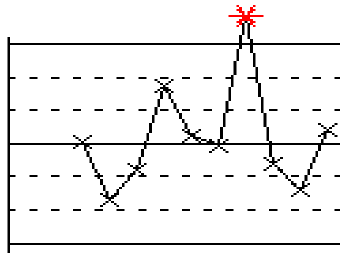
Types of Control Charts

Selecting Control Charts



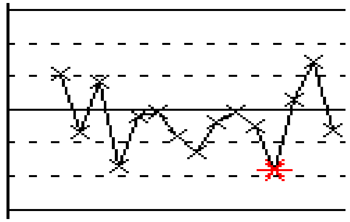
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Identifying Special Cause Variations



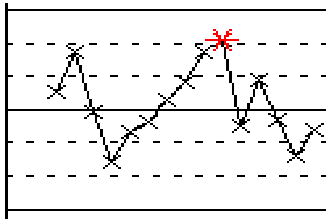
Test 1. Points beyond Control Limits:

Points beyond control limits are isolated high or low points. 1 point more than 3σ from center line



Test 2. Points on one side of the center line

9 points in a row on same side of center line

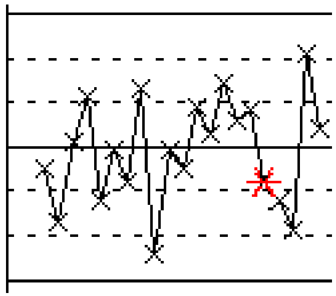


Test 3. Trend

Trends will continue up or down without a well defined end. Six points in a row, all increasing or all decreasing

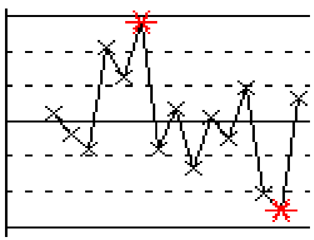
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Identifying Special Cause Variations



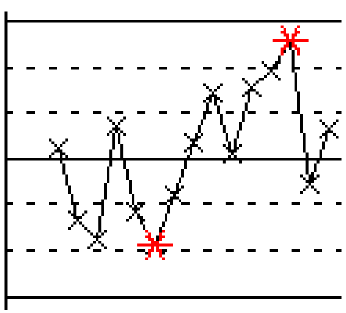
Test 4. Variation pattern

A cycle produces a pattern of up and down points, very much as if the values of the points were time dependent. Fourteen points in a row, alternating up and down



Test 5. Points on same side

Two out of three points more than 2s from the center line (same side)



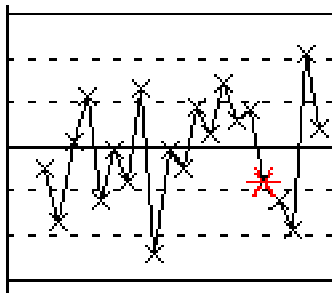
Test 6. Points on same side

Four out of five points more than 1s from the center line (same side)

KPMG

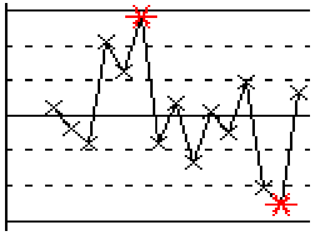
© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Identifying Special Cause Variations



Test 7. Identifies a pattern of variation

Fifteen points in a row within 1s of center line (either side)



Test 8. Mixture Pattern

Eight points in a row more than 1s from center line (either side)

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

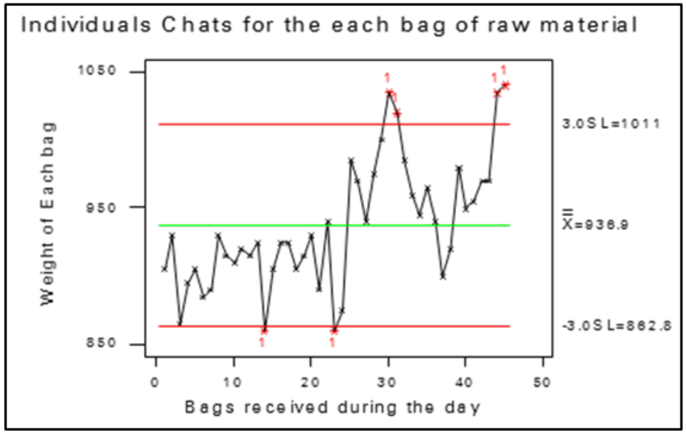
Control Limits - Continuous Data

	LCL	UCL
Xbar chart	$\bar{\bar{X}} - 3\frac{\bar{R}}{d_2}$	$\bar{\bar{X}} + 3\frac{\bar{R}}{d_2}$
R chart	$D_3\bar{R}$	$D_4\bar{R}$
I chart	$\bar{X} - 3\frac{\bar{R}}{d_2}$	$\bar{X} + 3\frac{\bar{R}}{d_2}$
MR chart	0	$D_4 M\bar{R}$

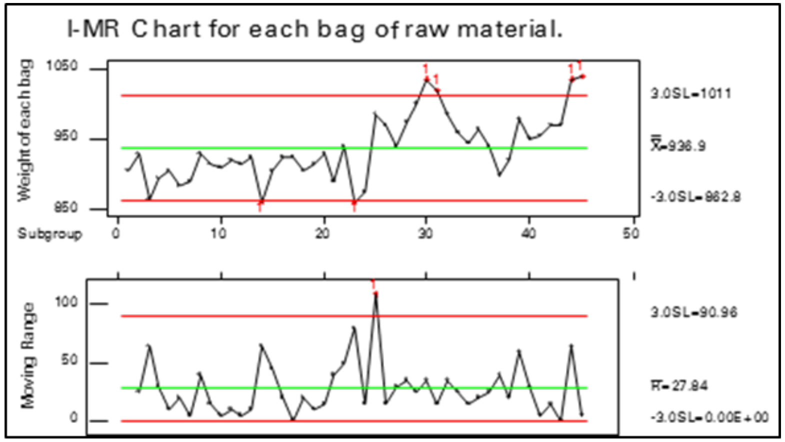
n (Sample Size)	d_2	D_3	D_4
2	1.128	–	3.268
3	1.693	–	2.574
4	2.059	–	2.282
5	2.326	–	2.114
6	2.534	–	2.004
7	2.704	0.076	1.924
8	2.847	0.136	1.864
9	2.970	0.184	1.816
10	3.078	0.223	1.777
11	3.173	0.256	1.744
12	3.258	0.283	1.717
13	3.336	0.307	1.693
14	3.407	0.328	1.672
15	3.472	0.347	1.653

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

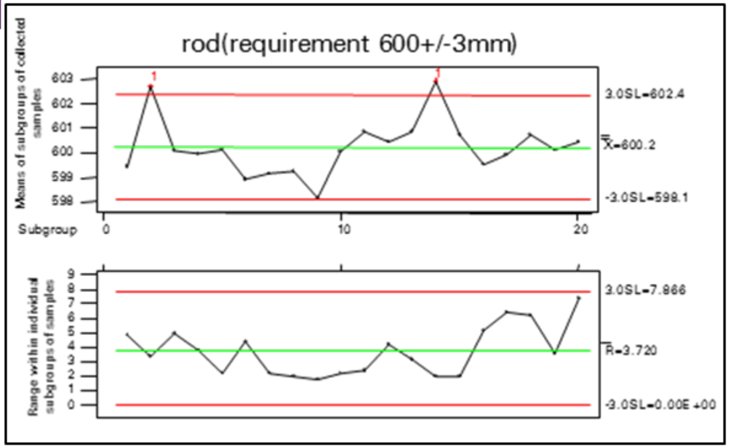
Control Charts - Variable / Continuous Data



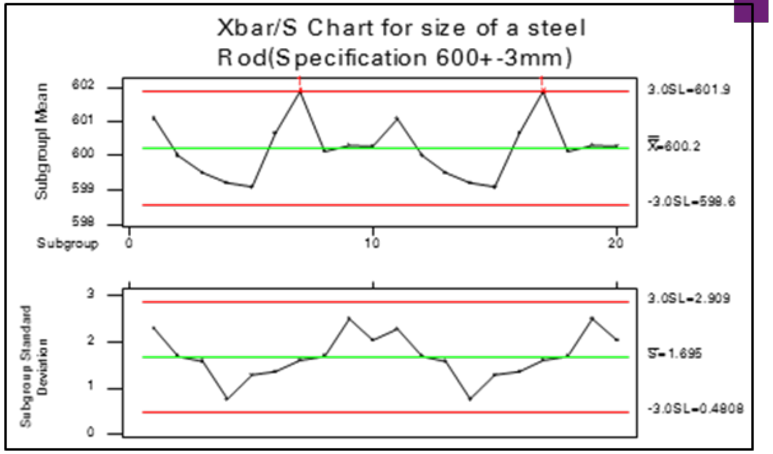
Individual-Chart



I MR -Chart



X bar R-Chart



X bar S-Chart

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Control Limits - Attribute Data

Defective Data: p Chart

$$UCLp = \overline{p} + 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}}$$

$$LCLp = \overline{p} - 3\sqrt{\frac{\overline{p}(1-\overline{p})}{n}}$$

Defective Data: np Chart

$$UCLnp = n\overline{p} + 3\sqrt{n\overline{p}(1-\overline{p})}$$

$$LCLnp = n\overline{p} - 3\sqrt{n\overline{p}(1-\overline{p})} \quad \text{where } \overline{p} = \frac{\sum np}{\sum n}$$

Defect Data: U Chart

$$UCLu = \overline{u} + 3\sqrt{\frac{\overline{u}}{n}}$$

$$LCLu = \overline{u} - 3\sqrt{\frac{\overline{u}}{n}}$$

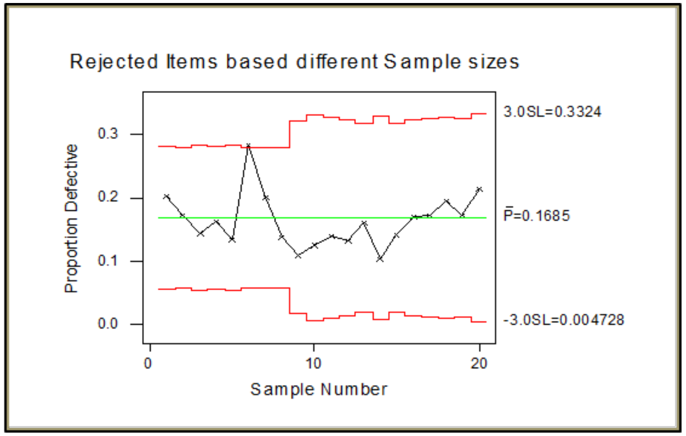
Defect Data: C Chart

$$UCLc = \overline{c} + 3\sqrt{\overline{c}}$$

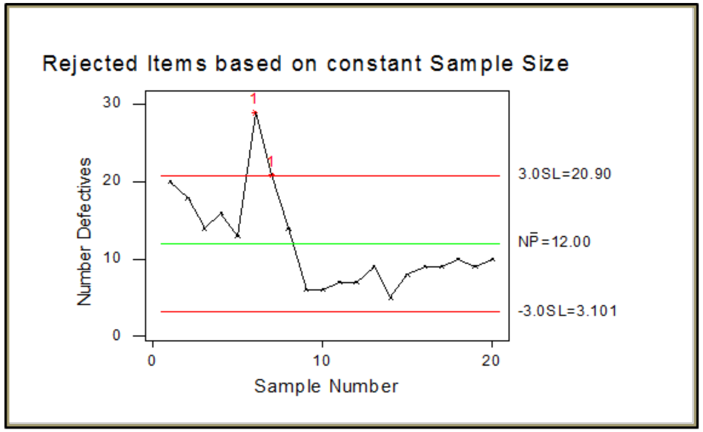
$$LCLc = \overline{c} - 3\sqrt{\overline{c}}$$

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

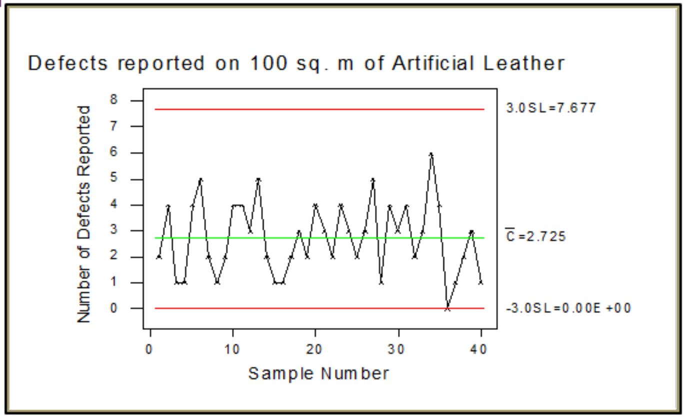
Control Charts - Attribute / Discrete Data



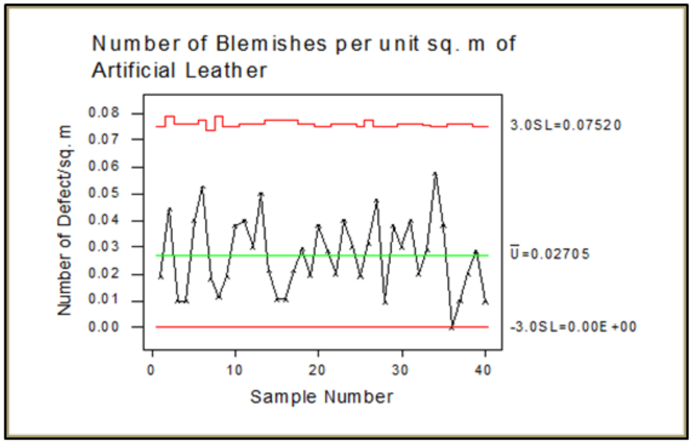
p-Chart



np-Chart



c-Chart



u-Chart

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Benefits of Control Charts

- Control Charts reduces defects by keeping processes centered
- Improves overall quality by reducing chances of quality deviations
- Aids in timely troubleshooting
- Serves as a communication tool for changes in CTQs
- Sustain improvements

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Response Plan and Documentation

Project Closure

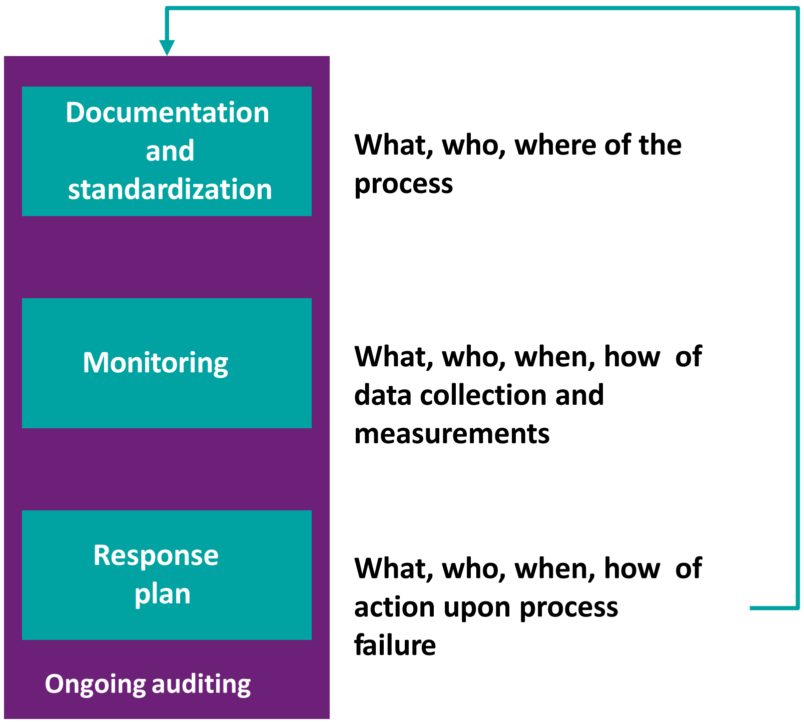
Essentials of project closure:

- Document and communicate results
- Recommend translation opportunities
- Evaluate team success-results, process and relationships
- Celebrate
- Disband project team

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Process Management System

New special cause variations identified



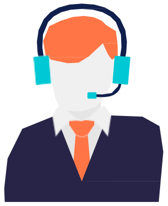
- Deployment flowchart
- Activity details

- Key measures
- Standards
- Measurement and process

- Procedure for adjustment
- Containment
- Procedure for improvement

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Process Management System



Documentation		Monitoring			Response plan		
The plan for doing the work		Checking the work			The response to special causes		
Deployment flowchart	Details on key tasks	Key process and output measures	Monitoring standards	Method for recording data	Containment	Procedure for process adjustment	Procedure for process improvement

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Documentation

Documentation process is an activity that help ensure that the knowledge gained by the project team is:

- Implemented successfully
- Retained by the business
- Used to standardize the business processes
- Future resources can be trained easily
- Improvements can be shared and translated across
- Institutionalized

Documentation Levels

Process Management System	Flowcharts	Procedures	Checklist

Increasing level of detail →

Core Process — Sub Process — Micro Process

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Inference from Control Phase

- Develop and implement control plans.
- Identify the appropriate KPIVs to be measured.
- Identify the appropriate control chart.
- Transfer knowledge and responsibility for process control to appropriate position(s).

© 2024 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.

Thank You

Vijay Gogoi - KPMG
vijaygogoi@kpmg.com
www.in.kpmg.com

For Feedbacks mail us at in-fmsixsigmafdbk@kpmg.com

© 2020 KPMG, an Indian Partnership and a member firm of the KPMG network of independent member firms affiliated with KPMG International Cooperative ("KPMG International"), a Swiss entity. All rights reserved.